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Silicon to Software™

CTS

Synopsys University Education Program

WEEK 9 Introduction to CTS
WEEK 10 CTS & PSYN
WEEK 11 Homework

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WEEK 9 : Introduction to CTS

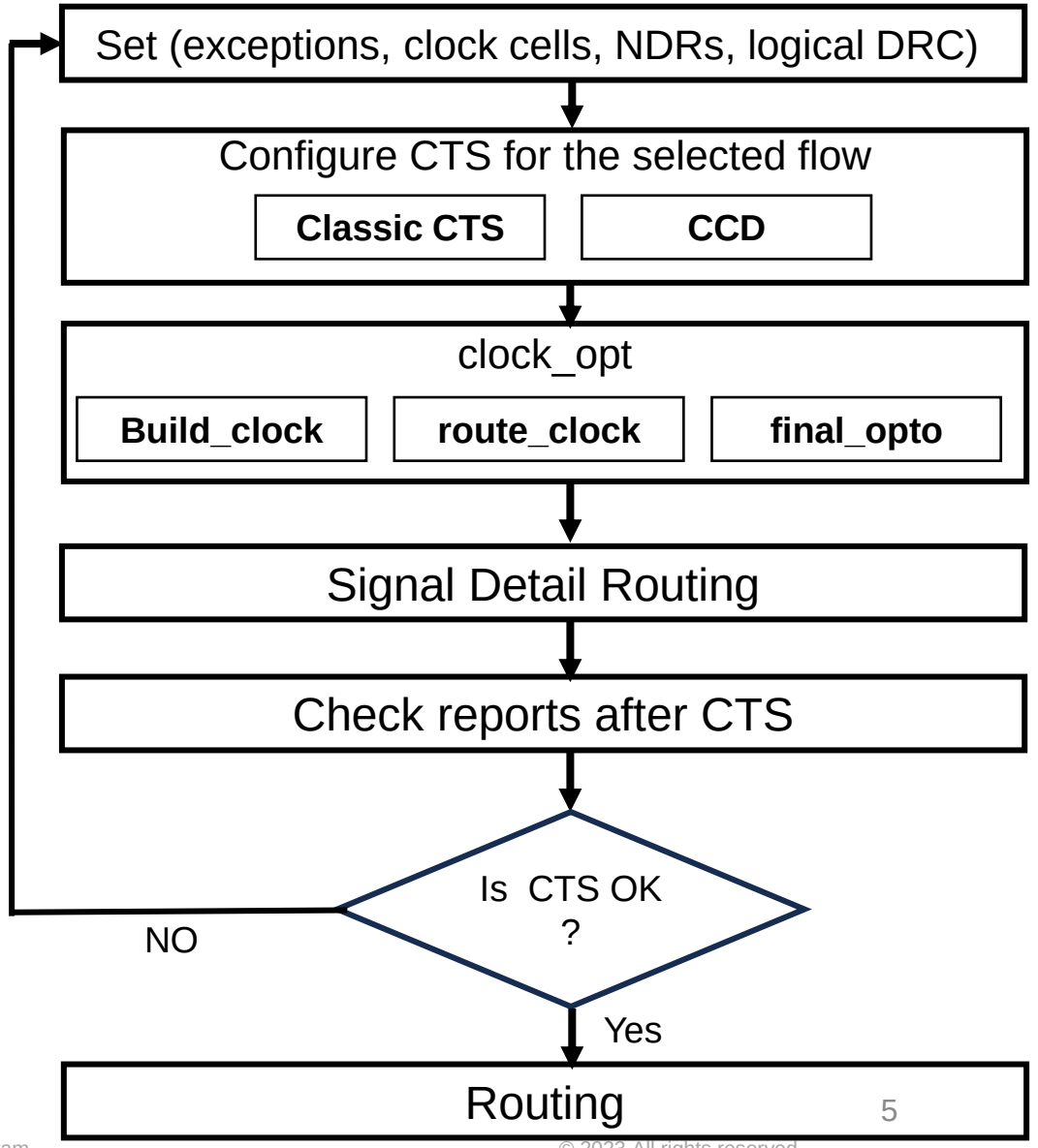
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WEEK 9-1 : Overview to CTS

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Introduction to CTS

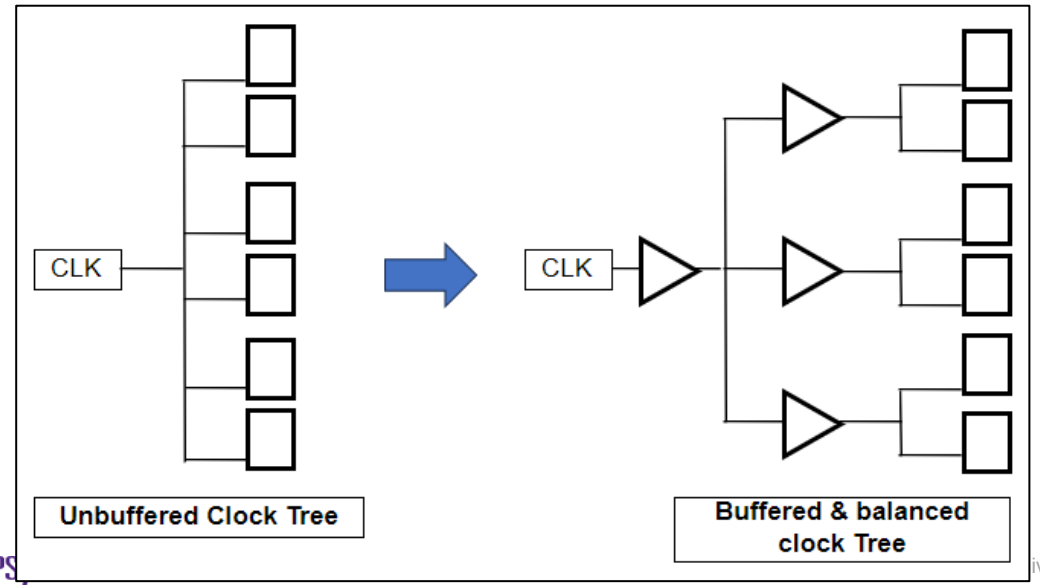
- CTS Overview
  - This module provides an overview of some of the key steps in C TS :
  - **Defining**
    - Tech file (.tf)
    - Netlist
    - Library files (.lib & .lef) & TLU+ file
    - Placement DEF file
    - Insertion delay, skew, clock transition, clock cells, NDR, CTS exceptio ns, list of buffers/inverters … files
  - **Overview to CTS (directory, input files, script)**
  - **CTS**



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Introduction to CTS : Overview to CTS

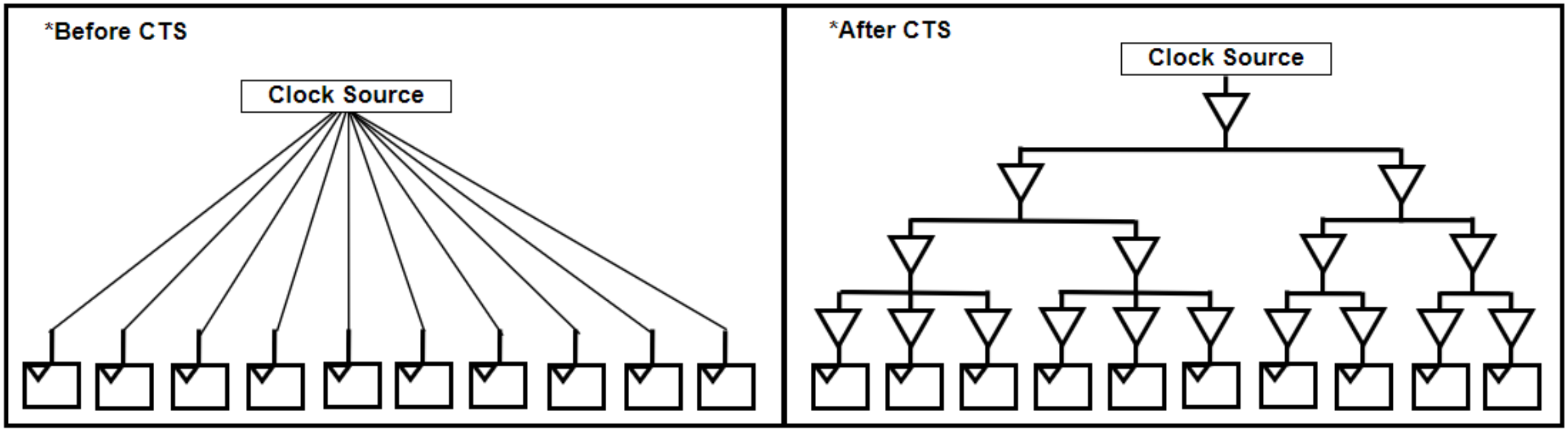
- **What is CTS**
  - CTS(Clock Tree Synthesis) is one of the most important stages in P&R
  - CTS QoR decides timing convergence & power (In most of the ICs clock consumes 30~40% of total power, so efficient clock architecture, clock gating & clock tree implementation helps to reduce power)
  - The Process of distributing the clock and balancing the load is called CTS (Basically, delivering the clock to all sequential ele ments)
  - CTS is the process of insertion of buffers or inverters along the clock paths of ASIC design in order to achieve zero/minimu m skew or balanced skew
  - Before CTS, all clock pins are driven by a single clock source
  - CTS starting point is clock source and CTS ending point is clock pins of sequential cells



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Introduction to CTS : Overview to CTS

- **Compare Before CTS & After CTS**



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Introduction to CTS : CTS Goal and Flows

- **Placement should be completed**
  - Acceptable congestion, setup timing, and logical DRCs
  - High Fanout signal nets (reset, scan_enable, etc) are buffered

- **The goal of the CTS(Clock Tree Synthesis) phase is to:**
  - Build the clock tree buffer structure
  - Detail-route the clock nets
  - Global-route and optimize data-path logic for setup and hold timing, power and logic DRCs

- **Two CTS flows are supported:**
  - Classic CTS flow: CTS first, followed by data path optimization
  - Concurrent Clock & Data flow (CCD): CTS and data path optimization performed concurrently
    - ❖ CCD is enabled by default during clock_opt
    - ❖ CCD is also available during route_opt

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Introduction to CTS : Overview to CTS

- **Difference between HFS (High Fanout Synthesis) & CTS (Clock Tree Synthesis)**
  - Buffers and clock inverter with equal rise and fall times are used (HFS uses buffers and inverters with a relaxed rise and fall times)
  - HFS are used mostly for reset, scan enable and other static signals having fanouts (There is not stringent requirement of bal ancing and power reduction)
  - Clock Tree power is given special attention as it is a constantly switching signal (HFS are mostly performed for static signals and hence not much attention to power is needed)
  - NDR (Non Default Rules) are used for clock net routing

- **Difference between clock buffer and normal buffer**
  - Clock buffer have equal rise time and fall time, therefore pulse with violation is avoided
  - Normal buffer may not have equal rise and fall time
  - Clock buffers are usually designed such that an input signal with 50% duty cycle produces an output with 50% duty cycle

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Introduction to CTS : Overview to CTS

- **Sanity Checks need to be done before CTS**
  - Check legality
  - Check Power stripes, STD rails & also verify PG connections (after Powerplan)
  - Timing QOR (Setup should be under control)
  - Timing DRVs
  - High Fanout nets (scan enable or any static signal nets)
  - Congestion (running CTS on congestion design, design with congestion hotspot can create more congestion and other issues)
  - Remove dont_use attribute on clock buffers and inverters
  - Check Whether all pre-existing cells in clock path are balanced cells (CK* cells)
  - Check & qualify dont_touch, dont_size attributes on clock components

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- **Type of Latency**
- **Source Latency**
  - Source latency is defined as the time taken by the clock signal in traversing from clock source (Oscillator, PLL, some other s ource) to the clock definition point
  - It is also known as source insertion delay

- **Network Latency**
  - Network Latency is defined as the time taken by the clock signal in traversing from clock definition point to the sinks of the cl ock
  - Each sink of the clock has a different network latency

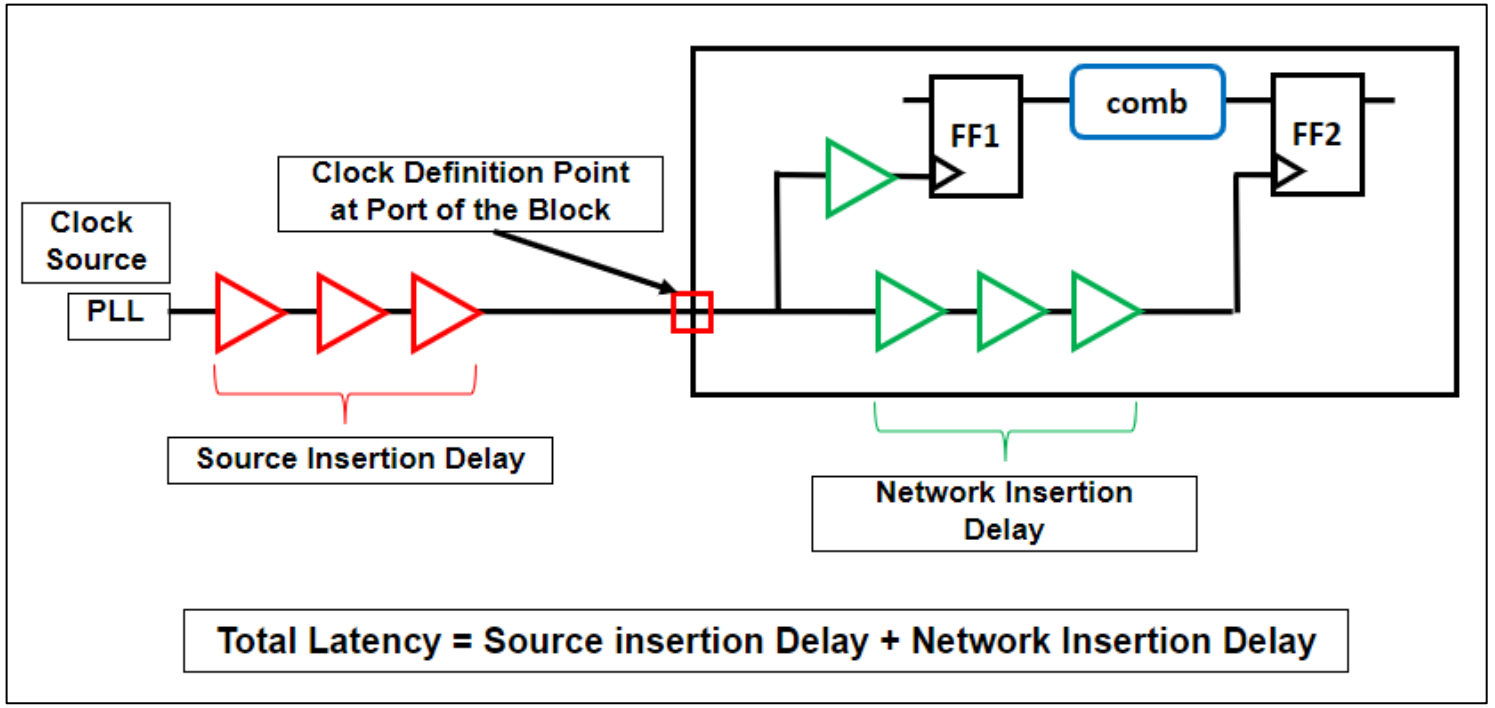
- **What is Clock sink?**

  - Sinks are nodes that receive the clock signal
  - Sinks are usually the clock pins of synchronous elements like flip-flips
  - Sinks are used for delay balancing, which is the process of minimizing the difference among the delays from the clock source to clock sinks

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Introduction to CTS : Overview to CTS

- **Clock Latency**



- What is Clock Tree leaf?

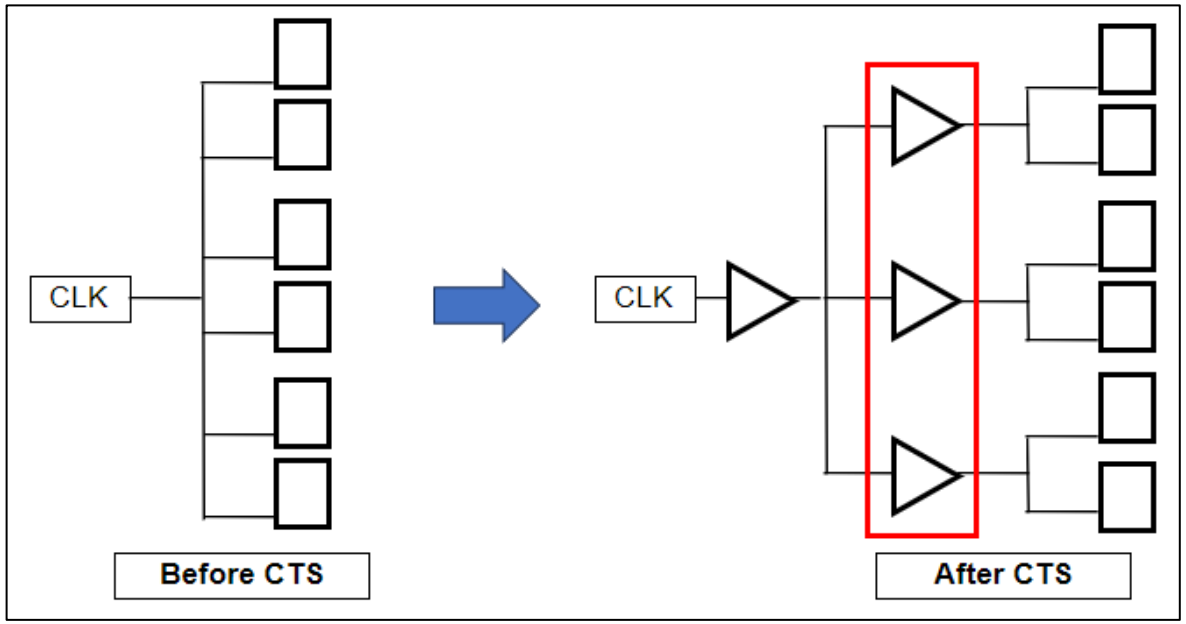
The clock source is the starting point for CTS, and the clock pins of subsequent cells are the ending point for CTS. The path from the Clock Source (Root) to the Clock Sinks is known as the Clock Tree (Leaf)

- Total time taken by the clock signal to reach the input of the reg source latency is the time between clock sources to clock definition ports
- Network latency is the time between clock definition ports to clock leaf cells in the design

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Introduction to CTS : Overview to CTS

- **Insertion Delay**



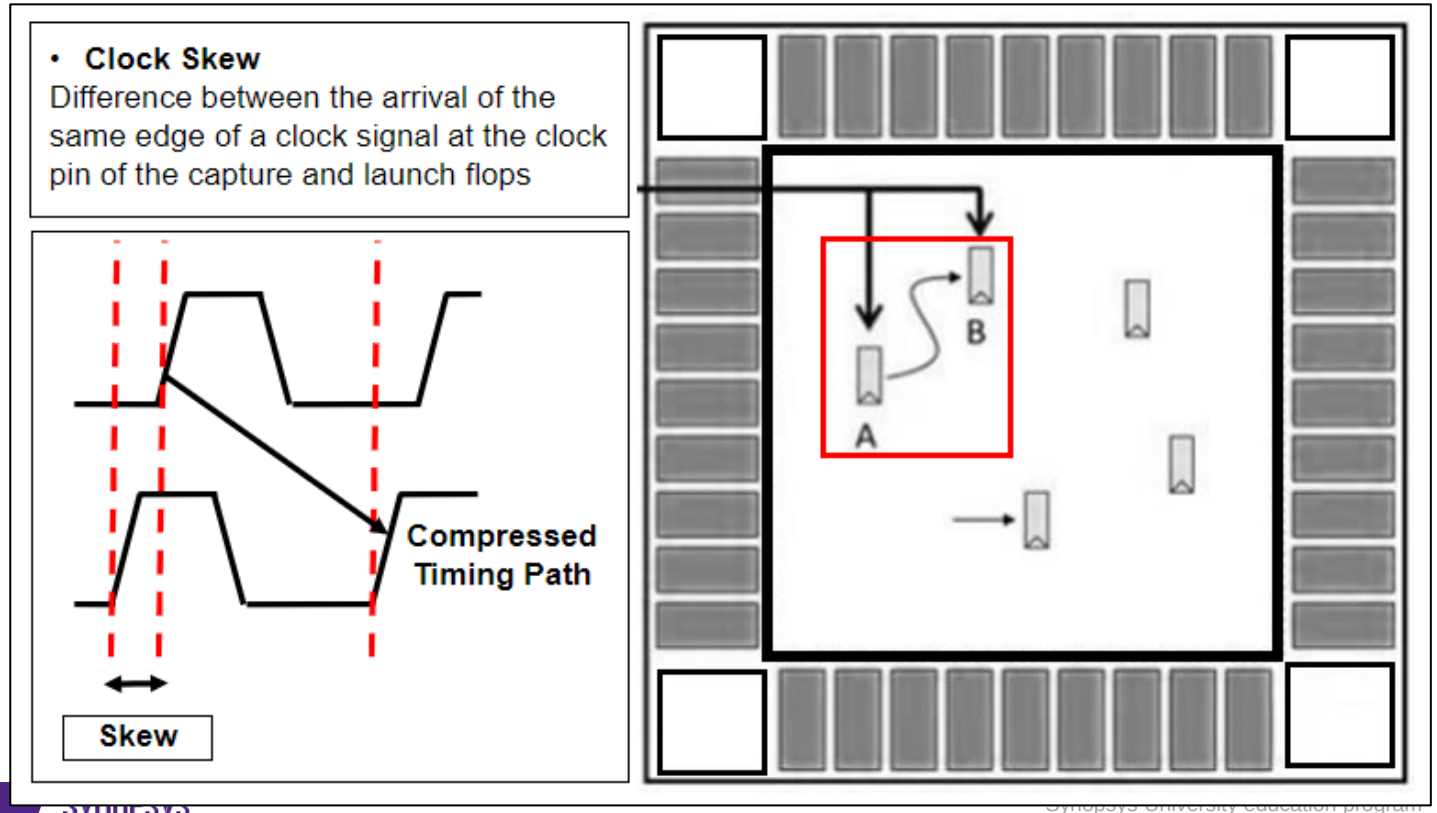
- Insertion Delay is the clock latency, but after Clock Tree is synthesized
- Insertion Delay is the physical delay and Clock Latency is the virtual delay
- Latency is a target given to the tool through SDC or clock tree attribute file and Insertion Delay is the achieved delay value after CTS
- Source and Network Latency (Original Clock & Generated Clock)

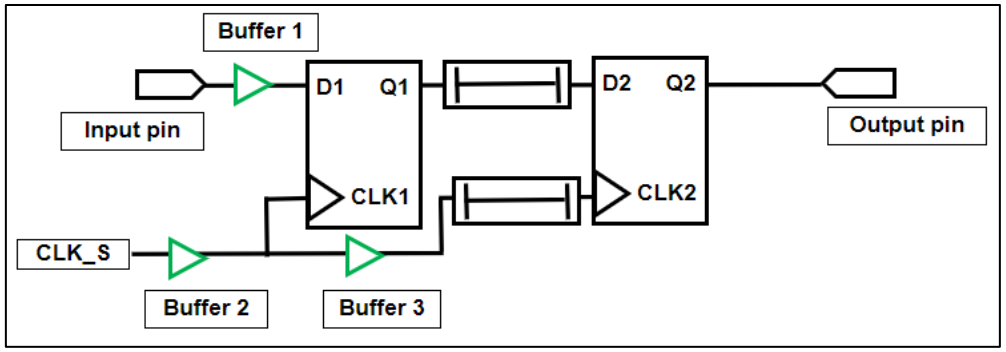
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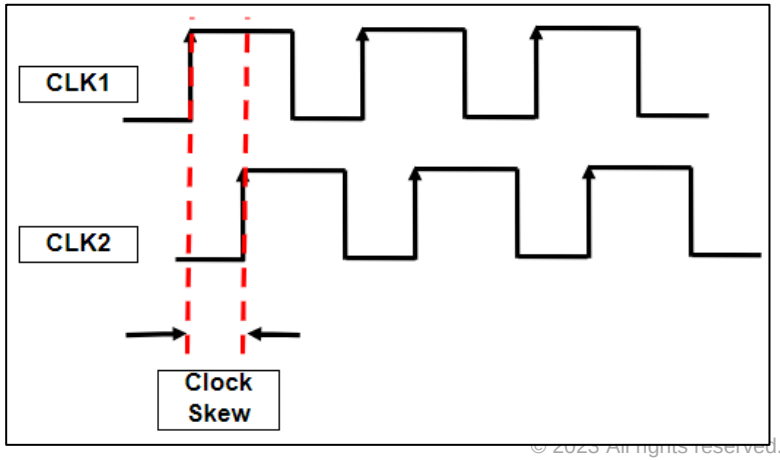
Introduction to CTS : Overview to CTS

- **Clock Uncertainty**
  - Clock uncertainty is the time difference between the arrivals of clock signals at registers in one clock domain or between do mains
  - Uncertainties include Clock Skew, Clock Jitter and Clock Margin

- **Clock Skew: Spatial Clock Variation**





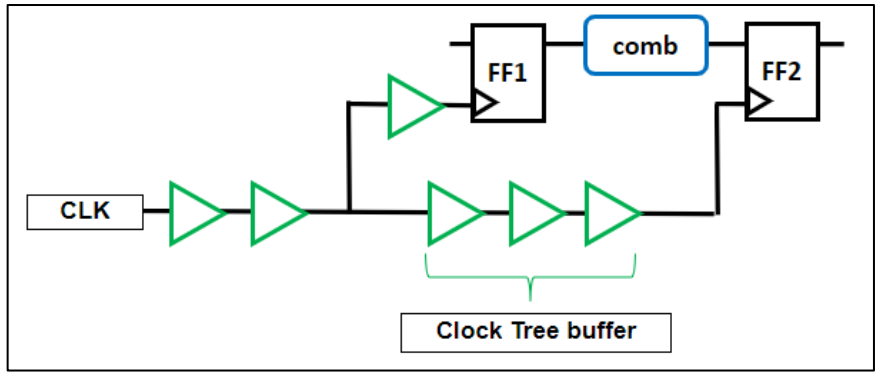


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Introduction to CTS : Overview to CTS

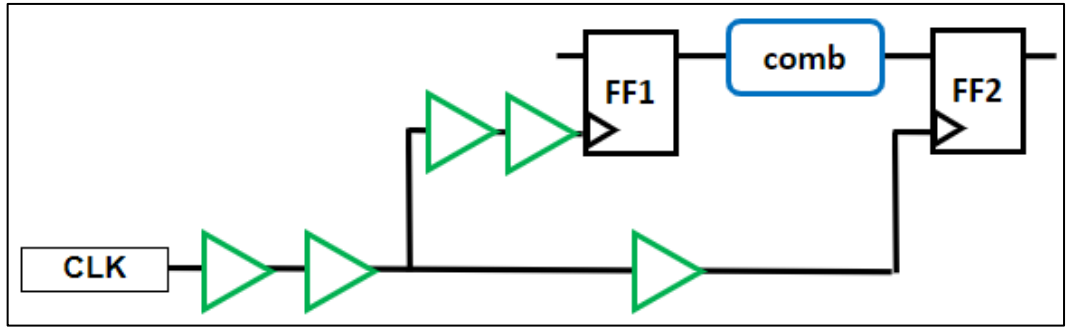
- **Positive Skew**

  If capture clock corners late than launch clock then it is called positive clock



- **Negative Skew**

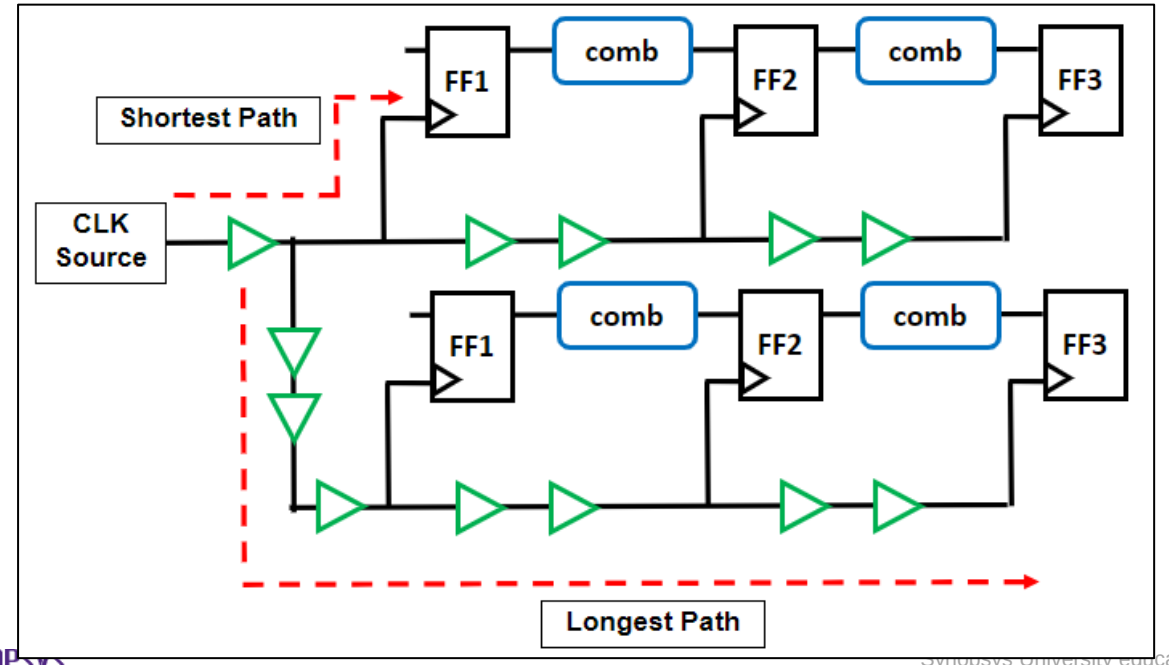
  If capture clock corners Early than launch clock then it is called negative clock



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Introduction to CTS : Overview to CTS

- **Local Skew**
  - Local skew refers to the disparity in latency between two related flops in a design
  - It is the difference in arrival of clock at two consecutive pins of a sequential element
- **Global Skew**
  - Global skew is the difference in clock delay between two unrelated flops
  - Global skew is defined as the difference between max insertion delay and min insertion delay of any flops
  - It is also defined as the difference between shortest clock path delay and longest clock path delay reaching two sequential el ements

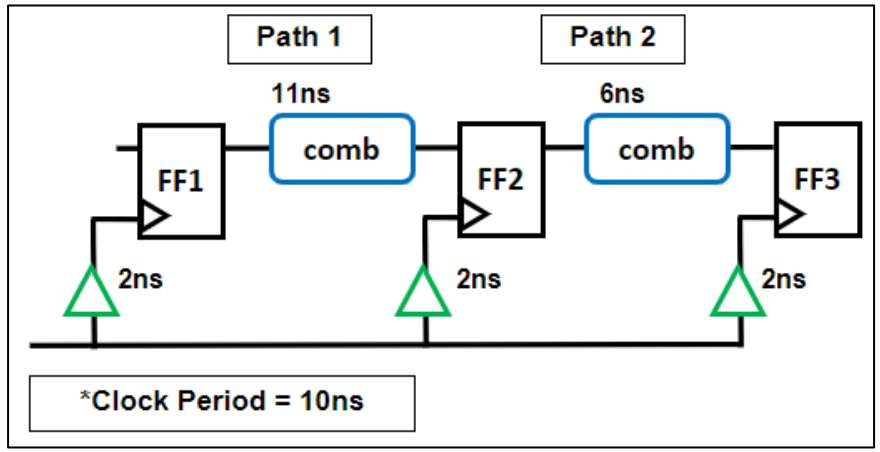


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Introduction to CTS : Overview to CTS

- **Boundary Skew**
  - Boundary skew defined as the difference between max insertion delay and the min insertion delay of boundary flops
- **Useful Skew**
  - If clock is skewed intentionally to resolve violations (it is called useful skew)
  - Useful skew is a technique of intentionally adding clock skew to resolve timing violations in VLSI design
  - For example, if there is a setup violation between two flip-flops, we can add some skew along the clock path to increase the r equired time and eliminate the violation (Useful skew can improve the performance and reliability of the circuit)

❖ **Example of Useful Skew**



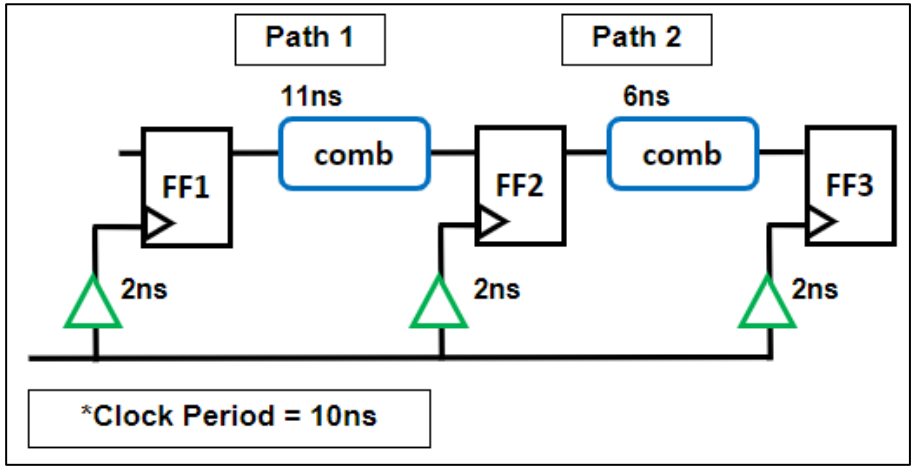
Clock Period of the circuit as 10ns, with 2ns insertion delay to clock pin of each register (also assume that the setup and hold times of the flops are zero for this example)

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❖ **Example of Useful Skew**

The path 1 combinational delay is 11ns and path2 is 5ns in the picture



The setup timing requirement of path 1 :

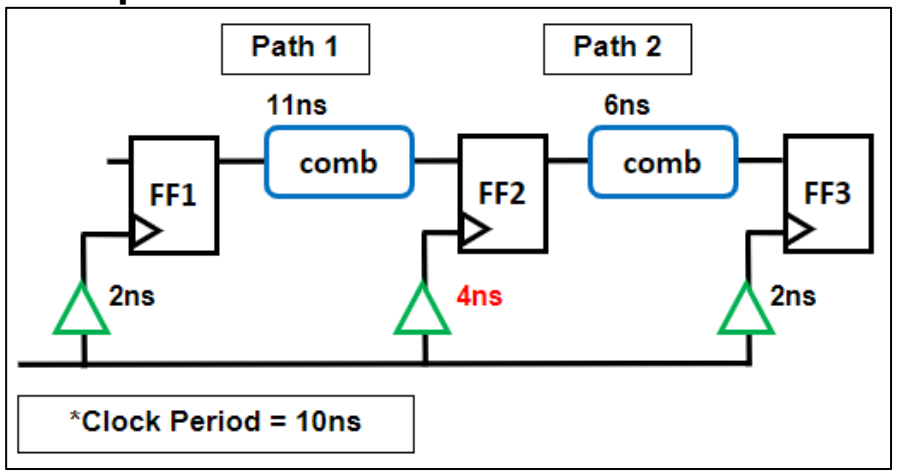
2ns (Clk path delay till FF1/CP)+11ns < 2ns (Clk path delay till FF2/CP)+10ns(Clk period) = 13ns < 12 ns (VIOLATED)

- If we keep the clock network intact, we need to fix this timing violation by reducing the combinational path delay of Path1
- However, Path2 timing is met with considerable margin
- Path 2 -> 2ns+6ns < 2ns+10ns = 8ns < 12ns (MET)
- So, ideally we could borrow some cycle time from path2, if no other paths are affected by it

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❖ **Example of Useful Skew**



- **The setup timing requirement of path 1 :**

  2ns (Clk path delay till FF1/CP)+11ns < 4ns (Clk path delay till FF2/CP)+10ns(Clk period) = 13ns < 14 ns (MET)

- **Clock skew of 2ns between these registers, and the Path2 timing has now changed to :**

  4ns (Clk path delay till FF2/CP)+6ns < 2ns+10ns = 10ns < 12ns (MET)

  ❖ Still not violated, even though margin is now less

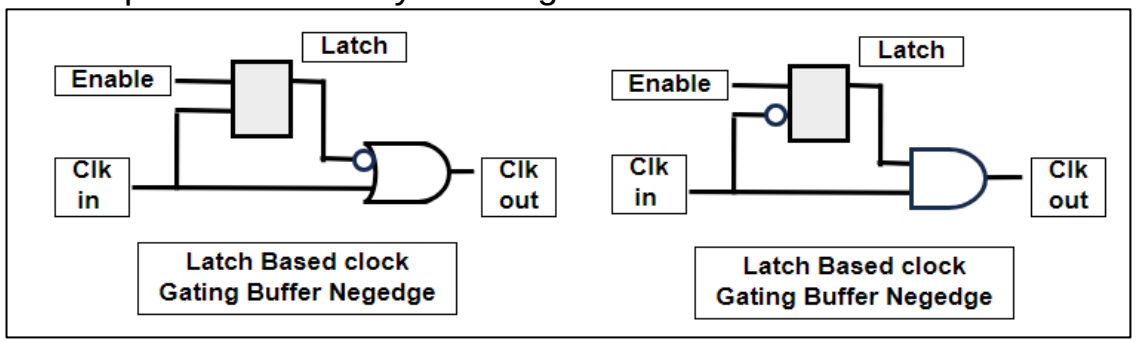
- P&R tools when enabled 'useful skew' can change the clock network and introduce intentional skew to meet timing with smaller cost

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Introduction to CTS : ICG(Integrated Clock Gating) cell

- What is ICG cells
  - ICG cell is a standard cell used in one of low power techniques in ASIC design
  - It is used to reduce the dynamic power in SoC Design
  - It is used to stop the clock signal propagation to a big group of logic cells when the group is not required to operate
  - The ICG cell stops the clock propagation through it when a low clock enable signal is applied to it, a phenomenon known as clock g ating

- Types of Clock Gating cells
  - Latch Based Clock Gating Buffer for Neg-edge
    - The circuit employs a latch and OR gate with one input inverted
    - The output clock is always clock gated low when Enable is low
  - Latch Based Clock Gating Buffer for Pos-edge
    - The circuit employs a latch with inverted clock input and a AND gate
    - The output clock is always clock gated HIGH when Enable is low



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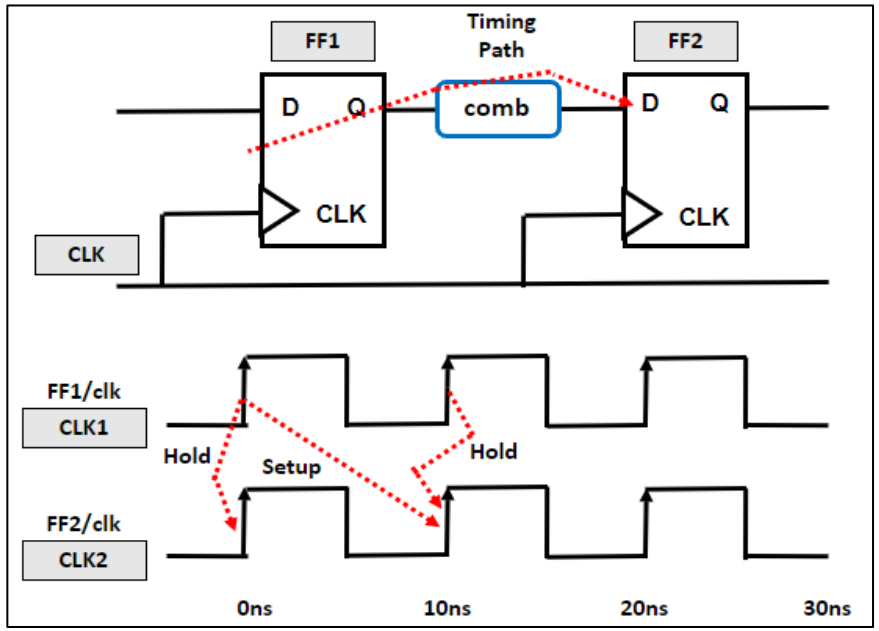
Introduction to CTS : Delay cell

- Delay cell
  - What is Delay cell
  - Delay cell insertion is the conventional way to fix hold time violation tends to penalized in area percentage increment
  - Lesser number of delay cells are required for hold time fixing as compared to buffers but it will have area much greater than normal b uffers
  - Increasing gate width reduces gate capacitance hence reduces delay, but results in higher leakage
  - Depending on the delay of the cell, pair of inverter and pass transistor can be repeated multiple times
  - What is Delay
    - Delay = Cell Delay + Net Delay
  - What is Cell Delay
    - Cell delay is the timing delay between input and output pin of a cell.
    - It is the time required to charge, discharge the load capacitance
  - What is Net Delay
    - Net means a wire, and Net Delay means a delay that occurs as it passes through the wire.

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Introduction to CTS : Overview to CTS

- **Setup & Hold Violation**



- Data is launching from FF1/D pin to FF1/Q at the positive clock edge at FF1/CLK pin
- At FF2/D pin, input data is coming from FF1/Q through a combinational logic
- Data is capturing at FF2/D, at the positive clock edge at FF2/CLK pin
- Launching FF = FF1
- Capturing FF = FF2
- Data Path = FF1/CLK > FF1/Q > FF2/D

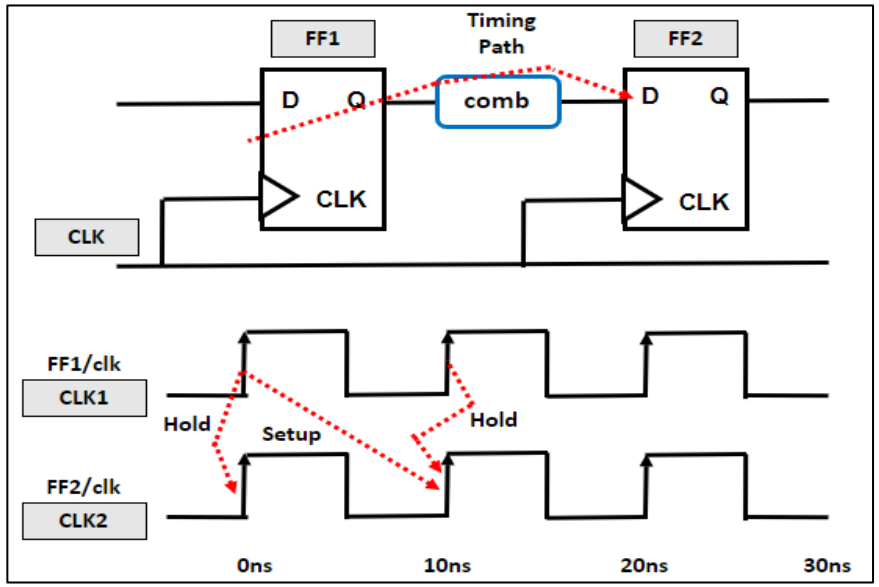
- **Setup violation at FF2**
  - If setup Time = 0ns, data launched from FF1 at time 0ns should arrive at FF2/D before or at time 10ns
  - If data takes too long (greater then 10ns) to arrive, it is reported as setup violation

  - If setup Time =2ns, then, data launched from FF1 at time 0ns, should arrive at FF2/D before at time = (10ns -2ns) 8ns
  - If data takes too long (greater then 8ns) to arrive, it is reported as setup violation
  - Slack = Data Required Time – Data Arrival time < 0

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Introduction to CTS : Overview to CTS

- **Setup & Hold Violation**



- Data is launching from FF1/D pin to FF1/Q at the positive clock edge at FF1/CLK pin
- At FF2/D pin, input data is coming from FF1/Q through a combinational logic
- Data is capturing at FF2/D, at the positive clock edge at FF2/CLK pin
- Launching FF = FF1
- Capturing FF = FF2
- Data Path = FF1/CLK > FF1/Q > FF2/D

- **Hold violation at FF2**
  - ❖ *There should not be any change in the input data at FF2/D pin between positive edge of clock at FF2 at Time=10ns and Time=10ns+Hold Time of FF2*
  - To satisfy the Hold Condition at FF2 for the Data launched by FF1 at 0ns, the data launched by FF1 at 10ns should not reach at FF2/D pin before 10ns + Hold Time of FF2
  - If Hold Time of FF2 = 0.3ns, then we can say that the data launched from FF1 at time 10ns does not get propagated so soon that it reaches at FF2 before time (10ns+0.3ns) = 10.03ns
  - If data arrive soon, its reported Hold violation
  - Slack = Data Arrival Time – Data Required time < 0

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Introduction to CTS : Overview to CTS

- **Fix Setup & Hold Violation**
  - **Setup Violation**
    - **Reduce the amount of buffering in the path**
      - It will reduce the cell delay but increase the wire delay
    - **Replace buffers with 2 inverters place farther apart**
      - Insert inverter pair decreases the transition time 2 times then the existing buffer gate (RC delay of the wire decreases)
      - Dealy of 1 buffer = Delay of 2 inverter
    - **VTH swap (change HVT cells)**
    - **Increase Driver Size or increase Drive strength (Upsize Cell)**
      - Ex) X1 inverter has the minimum size, low speed, low power operations
      - Ex) X2 inverter uses two devices X1 inverters, in parallel (The resulting circuit is an inverter with twice the current capabilities)
    - **Insert Buffer or repeaters**
      - Inserting buffer to decrease over all delay in case of long wire
      - Inserting buffer decreases the transition time, which decreases the wire delay
    - **Adjust cell position in layout**
    - **Minimize long routing wires**

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Introduction to CTS : Overview to CTS

- **Fix Setup & Hold Violation**
  - **Hold Violation**
    - **Introduce delay in data path by adding buffers in data path**
    - **VTH swap (change HVT cells)**
    - **decrease Driver Size or decrease Drive strength (Downsize Cell)**
    - **Insert Buffer / Inverter pairs / delay cells**
      - Inserting buffer to decrease over all delay in case of long wire
      - Inserting buffer decreases the transition time, which decreases the wire delay
    - **Increase the width of wire**

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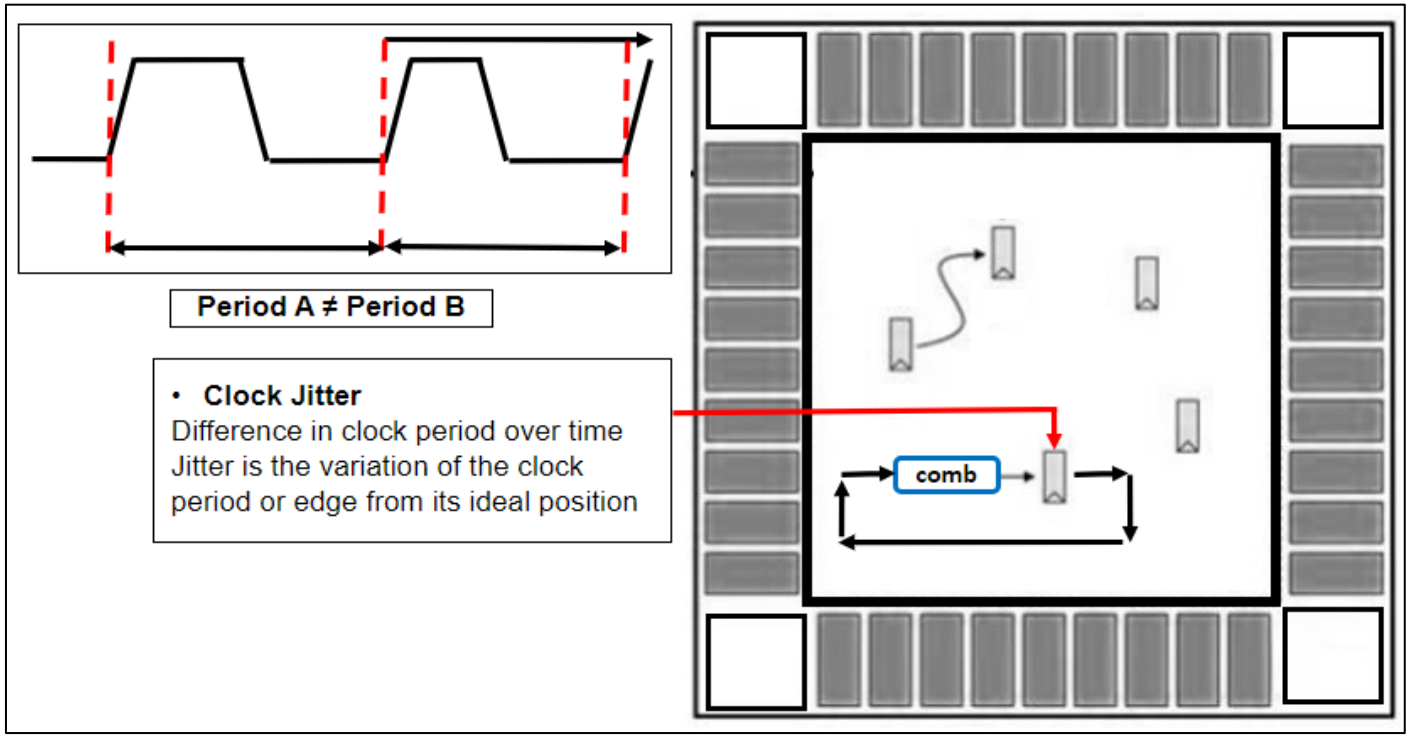
- **Max Capacitance Violation**
  - The maximum capacitance is a pin-level attribute used to define the maximum total capacitive load that an output pin can drive
  - The pin cannot connect to a net that has a total capacitance (load pin capacitance and interconnect capacitance) greater than or equal to the maximum capacitance defined at the pin
  - Max capacitance violation is a situation where the total capacitance on a net exceeds the max capacitance defined at the pin (This can affect the performance and reliability of the device)

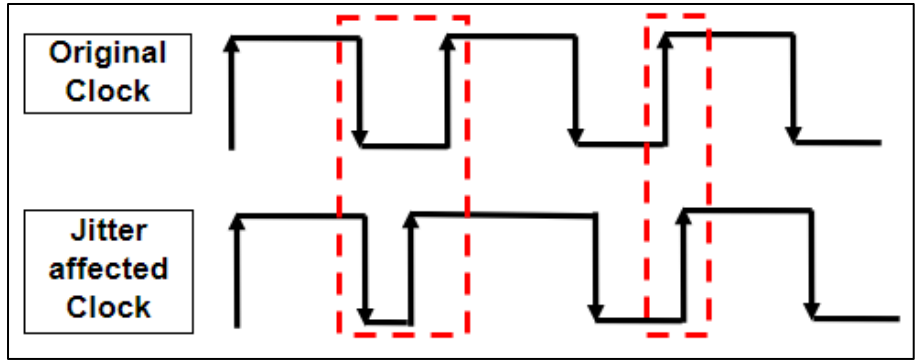
- **How to fix Max Capacitance Violation**
  - Increase the drive strength of the cell
  - Buffer some of the fan-out paths to reduce the capacitance send by the output pin
  - Upsize the driver cell
  - Insert Buffers
  - Decrease the net length by moving cells nearer or reduced long net (routed)

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Introduction to CTS : Overview to CTS

- **Clock Jitter : Temporal Clock Variation**





- Jitter is the short-term variations of a signal with respect to its ideal position in time
- Factors causing Jitter includes imperfections in Clock oscillator, supply voltage variations, Temperature variations, Crosstalk
- Jitter can affect the performance and reliability of VLSI circuits, as it can introduce timing errors or glitches
- Jitter can be modeled by adding uncertainty regions around the clock edges

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Introduction to CTS : Overview to CTS

- **CTS Pre-requisites**
  - Legally placed and optimized with acceptable congestion timing should be good
  - No DRC violations Power/Ground nets are Pre-routed "HFS" done
  - Logical & Physical Library should have special Clock Cells

- **Objects during CTS**
  - The timer starts from every Clock source and traces forward over Combinational Arcs until it reaches the Clock Pin of a flop or another Clock source
  - All pins & Timing Arcs in the forward trace before a valid Leaf are considered to be in the clock network Pin or Combinationa l Timing Arcs that trace to a non-clock pin are not part of Clock tree network (ex D pin of Flip-Flop)
  - Sequential elements are traced through if it is a source of the Generated Clock
  - Clock tracing should be Mode aware
  - Inverters are added in Clock Tree for better Duty Cycle Limit the buffer/inverter list to just 3 or 4 buf & inv sizes

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Introduction to CTS : Overview to CTS

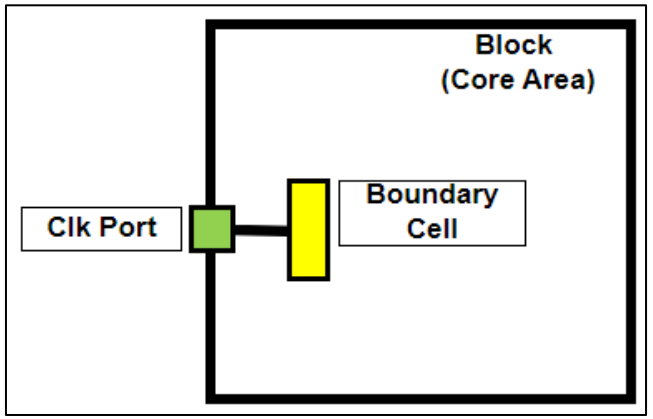
- **CTS Flow**
  - Check & fix Macro location before placement
  - Read SDC : Clock Tree begins at SDC defined clock pin and ends at stop pin of the flop Generate
  - ❖ **CTS Specification file (Include in CTS script & setup file)**
    - Max. Skew
    - Min, Max. Insertion Delay
    - Max. Transition, Capacitance, Fanout No. Buffer, Buffer/Inverter list
    - Clock Tree Routing Metal Layers
    - Clock Tree Leaf Pin, Root Pin, Preserve Pin, Through Pin and Exclude Pin

- **Clock Tree Reference**
  - By default, each clock tree referenced list contains all the clock buffers and clock inverters in the logic library
  - ❖ **Clock Tree reference list :**
    - Clock tree synthesis Boundary cell insertions sizing
    - Delay Insertion

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Introduction to CTS : Overview to CTS

- **Boundary cell insertions:**

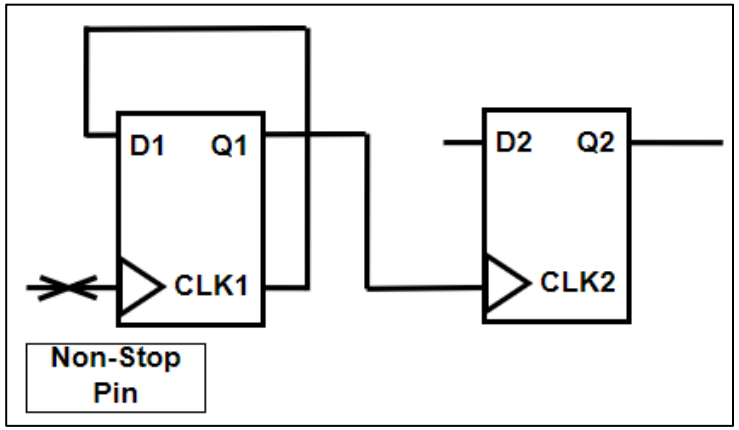


  - When you are working on a block-level design (sub block), you might want to preserve the boundary conditions of the block's clock ports (the boundary clock pins)
  - A boundary cell is a fixed buffer that is inserted immediately after the boundary clock pins to preserve the boundary conditions of the clk pin
  - When boundary cell insertion is enabled, buffer is inserted from the clock tree reference list immediately after the boundary clk pins
  - For Multi-voltage designs, buffers are inserted at the boundary in the default voltage area
  - The boundary Cells are fixed for clock tree synthesis after insertion; it can’t be moved of sized
  - In addition, no cells are inserted between a clock pin and its boundary cell
- **Delay cell insertions:**
  - If the delay is more, instead of adding many buffers we can just add a delay cell of particular delay value
  - Advantage is the size and also power reduction
  - Delay cell has high variation, so usage of delay cells in clock tree is not recommended

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- **Clock Tree Exceptions:**
  - **Non-stop pin**

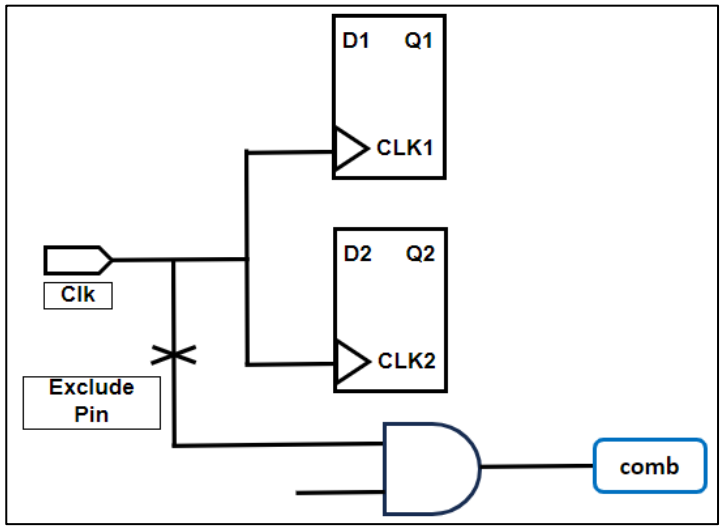


  - Non-stop pins trace through the endpoints that are normally considered as endpoints of the clock tree
  - The clock pin of sequential cells driving generated clock are implicit non-stop pins
  - Clock pins of ICG cells are also non-stop pins
  - They are used to clock tree trace the continuous against the default behavior

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Introduction to CTS : Overview to CTS

- **Clock Tree Exceptions:**
  - **Exclude pin**



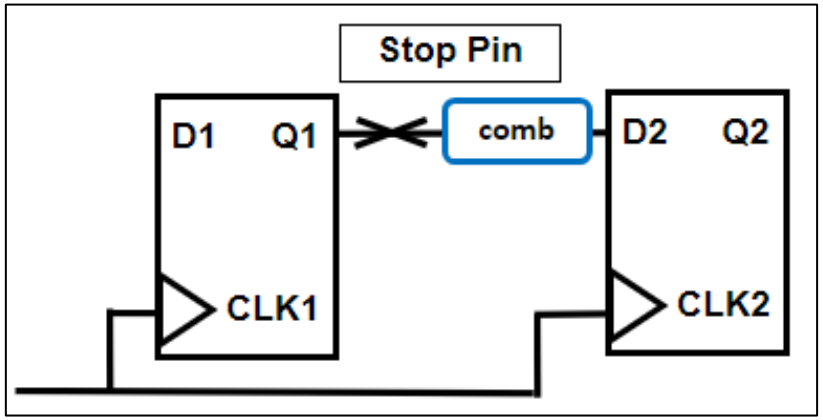
- Exclude pin are clock tree endpoints that are excluded from clock tree timing calculation and optimization
- The tool considers exclude pins only in calculation and optimizations for design rule constraints
- During CTS, the tool isolates exclude pins from the clock tree by inserting a guide buffer before the pin or these pins are need not to be consid ered during the clock tree propagation

- In the above figure, beyond the exclude pin the tool never perform skew or insertion delay optimization but does perform design rule fixing

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Introduction to CTS : Overview to CTS

- **Clock Tree Exceptions:**
  - **Stop pin**

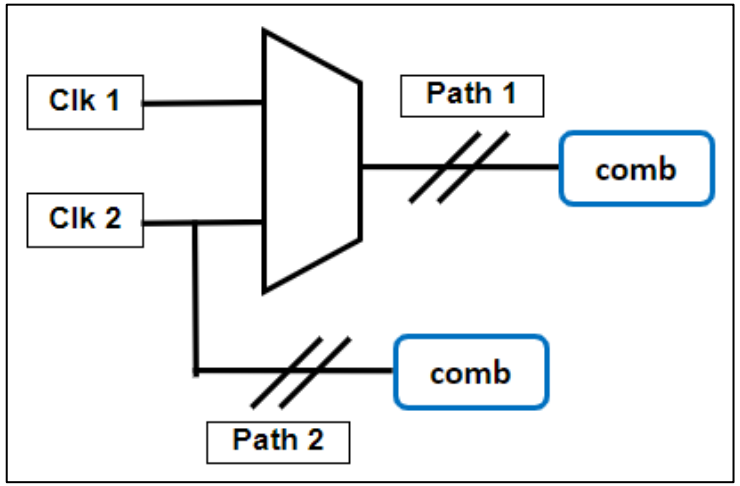


- Stop Pins are the endpoints of clock tree that are used for delay balancing
- In CTS, the tool uses stop pins in calculation & optimization for both DRC and Timing (Clock sink are implicit stop pins)
- The optimization is done only up to the stop pin as shown in the above fig
- The clock signal should not propagate after reaching the stop & sync
- This pin needs to be considered for building the clock tree

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Introduction to CTS : Overview to CTS

- **Clock Tree Exceptions:**
  - **Dont touch Sub-tree**



- If preserve a portion of an existing clock tree, we put dont_touch exception on the sub-tree

- CLK1 is the pre-existing clock and path1 is optimized with respect to CLK1
- CLK2 is the new generated clock
- For example if path1 is 300ns and paht2 is 200ns. During balancing delay are added in path2
- If path1 is 200ns and path2 is 300ns, during balancing delay cannot be added on path1 because on path1 dont_touch attribute is set and we get violation

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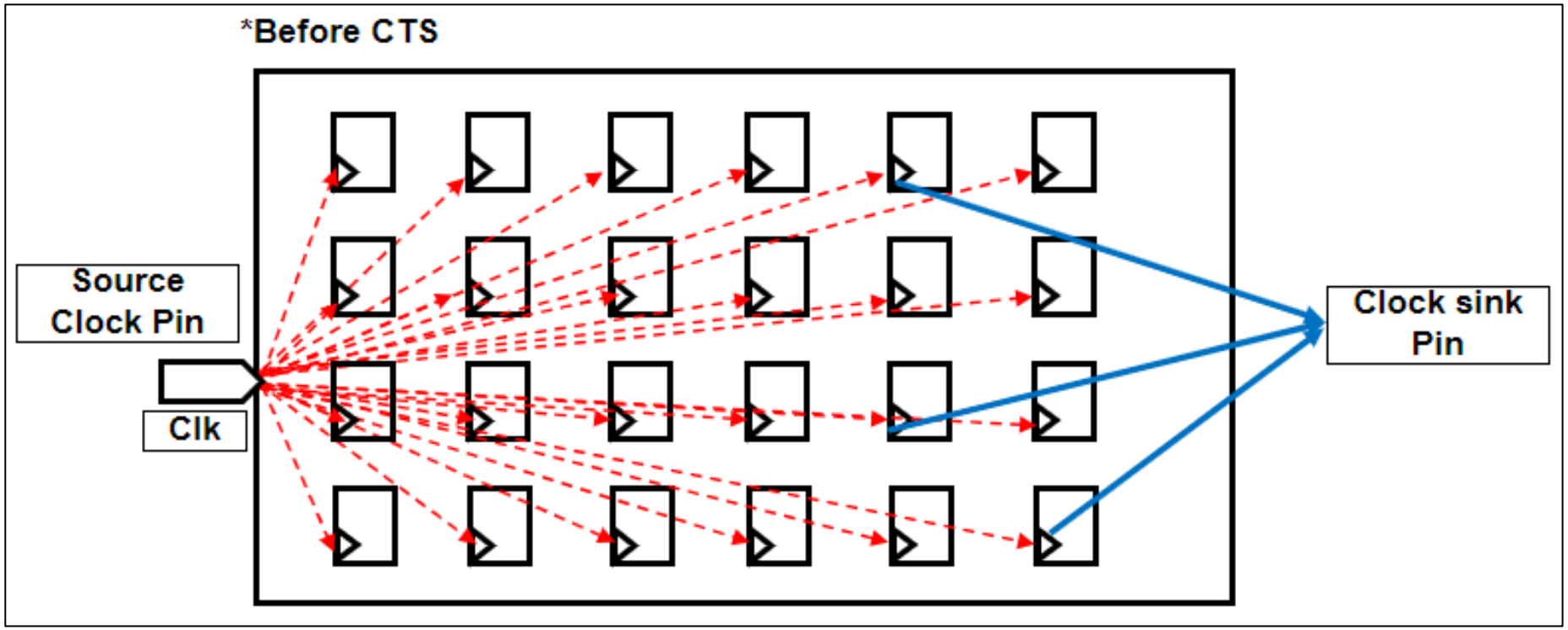
Introduction to CTS : Overview to CTS

- **Clock Tree Exceptions:**
  - **Dont Buffer Net**
    - It is used in order to improve the results, by preventing the tool from buffering certain nets
    - Don’t buffer nets have high priority than DRC
    - CTS do not add buffers on such nets (If the path is a false path, then no need of balancing the path)
  - **Dont Size cell**
    - To prevent sizing of cells on the clock path during CTS and optimization, we must identify the cell as don’t size cells
  - **Specifying Size-Only Cells**
    - During CTS & optimization, size only cells can only be sized not moved or split
    - After sizing, if the cells overlap with an adjacent cell after sizing, the size-only cell might be moved during the legalization step

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Introduction to CTS : Overview to CTS

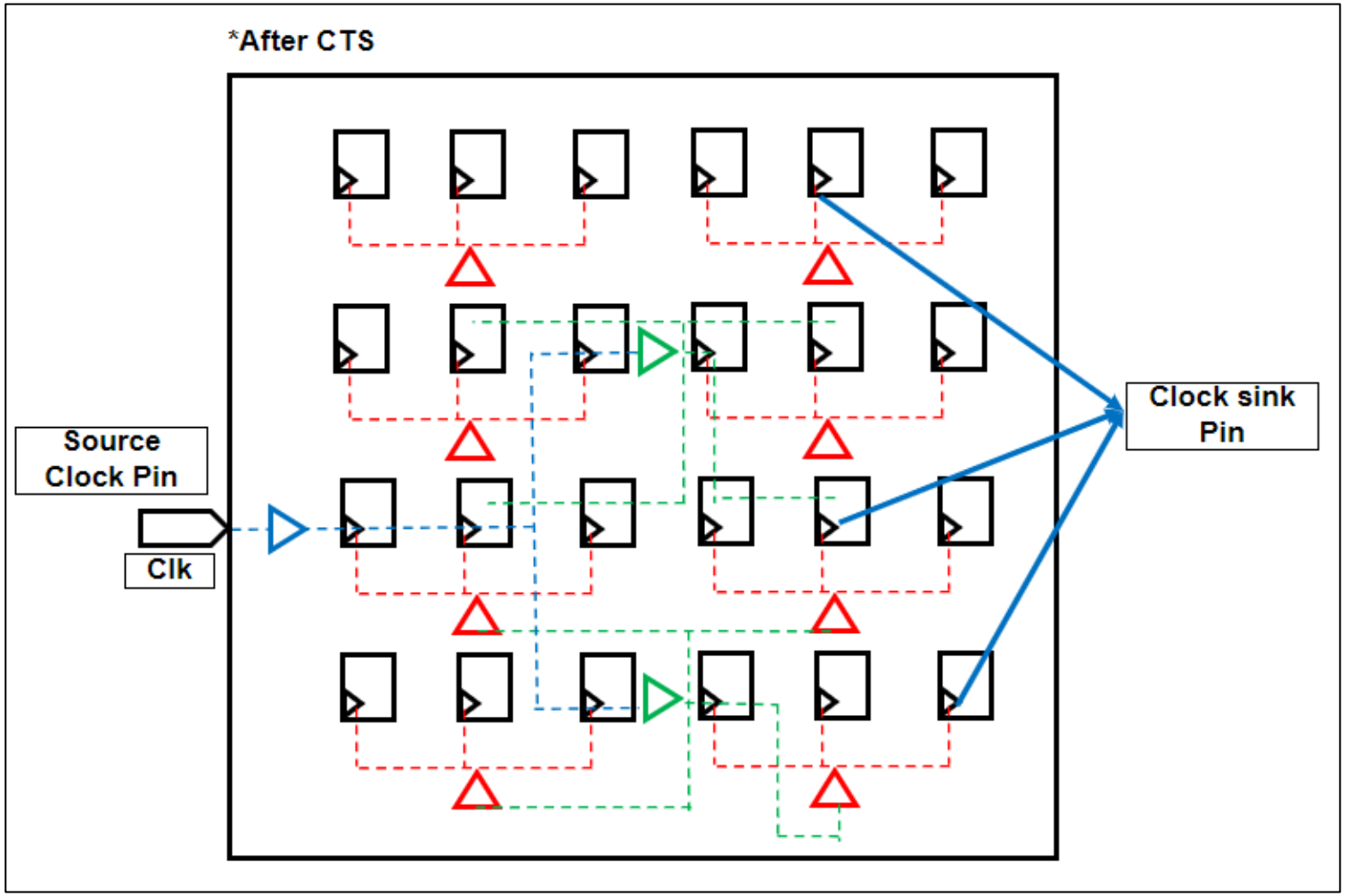
- **CTS Algorithms**
  - **Before CTS all Clock Pins are driven by a single Clock Source**



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Introduction to CTS : Overview to CTS

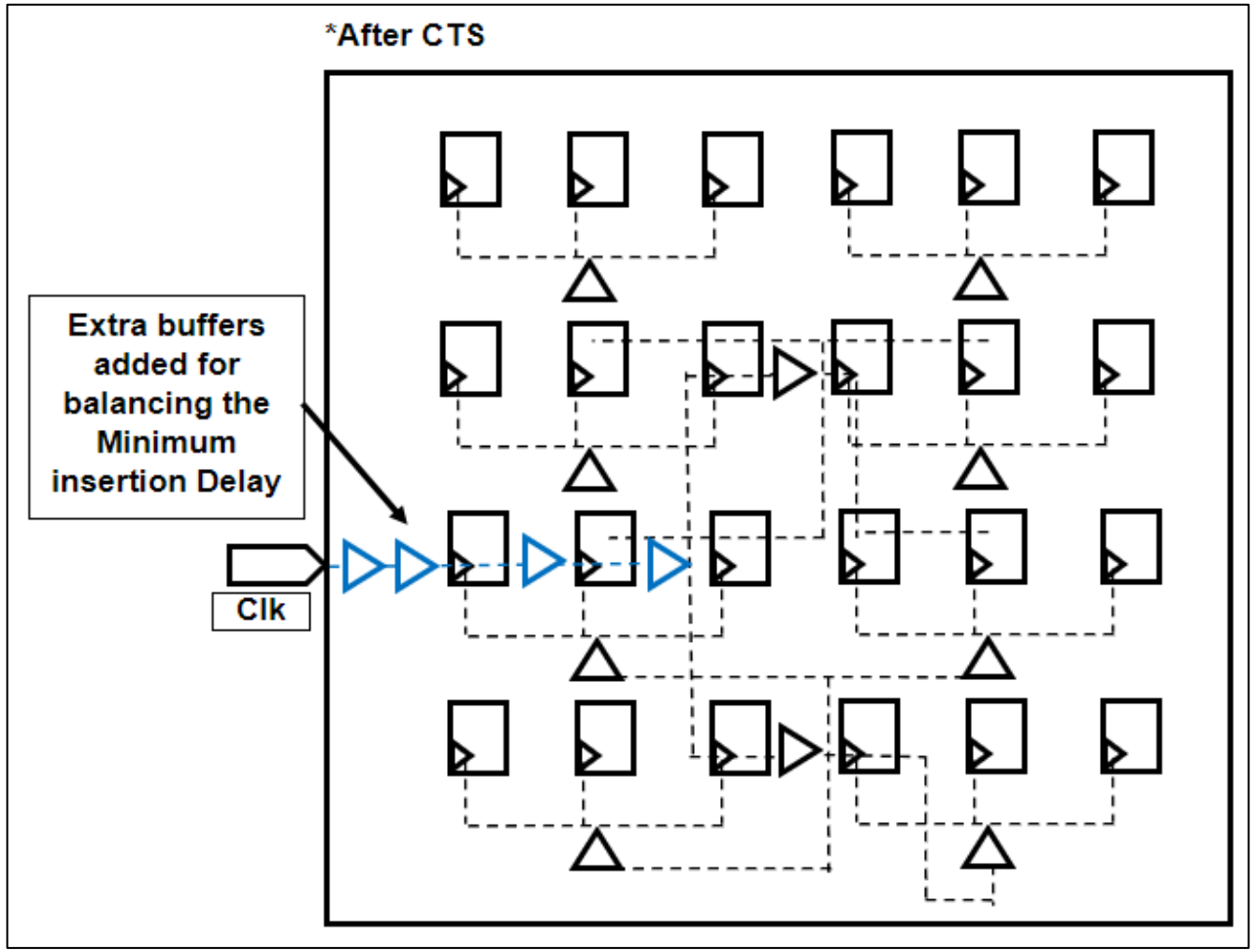
- **CTS Algorithms**
  - **After CTS, the buffer tree is built to balance the loads and minimize the skew**



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Introduction to CTS : Overview to CTS

- **CTS Algorithms**
  - **After CTS a "delay line" is added to meet the minimum insertion Delay**



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Introduction to CTS : Overview to CTS

- **Analyze list after CTS**
  - Report Timing (setup & hold)
  - If timing not met then check clocks be grouped (balanced together)
  - Report Insertion Delay & Skew and verify that the targets are achieved
  - Report DRV targets (Fanout split, Capacitance and Transition)
  - Check the intended Leaf Cell (Clock Sinks) is reached
  - Check the Clock Tree Exceptions are not in the Clock Tree
  - Report the pre-existing cells, such as Clock Gating Cells
  - Quality-of-Report (QoR summary)
  - Check Clock Tree converges either with itself or with another Clock Tree
  - Clock Tree has timing relationship with other Clock Trees for inter Clock Skew balancing
  - Check Design Rule Constraints
  - Check Routing Constraints
  - Report power and Area

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Introduction to CTS : Overview to CTS

- **Post CTS Optimization**
  - Optimization with useful skew
  - Optimization with TNS (Total Negative Slack)
  - Fine Grid spacing
  - Post CTS Optimization(PSYN) Techniques
    - Net shielding
    - Cell sizing
    - Buffer relocation
    - Level adjustment
  - Optimize the design for Hold time
  - Hold violations should be fixed first in best corner and then in worst corner
  - Area optimization
- **Output of CTS**
  - Timing report
  - Congestion report
  - Skew report
  - Insertion delay report
  - CTS DEF file

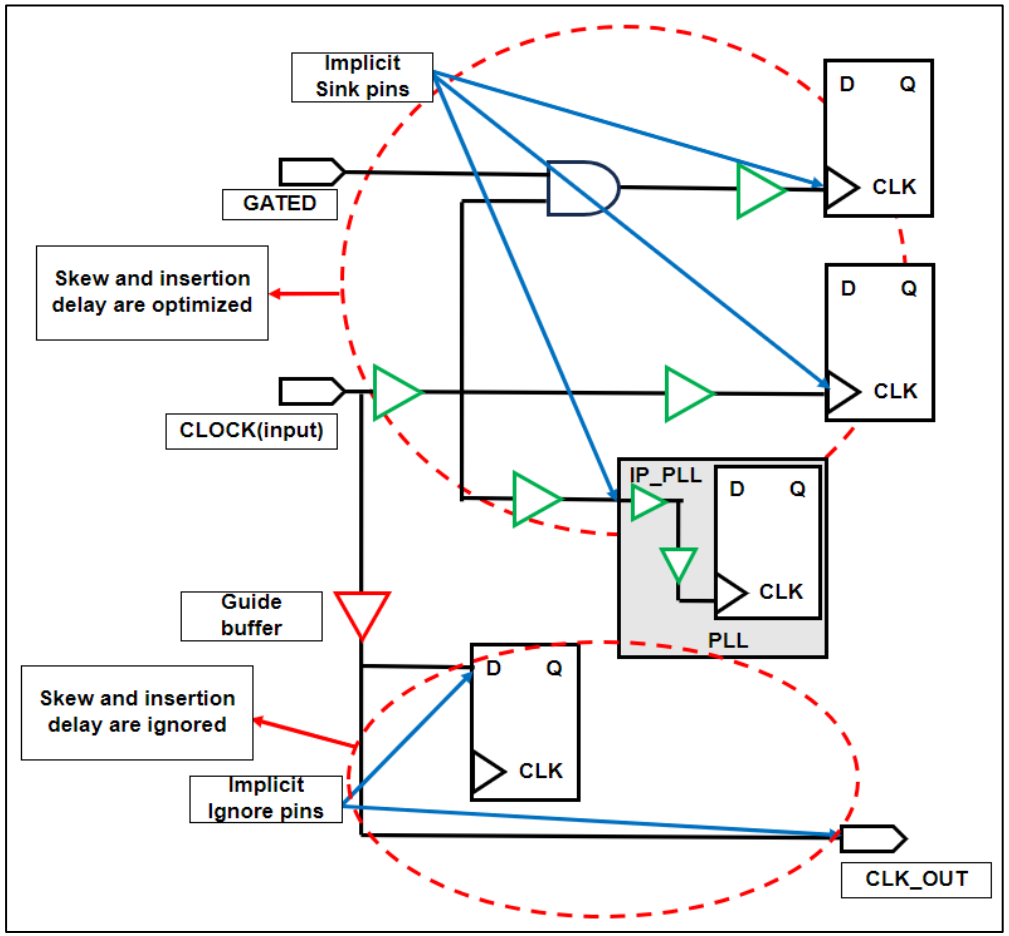
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WEEK 9-2 : CTS Setup before running

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Introduction to CTS : Sink & Ignore pins

- Sink & Ignore pins



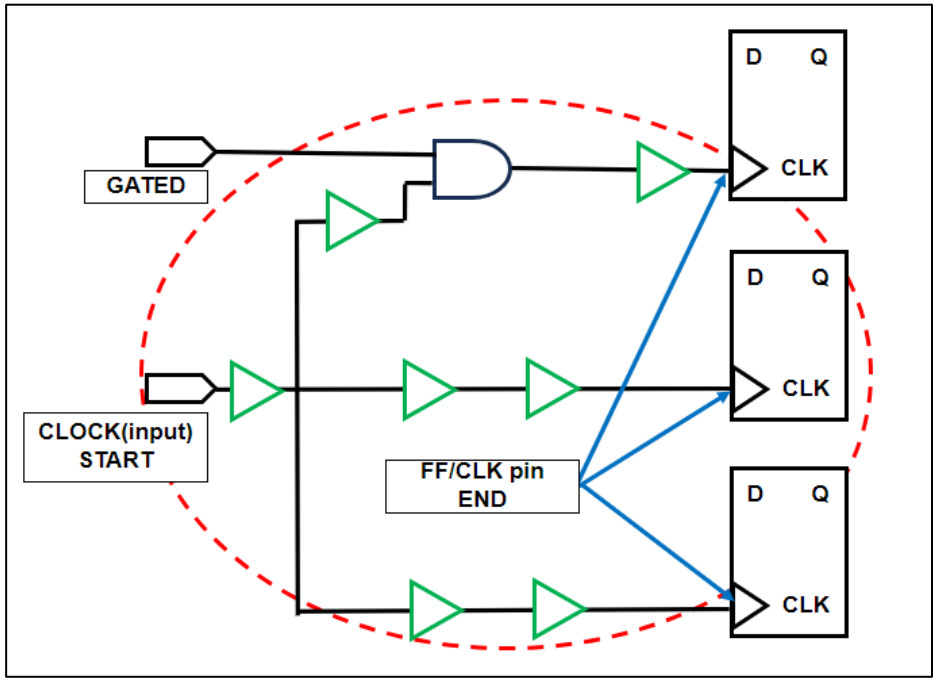
- **Sink Pins**
  - CTS optimizes for DRC and clock tree targets (skew, insertion delay)
  - Considers internal insertion delays
- **Ignore Pins**
  - CTS ignores skew and insertion delay targets
  - CTS inserts a guide buffer (only DRCs fixed after guide buffer)

- **Implicit sink pins are automatically defined by CTS on:**
  - Clock pins of sequential cells (Flip Flop and Latches)
  - Clock pins of Macros
- **Implicit ignore pins are automatically defined by CTS on:**
  - Non-clock input pins of sequential cells (ex) D, set, Reset pins)
  - Output ports
  - Pins of combinational cells or integrated clock-gating cells that do not have any fanout or that do not have any enabled timing arcs
  - Select/Control pin of mux used in the data path
  - Input pin of pre-existing gate, if all pins in its fanout are exclude pins
  - Incorrectly defined clock pins (missing or incorrect pin definition in the standard cell or macro cell definition)
  - Buffer or inverter input pins that are held constant (by using set_case_analysis in SDC)

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Introduction to CTS : Begin & End of the Clock Tree

- Start & End point of the Clock Tree



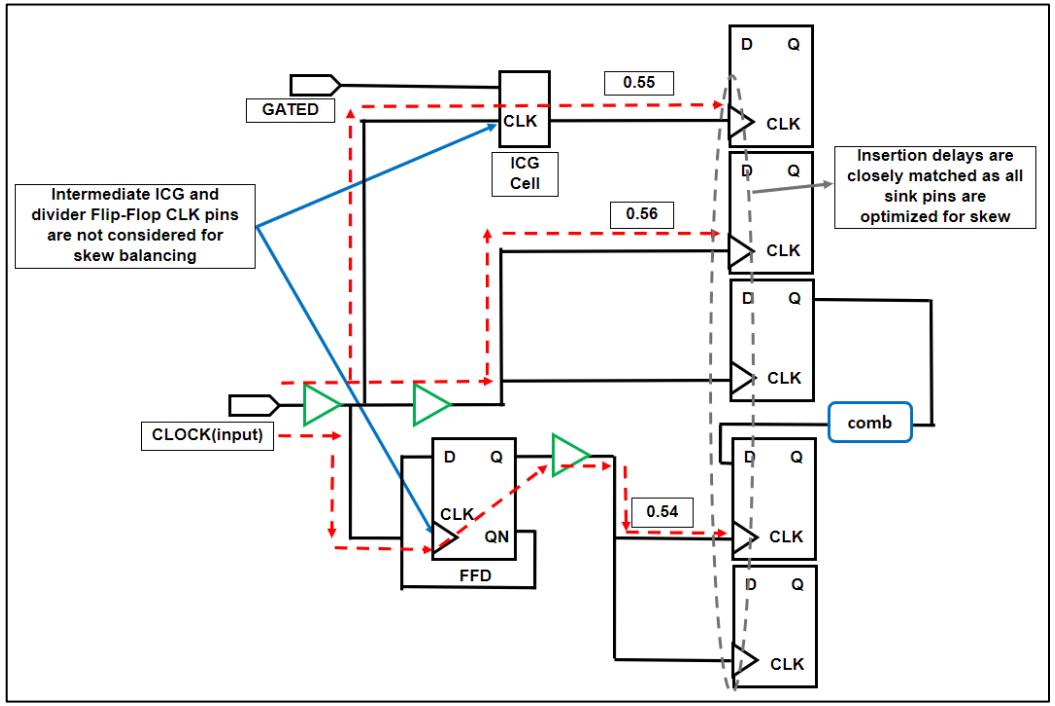
- Clock trees start at their "source" defined by "**create_clock**"
- Synthesizable clock tress end on clock pins of registers or Macros
- You can check source of a clock use "get_attribute" command

  ex) get_attribute [get_clocks PCI_CLK] sources

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Introduction to CTS : Generated & Gated Clocks

- Generated & Gated Clocks



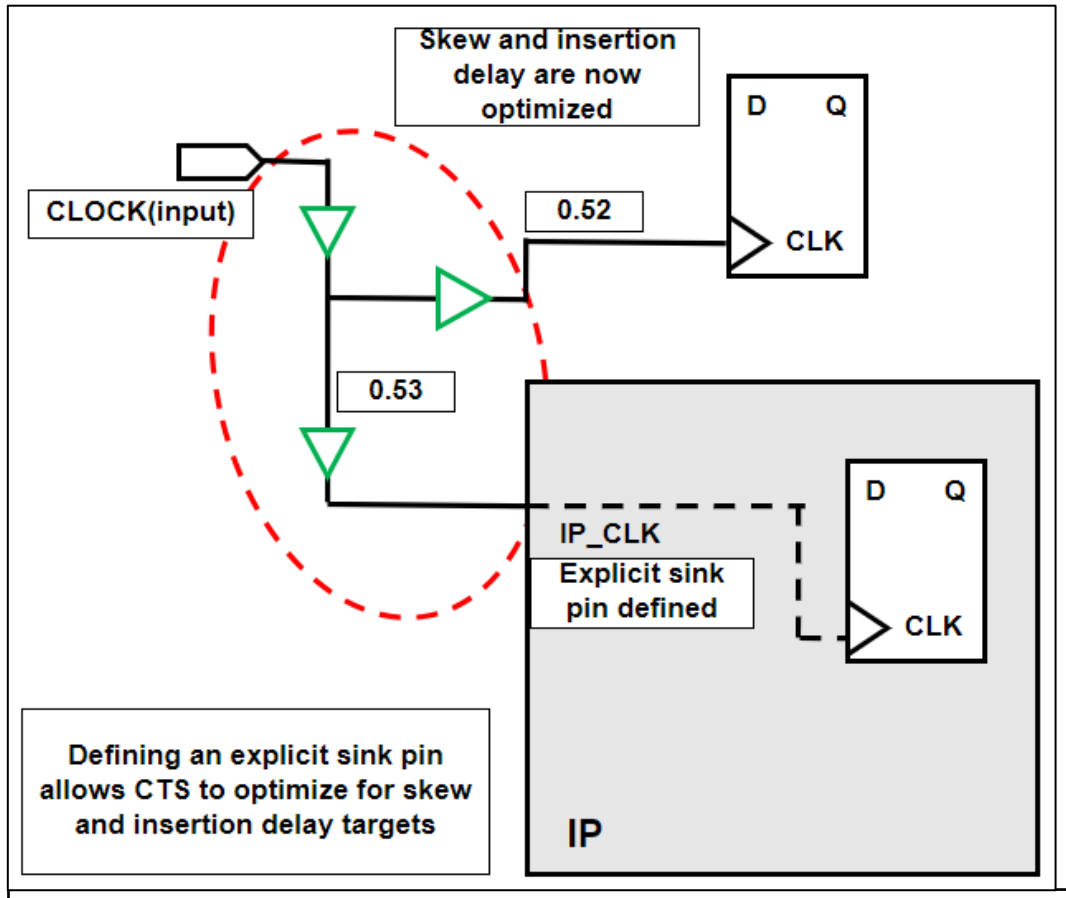

```
create_clock [get_ports pclk]  -period 8  -waveform {0 4}
create_generated_clock [get_ports sd_CK]  -name SD_DDR_CLK  -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK_1X]  -divide_by 1
```

- By default, CTS will not balance the skew of FFs that cross clock domains; generated clocks are considered a part of the master clock domain
- By default, CTS will balance the skew 'globally' per clock (This means that the skew of all sequential devices in the same clock domain, related or not, will be balanced)
- Two sequential devices are related if there is a data path connection between them

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Introduction to CTS : Defining an Explicit Sink Pin

- Defining an Explicit Sink Pin



- CTS has no knowledge of the IP-internal clock delay(it can only "see" up to the sink pin)

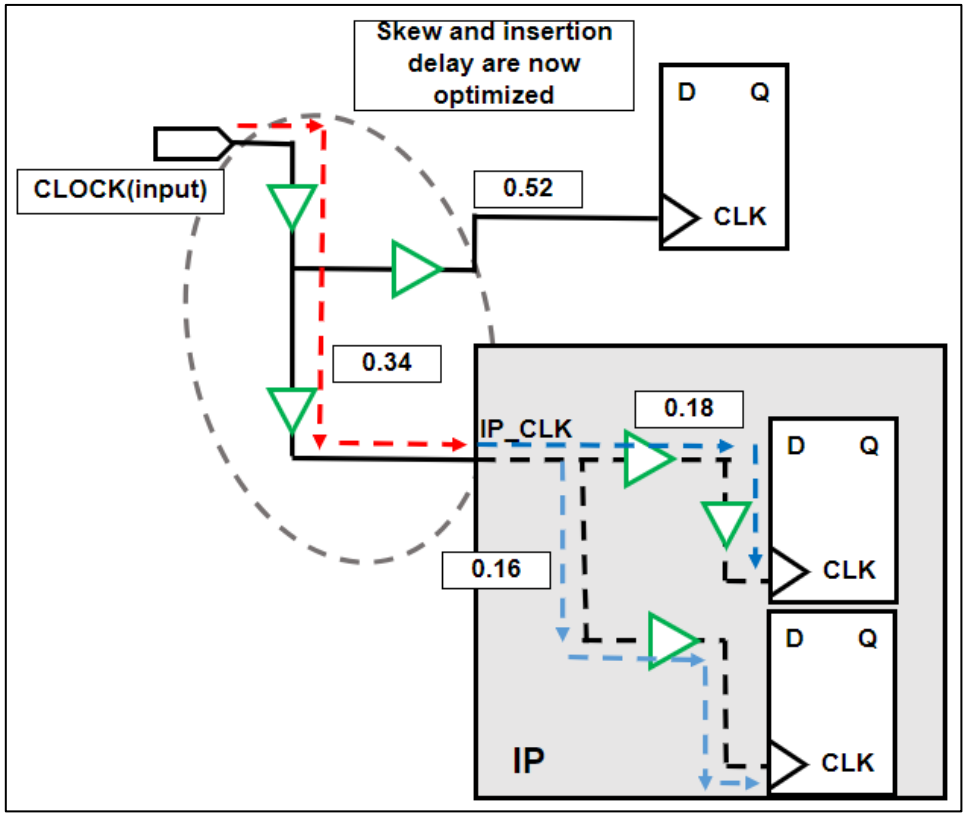
```
Set_clock_balance_points \
      -balance_points [get_pins IP/IP_CLK]
```

- In the above example, CTS is doing the right thing –balancing the insertion delays up to all stop pins
- CTS doesn't know that internal to the IP, there is a 0.18 delay to the actual clock pins of the flip flops

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Introduction to CTS : Defining an Explicit Sink Pin with Delay

- Defining an Explicit Sink Pin with Delay



- Defining an explicit sink pin with delay allows CTS to adjust the insertion delays based on the specification

```
set_clock_balance_points \
        -balance_points [get_pins IP/IP_CLK] \
        -delay 0.18 -late \
        -corner ss125c
```

- Specifying a positive delay prepones, specifying a negative delay postpones the clock pin
- This balance point constraint (with –delay option) can be used to manipulate the placement of the macro (If a macro has internal clock delays, the balance point delay values override the macro model's values)

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Introduction to CTS : Control CTS Cell Selection

- Control CTS Cell Selection
  - In the following example, only $cts_libcells are used to build the clock trees

```
set cts_libcells [get_lib_cells "*INV*_RVT */BUFX2_RVT */BUFX4_RVT \
                                */MUX*RVT */AO* ..."

set_lib_cell_purpose -exclude cts [get_lib_cells] ;#Specifies valid purposes for library cells
set_lib_cell_purpose -include cts $cts_libcells

set_dont_touch $cts_libcells false ;#set dont_touch attribute on cells and nets in the current design or on designs and library cells
```

  - In addition to buffers/inverters, include logically equivalent cells of any gates (MUX, AND, ICG, etc) and FF in the clock t ree
  - Always-on buffers should be added, to allow always-on CTS for MV(Multi Voltage) designs

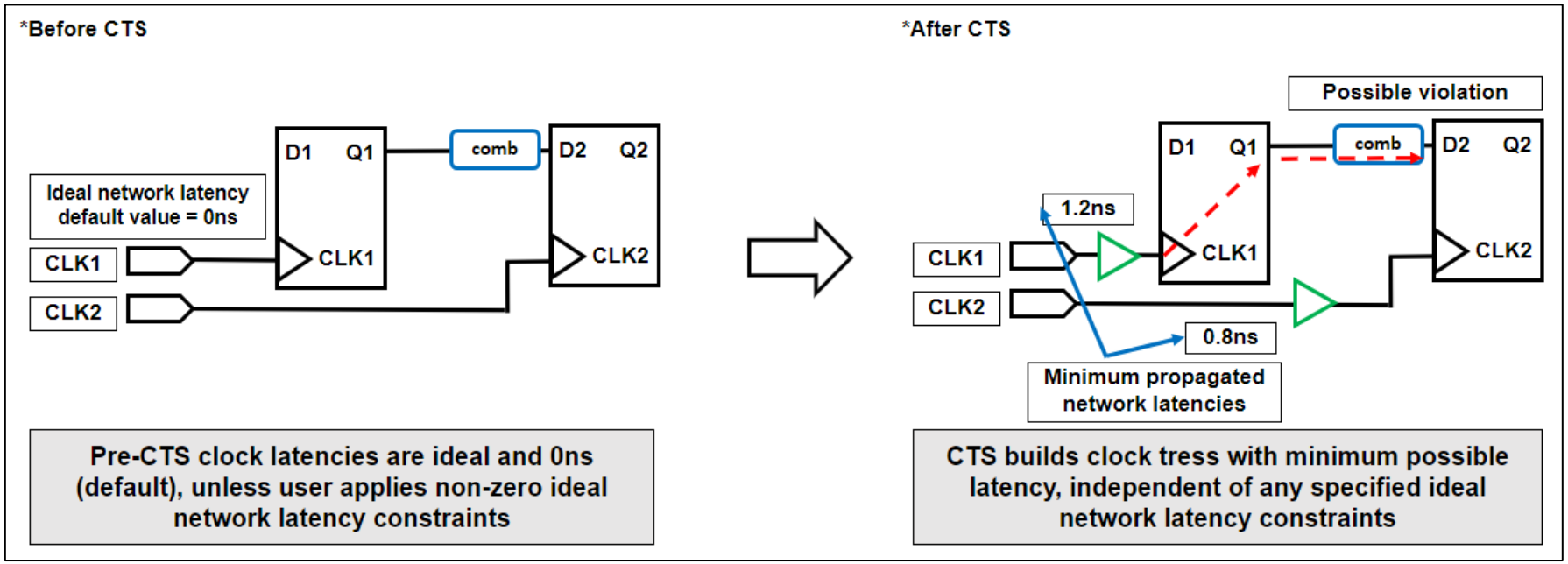
    (If a clock branch needs to feed through a shut-down voltage area, CTS inserts AO buffers otherwise, must go around it)

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Introduction to CTS : Default Clock Tree Targets

- Default Clock Tree Targets
  - The default target skew and target latency is 0ns for each clock domain

    (SDC uncertainty and network latency constraints are ignored)
  - Relax clock skew targets for non-timing-critical clocks (reduces overall buffer count, power and runtime)
  - Specify network latency targets to help Post CTS timing



- CLK2 arrives at 0.8ns latency, which is 0.4 earlier than CLK1, there may be a violation on timing paths going from CLK1 to CLK2 (If there are no violations, then there is no issue and no further steps need to be taken)

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Introduction to CTS : Control CTS Cell Selection

- Control CTS Cell Selection
  - You need to ensure that $cts_libcells includes LEQs of existing cells and ICGs, otherwise no resizing takes place
  - To help with identifying existing clock cells, use (derive_clock_cell_references)

```
set_lib_cell_purpose -include cts $cts_libcells ;# ex) buf, inv, ff, ao...

derive_clock_cell_references -output cts_leq_set.tcl ;# This command identifies the clock gates and ICGs in the design
```

  - The identified LEQs will be listed in the file cts_leq_set.tcl

```
# References enabled for CTS:

# References already available for CTS:
#set_lib_cell_purpose -include cts {gtech/GTECH_OA21}
#set_lib_cell_purpose -include cts {gtech/GTECH_AO22}
#set_lib_cell_purpose -include cts {gtech/GTECH_MUX2}
#set_lib_cell_purpose -include cts {gtech/GTECH_FD1}
#set_lib_cell_purpose -include cts {gtech/GTECH_FD1S}
#set_lib_cell_purpose -include cts {gtech/GTECH_FD2S}
#set_lib_cell_purpose -include cts {gtech/GTECH_FD4S}
#set_lib_cell_purpose -include cts {gtech/GTECH_LD1}


# Disabled references in CTS
# 1. List of references disabled due to dont_touch set on them:

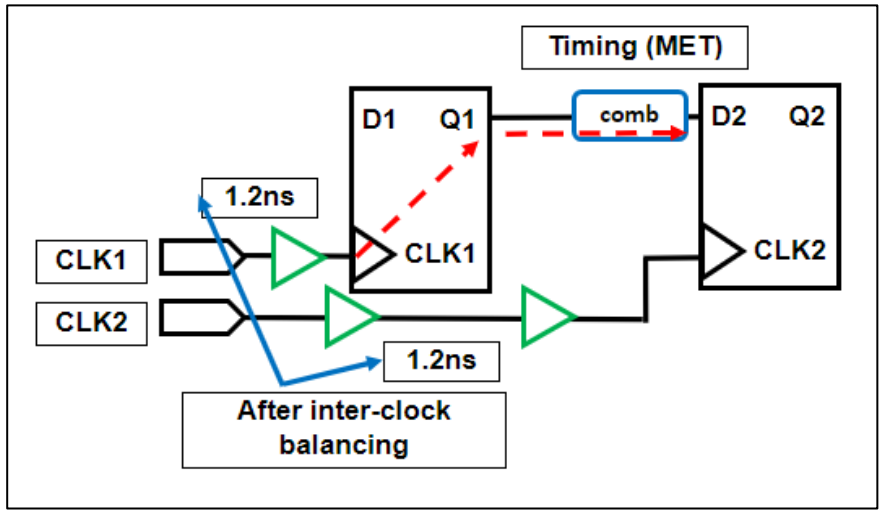
# 2. List of references disabled due to user driven exclude set for CTS purpose:
#    set_lib_cell_purpose -include cts [get_lib_cell saed32hvt/AO22X1_HVT]
#    set_lib_cell_purpose -include cts [get_lib_cell saed32hvt/AO22X2_HVT]
```


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Introduction to CTS : Inter-Clock Skew Balancing

- Inter-Clock Skew Balancing
  - Instead of specifying target latencies for each clock, it may be easier to balance clock domains instead
  - Create_clock_balance_group creates a balancing group in the current mode

```
create_clock_balance_group -name group1 -objects [get_clocks "CLK1 CLK2"]
#This command creates a new clock balance group in which all clocks have the similar insertion delay or
#differences by offset values if '-offset_latencies' is specified
```



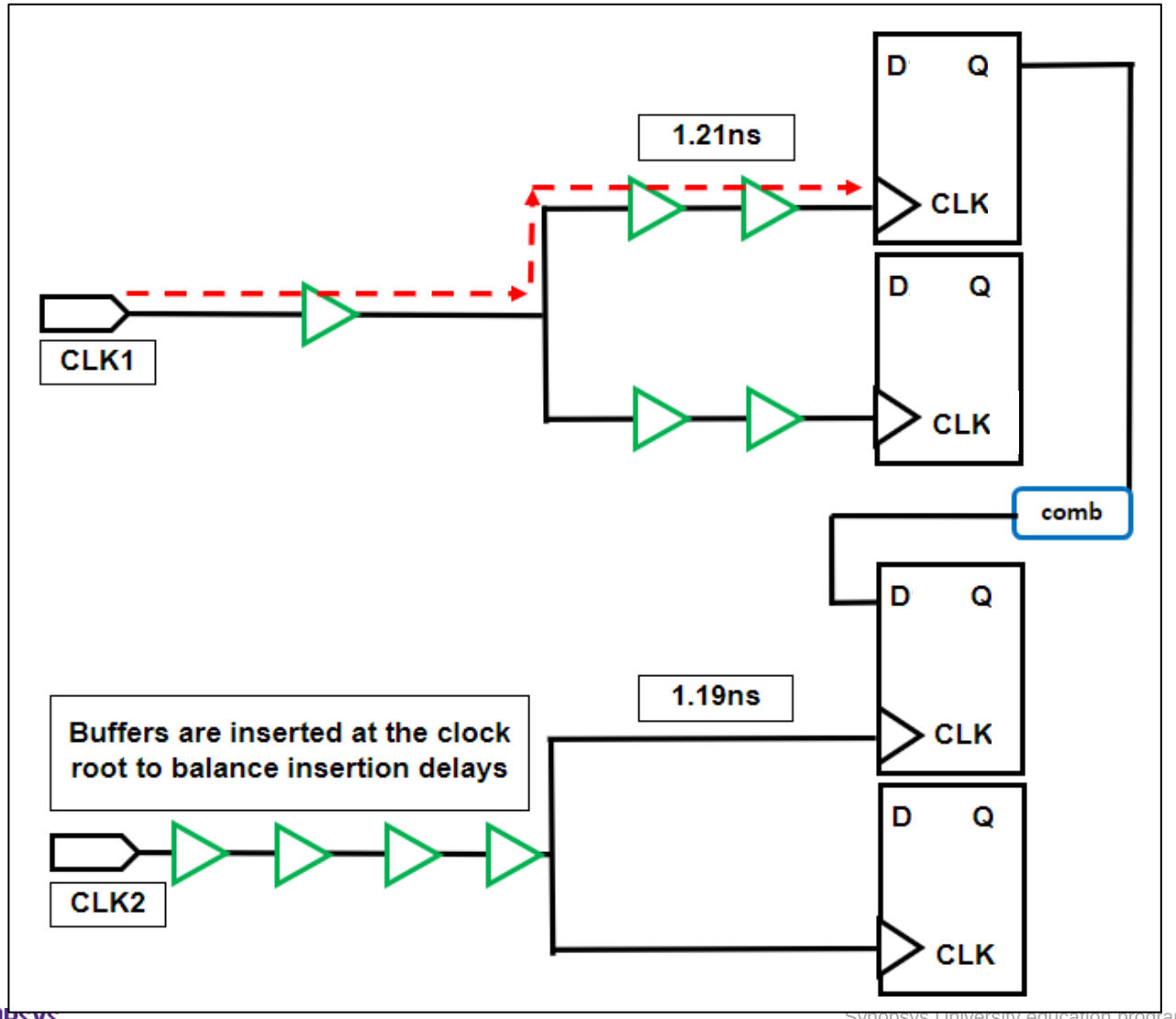
  - You can let the tool determine the clock groups that need to be balanced :

```
derive_clock_balance_constraints -slack_less_than -0.2
#This command collects a set of clocks which have timing slack smaller than the threshold
report_clock_balance_groups
#report clock balance groups
```


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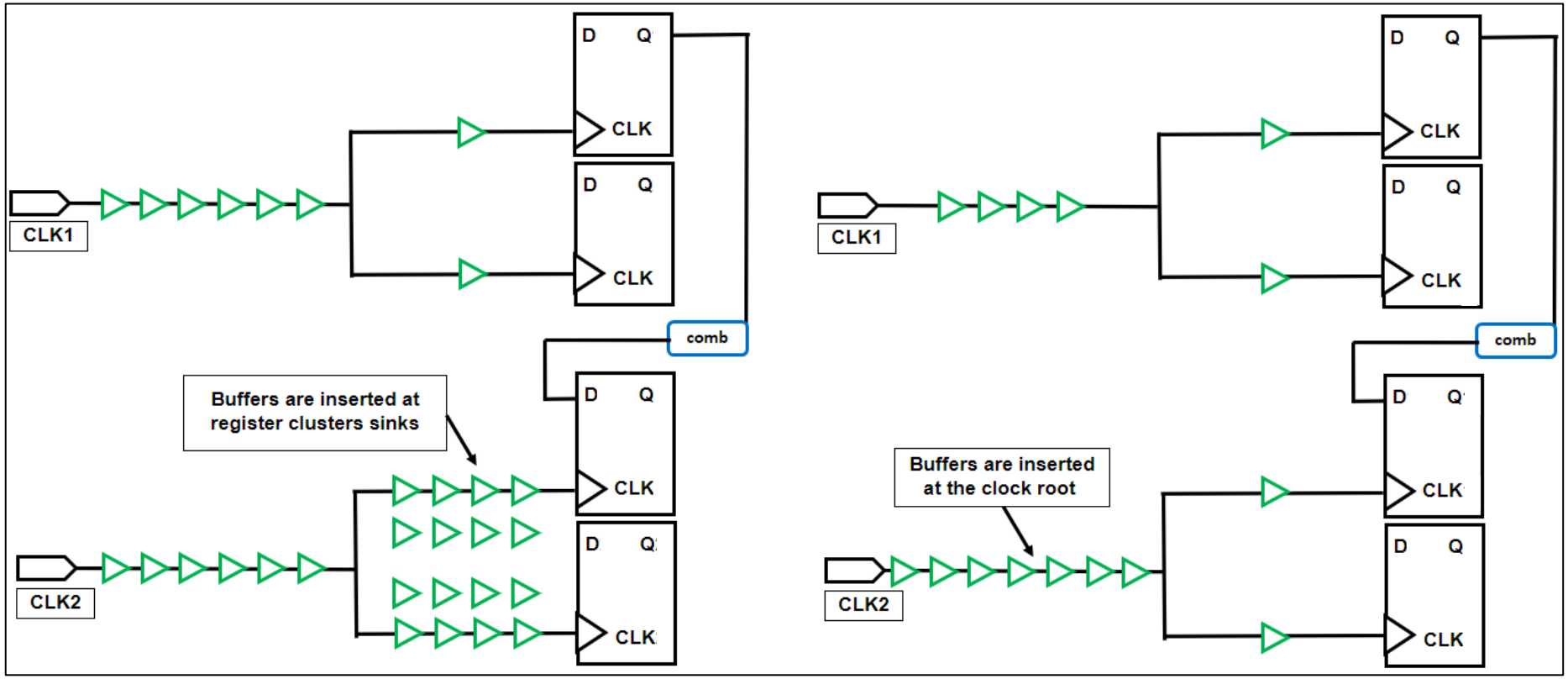
Introduction to CTS : Balancing occurs during "clock_opt"

- Balancing occurs during the "build_clock" stage



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Introduction to CTS : Inter-Clock Balancing is important for CCD



- Without inter-clock balancing, CCD will try to meet timing by skewing individual sinks
  - Longer runtime, many more buffers

- With inter-clock balancing, CTS inserts buffers at the root
  - CCD fine-tunes individual sinks afterwards
  - Shorter runtime, fewer buffers

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Introduction to CTS : Propagation Limitation

- Propagation Limitation of "dont_touch" on Clock Networks
  - The dont_touch attribute will not propagate past filp-flops downstream from the set_dont_touch_network point
  - If downstream propagation is desired, apply additional "dont_touches" :

```
set_dont_touch [get_cells "*mega_shift_reg*"]
#This command sets the dont_touch attribute on cells and nets in the current
#design or on designs and library cells
set_dont_touch_network -clock_only [get_pins "*mega_shift_reg*/D"]
#This command sets the clocks, pins, or ports in the current  design  as
#source of network  dont_touch
```

  - dont_touch propagates through the entire transitive fanout of the specified clocks :

```
mark_clock_trees -dont_touch -clocks [get_clocks "sys_clk"]
#This command marks clock attributes on clock  objects  to  prevent  the
#subsequent optimization programs to modify the built clock trees
```

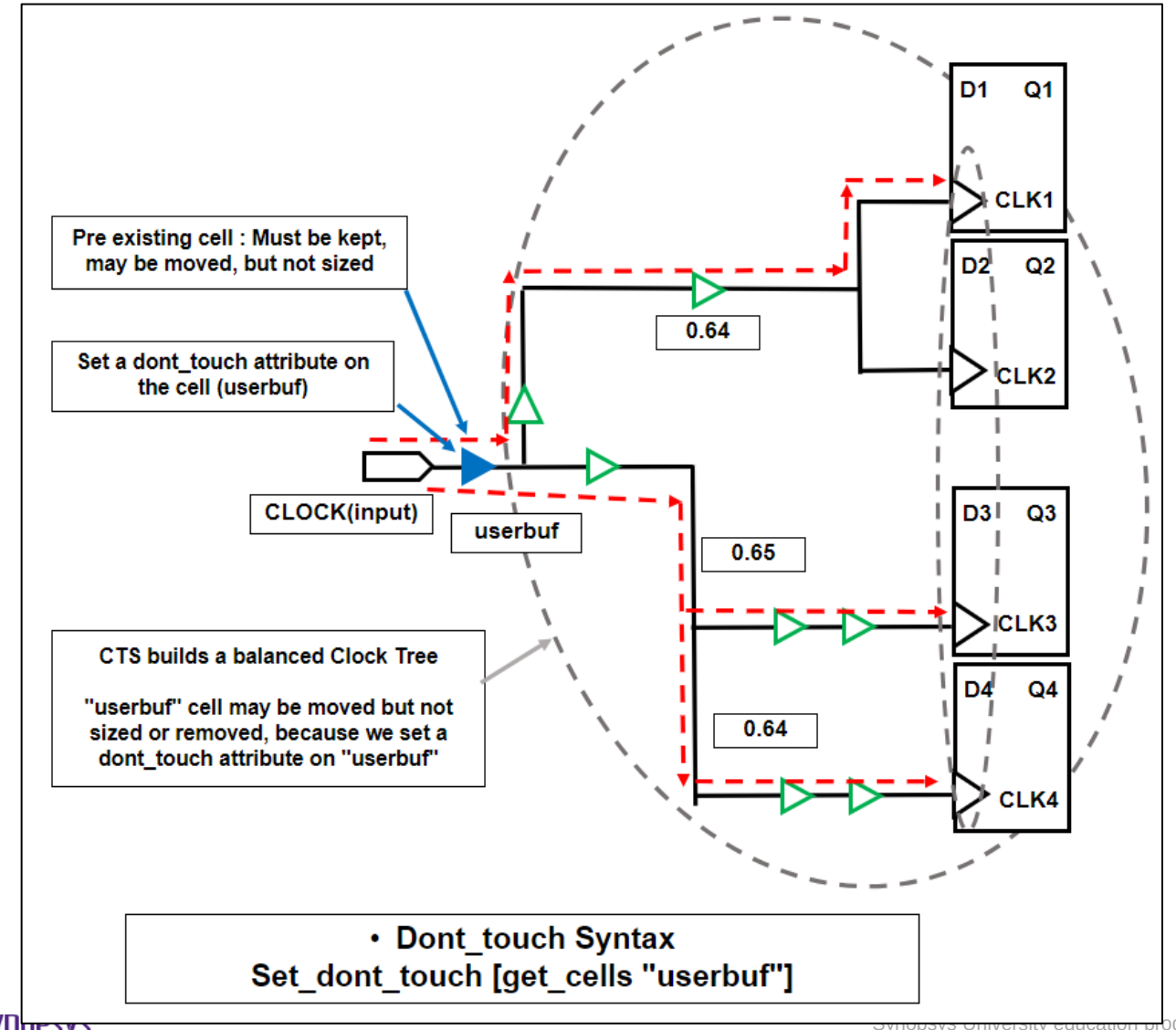
    - ❖ mark_clock_trees can only be applied to an entire clock domain, not to a pin

```
mark_clock_trees
               [-clock clock_list]
               [-clear]
               [-dont_touch]
-clock clock_list
   Marks only the clock trees specified in clock_list. By default, the command marks for all currently defined
   clocks in all active modes.
-clear
   Removes  the  clock  attributes on clock objects. If the "-dont_touch" option is specified, the tool
   removes the dont_touch attribute on clock objects.
-dont_touch
   Sets dont_touch attribute on clock objects, including cells and nets in the clock trees.
```


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Introduction to CTS : Preserving Pre-Existing Cells

- dont_touch cells

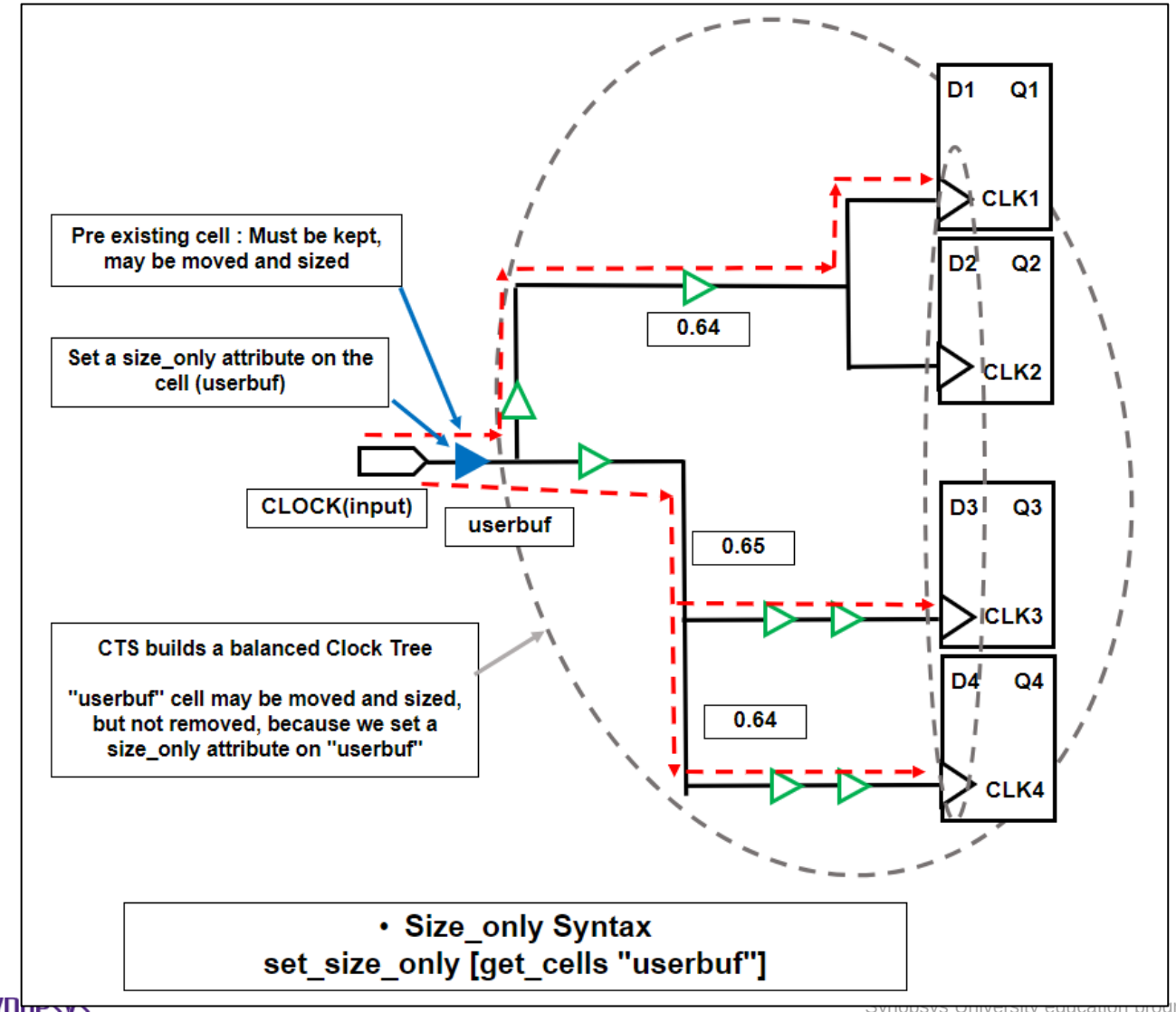


- set_dont_touch command sets the dont_touch attribute on cells and nets in the current design or on designs and library cells (dont_touch attribute prevents modification or replacement of these objects during optimization)
- Setting the dont_touch attribute on a hierarchical cell implies a dont_touch attribute on all cells below it
- Setting the dont_touch attribute on a library cell implies a dont_touch attribute on all instances of that cell

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Introduction to CTS : Preserving Pre-Existing Cells

- set_size_only cells



- Set_size_only command applies size_only settings to the specified leaf cells
- When you apply a size-only setting of true to leaf cell, optimization can perform only sizing operations on that cell
- A setting of false removes the size-only settings for the specified leaf cells, unless the presence of constraints on the cell is also causing it to be size_only

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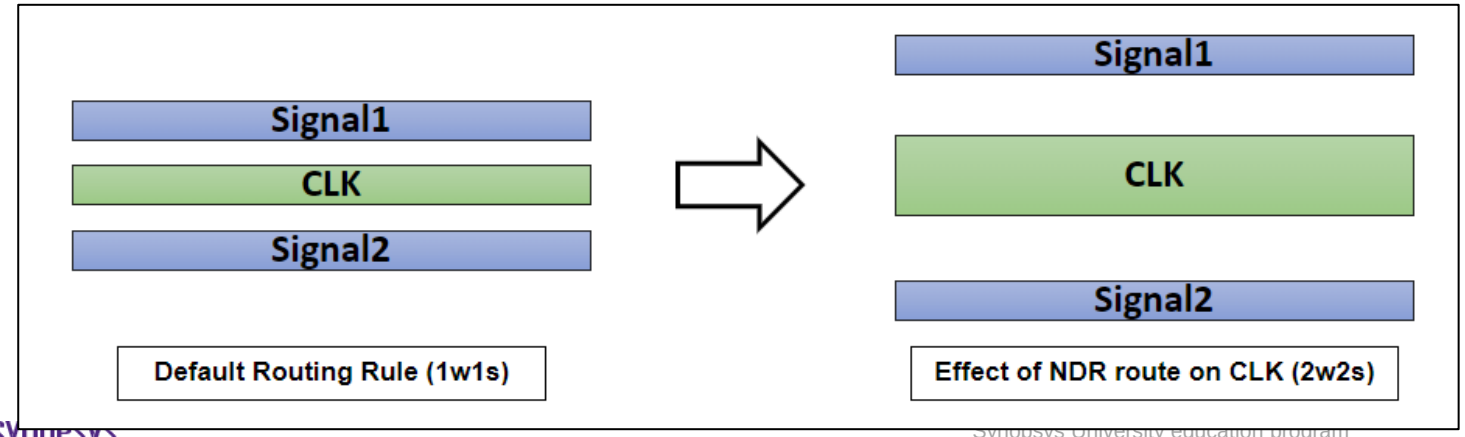
Introduction to CTS : NDR(Non-Default Routing Rules)

- **Clock nets can be routed using non-default routing (NDR) rules, e.g. double-spacing, double-width, net shielding**
  - NDR rules are treated as physical DRCs – reported as violations if not MET

- **NDR rules are often used to "harden" the clock, e.g. to make the clock routes less sensitive to cross-talk and electro-migration (EM) effects**

- **Crosstalk**
  Crosstalk is any phenomenon in electronics that occurs when a signal carried on one circuit or channel of a transmission system causes an undesirable effect in another circuit or channel

- **Electro-Migration (EM)**
  EM in VLSI is the movement of atoms caused by a current flowing through a substance (it can cause open-circuit and short-circuit failures in the interconnections of an integrated circuit)

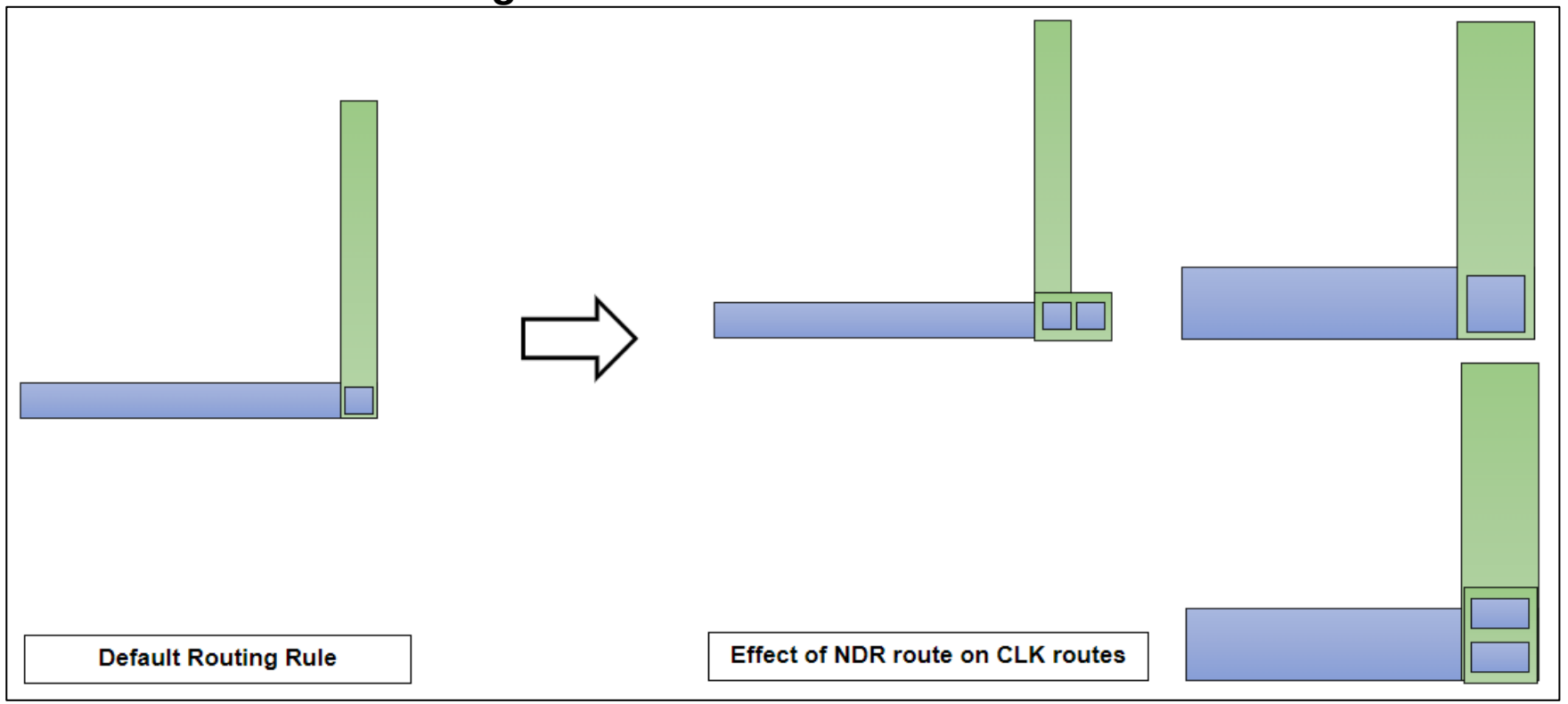


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Introduction to CTS : NDR(Non-Default Routing Rules)

- **Clock nets are hardened further by often requiring 100% via optimization to enhance reliability**
  - Minimum size single cut vias are replaced with:
    - Multiple-cut via arrays
    - Larger "square" or "bar" cuts

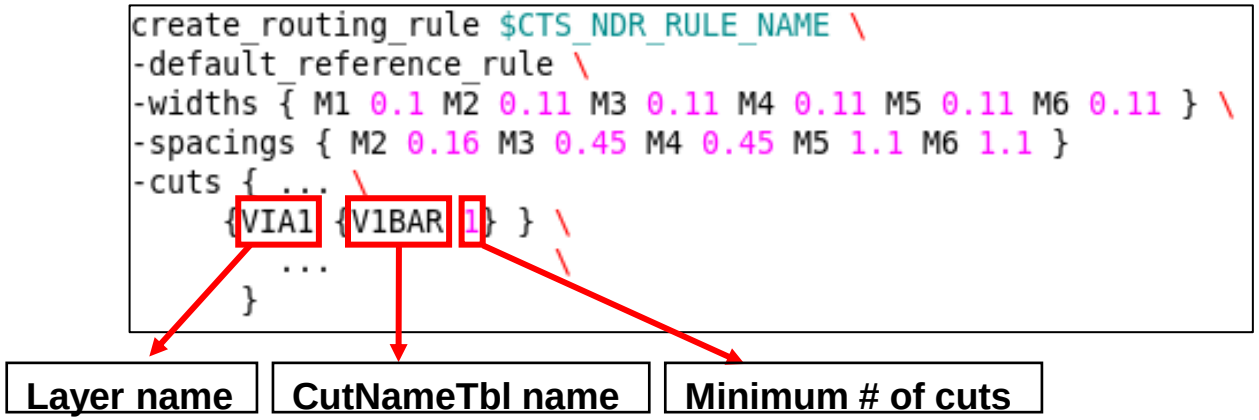
- **This is also defined using clock NDR rules**



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Introduction to CTS : Defining NDR & Via Rules

- **Define the NDR rules**



- With the –cuts option you use "symbolic" via names defined in the tech file as cutNameTb1
- This simplifies the NDR definition because one symbolic via can match many specific vias and arrays
- Typically, the spacing option uses 2s_2w (2spacing_2width)

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Introduction to CTS : Defining NDR & Via Rules (Tech file)

```
Layer           "VIA1" {
                layerNumber                     = 12
                maskName                        = "via1"
                isDefaultLayer                  = 1
                visible                         = 1
                selectable                      = 1
                blink                           = 0
                color                           = "cyan"
                lineStyle                       = "solid"
                pattern                         = "rectangleX"
                /*pitch                         = 0.36*/
                defaultWidth                    = 0.05
                minWidth                        = 0.05
                minSpacing                      = 0.07
                fatTblDimension                 = 2
                fatTblThreshold                 = (0.0, 0.16)
                fatTblDimension2                = 2
                fatTblThreshold2                = (0.0, 0.16)
                fatTblFatContactNumber          = (" 1, 5", " 3, 7, 1, 5", " 3, 7, 1, 5", " 4, 8, 3, 7, 1, 5")
                fatTblFatContactMinCuts         = (" 1, 1", " 1, 1, 2, 2", " 1, 1, 2, 2", " 1, 1, 2, 2, 4, 4")
                cutTblSize                      = 3
                cutNameTbl                      = (V1SQ, V1BAR, V1LG)
                cutWidthTbl                     = (0.05, 0.05,  0.1)
                cutHeightTbl                    = (0.05, 0.1,   0.1)
                sameSegAlignedUpperWireMaxSpacingThreshold = 0.08
                sameSegAlignedLowerWireMaxSpacingThreshold = 0.08
                sameSegAlignedCutMinSpacing = 0.07
    onWireTrack = 1
    onGrid = 1
}
```

```
ContactCode     "VIA12SQ_C" {
                contactCodeNumber               = 1
                cutLayer                        = "VIA1"
                lowerLayer                      = "M1"
                upperLayer                      = "M2"
                isDefaultContact                = 1
                cutWidth                        = 0.05
                cutHeight                       = 0.05
                upperLayerEncWidth              = 0.005
                upperLayerEncHeight             = 0.03
                lowerLayerEncWidth              = 0.03
                lowerLayerEncHeight             = 0.005
                minCutSpacing                   = 0.07
                maxNumRowsNonTurning            = 4
                unitMinResistance               = 0.0008
                unitNomResistance               = 0.0016
                unitMaxResistance               = 0.0024
}
```

- You can check ContactNumber, ContactMinCuts value in tech file
- You can check & use cutNameTbl, Layer Number, MinCuts value to define VIA Rules for CTS

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Introduction to CTS : Defining NDR & Via Rules

- **Defining Via NDRs with –vias**
  - **You may also use the –vias option**
  - **This will require you to specify the exact "ContactCode" names defined in the Tech File**
  - **This example is the equivalent of using –cuts option shown two pages earlier**

```
create_routing_rule $CTS_NDR_RULE_NAME \
-default_reference_rule \
-widths { M1 0.1 M2 0.11 M3 0.11 M4 0.11 M5 0.11 M6 0.11 } \
-spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1 }
-vias {  \
{VIAHH 1x2 NR} {VIAHH 1x2 R} {VIAHH 2x1 NR} {VIAHH 2x1 R} \
}
```

- ❖ The site count portion of the triplet is in the format m x n (rows x columns), where m is the horizontal site count (i.e., the number of columns in the via) and n is the vertical site count (i.e., the number of rows in the via), separated by the x character
- ❖ The rotation portion of the triplet is specified by r or nr, where r indicates a rotated via and nr indicates an unrotated via
- ❖ Only those modes and rotations that are explicitly specified are allowed by the non-default routing rule

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Introduction to CTS : Defining NDR & Via Rules

- **Define the NDR rules**

```
create_routing_rule $CTS_NDR_RULE_NAME ...

set_clock_routing_rules -rules $CTS_NDR_RULE_NAME \
-min_routing_layer M4 \
-max_routing_layer M6
}
```

- **Automatically tapers NDR (Non Default Rule) widths to connect to pins**
- **The lower metal layer in this list is a soft rule – will switch to lower metal layers if needed**

- **Note that the NDR was specified also for M1-M3**
  - Although the clocks will be primarily routed on M4 to M6 as shown above, they need to route on lower layers to connect to the standard cell pins, therefore should be covered by an NDR as well

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Introduction to CTS : Defining CTS-Specific DRC Values

- **Max transition and max capacitance design rules can be specified in two ways (Library & SDC)**
  - **Tool will always use the smallest value**

- **You can define your own CTS-specific DRC values:**

```
set_max_transition 0.5 <default value> -clock_path [all_clocks] ;# 0.5ns
set_max_capacitance 0.6 <default value> -clock_path [all_clocks] ;#0.6pF
```

- **Design rule constraints can be selectively applied per clock and per scenario**

  (applies to current scenario, by default);

```
set_max_transition 0.2 -clock_path \
                -scenarios "func.ss_n40c func.ff_n40c" \
                [get_clocks SYS_CLK]
```

❖ **Be careful not to over-constrain these clock tree DRC option settings, otherwise ICC2 may use more buff ers than necessary, and/or may not be able to meet all DRCs**

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Introduction to CTS : Remove Skew from Uncertainty

- **SDC constraints usually include**

  set_clock_uncertainty -setup | -hold <number>

  Applied to each clock

  ❖ Models estimated effects of clock skew, jitter and additional timing margin on setup/hold timing, pre-CTS

- **To avoid pessimistic timing analysis during clock_opt, remove or reduce uncertainty by the estimated sk ew before you run clock_opt :**

```
set_clock_uncertainty -scenarios "func.ff_n40c" -setup or hold <Jitter+Margin value> CLK1
set_clock_uncertainty -scenarios "func.ss_n40c func.ff_n40c" -setup or hold <Jitter+Margin value> CLK2
```

❖ If the uncertainty constraint values include jitter and/or margin, you can continue to take these effects into account, while also al lowing the tool to calculate the actual skew (by calculating actual propagated insertion delays and seeing the difference in arriv al time at register clock pins), by reducing the uncertainty constraint values by the estimated skew portion. Assuming, for exam ple, that the pre-CTS uncertainty constraints were

```
set_clock_uncertainty -setup 0.7 [get_clocks MY_CLK] (Skew 0.4; Jitter 0.2; Margin 0.1)
set_clock_uncertainty -hold  0.5 [get_clocks MY_CLK] (Skew 0.4; Jitter 0.0; Margin 0.1)
```

❖ It is recommended to change the constraints - before running CTS - as follows, to keep the same jitter/margin:

```
set_clock_uncertainty -scenarios [all_scenarios] -setup 0.3 [get_clocks MY_CLK]
set_clock_uncertainty -scenarios [all_scenarios] -hold  0.1 [get_clocks MY_CLK]
```

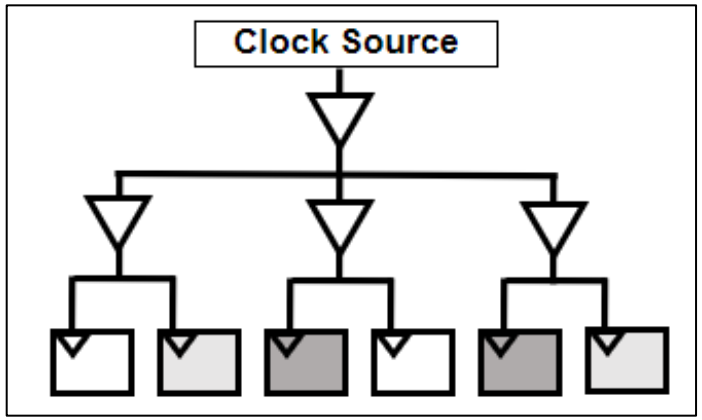

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WEEK 9-3 : Running CTS after Setup

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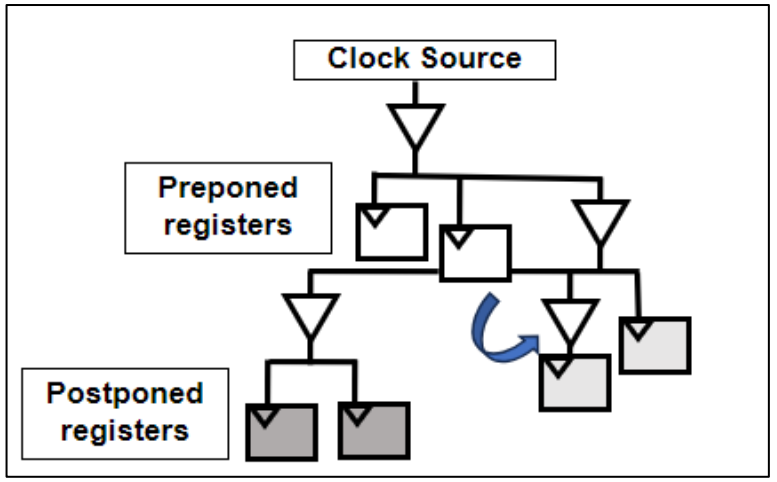
Introduction to CTS : Comparing Classic CTS vs CCD

Classic CTS



- Clock tree is built while ignoring the data paths
- Goal: Minimize skew (Allows full clock-cycle reg-to-reg timing)
- Data path is optimized post-CTS-Clock tree remains untouched

CCD (Concurrent Clock & Data)



- Clock tree is built with full knowledge of data path timing
- Goal : Meet setup/hold timing (reg-to-reg timing may be larger or smaller than a clock-cycle)
- Data path is optimized along with incremental clock tree modifications

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Introduction to CTS : Setup for Classic CTS and CCD flows

- **The classic CTS and CCD flows require common setup steps prior to executing CTS and optimization**
  - Controlling clock tree balancing
  - Handling pre-existing clock tree elements
  - Specifying non-default routing and via rules
  - Specifying clock timing and DRC constraints

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Introduction to CTS : Check the Design Ready for CTS

check_design –checks pre_clock_tree_stage

```
Running mega-check 'pre_clock_tree_stage' :
        Running atomic-check 'design_mismatch'
        Running atomic-check 'scan_chain'
        Running atomic-check 'mv_design'
        Running atomic-check 'legality'
        Running atomic-check 'design_extraction'
        Running atomic-check 'timing'
        Running atomic-check 'clock_trees'
```

- Creates an EMS database (Use the Message Browser to examine)
- The clock_trees check runs the atomic command check_clock_trees

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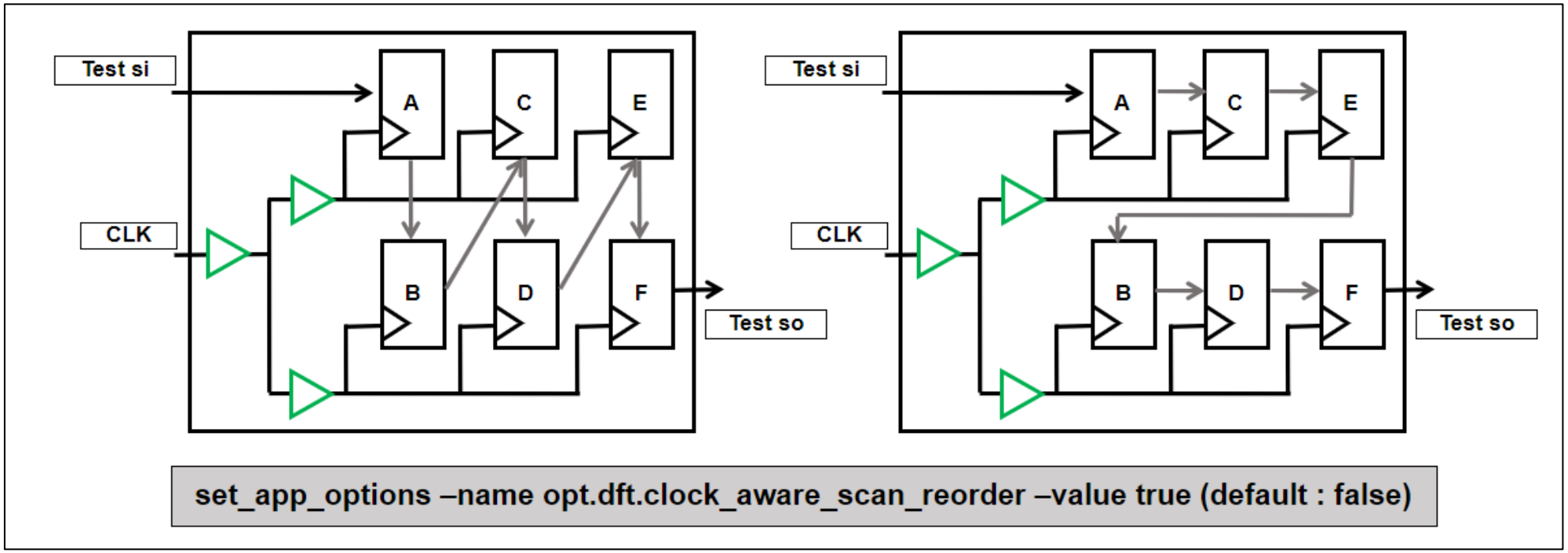
Introduction to CTS : CTS Goal and Flows

- **"check_design" checks for, and suggests solutions to address:**
  - **Clock Definitions and Propagation**
    - ❖ Generated clocks cannot be reached by master clock,etc
  - **Reference Cells**
    - ❖ Clock references with dont_touch, etc
  - **Skew Balancing**
    - ❖ Balancing conflicts between clocks, etc
  - **Multi-Voltage**
    - ❖ Clock nets with MV violations, no AON (Always ON) bufs or invs for CTS etc
  - **Capacitance and Transition Constraints**
  - **Other Issues: dont_touch nets in clock network, etc**

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Introduction to CTS : Minimize Hold Time Violations in Scan Paths

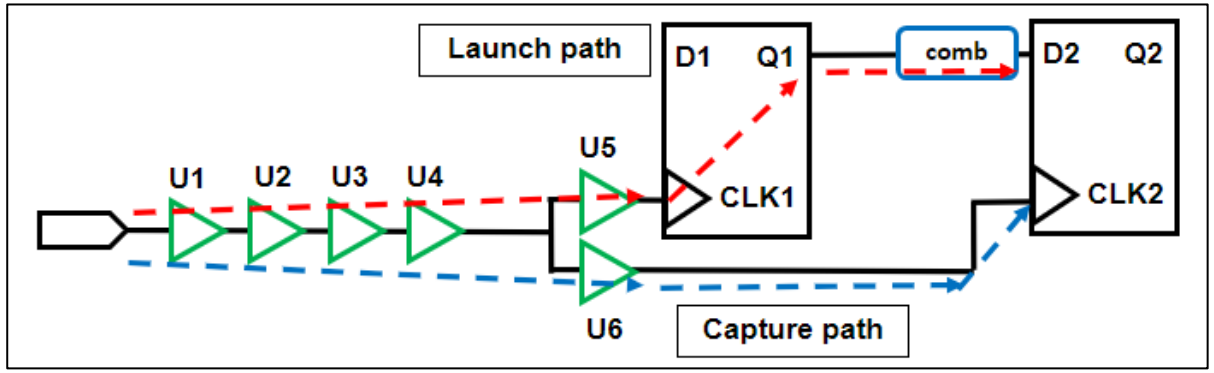
- **Enable scan chains reordering to minimize branch crossings**
- **Can reduce hold time violations in the scan chain**



- This app option should never be enabled before actual clock trees have been built
- If enabled earlier, you are instructing the tool to optimize scan chains for hold reduction, which should not be the case before the clock trees have been inserted (Therefore, make sure this is set to false prior to clock_opt)

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Introduction to CTS : Clock Reconvergence Pessimism



- **To reduce OCV effects, clock trees try to share as many buffers as possible**
- **Remove the pessimism from the timing calculation caused by shared clock paths (late/early for launch/capture calculation)**
  - ❖ **Otherwise, false data path timing will lead to pessimistic clock trees (especially for CCD)**

```
set_app_options -name time.remove_clock_recovergence_pessimism \
                -value true ;# Default : false
```

- By default, to account for OCV timing, the tool calculates delays of the launch versus the capture paths using late and early arrival, even for the common cells (U1-U4 above)
- The delays through the common cells can never be different when calculated as part of the launch versus the capture path (assuming the launch and capture clock triggering edges are the same - both positive or both negative)
- The artificial delay difference that is calculated for these shared cells creates pessimistic timing, and is called “clock reconvergence pessimism” (CRP)
- The amount of CRP is defined as the difference between the latest and earliest arrival times, through the shared or common cells, to the “common point” (For realistic timing analysis, the effect of CRP should be removed)

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Introduction to CTS : Clock Reconvergence Pessimism Removal

- **The pessimism is removed for setup timing by adding the CRP in the capture path, and by subtracting for hold**

Example *setup* timing report: Added *CRP* in capture path

```
clock SYS_CLK (rise edge)                          5.00     5.00
clock network delay (propagated)                   1.76     6.76
clock reconvergence pessimism                      0.08     6.84
clock uncertainty                                 -0.10     6.74
I_BLENDER_0/s4_op1_reg[30]/CLK (SDFFX1_LVT)        0.00     6.74 r
library setup time                                -0.26     6.58
data required time                                          6.58
```

For a *hold* report this would be negative

- **Here are default values of more CRP-related variables**

Time.crpr_remove_clock_to_data_crp = false

When this application option is set to true then CRP will be removed for all paths that fan out directly from clock source pins to the data pins of sequential devices.

When this option is set to false, only the CRP up to the clock source pin which fans out to the data pin of the sequential device would be removed. This is because the path up to a clock source pin is considered to be a clock path.

Time.clock_reconvergence_pessimism = normal

Determines how the value of the clock reconvergence pessimism removal(CRPR) is computed with respect to transition sense. Allowed values are as follows:

- normal (default value)
- same_transition

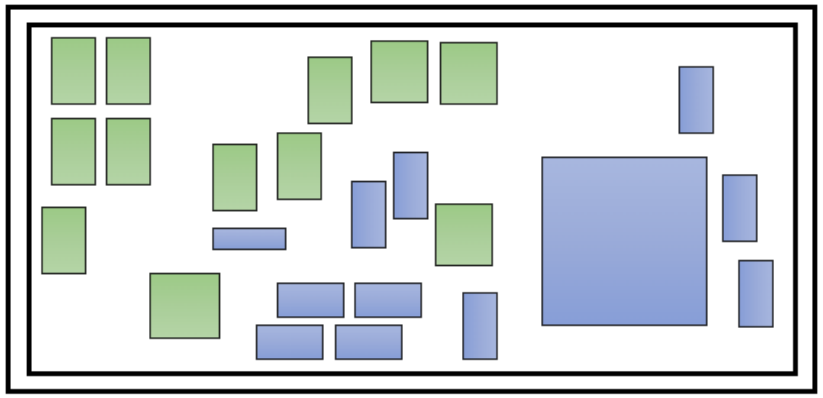
When set to **normal**, the CRPR value is computed even if the clock transitions to the source and destination latches are in different directions on the common clock path. It is computed separately for rise and fall transitions and the value with smaller absolute value is used.

When set to **same_transition**, the CRPR value is computed only when the clock transition to the source and destination latches have a common path and the transition is in the same direction on each pin of the common path. If the source and destination latches are triggered by different edge types, CRPR is computed at the last common pin at which the launch and capture edges match.

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Introduction to CTS : Congestion-Aware Initial CTS

- **Initial clock tree synthesis (build_clock) is not congestion aware (Based on virtual routing. By default)**
- **On large complex-floorplan designs, this may lead to:**
  - Pre- vs. Post-route clock skew, latency, and logical DRC degradation
  - Congestion hotspots
  - Route DRCs

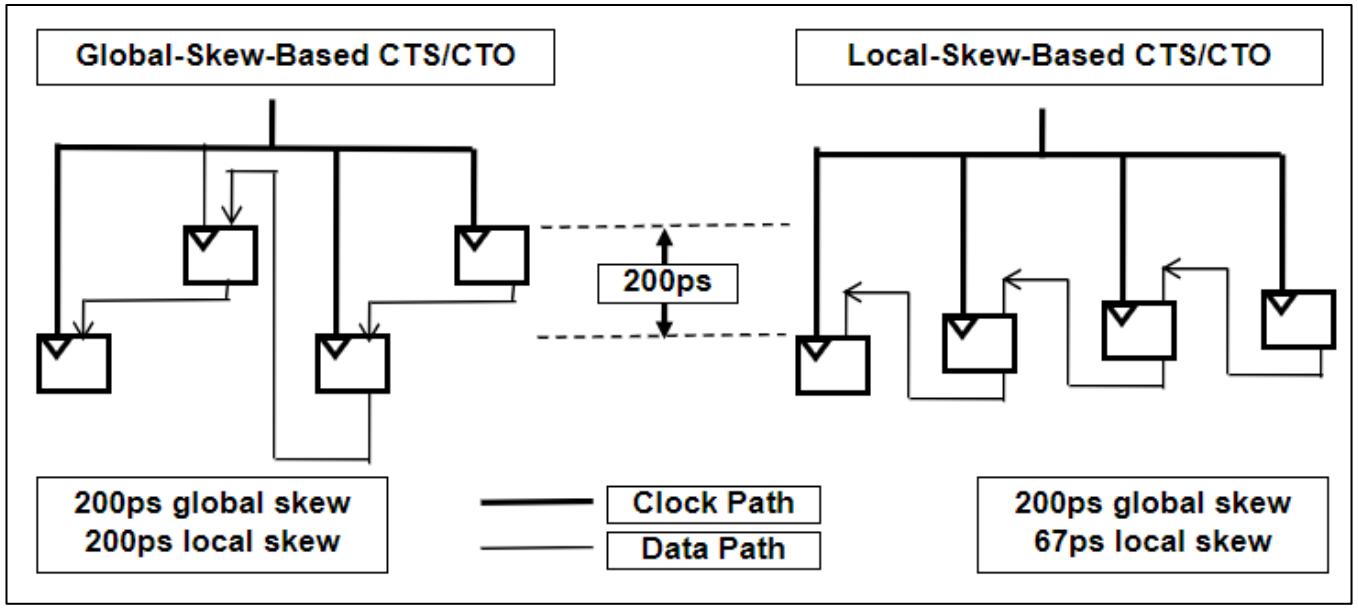


- **Enable global routing for congestion estimation and congestion-aware clock tree construction during the build_clock stage:**

```
set_app_options -name cts.compile.enable_global_route \
                -value true ;# Default : false
```


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Introduction to CTS : Local Skew Optimization



- **Performs timing aware clustering (as opposed to location based)**
- **Optimizes local skew directly for setup and hold timing critical pairs**
- **Can relax skew target (if slack is positive) to save clock buffer area**

- **Timing aware clustering:**
  Clusters timing-related clock tree nodes together to improve path sharing without degrading latency
- **Automatic derivation of target skew:**
  - ❖ Derives target skew for clocks based on timing QoR seen with early estimation of timing QoR. As a safe guard, derived value is limited to within 3% to 10% of clock period
  - ❖ CTS works to meet derived skew instead of trying to achieve “0” skew and thus reducing clock buffer/cell area
  - ❖ Local skew optimization ensures that timing related registers have minimal skew
- **Local skew optimization during CTO:**
  - ❖ After (traditional) global skew optimization stage in CTO, it now looks at timing QoR, and optimizes local skew for timing critical pairs to improve timing QoR.
  - ❖ Makes sure that skew/latency/DRCs across scenarios are not degraded.

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Introduction to CTS : Enable Local Skew CTS and CTO

- **To use local skew for classic CTS, you need to enable it specifically:**

```
set_app_options -name cts.compile.enable_local_skew \
                -value true ;# Default : false
set_app_options -name cts.optimize.enable_local_skew \
                -value true ;# Default : false
```

- **When using CCD, local skew optimization is always used**
  - ❖ **Not user-configurable**

- **set_app_options –name cts.compile.enable_local_skew –value true**

  When this option is true, clock tree synthesis will construct local skew friendly clock trees by considering timing interactions between the clock tree sinks

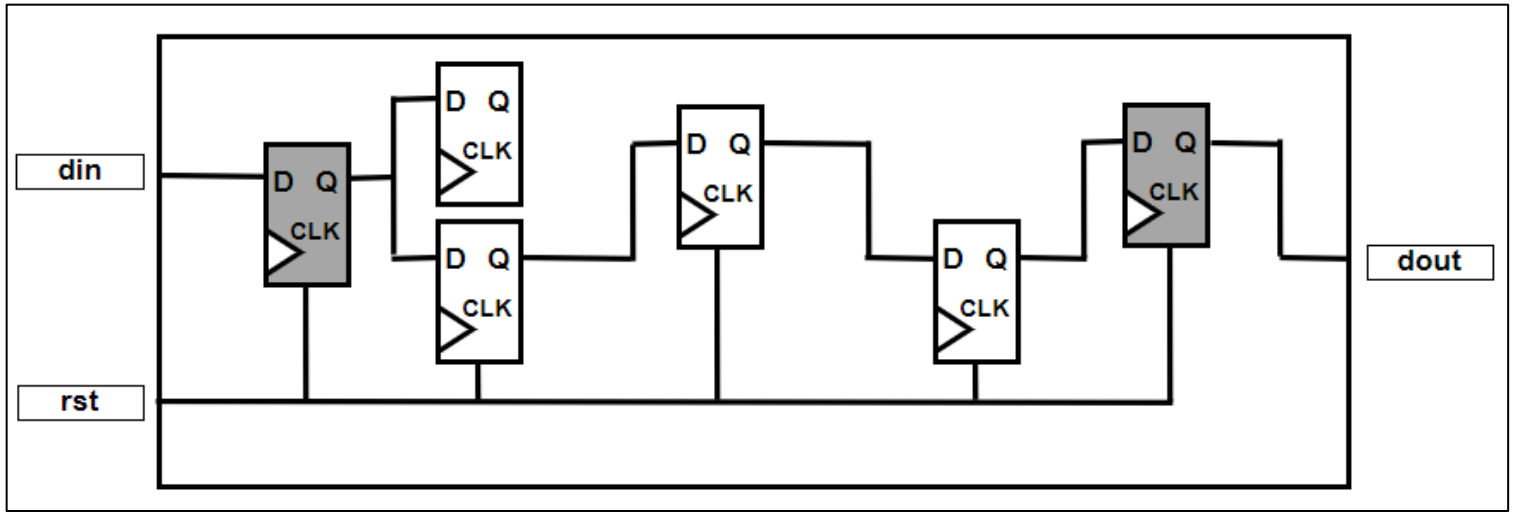
  By default this option is false and clock tree synthesis considers only the traditional metrics like latency, skew and area (In clock_opt, this option will be turned on automatically when clock_opt.flow.enable_ccd is true)

- **set_app_options –name cts.optimize.enable_local_skew –value true**

  When this option is true, clock tree optimization will minimize local skew of the clock trees. By default, this option is false and clock tree optimization targets global skew only (In clock_opt, this option will be turned on automatically when clock_opt.flow.enable_ccd is true)

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Introduction to CTS : CCD Optimization and Boundary Registers



- **It is discouraged to exclude all boundary registers, as this may severely impact CCD's optimization potential**
- **Use only when absolutely necessary**

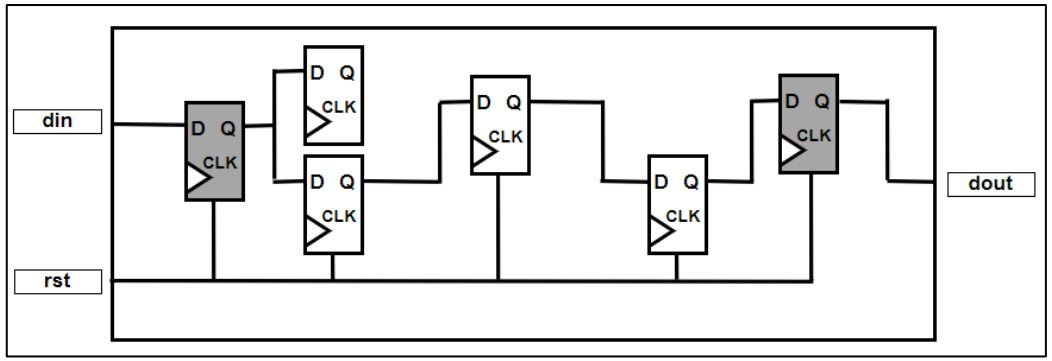
```
ccd.optimize_boundary_timing ;#set to false (Default : true)
```

- **By default, CCD skews all registers (including boundary FFs) for timing**
- **To prevent latency adjustment on boundary registers, use:**

- **When the option is false:**
  - **Only internal flip-flops are available for CCD**
    - The tool performs boundary identification to differentiate between the boundary and internal flip-flops
    - If your design contains ports that connect to a large number of flip-flops, such as scan enable and reset ports, you might want to exclude these ports from the boundary identification; otherwise, most of the flip-flops are considered boundary flip-flops and the CCD scope is very limited
  - **Latencies of the boundary flip-flops will not be adjusted**
  - **The clock tree fan-in to the boundary flip-flops can be optimized if the clock path is shared with an internal flip-flop**

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Introduction to CTS : CCD Optimization and Boundary Registers



- **Primary ports are used to differentiate between boundary and internal FFs**
- **Ignore scan and reset ports by using:**

```
ccd.ignore_scan_reset_for_boundary_identification ;#set to true (Default : false)
#The ccd.ignore_scan_reset_for_boundary_identification application option is used
#to ignore the connectivities on scan/reset pins when concurrent clock and data optimization
#(CCD) performs boundary flop identification
```

❖ **To ignore specific ports for boundary identification, use:**

```
set_app_options \
        -name ccd.ignore_ports_for_boundary_identification
        -value {scan_en ...}

#The ccd.ignore_ports_for_boundary_identification application option
#specifies the ports to exclude when concurrent clock and data optimization
#(CCD) performs boundary identification
```

- **If you specify the "ccd.ignore_scan_reset_for_boundary_identification" option, is used to ignore the connectivities on scan/reset pins when concurrent clock and data optimization (CCD) performs boundary flop identification**
- **The reset/scan ports are identified based on their connectivity to scan/reset pins on the flip-flops**

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Introduction to CTS : CCD Optimization and Boundary Registers

- **Although CCD setup optimizations are hold-aware, hold violations can be introduced to improve setup ti ming**
  - Usually, data path optimizations are able to address these hold violations
- **You may want to prioritize hold during CCD if**
  - Fixing hold in your design is difficult/ not possible
  - Your design is area-critical, and the hold-buffer area increase is to be avoided
- **To specify hold criticality during CCD, use:**

```
ccd.hold_control_effort ;#Default : low
```

  - Set to medium, high or ultra to reduce hold degradation
  - Use only if hold timing is critical, since setup optimizations may degrade
  - Only affects "final_opto", as well as CCD optimization during "route_opt"

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Introduction to CTS : Classic CTS and CCD Execution

- **Clock tree synthesis, clock tree detail routing, data path optimization and signal global routing are all exe cuted by:**

```
clock_opt
```

  - ❖ CCD is enabled by default (clock_opt.flow.enable_ccd)

- **The three stages of clock_opt are:**
  - The three stages of clock_opt are:



- **You can control which stages are executed with -from/-to**
  - ❖ **For example, to perform only clock tree synthesis (CTS+CTO):**

```
clock_opt –to build_clock
```


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Introduction to CTS : CCD Flow Stages

```
set_app_options -name clock_opt.flow.enable_ccd -value true ;# Default : true
```

Command	What does it do?
clock_opt -to build_clock	• Builds timing-driven GR clock trees, and performs inter-clock balancing, to improve setup/hold timing and DRCs • Updates I/O latencies
Clock_opt \ -from route_clock \ -to route_clock	• Detail-routes the clock nets
Clock_opt -from final_opto	• Performs global routing of all signal nets (GRE) • Optimizes the data path logic and simultaneously optimizes clock logic to further improve setup/hold timing and DRCs

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Introduction to CTS : Classic CTS Flow Stages

- **To enable classic CTS, disable CCD**

```
set_app_options -name clock_opt.flow.enable_ccd -value false
```

Command	What does it do?
clock_opt -to build_clock	• Builds GR clock trees, and performs inter-clock balancing, for minimum skew • Updates I/O latencies
Clock_opt \ -from route_clock \ -to route_clock	• Detail-routes the clock nets
Clock_opt -from final_opto	• Performs global routing of all signal nets (GRE) • Performs data path optimization to meet timing and DRCs

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Introduction to CTS : Post-CTS(Post-route Clock Tree Optimization)

- **Problems:**
  - Possibility of clock tree QoR degradation (skew, latency, logical DRC) due to miscorrelation between global route (build_clock) and detail route (route_clock)
  - Clock tree QoR can further degrade due to coupling capacitance and crosstalk effects

- **Solution: Post-route clock tree optimization**
  - Works on detail-routed clock trees to address these degradations
  - Optimizations performed include on-route buffer insertion, buffer sizing, and gate sizing

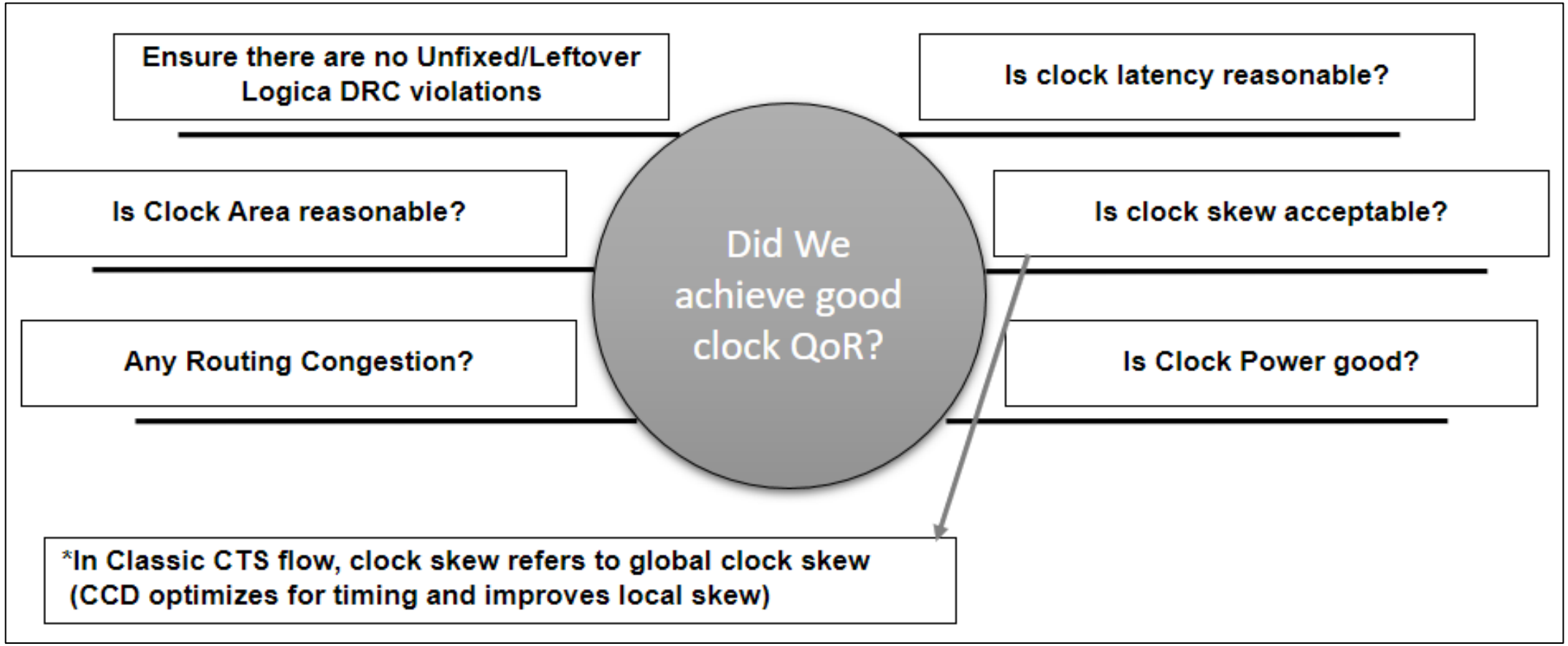
```
synthesize_clock_trees -postroute
```

- **Post-route CTO can also be performed after signal routing**
- **Do not use if you have used CCD**

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Introduction to CTS : Analyzing the Clock Tree

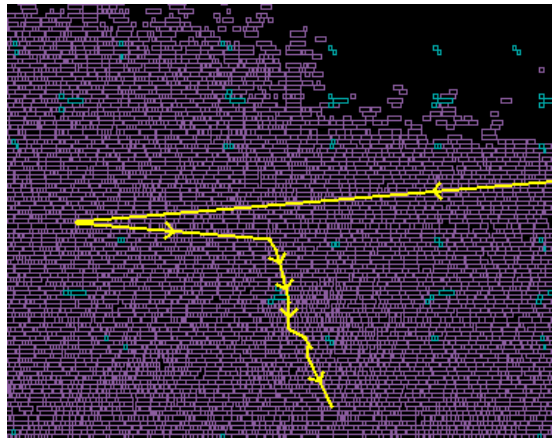
- **After clock_opt build_clock, analyse the clock tree before proceeding to data path optimization and routi ng stages**



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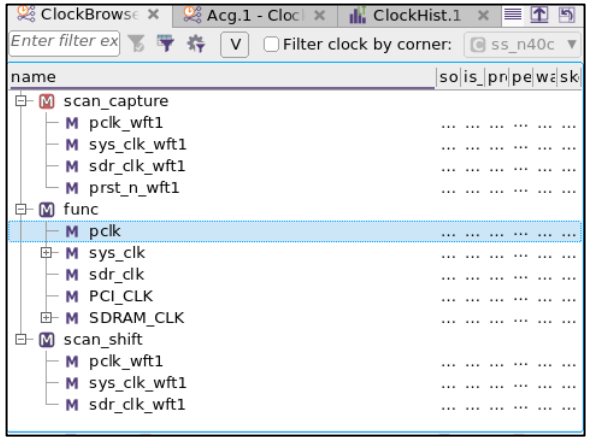
Introduction to CTS : Analysis Using the CTS GUI

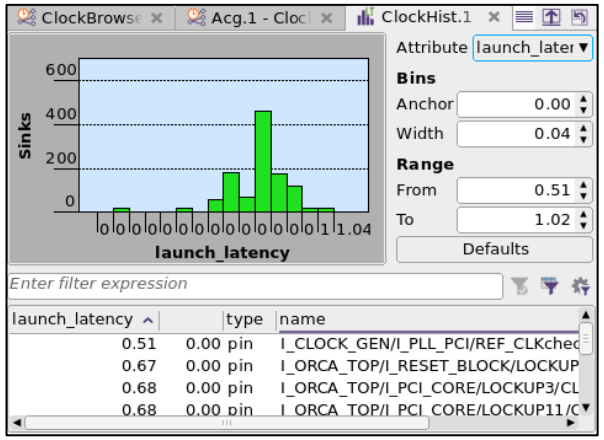
- **CTS browser**
  - **Identify clock tree object properties and attributes**
  - **Traverse clock tree levels**

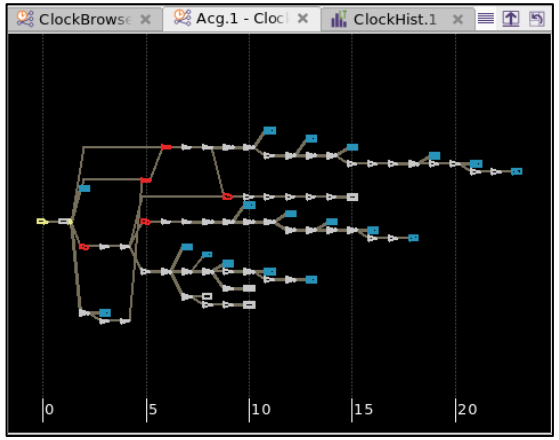


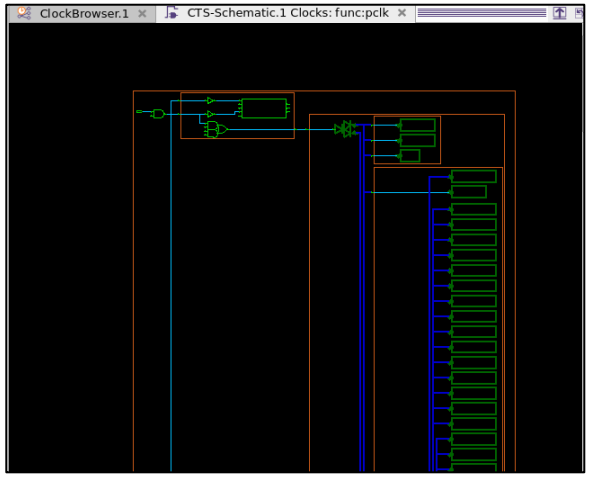
Highlight Longest Path From Source
Highlight Shortest Path From Source
Highlight Fanout in Layout

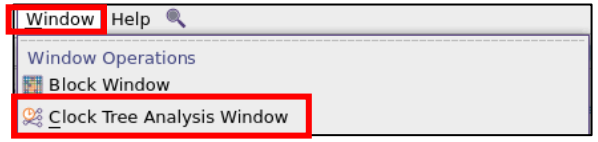
- **CTS schematic**
  - **Trace clock paths**
  - **Cross-probe with layout view**











Launch Latency Histogram of Selected Corner
Capture Latency Histogram of Selected Corner

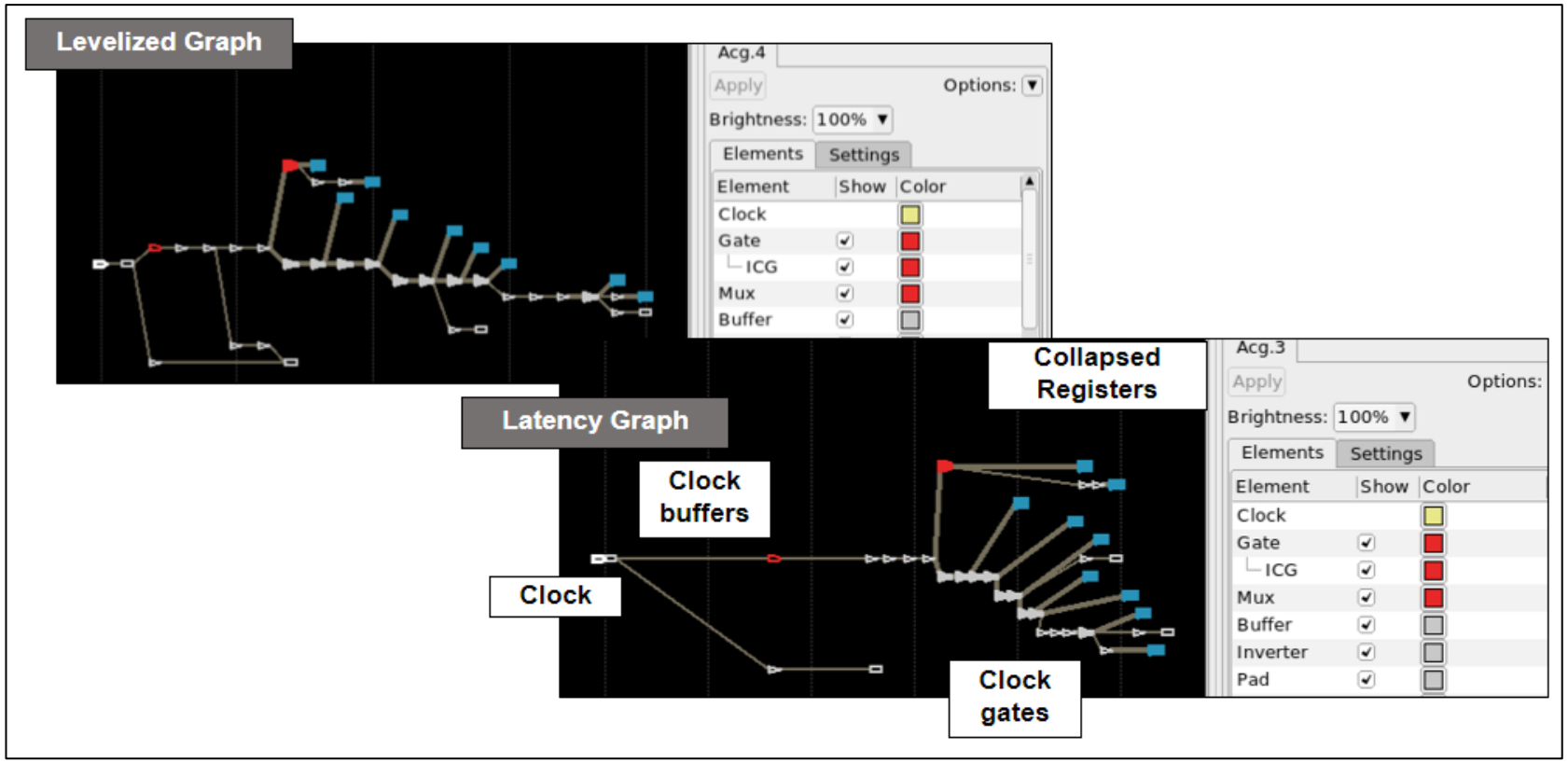
Clock Tree Levelized Graph

Clock Tree Schematic

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Introduction to CTS : GUI Analysis

- **Clock Tree Levelized and Latency Graphs**
  - **Configure the clock objects you want to see to focus on what you need**

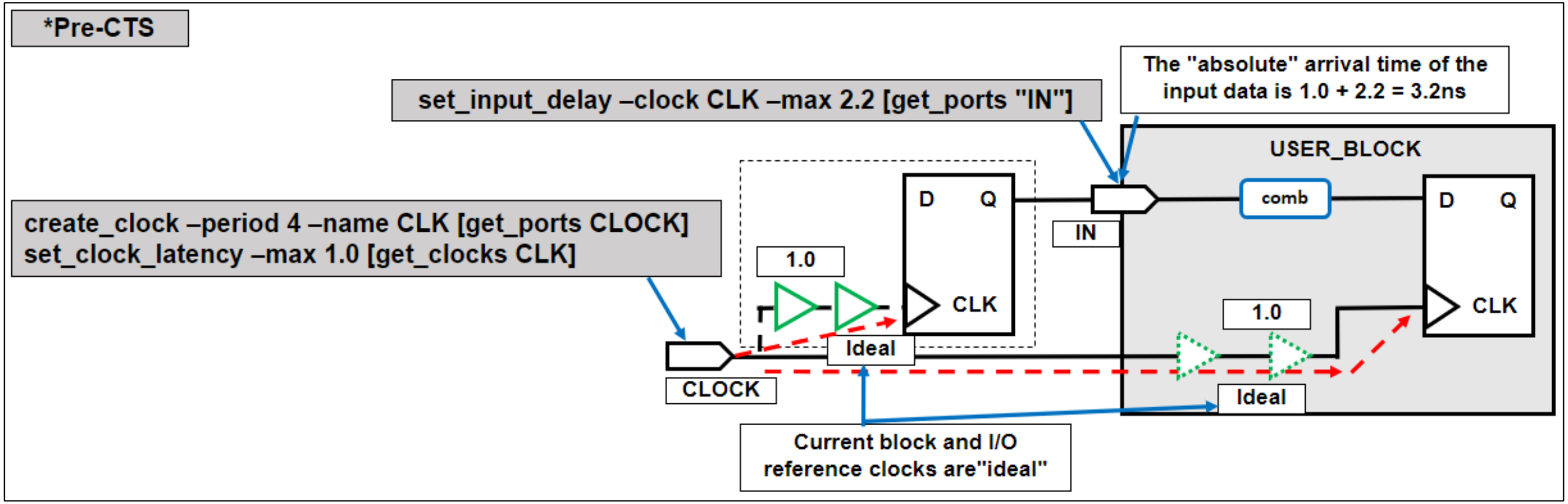


- **On the bottom, you see that gates, buffers and muxes will be colored the same**
- **You can choose difference colors to make them easier to distinguish**
- **Abstract clock tree graph works on the current scenario**

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Introduction to CTS : Ideal Clock Latencies (Pre-CTS)

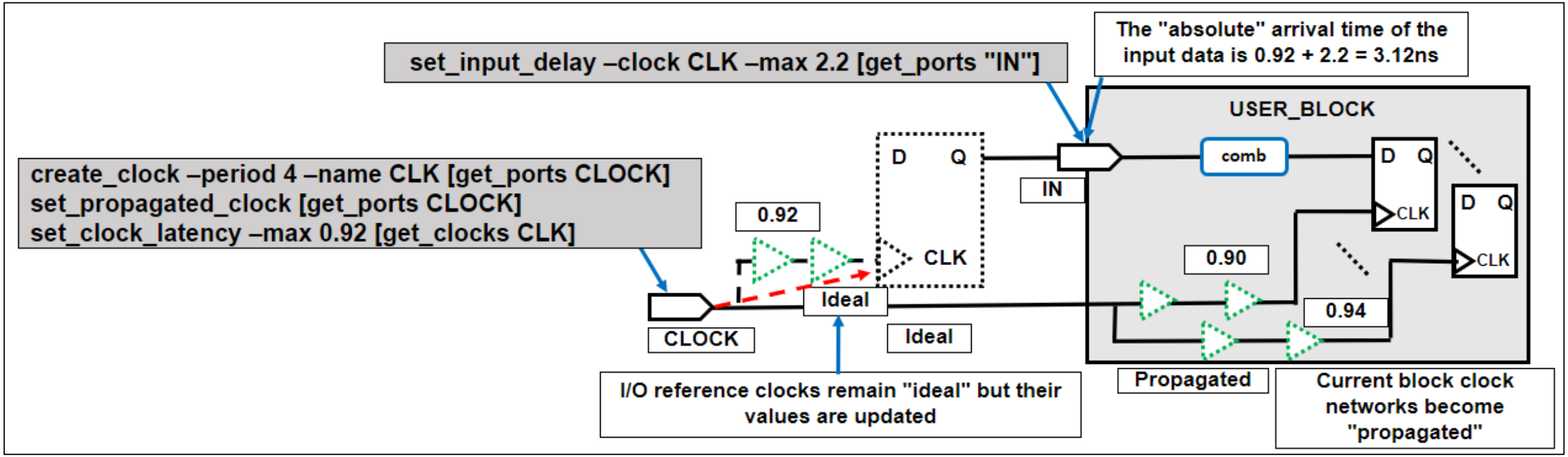
- **Prior to CTS, clock networks are "ideal":**
  - **Clock latencies are 0ns, by default**
  - **Timing analysis uses ideal clock latency constraint values ("set_clock_latency")**
  - **Applies to clock latencies inside the current block and to I/O reference clocks (defined by "set_input_delay" / "set_output_delay")**



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Introduction to CTS : Propagated Clock Sources/Updated Latencies (Post-CTS)

- **Clock_opt's "build_clock" stage applies set_propagated_clock to clock sources (ports/pins) and automatically updates clock network latencies:**
  - Actual propagated clock network latencies calculated inside the current block
  - Updated ideal clock network latencies apply to I/O reference clocks
    - ❖ Updated to the median value of the block's propagated latencies



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Introduction to CTS : Update Latencies in ALL Scenarios (Post-CTS)

- **Only clocks in active scenarios are propagated and get updated latencies**
- **If not all scenarios were active during CTS, explicitly update clock network latencies in all scenarios imm ediately after CTS**
  - ❖ This also makes any non-propagated clock sources propagated

```
clock_opt -to route_clock

set_scenario_status -active true [all_scenarios]

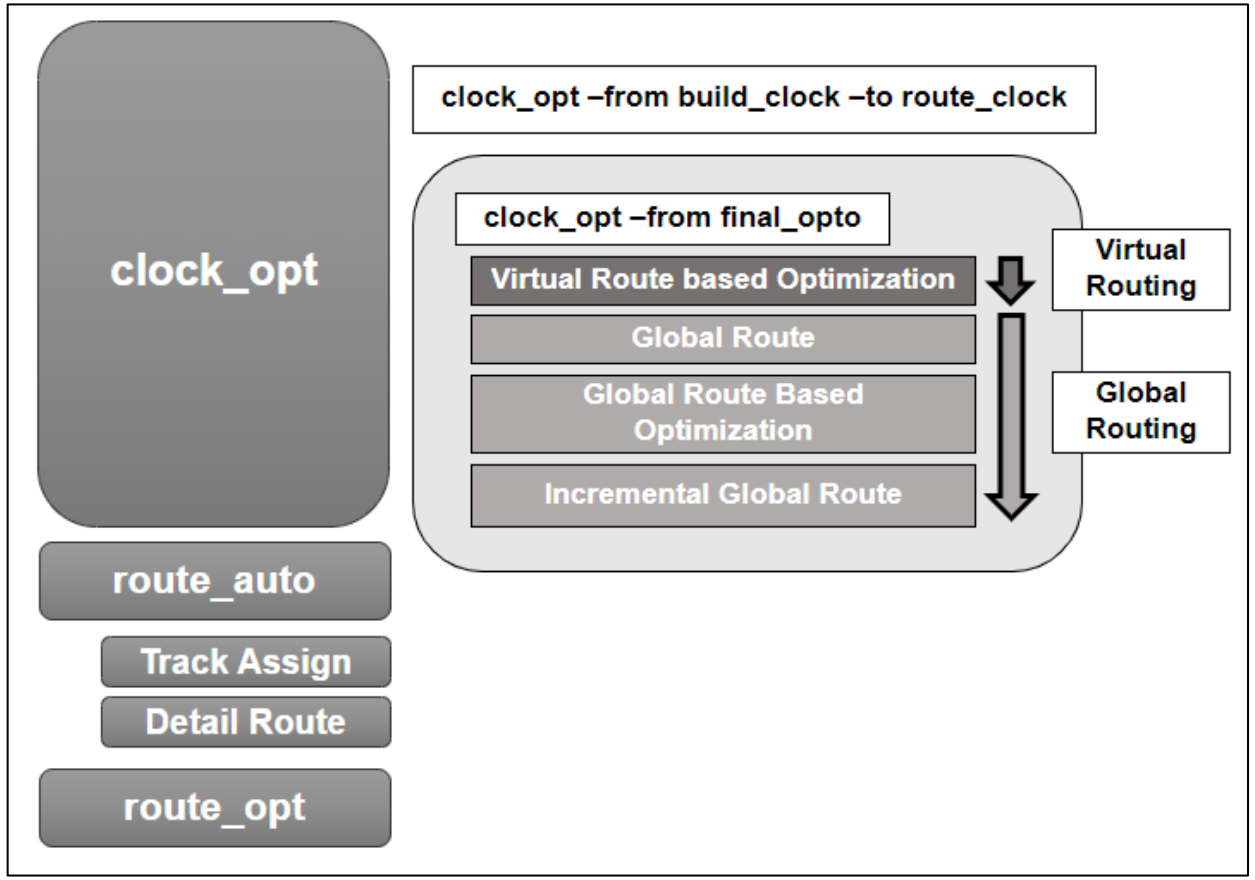
#To make clocks propagated without latency updating:
#synthesize_clock_trees -propagate_only

compute_clock_latency

#This command is used primarily to correct your block's timing constraints after clock tree synthesis
```


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Introduction to CTS : Post-CTS Optimization with GRE

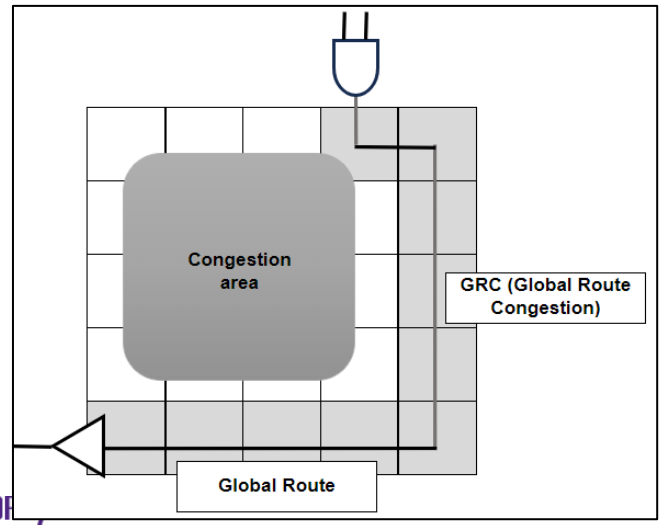


- After the "final_opto" stage, all signal nets are global-routed and data paths are optimized for timing, power and logic DRC
- With CCD enabled, data path and clock logic are optimized simultaneously to further improve setup/hold timing and DRCs

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Introduction to CTS : Global Route (GR)

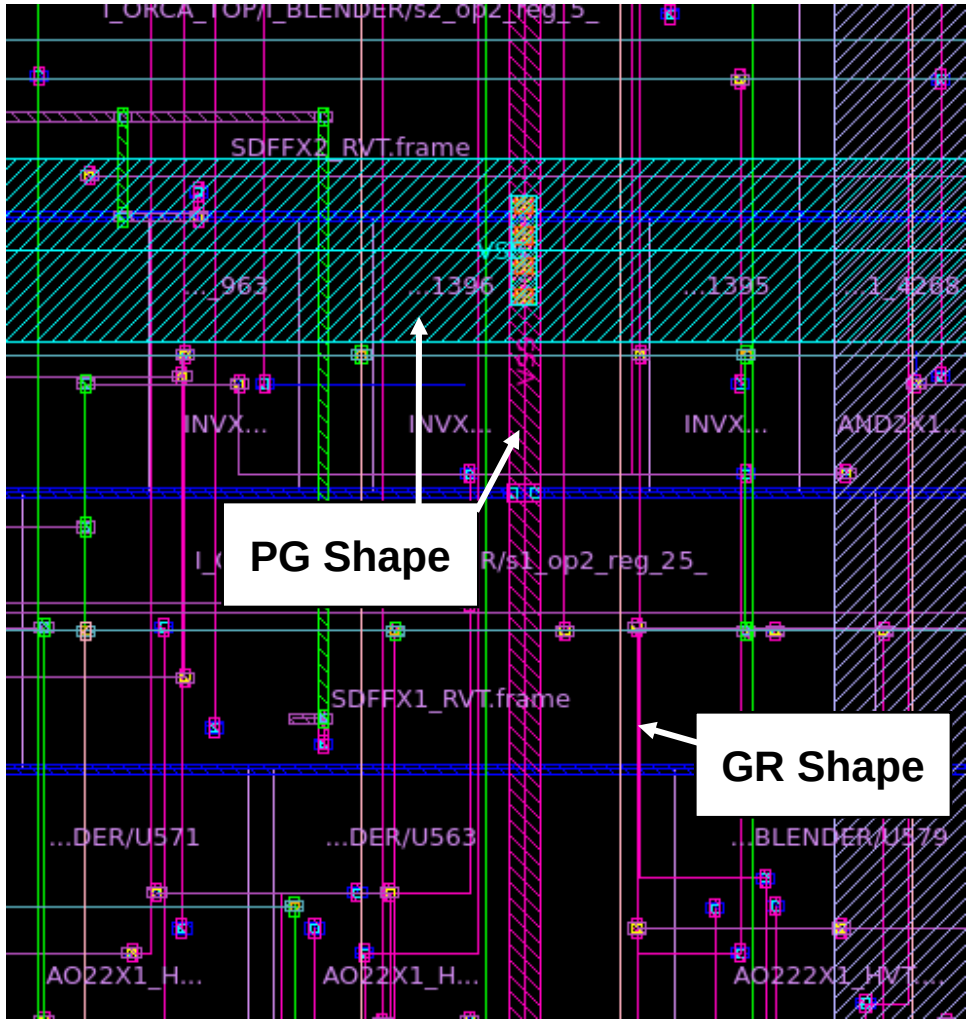
- **Global Routing (GR) assigns nets to specific metal layers (in the preferred routing direction) and through specific global routing cells (GRC or Gcells)**
- **No metal traces are created**
  - Uses colored zero-width lines to indicate global routes and assigned metal layers
- **GR tries to avoid congested Gcells while minimizing detours**
  - Congestion exists when more tracks are needed than available
  - Detours increase wire length (delay)
- **GR also avoids:**
  - P/G (rings/straps/rails/MESH)
  - Routing blockages



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Introduction to CTS : Global Route (GRE)

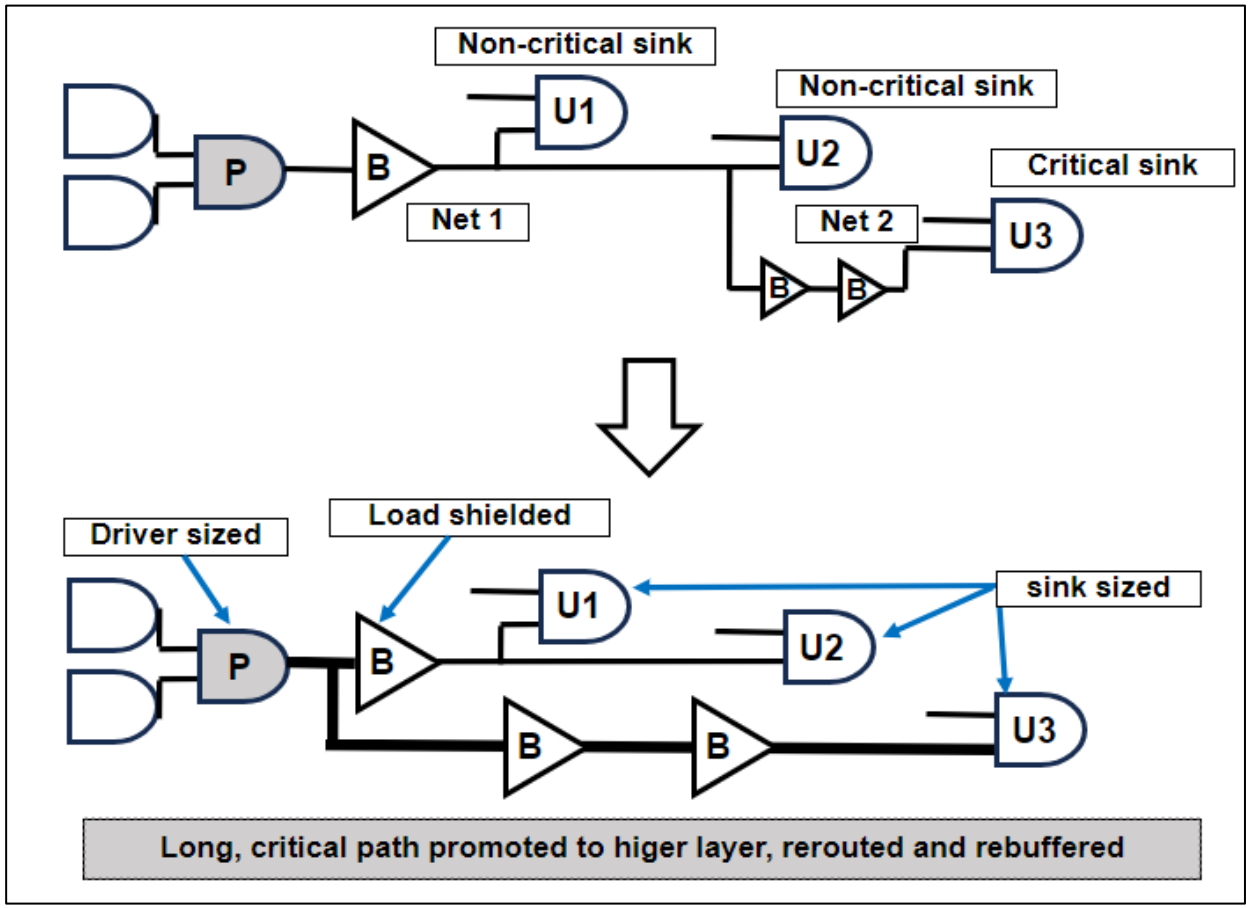
- **Global routing does not create metal shapes**
  - You can select and interact with global routing in the GUI



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Introduction to CTS : GRE Optimization

- **Global routing does not create metal shapes**
  - You can select and interact with global routing in the GUI



- **Good correlation between PSYN and Post route optimization**
- **Simultaneously optimize for timing and congestion**
  - Timing is global routing based
  - Optimization is congestion aware
  - Routing is Timing driven
  - All optimization techniques supported
  - Layer promotion, rerouting and rebuffering

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WEEK 10 : Run CTS & PSYN

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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/05_cts/script
  < user@linux script >> ls
01_1_design_setup_for_cts.tcl       05_setup_for_skew_and_latency.tcl  09_1_run_PSYN_and_save.tcl
01_2_design_setup_for_PSYN.tcl      06_1_app_option_for_cts.tcl        09_2_run_PSYN_ccd_and_save.tcl
01_3_design_setup_for_cts_ccd.tcl   06_2_app_option_for_cts_ccd.tcl    10_1_report_PSYN_and_exit.tcl
01_4_design_setup_for_PSYN_ccd.tcl  07_1_run_cts_and_save.tcl          10_2_report_PSYN_ccd_and_exit.tcl
02_mcmm.tcl                         07_2_run_cts_ccd_and_save.tcl      full_script
03_read_sdc.tcl                     08_1_report_cts_and_exit.tcl
04_setup_for_ndr_and_cells.tcl      08_2_report_cts_ccd_and_exit.tcl
```

  - **Review 01_*_design_setup_for_(cts, PSYN, cts_ccd, PSYN_ccd).tcl**

```
#set current step & before step
set step cts_ccd
set before_step refine_opt

#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#source timer.rpt script - report runtime for placement
source ./icc2_common_env/sec_runtime.tcl

sec_StartTimer copy_lib

#set_host_options -max_cores 16

#open lib
open_lib ../04_placement/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../04_placement/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind

sec_ReportResourceUtil copy_lib
```

The name of the step is different for each script

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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review 02_mcmm.tcl**

```
## Setup
###################################
#remove scenarios modes corners before setting

remove_scenarios -all
remove_modes -all
remove_corners -all

# Read TLU+ files
# TLU+ file is given by Foundary Company
# This file contains Metal capacitance at different spacing and width in the form of a lookup -
# talbe which provides high accuracy and runtime (This file can be used to extract RC value from interconnects)

read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)

# source mcmm(multi corners multi modes).tcl

source -e ./icc2_common_env/mcmm.tcl
```

  - **Review mcmm.tcl**

```
#set scenarios modes corners

#mode
set m_constr(func) func
set m_constr(scan_shift) scan_shift
set m_constr(scan_capture) capture

#corners
set c_constr(ss_125c) ss_125c
set c_constr(ss_n40c) ss_n40c
set c_constr(ff_125c) ff_125c
set c_constr(ff_n40c) ff_n40c
```

```
#scenarios
set s_constr(func.ss_125c) func.ss_125c
set s_constr(func.ss_n40c) func.ss_n40c
set s_constr(func.ff_125c) func.ff_125c
set s_constr(func.ff_n40c) func.ff_n40c

set s_constr(scan_shift.ss_125c) scan_shift.ss_125c
set s_constr(scan_shift.ss_n40c) scan_shift.ss_n40c
set s_constr(scan_shift.ff_125c) scan_shift.ff_125c
set s_constr(scan_shift.ff_n40c) scan_shift.ff_n40c

set s_constr(scan_capture.ss_125c) scan_capture.ss_125c
set s_constr(scan_capture.ss_n40c) scan_capture.ss_n40c
set s_constr(scan_capture.ff_125c) scan_capture.ff_125c
set s_constr(scan_capture.ff_n40c) scan_capture.ff_n40c
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review mcmm.tcl**

```
# scenario mode corner creation

foreach m [array names m_constr] {
create_mode $m
}

foreach c [array names c_constr] {
create_corner $c
}

foreach s [array names s_constr] {
lassign [split $s "."] m c
create_scenario -name $s  -mode $m  -corner $c
}

#check_modes after create modes
report_modes

#check corners after create scenarios
report_corners

#check scenarios after create scenarios
report_scenarios
```

```
# set parasitic_parameters after read TLU+ files

#set ff_n40c corner
current_corner ff_n40c
set_parasitic_parameters -corners ff_n40c -library ORCA_${step}.nlib -late_spec minTLU -early_spec min
TLU -late_temperature -40 -early_temperature -40

#set ss_n40c corner
current_corner ss_n40c
set_parasitic_parameters -corners ss_n40c -library ORCA_${step}.nlib -late_spec maxTLU -early_spec max
TLU -late_temperature -40 -early_temperature -40

#set ff_125c corner
current_corner ff_125c
set_parasitic_parameters -corners ff_125c -library ORCA_${step}.nlib -late_spec minTLU -early_spec min
TLU -late_temperature 125 -early_temperature 125

#set ss_125c corner
current_corner ss_125c
set_parasitic_parameters -corners ss_125c -library ORCA_${step}.nlib -late_spec maxTLU -early_spec max
TLU -late_temperature 125 -early_temperature 125
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review mcmm.tcl**

```
#set pvt value for all corners

set_process_number 1.01 -corners ff_n40c
set_temperature -40 -min -40 -corners ff_n40c
set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets VDD]
set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets AVDD]
set_voltage 1.16 -corner ff_n40c -object_list [get_supply_nets DVDD]
set_voltage 2.75 -corner ff_n40c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ff_n40c -object_list [get_supply_nets AVSS]

set_process_number 0.99 -corners ss_n40c
set_temperature -40 -min -40 -corners ss_n40c
set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets VDD]
set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets AVDD]
set_voltage 0.95 -corner ss_n40c -object_list [get_supply_nets DVDD]
set_voltage 2.25 -corner ss_n40c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ss_n40c -object_list [get_supply_nets AVSS]
```

```
set_process_number 1.01 -corners ff_125c
set_temperature 125 -min 125 -corners ff_125c
set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets VDD]
set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets AVDD]
set_voltage 1.16 -corner ff_125c -object_list [get_supply_nets DVDD]
set_voltage 2.75 -corner ff_125c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ff_125c -object_list [get_supply_nets AVSS]

set_process_number 0.99 -corners ss_125c
set_temperature 125 -min 125 -corners ss_125c
set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets VDD]
set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets AVDD]
set_voltage 0.95 -corner ss_125c -object_list [get_supply_nets DVDD]
set_voltage 2.25 -corner ss_125c -object_list [get_supply_nets VDDIO]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSS]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets VSSIO]
set_voltage 0.00 -corner ss_125c -object_list [get_supply_nets AVSS]

#check corner after set PVT values

report_corners
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review 03_read_sdc.tcl**

```
#read_sdc after create mcmm

#read_sdc(synopsys design constraints)
#SDC contain information such as clock definitions, input and output delays, false paths, and timing exeptions

#scan_shift mode
current_mode scan_shift
read_sdc $data(shift_sdc) ;# ${libdir}/SDC/ORCA_shift.sdc

#func mode
current_mode func
read_sdc $data(func_sdc) ;# ${libdir}/SDC/ORCA_func.sdc

#scan_capture mode
current_mode scan_capture
read_sdc $data(capture_sdc) ;# ${libdir}/SDC/ORCA_capture.sdc
```

  - **Review 04_setup_for_ndr_and_cells.tcl**

```
#source cts_ex_ndr.tcl before cts

source -e ./icc2_common_env/cts_ex_ndr.tcl

get_scenario  -filter active&&hold
report_scenarios

#set hold & optimization cells at CTS
set_lib_cell_purpose -exclude hold [get_lib_cells]
set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]

set_lib_cell_purpose -exclude optimization [get_lib_cells]
set_lib_cell_purpose -include optimization [get_lib_cells "*/NBUFFX2_*VT */NBUFFX4_*VT */NBUFFX8_*VT */NBUFFX16_*VT"]
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review cts_ex_ndr.tcl**

```
# Clock Tree Balance Points

set all_modes "func scan_shift scan_capture"

#This command adds constraints that are used during clock tree synthesis for either excluding or explicitly balancing ports/
pins with a specific delay value (A port/pin can only be an exclude point or a balancing point)

foreach mode $all_modes {
   current_mode $mode
   set_clock_balance_points \
      -consider_for_balancing true \
      -balance_points [get_pins "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*/S0"]
}

# set Size only cells

set_size_only [get_cells "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*"] ;# This command applies size-only settings to the specified leaf
 cells
set_size_only [get_cells "I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_ff_reg"]

## CTS Cell Selection

set CTS_LIB_CELL_PATTERN_LIST "*/INVX2_RVT */INVX4_RVT */INVX8_RVT */INVX16_RVT"
set CTS_CELLS [get_lib_cells $CTS_LIB_CELL_PATTERN_LIST]
set_dont_touch $CTS_CELLS false
suppress_message ATTR-12
set_lib_cell_purpose -exclude cts [get_lib_cells] ;#Defines the valid purposes for the specified library cells(exclude CTS)
set_lib_cell_purpose -include cts $CTS_CELLS ;#Defines the valid purposes for the specified library cells(include CTS)
unsuppress_message ATTR-12
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review cts_ex_ndr.tcl**

```
## CTS NDRs

source -e ./icc2_common_env/ndr.tcl

## Timing and DRC Setup

# Change the uncertainty for all clocks in all scenarios
current_mode func

foreach_in_collection scen [all_scenarios] {
current_scenario $scen
set_clock_uncertainty 0.3 -setup [all_clocks] ;#Specifies the clock uncertainty (skew characteristics) of the specified clock networks
set_clock_uncertainty 0.1 -hold [all_clocks]
}

current_mode scan_capture

foreach_in_collection scen [all_scenarios] {
current_scenario $scen
set_clock_uncertainty 0.3 -setup [all_clocks]
set_clock_uncertainty 0.1 -hold [all_clocks]
}

current_mode scan_shift

foreach_in_collection scen [all_scenarios] {
current_scenario $scen
set_clock_uncertainty 0.3 -setup [all_clocks]
set_clock_uncertainty 0.1 -hold [all_clocks]
}
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review cts_ex_ndr.tcl**

```
# Set a max transition for the clocks in func mode only
current_mode func
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_capture
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_shift
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

# END
return 1
```

  - **Review *ndr.tcl* in cts_ex_ndr.tcl**

```
#set routing Layer
set CTS_NDR_MIN_ROUTING_LAYER "M4"
set CTS_NDR_MAX_ROUTING_LAYER "M6"
set CTS_LEAF_NDR_MIN_ROUTING_LAYER "M1"
set CTS_LEAF_NDR_MAX_ROUTING_LAYER "M6"
set CTS_NDR_RULE_NAME "cts_w2_s2_vlg"
set CTS_LEAF_NDR_RULE_NAME "cts_w1_s2"
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *ndr.tcl* in cts_ex_ndr.tcl**

```
#remove_routing rules
if {$CTS_NDR_RULE_NAME != ""} {
remove_routing_rules $CTS_NDR_RULE_NAME > /dev/null

#create_routing rules for routing layers
create_routing_rule $CTS_NDR_RULE_NAME \
-default_reference_rule \
-widths { M1 0.1 M2 0.11 M3 0.11 M4 0.11 M5 0.11 M6 0.11 } \
-spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1 }

#set min & max clock routing rules
#This command assigns the routing rules used in clock tree synthesis. You can associate a rule with specific
 clocks or nets. If a net already has a nondefault routing rule, this rule is also honored
set_clock_routing_rules -rules $CTS_NDR_RULE_NAME \
-min_routing_layer $CTS_NDR_MIN_ROUTING_LAYER \
-max_routing_layer $CTS_NDR_MAX_ROUTING_LAYER

}
```

```
#create routing rules for leaf
if {$CTS_LEAF_NDR_RULE_NAME != ""} {
remove_routing_rules $CTS_LEAF_NDR_RULE_NAME > /dev/null

create_routing_rule $CTS_LEAF_NDR_RULE_NAME \
-default_reference_rule \
-spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1}

#set min & max clock routing rules
#This command assigns the routing rules used in clock tree synthesis. You can associate a rule with specific
 clocks or nets. If a net already has a nondefault routing rule, this rule is also honored

set_clock_routing_rules -net_type sink -rules $CTS_LEAF_NDR_RULE_NAME \
-min_routing_layer $CTS_LEAF_NDR_MIN_ROUTING_LAYER \
-max_routing_layer $CTS_LEAF_NDR_MAX_ROUTING_LAYER
}
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review 05_setup_for_skew_and_latency.tcl**

```
# Clock Tree Routing and IO latency update
###################################

# all about ss corner
set_scenario_status {*.ss*} -setup true -hold true \
                            -active true -max_transition true \
                            -max_capacitance true \
                            -min_capacitance false \
                            -leakage_power true \
                            -dynamic_power true
# all about ff corner
set_scenario_status {*.ff*} -setup true -hold true \
                            -active true -max_transition true \
                            -max_capacitance true \
                            -min_capacitance false \
                            -leakage_power true \
                            -dynamic_power true

# remove ideal network before cts
source -e ./user_script/remove_ideal_network_before_cts.tcl

# set_disable_timing
source -e ./user_script/set_disable_timing.tcl
```

```
#set target skew & latency constraints for clock trees
set mode "func scan_shift scan_capture"

foreach MODE $mode {
current_mode $MODE

echo "current mode is $MODE"
        switch $MODE {
                func {
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*] ;# Specifies the required value for maximum skew
for the specified clock trees (The default is 0)
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*] ;# Specifies the minimum early insertion delay
 constraint for the specified clock trees (There is no default for this option)
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
                }
                scan_shift {
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
                }

                scan_capture {
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
                        set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
                        set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
                }
        }
}
```

```
#report_clock_tree_options & check setting value

report_clock_tree_options
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *remove_ideal_network_before_cts.tcl* (remove all ideal network set in the placement step)**

```
set mode "func scan_shift scan_capture"

foreach MODE $mode {
current_mode $MODE
echo "current mode is $MODE"
        switch $MODE {
                func {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        set_propagated_clock [all_clocks]
                }
                scan_shift {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        set_propagated_clock [all_clocks]
                }
                scan_capture {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        set_propagated_clock [all_clocks]
                }
        }
}
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review set_disable_timing*.tcl***

```
#set current_scenarios

current_scenario "func.ss_125c"

#set_disable_timing SDRAM_DQ_Bus_1__sdram_DQ_iopad ~ SDRAM_DQ_Bus_15__sdram_DQ_iopad from EN pin to PADIO pin

set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_1__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_1__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_2__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_2__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_3__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_3__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_4__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_4__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_5__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_5__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_6__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_6__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_7__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_7__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_8__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_8__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_9__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_9__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_10__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_10__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_11__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_11__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_12__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_13__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_13__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_14__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_14__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_15__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_15__sdram_DQ_iopad/PADIO } ]
```

  - **Review 06_1_app_option_for_cts.tcl**

```
#report_clock_tree_options & check setting value

report_clock_tree_options

#setup for Non-ccd flow

#Non-ccd flow
set_app_options -name clock_opt.flow.enable_ccd -value false ;#;# This app option controls whether the clock_opt command performs concurrent
 clock and data (CCD) optimization (default : true)

#common setup for CTS
set_app_options -name opt.dft.clock_aware_scan_reorder -value true ;#This app option check hold violation status in scan chain first, and if
 there exists hold violations along scan path, clock driver aware reordering only will be employed to reduce the hold violations along scan
path witin each scan chain to save the hold fixing resource on scan path
set_app_options -name time.remove_clock_reconvergence_pessimism -value true
# When this option is set to true, the tool removes clock reconvergence pessimism from the slack calculation

#set max_fanout for CTS step
set_app_options -list {opt.tie_cell.max_fanout {16}}
set_app_options -list {cts.common.max_fanout {16}}
set_app_options -list {opt.common.max_fanout {16}}
```

```
#check before CTS

#check_clock_trees
check_clock_trees

#report_ideal_networks
report_ideal_network -modes {func scan_capture scan_shift}
;#if ideal network exist, you must remove them
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review 06_2_app_option_for_cts_ccd.tcl**

```
#report_clock_tree_options & check setting value

report_clock_tree_options

#setup for CCD flow
set_app_options -name ccd.optimize_boundary_timing -value false ;#This app option controls whether CCD performs optimization on boundary timing paths
(default : true)
set_app_options -name ccd.ignore_scan_reset_for_boundary_identification -value true ;#This app option used o ignore the connectivities on scan/reset p
ins when concurrent clock and data optimization (CCD) performs boundary flop identification (default : false)
set_app_options \
        -name ccd.ignore_ports_for_boundary_identification \
        -value {scan_en prst_n} ;# This app option specifies the ports to exclude when concurrent clock and data optimization (CCD) performs boundary
identification (Typically you would specify high-fanout ports, such as scan enable and reset ports, to prevent large numbers of flip-flops from being
identified as boundary flip-flops)
set_app_options -name ccd.hold_control_effort -value high ;# This app option specifies the effort level for hold during CCD optimization
set_app_options -name clock_opt.flow.enable_ccd -value true ;# This app option controls whether the clock_opt command performs concurrent clock and da
ta (CCD) optimization

#common setup for CTS
set_app_options -name opt.dft.clock_aware_scan_reorder -value true ;#This app option check hold violation status in scan chain first, and if there exi
sts hold violations along scan path, clock driver aware reordering only will be employed to reduce the hold violations along scan path witin each scan
 chain to save the hold fixing resource on scan path
set_app_options -name time.remove_clock_reconvergence_pessimism -value true ;# When this option is set to true, the tool removes clock reconvergence p
essimism from the slack calculation
set_app_options -name clock_opt.hold.effort -value high ;#The hold fixing effort can be: none, low, medium, or high (default : medium)


#set max_fanout for CTS step
set_app_options -list {opt.tie_cell.max_fanout {16}}
set_app_options -list {cts.common.max_fanout {16}}
set_app_options -list {opt.common.max_fanout {16}}
```

```
#check before CTS

#check_clock_trees
check_clock_trees

#report_ideal_networks
report_ideal_network -modes {func scan_capture scan_shift}
;#if ideal network exist, you must remove them
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *07_1_run_cts_and_save.tcl* & *07_2_run_cts_ccd_and_save.tcl***

< 07_1_run_cts_and_save.tcl >

```
########################
#Running CTS
########################

sec_StartTimer cts

# pre-CTS
clock_opt -to route_clock

sec_ReportResourceUtil cts

save_lib -all

#report after CTS

#update timing
update_timing -full ;#Updates timing information on the current design

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#Qor summary
report_qor -summary ;#Displays QoR information and statistics for the current design
```

< 07_2_run_cts_ccd_and_save.tcl >

```
########################
#Running CTS-CCD
########################

sec_StartTimer cts_ccd

# pre-CTS
clock_opt -to route_clock ;# CTS to route_clock

sec_ReportResourceUtil cts_ccd

#save CTS nlib
save_lib -all

#report after CTS

#update timing
update_timing -full ;#Updates timing information on the current design

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#Qor summary
report_qor -summary ;#Displays QoR information and statistics for the current design
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *08_1_report_cts_and_exit.tcl & 08_2_report_cts_ccd_and_exit.tcl***

```
############################################################
#report Timing & constraints for CTS
############################################################

#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of placement at ${rundir}/reports/cts/* directory
source ./user_script/cts_rpt.tcl

sec_ReportResourceUtil report


exit
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *09_1_run_PSYN_and_save.tcl & 09_2_run_PSYN_ccd_and_save.tcl***

< 09_1_run_PSYN_and_save.tcl >

```
########################
#Running PSYN
########################

sec_StartTimer psyn

# PSYN(optimization)
clock_opt -from final_opto

sec_ReportResourceUtil psyn

save_lib -all

#report after PSYN

#update timing
update_timing -full ;#Updates timing information on the current design

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#QoR summary
report_qor -summary ;#Displays QoR information and statistics for the current design

#report clock QoR
report_clock_qor -type structure

#report_clocks
report_clocks
report_clocks -skew -nosplit -significant_digits 3
```

< 09_2_run_PSYN_ccd_and_save.tcl >

```
########################
#Running PSYN-CCD
########################

sec_StartTimer psyn_ccd

# PSYN(optimization)
clock_opt -from final_opto ;# optimization after finish route_clock

sec_ReportResourceUtil psyn_ccd

#save PSYN nlib
save_lib -all

#report after PSYN

#update timing
update_timing -full ;#Updates timing information on the current design

#timing
report_timing -significant_digits 5 ;#Displays timing information about a design

#QoR summary
report_qor -summary ;#Displays QoR information and statistics for the current design

#report clock QoR
report_clock_qor -type structure

#report_clocks
report_clocks
report_clocks -skew -nosplit -significant_digits 3
```


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CTS : Introduction to CTS & PSYN.tcl

- Review and check the variables in each scripts
  - **Review *10_1_report_PSYN_and_exit.tcl & 10_2_report_PSYN_ccd_and_exit.tcl***

```
############################################################
#report Timing & constraints for CTS after PSYN
############################################################

#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of psyn at ${rundir}/reports/psyn/* directory
source ./user_script/psyn_rpt.tcl

sec_ReportResourceUtil report

exit
```

The name of the step is different for each script

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WEEK 10-1 : Run CTS

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CTS : Use ICC2 (IC Compiler II tool)

1. **Invoke tool icc2_shell**
   - **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./05_cts.log ;# (| tee -I ./05_cts.log) is the command to output log

```
 < user@linux 05_cts >> icc2_shell -gui | tee -i ./05_cts.log



                            IC Compiler II (TM)

                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
icc2_shell>
```

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_1_design_setup_for_cts.tcl

- ❖ **If you want ccd_flow, source 01_3_design_setup_for_cts_ccd.tcl**

- **set step & source design_setup.tcl**

```
icc2_shell> set step cts
cts
icc2_shell> set before_step refine_opt
refine_opt
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

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CTS : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_1_design_setup_for_cts.tcl

- **If you want ccd_flow, source 01_3_design_setup_for_cts_ccd.tcl**

- **Open_lib & copy lib**

```
icc2_shell> source ./icc2_common_env/sec_runtime.tcl
icc2_shell> sec_StartTimer copy_lib
1
icc2_shell> open_lib ../04_placement/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/04_placement/nlib/ORCA_refine_opt.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_refine_opt.nlib}
icc2_shell>
icc2_shell> copy_lib -from ../04_placement/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_
${step}.nlib
Information: Copying from library "ORCA_refine_opt.nlib" to "./nlib/ORCA_cts.nlib".
{ORCA_cts.nlib}
```


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CTS : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_1_design_setup_for_cts.tcl

- **If you want ccd_flow, source 01_3_design_setup_for_cts_ccd.tcl**

- **Close reference lib & open & link current block**

```
icc2_shell> close_lib -all
Closing all libraries...
1
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/05_cts/nlib/ORCA_cts.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_cts.nlib}
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
```

```
icc2_shell> link_block
Using libraries: ORCA_cts.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32rvt saed32s
ram
Visiting block ORCA_cts.nlib:ORCA.design

Design 'ORCA' was successfully linked.
1
icc2_shell> link_block -rebind
Warning: Block 'ORCA_cts.nlib:ORCA.design' is already linked. (LNK-067)
0
```


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CTS : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/02_mcmm.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> remove_scenarios -all
12
icc2_shell> remove_modes -all
3
icc2_shell> remove_corners -all
4
```

- **Read TLU+ files**

```
icc2_shell> read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
1
icc2_shell> read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
1
```

- **Source mcmm.tcl (create mcmm, set parasitic_parameters, set_voltage)**

```
icc2_shell> source -e ./icc2_common_env/mcmm.tcl
```

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/03_read_sdc.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> read_sdc $data(shift_sdc)
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> read_sdc $data(capture_sdc)
```

```
icc2_shell> current_mode func
{func}
icc2_shell> read_sdc $data(func_sdc)
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set_clock_balance_points**

```
icc2_shell> set all_modes "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell> foreach mode $all_modes {
?    current_mode $mode
?    set_clock_balance_points \
?       -consider_for_balancing true \
?       -balance_points [get_pins "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*/S0"]
? }
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
```

- **Set_size_only_cells**

```
icc2_shell> set_size_only [get_cells "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*"]
1
icc2_shell> set_size_only [get_cells "I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_ff_reg"]
1
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set CTS_cell**

```
icc2_shell> set CTS_LIB_CELL_PATTERN_LIST "*/INVX2_RVT */INVX4_RVT */INVX8_RVT */INVX16_RVT"
*/INVX2_RVT */INVX4_RVT */INVX8_RVT */INVX16_RVT
icc2_shell> set CTS_CELLS [get_lib_cells $CTS_LIB_CELL_PATTERN_LIST]
{saed32rvt/INVX2_RVT saed32rvt/INVX4_RVT saed32rvt/INVX8_RVT saed32rvt/INVX16_RVT}
icc2_shell> set_dont_touch $CTS_CELLS false
1
icc2_shell> suppress_message ATTR-12
icc2_shell> set_lib_cell_purpose -exclude cts [get_lib_cells]
icc2_shell> set_lib_cell_purpose -include cts $CTS_CELLS
1
icc2_shell> unsuppress_message ATTR-12
```

- **Set ndr_rule (source ./icc2_common_env/ndr.tcl)**

```
icc2_shell> set CTS_NDR_MIN_ROUTING_LAYER "M4"
M4
icc2_shell> set CTS_NDR_MAX_ROUTING_LAYER "M6"
M6
icc2_shell> set CTS_LEAF_NDR_MIN_ROUTING_LAYER "M1"
M1
icc2_shell> set CTS_LEAF_NDR_MAX_ROUTING_LAYER "M6"
M6
icc2_shell> set CTS_NDR_RULE_NAME "cts_w2_s2_vlg"
cts_w2_s2_vlg
icc2_shell> set CTS_LEAF_NDR_RULE_NAME "cts_w1_s2"
cts_w1_s2
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set ndr_rule (source ./icc2_common_env/ndr.tcl)**
- **Routing rules for routing layers**

```
icc2_shell> if {$CTS_NDR_RULE_NAME != ""} {
? remove_routing_rules $CTS_NDR_RULE_NAME > /dev/null
?
? create_routing_rule $CTS_NDR_RULE_NAME \
? -default_reference_rule \
? -widths { M1 0.1 M2 0.11 M3 0.11 M4 0.11 M5 0.11 M6 0.11 } \
? -spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1 }
?
? set_clock_routing_rules -rules $CTS_NDR_RULE_NAME \
? -min_routing_layer $CTS_NDR_MIN_ROUTING_LAYER \
? -max_routing_layer $CTS_NDR_MAX_ROUTING_LAYER
?
? }
Information: The command 'set_clock_routing_rules' cleared the undo history. (UNDO-016)
1
```

- **Routing rules for leaf**

```
icc2_shell> if {$CTS_LEAF_NDR_RULE_NAME != ""} {
? remove_routing_rules $CTS_LEAF_NDR_RULE_NAME > /dev/null
?
? create_routing_rule $CTS_LEAF_NDR_RULE_NAME \
? -default_reference_rule \
? -spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1}
?
? set_clock_routing_rules -net_type sink -rules $CTS_LEAF_NDR_RULE_NAME \
? -min_routing_layer $CTS_LEAF_NDR_MIN_ROUTING_LAYER \
? -max_routing_layer $CTS_LEAF_NDR_MAX_ROUTING_LAYER
? }
Information: The command 'set_clock_routing_rules' cleared the undo history. (UNDO-016)
1
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**

- **Set clock uncertainty for all clocks in all scenarios**

```
icc2_shell> current_mode func
{func}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

- **Set max_transition for the clocks in all modes & corners**

```
icc2_shell> current_mode func
{func}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell> return 1
1
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_scenarios after create mcmm**

```
icc2_shell> get_scenario  -filter active&&hold
{func.ff_125c func.ff_n40c func.ss_125c func.ss_n40c scan_capture.ff_125c scan_capture.ff_n40c scan_capture.ss_125c scan_captu
re.ss_n40c scan_shift.ff_125c scan_shift.ff_n40c scan_shift.ss_125c scan_shift.ss_n40c}
```

```
icc2_shell> report_scenarios
****************************************
Report : scenario
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 11:56:19 2023
****************************************
```

```
                                                                Leakage  Dynamic                              Cell  Signa
l
Name            Mode            Corner          Active  Setup  Hold  Power    Power    Max_tran  Max_cap  Min_cap  EM    EM
-------------------------------------------------------------------------------------------------------------------------
-
func.ff_125c *  func            ff_125c         true    true   true  true     true     true      true     true     false false
func.ff_n40c *  func            ff_n40c         true    true   true  true     true     true      true     true     false false
func.ss_125c *  func            ss_125c         true    true   true  true     true     true      true     true     false false
func.ss_n40c *  func            ss_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_125c *
                scan_capture    ff_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_n40c *
                scan_capture    ff_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_125c *
                scan_capture    ss_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_n40c *
                scan_capture    ss_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_125c *
                scan_shift      ff_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_n40c *
                scan_shift      ff_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_125c *
                scan_shift      ss_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_n40c *
                scan_shift      ss_n40c         true    true   true  true     true     true      true     true     false false

1
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
report_clock_balance_points
****************************************
Report : clock tree balance points
Design : ORCA
Mode   : scan_shift
Version: S-2021.06-SP5
Date   : Tue Oct 24 11:58:52 2023
****************************************
```

```
Clock Independent:
Balance points:              ------------- DELAY ---------------    ------------- OFFSET ---------------
                           Early-    Early-     Late-     Late-    Early-    Early-     Late-     Late-
Point                       Rise      Fall       Rise      Fall     Rise      Fall       Rise      Fall     Corner
-------------------------------------------------------------------------------------------------------------------
I_ORCA_TOP/I_SDRAM_IF/genblk1_15__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_14__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_13__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_11__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_10__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_9__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_8__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_7__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_6__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_5__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --     ss_125c
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
I_ORCA_TOP/I_SDRAM_IF/genblk1_4__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_3__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_2__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_1__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_0__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CKn/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c

Exclude points:
Point
--------------------------------------------------------------------------------------------------------------------
No exclude points.


1
```

- **Report_clock_routing_rules**

```
icc2_shell> report_clock_routing_rules
****************************************
 Report : clock routing rules
 Design : ORCA
 Date   : Tue Oct 24 12:01:49 2023
****************************************
   Design Base; Net Type: all;      Rule: cts_w2_s2_vlg; Min Layer: M4; Max Layer: M6
   Design Base; Net Type: sink;     Rule: cts_w1_s2; Min Layer: M1; Max Layer: M6
```


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CTS : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
icc2_shell> set_lib_cell_purpose -exclude hold [get_lib_cells]
```

```
icc2_shell> set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
```

```
icc2_shell> set_lib_cell_purpose -exclude optimization [get_lib_cells]
```

```
icc2_shell> set_lib_cell_purpose -include optimization [get_lib_cells "*/NBUFFX2_*VT */NBUFFX4_*VT */NBUFFX8_*VT */NBUFFX16_*VT"]
```

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set_scenarios_status**

```
icc2_shell> set_scenario_status {*.ss*} -setup true -hold true \
?     -active true -max_transition true \
?     -max_capacitance true \
?     -min_capacitance false \
?             -leakage_power true \
?             -dynamic_power true
Scenario func.ss_125c (mode func corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario func.ss_n40c (mode func corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ss_125c (mode scan_capture corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_capture.ss_n40c (mode scan_capture corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ss_125c (mode scan_shift corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
Scenario scan_shift.ss_n40c (mode scan_shift corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_capacitance analysis.
1
```


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CTS : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set_scenarios_status**

```
icc2_shell> set_scenario_status {*.ff*} -setup true -hold true \
?                              -active true -max_transition true \
?              -max_capacitance true \
?     -min_capacitance false \
?      -leakage_power true \
?                      -dynamic_power true
Scenario func.ff_125c (mode func corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario func.ff_n40c (mode func corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario scan_capture.ff_125c (mode scan_capture corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_capture.ff_n40c (mode scan_capture corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_shift.ff_125c (mode scan_shift corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
Scenario scan_shift.ff_n40c (mode scan_shift corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
1
```

- **Source remove_ideal_network_before_cts.tcl**

```
icc2_shell> source -e ./user_script/remove_ideal_network_before_cts.tcl
set mode "func scan_shift scan_capture"
foreach MODE $mode {
current_mode $MODE

echo "current mode is $MODE"
        switch $MODE {
                func {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
```


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CTS : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Source remove_ideal_network_before_cts.tcl**

```
                scan_shift {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
                scan_capture {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
        }
}
current mode is func
current mode is scan_shift
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
current mode is scan_capture
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
```


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CTS : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Source set_disable_timing.tcl**

```
source -e ./user_script/set_disable_timing.tcl
#set current_scenarios
current_scenario "func.ss_125c"
#set_disable_timing SDRAM_DQ_Bus_1__sdram_DQ_iopad ~ SDRAM_DQ_Bus_15__sdram_DQ_iopad from EN pin to PADIO pin
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_1__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_1__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_2__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_2__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_3__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_3__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_4__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_4__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_5__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_5__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_6__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_6__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_7__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_7__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_8__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_8__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_9__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_9__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_10__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_10__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_11__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_11__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_12__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_13__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_13__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_14__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_14__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_15__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_15__sdram_DQ_iopad/PADIO } ]
1
```

- **Set target_skew & latency constraints for clock trees**

```
set mode "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell>
icc2_shell> foreach MODE $mode {
? current_mode $MODE
?
? echo "current mode is $MODE"
? switch $MODE {
? func {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
```


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CTS : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set target_skew & latency constraints for clock trees**

```
? scan_shift {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
?
? scan_capture {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
? }
? }
current mode is func
current mode is scan_shift
current mode is scan_capture
```

- **Report clock_tree_options**

```
icc2_shell> report_clock_tree_options
****************************************
 Report : target skew and latency
 Design : ORCA
 Date   : Tue Oct 24 13:25:50 2023
****************************************

##Target Skew

Clock                         Corner                        Target
------------------------------------------------------------------
All clocks                    ss_125c                       0.3
All clocks                    ss_n40c                       0.3
All clocks                    ff_n40c                       0.3
All clocks                    ff_125c                       0.3
```

```
##Target Latency

Clock                         Corner                        Target
------------------------------------------------------------------
All clocks                    ss_125c                       0.5
All clocks                    ss_n40c                       0.5
All clocks                    ff_n40c                       0.5
All clocks                    ff_125c                       0.5
```


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CTS : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/06_1_app_option_for_cts.tcl

❖ **If you want ccd_flow, source 06_2_app_option_for_cts_ccd.tcl**

- **Set app_options for CTS**

Classic CTS

```
set_app_options -name clock_opt.flow.enable_ccd -value false ;
clock_opt.flow.enable_ccd false
icc2_shell> set_app_options -name opt.dft.clock_aware_scan_reorder -value true
opt.dft.clock_aware_scan_reorder true
icc2_shell> set_app_options -name time.remove_clock_reconvergence_pessimism -value true
time.remove_clock_reconvergence_pessimism true
icc2_shell> set_app_options -list {opt.tie_cell.max_fanout {16}}
opt.tie_cell.max_fanout 16
icc2_shell> set_app_options -list {cts.common.max_fanout {16}}
cts.common.max_fanout 16
icc2_shell> set_app_options -list {opt.common.max_fanout {16}}
opt.common.max_fanout 16
```

CCD_flow CTS

```
icc2_shell> set_app_options -name ccd.optimize_boundary_timing -value false
ccd.optimize_boundary_timing false
icc2_shell> set_app_options -name ccd.ignore_scan_reset_for_boundary_identification -value true
ccd.ignore_scan_reset_for_boundary_identification true
icc2_shell> set_app_options \
? -name ccd.ignore_ports_for_boundary_identification \
? -value {scan_en prst_n}
ccd.ignore_ports_for_boundary_identification {scan_en prst_n}
icc2_shell> set_app_options -name ccd.hold_control_effort -value high
ccd.hold_control_effort high
icc2_shell> set_app_options -name clock_opt.flow.enable_ccd -value true
clock_opt.flow.enable_ccd true
icc2_shell> set_app_options -name opt.dft.clock_aware_scan_reorder -value true
opt.dft.clock_aware_scan_reorder true
icc2_shell> set_app_options -name time.remove_clock_reconvergence_pessimism -value true
time.remove_clock_reconvergence_pessimism true
icc2_shell> set_app_options -name clock_opt.hold.effort -value high
clock_opt.hold.effort high
icc2_shell> set_app_options -list {opt.tie_cell.max_fanout {16}}
opt.tie_cell.max_fanout 16
icc2_shell> set_app_options -list {cts.common.max_fanout {16}}
cts.common.max_fanout 16
icc2_shell> set_app_options -list {opt.common.max_fanout {16}}
opt.common.max_fanout 16
```


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CTS : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/06_1_app_option_for_cts.tcl

- **Report_ideal_network before CTS**

<func>

```
icc2_shell> report_ideal_network -modes {func scan_capture scan_shift}
****************************************
Report : ideal_network
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 13:42:54 2023
****************************************

Ideal network data for scenario 'func.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ss_n40c':
No ideal source pins or ports.
```

<scan_capture>

```
Ideal network data for scenario 'scan_capture.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ss_n40c':
No ideal source pins or ports.
```

<scan_shift>

```
Ideal network data for scenario 'scan_shift.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ss_n40c':
No ideal source pins or ports.

1
```


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CTS : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/07_1_run_cts_and_save.tcl

❖ **If you want ccd flow, source 07_2_run_cts_ccd_and_save.tcl**

- **Run CTS (to route clock)**

```
icc2_shell> sec_StartTimer cts
1
icc2_shell> clock_opt -to route_clock
```

❖ **check PVT mismatches in log**

```
Information: Corner ss_n40c: no PVT mismatches. (PVT-032)
Information: Corner ff_125c: no PVT mismatches. (PVT-032)
Information: Corner ss_125c: no PVT mismatches. (PVT-032)
Information: Corner ff_n40c: no PVT mismatches. (PVT-032)
```

- **Save libraries after CTS**

```
icc2_shell> sec_ReportResourceUtil cts
icc2_shell> save_lib -all
Saving all libraries...
1
```

- **Update timing before report Timing & QoR**

```
icc2_shell> update_timing -full
```


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CTS : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/07_1_run_cts_and_save.tcl

❖ **If you want ccd flow, source 07_2_run_cts_ccd_and_save.tcl**

- **Report_timing**

```
icc2_shell> report_timing -significant_digits 5
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 13:57:30 2023
****************************************
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sys_rst_n_buf_reg (rising edge-triggered flip-flop clocked by sys_clk)
  Endpoint: I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/empty_int_reg (rising edge-triggered flip-flop clocked by PCI_CLK)
  Mode: func
  Corner: ss_n40c
  Scenario: func.ss_n40c
  Path Group: PCI_CLK
  Path Type: max
```

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**

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CTS : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/07_1_run_cts_and_save.tcl

❖ **If you want ccd flow, source 07_2_run_cts_ccd_and_save.tcl**

- **Report_timing**

DAT Path

```
Point                                                  Incr       Path
-----------------------------------------------------------------------------
clock sys_clk (rise edge)                           4.00000    4.00000
clock network delay (propagated)                    1.02564    5.02564

I_ORCA_TOP/I_RESET_BLOCK/sys_rst_n_buf_reg/CLK (SDFFARX1_LVT)
                                                    0.00000    5.02564 r
I_ORCA_TOP/I_RESET_BLOCK/sys_rst_n_buf_reg/Q (SDFFARX1_LVT)
                                                    0.16315    5.18879 f
I_ORCA_TOP/I_RESET_BLOCK/U13/Y (AND2X1_HVT)         0.10271    5.29150 f
I_ORCA_TOP/I_RESET_BLOCK/C56/Y (OR2X4_HVT)          0.18887    5.48037 f
I_ORCA_TOP/HFSINV_2602_2334/Y (INVX4_HVT)           0.09554    5.57591 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/HFSINV_1673_2330/Y (INVX4_HVT)
                                                    0.08596    5.66187 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/syncff_reg_5_/Q (SDFFARX1_HVT)
                                                    0.13302    5.79489 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U35/Y (XOR2X1_HVT)
                                                    0.24645    6.04134 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U34/Y (XOR2X1_HVT)
                                                    0.26085    6.30219 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U33/Y (XOR2X1_HVT)
                                                    0.24063    6.54282 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U32/Y (XOR2X1_HVT)
                                                    0.25679    6.79961 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U31/Y (XOR2X1_HVT)
                                                    0.25602    7.05563 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U5/Y (INVX0_HVT)
                                                    0.06254    7.11817 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U9/Y (NAND2X0_HVT)
                                                    0.07875    7.19693 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U2_1/CO (FADDX1_HVT)
                                                    0.19802    7.39495 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U2_2/CO (FADDX1_HVT)
                                                    0.19718    7.59213 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U2_3/CO (FADDX1_HVT)
                                                    0.19600    7.78813 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U2_4/CO (FADDX1_HVT)
                                                    0.19770    7.98582 r
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U3_U2_5/S (FADDX1_HVT)
                                                    0.29879    8.28461 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U46/Y (OR2X1_HVT)
                                                    0.13136    8.41597 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U45/Y (OR3X1_HVT)
                                                    0.12251    8.53848 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/U44/Y (OR3X1_HVT)
                                                    0.11320    8.65168 f
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/empty_int_reg/D (SDFFARX1_HVT)
                                                    0.00000    8.65168 f
data arrival time                                              8.65168
```

```
clock PCI_CLK (rise edge)                           8.00000    8.00000
clock network delay (propagated)                    0.47949    8.47949
clock reconvergence pessimism                       0.00000    8.47949
I_ORCA_TOP/I_PCI_WRITE_FIFO/PCI_WFIFO_CTL/U2/empty_int_reg/CLK (SDFFARX1_HVT)
                                                    0.00000    8.47949 r
clock uncertainty                                  -0.30000    8.17949
library setup time                                 -0.21622    7.96326
data required time                                             7.96325
-----------------------------------------------------------------------------
data required time                                             7.96325
data arrival time                                             -8.65168
-----------------------------------------------------------------------------
slack (VIOLATED)                                              -0.68843
```

DRT Path

DRT – RAT = Slack

❖ **When analyzing this Timing report, Timing violation(Slack) occurs because the length of the DRT path is shorter than that of the DAT path**

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

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CTS : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/07_1_run_cts_and_save.tcl

❖ **If you want ccd flow, source 07_2_run_cts_ccd_and_save.tcl**

- **Report_qor -summary**

```
Information: Timer using 'CRPR'. (TIM-050)

Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)             0.90            0.00               0
func.ff_n40c       (Setup)             1.08            0.00               0
func.ss_125c       (Setup)            -0.14           -1.10              19
func.ss_n40c       (Setup)            -0.69          -15.35              78
scan_capture.ff_125c
                   (Setup)             9.27            0.00               0
scan_capture.ff_n40c
                   (Setup)             9.33            0.00               0
scan_capture.ss_125c
                   (Setup)             8.65            0.00               0
scan_capture.ss_n40c
                   (Setup)             8.46            0.00               0
scan_shift.ff_125c (Setup)             9.58            0.00               0
scan_shift.ff_n40c (Setup)             9.60            0.00               0
scan_shift.ss_125c (Setup)             9.38            0.00               0
scan_shift.ss_n40c (Setup)             9.28            0.00               0
Design             (Setup)            -0.69          -15.40              81

func.ff_125c       (Hold)             -0.25         -337.88            3918
func.ff_n40c       (Hold)             -0.23         -276.95            4117
func.ss_125c       (Hold)             -0.66         -111.61            1472
func.ss_n40c       (Hold)             -0.31          -29.13             261
scan_capture.ff_125c
                   (Hold)             -0.08           -6.16             220
scan_capture.ff_n40c
                   (Hold)             -0.08           -7.52             308
scan_capture.ss_125c
                   (Hold)             -0.07           -0.54               8
scan_capture.ss_n40c
                   (Hold)             -0.11           -0.90               8
scan_shift.ff_125c (Hold)             -0.11          -44.99            1997
scan_shift.ff_n40c (Hold)             -0.11          -60.96            2747
scan_shift.ss_125c (Hold)             -0.13           -0.61              16
scan_shift.ss_n40c (Hold)             -0.18           -0.69              18
Design             (Hold)             -0.66         -427.47            6897
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                         748489.08
Cell Area (netlist and physical only):      1172489.08
Nets with DRC Violations:      434
1
```

❖ **You can check WNS(Worst Negative Slack) first, and check TNS(Total Negative Slack) with NVE**

❖ **You can check Cell Area and number of DRC violations**

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CTS : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/08_1_report_cts_and_exit.tcl

❖ **If you want ccd flow, source 08_2_report_ccd_cts_and_exit.tcl**

- **Update timing before report**

```
icc2_shell> update_timing -full
```

- **Report CTS & exit**

```
icc2_shell> source ./user_script/cts_rpt.tcl
icc2_shell> sec_ReportResourceUtil report
icc2_shell> exit
```

- **Analyze Reports about CTS**

```
  < user@linux cts >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/05_cts/reports/cts
  < user@linux cts >> ls
cts.clocks.rpt                      timing_report_max_cts_func.ff_n40c          timing_report_min_cts_func.ff_n40c
cts.clocks.vio.rpt                  timing_report_max_cts_func.ss_125c          timing_report_min_cts_func.ss_125c
cts.congestion.rpt                  timing_report_max_cts_func.ss_n40c          timing_report_min_cts_func.ss_n40c
cts.fanout.rpt                      timing_report_max_cts_scan_capture.ff_125c  timing_report_min_cts_scan_capture.ff_125c
cts.latnecy.rpt                     timing_report_max_cts_scan_capture.ff_n40c  timing_report_min_cts_scan_capture.ff_n40c
cts.local.skew.rpt                  timing_report_max_cts_scan_capture.ss_125c  timing_report_min_cts_scan_capture.ss_125c
cts.max_tran.rpt                    timing_report_max_cts_scan_capture.ss_n40c  timing_report_min_cts_scan_capture.ss_n40c
cts.qor.snap.rpt                    timing_report_max_cts_scan_shift.ff_125c    timing_report_min_cts_scan_shift.ff_125c
cts.runtime.rpt                     timing_report_max_cts_scan_shift.ff_n40c    timing_report_min_cts_scan_shift.ff_n40c
cts.structure.rpt                   timing_report_max_cts_scan_shift.ss_125c    timing_report_min_cts_scan_shift.ss_125c
cts.util.rpt                        timing_report_max_cts_scan_shift.ss_n40c    timing_report_min_cts_scan_shift.ss_n40c
timing_report_max_cts_func.ff_125c  timing_report_min_cts_func.ff_125c
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.clock.rpt (you can check for Hold & Setup Uncertainty for All clocks)**

Object	Min Rise Delay	Min Fall Delay	Max Rise Delay	Max Fall Delay	Hold Uncertainty	Setup Uncertainty	Related Clock	Scenario	Origin
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
pclk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
pclk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
pclk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
pclk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user

1

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.latency.rpt**

❖ *This .rpt file reports a summary table of latency and skew information on all clocks and skew groups, including global, local, boundary skew, minimum, maximum, median latency, and latency and skew targets*

❖ *The detailed section of the report shows sinks or balance points with their corresponding latencies*

❖ *In order to see the latency to a hierarchical balance pin, you might specify hierarchical pins as input to the -to option with this reporting*

```
Attributes
==========
M Master Clock
G Generated Clock
& Internal Generated Clock
U User Defined Skew Group
D Default Skew Group
* Generated Clock Balanced Separately

================================================
==== Latency Reporting for Corner ff_125c ====
================================================

================================================= Summary Table for Corner ff_125c =================================================
Clock /                              Attrs     Sinks    Target    Global    Target       Max       Min    Median   Latency  Boundary
Skew Group                                              Skew      Skew   Latency   Latency   Latency   Latency   Std Dev      Skew
---------------------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func.ff_125c
sdr_clk                               M,D      1204      0.30      0.14      0.50      0.44      0.30      0.38      0.03        --
SDRAM_CLK                             M,D      1203      0.30      0.09      0.50      0.30      0.21      0.27      0.02        --
pclk                                  M,D      1101      0.30      0.11      0.50      0.42      0.30      0.40      0.02        --
PCI_CLK                               M,D      1100      0.30      0.06      0.50      0.27      0.21      0.25      0.02        --
sys_clk                               M,D       786      0.30      0.28      0.50      0.51      0.23      0.46      0.04        --
 SYS_CLK                                G       519      0.30      0.13      0.50      0.25      0.12      0.22      0.02        --
 SYS_2x_CLK                             G       251      0.30      0.11      0.50      0.28      0.17      0.25      0.03        --
### Mode: scan_capture, Scenario: scan_capture.ff_125c
sdr_clk_wft1                          M,D      1204      0.30      0.14      0.50      0.44      0.30      0.38      0.03        --
pclk_wft1                             M,D      1101      0.30      0.11      0.50      0.42      0.30      0.40      0.02        --
sys_clk_wft1                          M,D       786      0.30      0.28      0.50      0.51      0.23      0.49      0.04        --
### Mode: scan_shift, Scenario: scan_shift.ff_125c
sdr_clk_wft1                          M,D      1204      0.30      0.14      0.50      0.44      0.30      0.38      0.03        --
pclk_wft1                             M,D      1101      0.30      0.11      0.50      0.42      0.30      0.40      0.02        --
sys_clk_wft1                          M,D       786      0.30      0.28      0.50      0.51      0.23      0.49      0.04        --
---------------------------------------------------------------------------------------------------------------------------------
All Clocks                                     3091      0.30      0.28      0.50      0.51      0.12        --        --        --


  & = Offset derived from max_clock_tree_path / min_clock_tree_path
  r = latency reported is for a rising edge triggered event at the sink
  f = latency reported is for a falling edge triggered event at the sink
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.latency.rpt**

```
Showing 5 largest and 5 smallest datapoints per clock / skew group (L=largest, S=smallest)
========================================= Details Table for Corner ff_125c ==========================================
Clock /                                 Sink                                 Launch     Capture        Late       Early
Skew Group                              Name                                 Latency    Latency      Offset     Offset
---------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func.ff_125c
sdr_clk
```

Largest datapoints

```
L   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_3_/CLK
                                        0.44 r      0.44 r         --          --
L   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_4_/CLK
                                        0.44 r      0.44 r         --          --
L   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_5_/CLK
                                        0.44 r      0.44 r         --          --
L   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_6_/CLK
                                        0.44 r      0.44 r         --          --
L   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_2_/CLK
                                        0.44 r      0.44 r         --          --
```

Smallest datapoints

```
S   I_ORCA_TOP/I_SDRAM_IF/DQ_out_1_reg_7_/CLK
                                        0.30 f      0.30 f         --          --
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_0__7_/CLK
                                        0.31 f      0.31 f         --          --
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__7_/CLK
                                        0.31 f      0.31 f         --          --
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_/CLK
                                        0.31 f      0.31 f         --          --
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__9_/CLK
                                        0.31 f      0.31 f         --          --
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.latency.rpt**

Largest Path per clock / skew group

```
------------------------------------------
Largest Path #1
Mode                : func
Corner              : ff_125c
Scenario            : func.ff_125c
Skew Group          : default_sdr_clk
Clock Fanout        : sdr_clk
Clock at Sink       : sdr_clk
Sink                : I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_3_/CLK
Latency             : 0.44
------------------------------------------

  Point                                          Fanout   Cap     Trans    Incr     Path
  ---------------------------------------------------------------------------------------
  source latency                                                             0.00    0.00
  sdr_clk (in)                                      1    2207.91   0.00     0.00    0.00 r
  sdr_clk_iopad/PADIO (I1025_NS)                                    0.00     0.00    0.00 r
  sdr_clk_iopad/DOUT (I1025_NS)                     2     31.55    0.14     0.21    0.21 r
  I_CLOCK_GEN/sdram_clk (CLOCK_GEN)                 2     31.55    0.14     0.00    0.21 r
  I_CLOCK_GEN/U9/A1 (AO22X1_RVT)                                    0.14     0.01    0.22 r
  I_CLOCK_GEN/U9/Y (AO22X1_RVT)                     2      7.59    0.06     0.07    0.29 r
  I_CLOCK_GEN/o_sdram_clk (CLOCK_GEN)               2      7.59    0.06     0.00    0.29 r
  I_ORCA_TOP/sdram_clk (ORCA_TOP)                   2      7.59    0.06     0.00    0.29 r
  I_ORCA_TOP/ZCTSINV_10599_7208/A (INVX8_RVT)                       0.06     0.00    0.29 r
  I_ORCA_TOP/ZCTSINV_10599_7208/Y (INVX8_RVT)       1     11.36    0.03     0.01    0.30 f
  I_ORCA_TOP/ZCTSINV_9646_7207/A (INVX16_RVT)                       0.03     0.00    0.30 f
  I_ORCA_TOP/ZCTSINV_9646_7207/Y (INVX16_RVT)       3     40.57    0.03     0.01    0.31 r
  I_ORCA_TOP/ZCTSINV_8966_7206/A (INVX16_RVT)                       0.03     0.01    0.32 r
  I_ORCA_TOP/ZCTSINV_8966_7206/Y (INVX16_RVT)       5     54.75    0.03     0.02    0.34 f
  I_ORCA_TOP/cto_inv_cln_8696/A (INVX16_RVT)                        0.03     0.00    0.34 f
  I_ORCA_TOP/cto_inv_cln_8696/Y (INVX16_RVT)        2     25.81    0.02     0.01    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/ZCTSNET_3 (SDRAM_RFIFO)
                                                    2     25.81    0.02     0.00    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/ZCTSNET_3 (SDRAM_RFIFO_DW_fifoctl_s2_sf_0)
                                                    2     25.81    0.02     0.00    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/ZCTSNET_1 (SDRAM_RFIFO_DW_FIFOCTL_IF_0)
                                                    2     25.81    0.02     0.00    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/clk_gate_this_addr_g_int_reg/CLK (SNPS_CLOCK_GATE_HIGH_SDRAM_RFIFO_DW_FIFOCTL_IF_0_0)
                                                    2     25.81    0.02     0.00    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/clk_gate_this_addr_g_int_reg/latch/CLK (CGLPPRX8_RVT)
                                                                     0.02     0.00    0.35 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/clk_gate_this_addr_g_int_reg/latch/GCLK (CGLPPRX8_RVT)
                                                   15     27.81    0.05     0.08    0.44 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/clk_gate_this_addr_g_int_reg/ENCLK (SNPS_CLOCK_GATE_HIGH_SDRAM_RFIFO_DW_FIFOCTL_IF_0_0)
                                                   15     27.81    0.05     0.00    0.44 r
  I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/count_int_reg_3_/CLK (SDFFARX2_HVT)
                                                                     0.05     0.00    0.44 r
  ---------------------------------------------------------------------------------------
  total clock latency                                                                0.44
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.local_skew.rpt**

❖ *This .rpt file reports the setup local skew and hold local skew information*

❖ *Setup local skew is calculated as "launch flop latency –capture flop latency –launch flop latency"*

❖ When setup=true, hold=false, only setup local skew is reported, when setup=false, hold=true, only hold local skew is reported and when setup=true, hold=true, both setup and hold local skews are reported

```
Attributes
==========
M Master Clock
G Generated Clock
& Internal Generated Clock
U User Defined Skew Group
D Default Skew Group
* Generated Clock Balanced Separately

================================================
==== Local Skew Reporting for Corner ff_125c ====
================================================

======================================== Summary Table for Corner ff_125c ========================================
Clock /                                Attrs     Sinks    Global       Max    Local Skew   Max Setup    Max Hold
Skew Group                                                 Skew   Latency    Pair Count  Local Skew  Local Skew
------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func.ff_125c
sdr_clk                                 M,D      1204    0.1367    0.4416         16560      0.1255      0.0828
SDRAM_CLK                               M,D      1203    0.0917    0.2995         16560      0.0803      0.0829
pclk                                    M,D      1101    0.1139    0.4189         17054      0.0591      0.0542
PCI_CLK                                 M,D      1100    0.0637    0.2693         17054      0.0589      0.0542
sys_clk                                 M,D       786    0.2774    0.5094          9817      0.2406      0.1214
 SYS_CLK                                  G       519    0.1301    0.2459          7720      0.0982      0.0518
 SYS_2x_CLK                               G       251    0.1098    0.2767          1486      0.0918      0.1094
### Mode: scan_capture, Scenario: scan_capture.ff_125c
sdr_clk_wft1                            M,D      1204    0.1367    0.4416         16415      0.1255      0.0828
pclk_wft1                               M,D      1101    0.1139    0.4189         15965      0.0591      0.0542
sys_clk_wft1                            M,D       786    0.2774    0.5094          9190      0.2406      0.1078
### Mode: scan_shift, Scenario: scan_shift.ff_125c
sdr_clk_wft1                            M,D      1204    0.1367    0.4416          1213      0.0698      0.0789
pclk_wft1                               M,D      1101    0.1139    0.4189          1097      0.0491      0.0538
sys_clk_wft1                            M,D       786    0.2774    0.5094          1135      0.2406      0.2124
------------------------------------------------------------------------------------------------------------------
All Clocks                                       3091    0.2774    0.5094                    0.2406      0.2124


  r = latency reported is for a rising edge triggered event at the sink
  f = latency reported is for a falling edge triggered event at the sink
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.local_skew.rpt**

```
Showing 5 largest and 5 smallest datapoints per clock / skew group (L=largest, S=smallest)
============================================== Details Table for Corner ff_125c ==============================================
Clock /                                 Launch Sink    Capture Sink      Launch     Capture        Late       Early       Local
Skew Group                              Name           Name             Latency     Latency      Offset      Offset        Skew
--------------------------------------------------------------------------------------------------------------------------------
### Mode: func, Scenario: func.ff_125c
sdr_clk
```

Setup Largest & Smallest datapoints

```
L   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__12_/CLK    0.4306 r    0.3051 f        --        --    0.1255
L   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__10_/CLK    0.4306 r    0.3051 f        --        --    0.1255
L   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__9_/CLK    0.4306 r    0.3051 f        --        --    0.1255
L   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_/CLK    0.4306 r    0.3051 f        --        --    0.1255
L   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__7_/CLK    0.4306 r    0.3051 f        --        --    0.1255
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__7_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --   -0.0828
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__9_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --   -0.0828
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__10_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --   -0.0828
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__12_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --   -0.0828
S   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --   -0.0828
```

Setup Largest & Smallest datapoints

```
(H) L   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --    0.0828
(H) L   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__12_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --    0.0828
(H) L   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__10_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --    0.0828
(H) L   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__9_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --    0.0828
(H) L   I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__7_/CLK I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_29__11_/CLK    0.3051 f    0.3879 f        --        --    0.0828
(H) S   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__12_/CLK    0.4306 r    0.3051 f        --        --   -0.1255
(H) S   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__10_/CLK    0.4306 r    0.3051 f        --        --   -0.1255
(H) S   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__9_/CLK    0.4306 r    0.3051 f        --        --   -0.1255
(H) S   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__7_/CLK    0.4306 r    0.3051 f        --        --   -0.1255
(H) S   I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__15_/CLK    0.4306 r    0.3051 f        --        --   -0.1255
```


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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.local_skew.rpt**

```
----------------------------------------
Largest Path #1
Mode                    : func
Corner                  : ff_125c
Scenario                : func.ff_125c
Skew Group              : default_sdr_clk
Clock Fanout            : sdr_clk
Clock at Launch Sink    : sdr_clk
Clock at Capture Sink   : sdr_clk
Launch Sink             : I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2
Capture Sink            : I_ORCA_TOP/I_SDRAM_IF/mega_shift_1_reg_30__12_/CLK
Setup Skew              : 0.1255
----------------------------------------

  ===== Launch Path =====

  Point                                        Fanout    Cap      Trans      Incr       Path
  ------------------------------------------------------------------------------------------
  source latency                                                              0.0000    0.0000
  sdr_clk (in)                                    1    2207.9141   0.0000    0.0000    0.0000 r
  sdr_clk_iopad/PADIO (I1025_NS)                                    0.0001    0.0000    0.0000 r
  sdr_clk_iopad/DOUT (I1025_NS)                   2     31.5517    0.1372    0.2113    0.2113 r
  I_CLOCK_GEN/sdram_clk (CLOCK_GEN)               2     31.5517    0.1372    0.0000    0.2113 r
  I_CLOCK_GEN/U9/A1 (AO22X1_RVT)                                    0.1374    0.0056    0.2170 r
  I_CLOCK_GEN/U9/Y (AO22X1_RVT)                   2      7.5890    0.0576    0.0719    0.2889 r
  I_CLOCK_GEN/o_sdram_clk (CLOCK_GEN)             2      7.5890    0.0576    0.0000    0.2889 r
  I_ORCA_TOP/sdram_clk (ORCA_TOP)                 2      7.5890    0.0576    0.0000    0.2889 r
  I_ORCA_TOP/ZCTSINV_10599_7208/A (INVX8_RVT)                       0.0576    0.0001    0.2890 r
  I_ORCA_TOP/ZCTSINV_10599_7208/Y (INVX8_RVT)     1     11.3551    0.0327    0.0120    0.3009 f
  I_ORCA_TOP/ZCTSINV_9646_7207/A (INVX16_RVT)                       0.0327    0.0002    0.3011 f
  I_ORCA_TOP/ZCTSINV_9646_7207/Y (INVX16_RVT)     3     40.5740    0.0277    0.0126    0.3137 r
  I_ORCA_TOP/ZCTSINV_8966_7206/A (INVX16_RVT)                       0.0289    0.0052    0.3189 r
  I_ORCA_TOP/ZCTSINV_8966_7206/Y (INVX16_RVT)     5     54.7482    0.0315    0.0172    0.3361 f
  I_ORCA_TOP/ZCTSINV_8070_7202/A (INVX16_RVT)                       0.0322    0.0032    0.3392 f
  I_ORCA_TOP/ZCTSINV_8070_7202/Y (INVX16_RVT)     3     34.4716    0.0241    0.0121    0.3513 r
  I_ORCA_TOP/ZCTSINV_7496_7201/A (INVX16_RVT)                       0.0241    0.0003    0.3516 r
  I_ORCA_TOP/ZCTSINV_7496_7201/Y (INVX16_RVT)     5     48.9152    0.0261    0.0161    0.3677 f
  I_ORCA_TOP/cto_inv_drc_8408/A (INVX4_RVT)                         0.0262    0.0006    0.3683 f
  I_ORCA_TOP/cto_inv_drc_8408/Y (INVX4_RVT)       2     10.4443    0.0211    0.0136    0.3818 r
  I_ORCA_TOP/cto_inv_drc_8412/A (INVX8_RVT)                         0.0211    0.0004    0.3822 r
  I_ORCA_TOP/cto_inv_drc_8412/Y (INVX8_RVT)       3     11.6105    0.0169    0.0105    0.3927 f
  I_ORCA_TOP/cto_inv_drc_8419/A (INVX2_RVT)                         0.0169    0.0003    0.3929 f
  I_ORCA_TOP/cto_inv_drc_8419/Y (INVX2_RVT)       8     10.7833    0.0243    0.0189    0.4118 r
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/ZCTSNET_3 (SDRAM_WFIFO)    8  10.7833   0.0243   0.0000   0.4118 r
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/macro_guide_buf_501/A (NBUFFX2_RVT)   0.0243   0.0001    0.4119 r
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/macro_guide_buf_501/Y (NBUFFX2_RVT)    1   0.4340   0.0138   0.0187   0.4306 r
  I_ORCA_TOP/I_SDRAM_WRITE_FIFO/SD_WFIFO_RAM/CE2 (SRAM2RW64x32)     0.0138    0.0000    0.4306 r
  ------------------------------------------------------------------------------------------
  total clock latency                                                                   0.4306
```

Largest Path per clock / skew group

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.structure.rpt**

```
****************************************
Report : report clock qor
         -type structure
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:09 2023
****************************************

C3 is corner ff_125c
C1 is corner ff_n40c
C2 is corner ss_125c
C4 is corner ss_n40c

Printing structure of pclk_wft1 (mode scan_capture) at root pin pclk:
(0) pclk [in Port] [Location (1437.90, 740.00)] [Net: pclk] [Fanout: 1]
 (1) pclk_iopad:PADIO->DOUT [Ref: saed32io_fc/I1025_NS] [Location (1440.00, 720.00)] [Net: net_pclk] [Fanout: 2]
  (2) I_CLOCK_GEN/macro_guide_buf_1:A->Y [Ref: saed32rvt/NBUFFX2_RVT] [Location (896.85, 724.18)] [Net: I_CLOCK_GEN/guide_buf_1] [Fanout: 1]
   (3) I_CLOCK_GEN/cto_dtrdly_8793:A->Y [Ref: saed32rvt/INVX2_RVT] [Location (895.03, 774.34)] [Net: I_CLOCK_GEN/cts15] [Fanout: 1]
    (4) I_CLOCK_GEN/cto_dtrdly_8798:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (893.36, 857.94)] [Net: I_CLOCK_GEN/cts18] [Fanout: 1]
     (5) I_CLOCK_GEN/I_PLL_PCI:REF_CLK->REF_CLKcheckpin1 [Ref: saed32pll/PLL] [Location (1014.46, 709.92)] [INTERNAL PIN] [SINK PIN]
  (2) I_CLOCK_GEN/U10:A1->Y [Ref: saed32rvt/AO22X1_RVT] [Location (895.48, 734.21)] [Net: I_CLOCK_GEN/o_pclk] [Fanout: 1]
   (3) cto_inv_drc_8579:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (896.85, 740.90)] [Net: cts1] [Fanout: 1]
    (4) cto_inv_drc_8578:A->Y [Ref: saed32rvt/INVX16_RVT] [Location (896.24, 737.56)] [Net: cts0] [Fanout: 6]
     (5) I_ORCA_TOP/ZCTSINV_6986_7062:A->Y [Ref: saed32rvt/INVX8_RVT] [Location (684.51, 750.93)] [Net: I_ORCA_TOP/ZCTSNET_6] [Fanout: 10]
      (6) I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSINV_6173_7037:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (718.86, 714.15)] [Net: I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSNET_8] [Fanout: 8]
       (7) I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSINV_6164_7036:A->Y [Ref: saed32rvt/INVX8_RVT] [Location (714.91, 714.15)] [Net: I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSNET_7] [Fanout: 7]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP14/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (710.67, 715.60)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP19/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.66, 715.60)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP15/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.66, 716.04)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP17/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.40, 718.94)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (716.75, 718.94)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP13/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (715.23, 716.04)] [SINK PIN]
```

- **The structure report is a very detailed representation of the clock tree structure**
- **The entire fanout of the clock tree is represented in this report, with indentation and numbering indicating different levels of the clock tree**
- **Clock balance points and ignore points and also represented in this report**
- **It can accept hierarchical pin as input to –to, or –through options**

❖ **Structure Level**

Number of buffers in clock tree divided from source

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.congestion.rpt (report about congestion)**

```
****************************************
Report : congestion
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:05 2023
****************************************

Layer     |    overflow     |              # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |     264 |     6 |     227  ( 0.02%) |       1
H routing |     155 |     6 |     130  ( 0.02%) |       1
V routing |     109 |     4 |      97  ( 0.01%) |       1

1
```

  - **Cts.fanout.rpt (report high fanout)**

```
****************************************
Report : net_fanout
        -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:05 2023
****************************************
1
```

❖ **There are no high fanout in this design**

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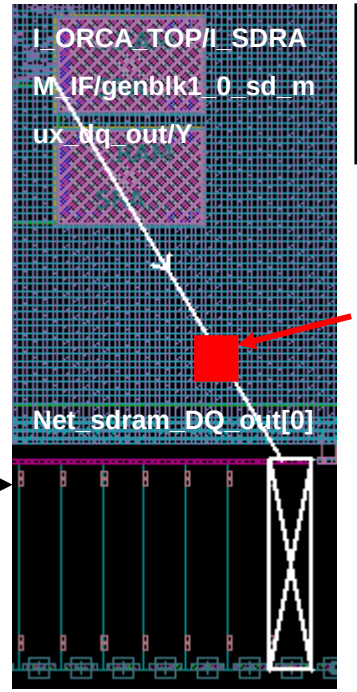
CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.max_tran.rpt (report about congestion)**

```
****************************************
Report : constraint
         -all_violators
         -max_transition
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:05 2023
****************************************


   Mode: func Corner: ff_n40c
   Scenario: func.ff_n40c
   max_transition
                             Required        Actual
   Net                      Transition     Transition        Slack  Violation
  -------------------------------------------------------------------------
   net_sdram_DQ_out[0]          0.19           0.30          -0.11  (VIOLATED)
     PIN : I_ORCA_TOP/I_SDRAM_IF/genblk1_0__sd_mux_dq_out/Y
                                0.19           0.30          -0.11  (VIOLATED)

   net_sdram_DQ_out[1]          0.19           0.28          -0.09  (VIOLATED)
     PIN : I_ORCA_TOP/I_SDRAM_IF/genblk1_1__sd_mux_dq_out/Y
                                0.19           0.28          -0.09  (VIOLATED)
```



❖ **This violation can be fixed by adding a buffer in front of the corresponding PAD pin**

Add buffer

❖ **Required Transition – Actual Transition = Slack**

❖ **This violation occurs because the net length between the PAD pin and the Mux Y pin is long**

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Cts.qor_snap.rpt**

```
Timing
------------------------------------------------------------------------------
Context                                 WNS           TNS           NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)            0.897         0.000             0
func.ff_n40c       (Setup)            1.076         0.000             0
func.ss_125c       (Setup)           -0.142        -1.096            19
func.ss_n40c       (Setup)           -0.688       -15.351            78
scan_capture.ff_125c (Setup)          9.265         0.000             0
scan_capture.ff_n40c (Setup)          9.335         0.000             0
scan_capture.ss_125c (Setup)          8.651         0.000             0
scan_capture.ss_n40c (Setup)          8.462         0.000             0
scan_shift.ff_125c (Setup)            9.582         0.000             0
scan_shift.ff_n40c (Setup)            9.596         0.000             0
scan_shift.ss_125c (Setup)            9.384         0.000             0
scan_shift.ss_n40c (Setup)            9.278         0.000             0
Design             (Setup)           -0.688       -15.401            81

func.ff_125c       (Hold)            -0.248      -337.878          3918
func.ff_n40c       (Hold)            -0.231      -276.950          4117
func.ss_125c       (Hold)            -0.661      -111.608          1472
func.ss_n40c       (Hold)            -0.314       -29.131           261
scan_capture.ff_125c (Hold)          -0.082        -6.164           220
scan_capture.ff_n40c (Hold)          -0.081        -7.522           308
scan_capture.ss_125c (Hold)          -0.069        -0.539             8
scan_capture.ss_n40c (Hold)          -0.114        -0.901             8
scan_shift.ff_125c (Hold)            -0.113       -44.992          1997
scan_shift.ff_n40c (Hold)            -0.107       -60.958          2747
scan_shift.ss_125c (Hold)            -0.135        -0.609            16
scan_shift.ss_n40c (Hold)            -0.183        -0.690            18
Design             (Hold)            -0.661      -427.473          6897
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                          748489.083
Cell Area (netlist and physical only):        1172489.083
Nets with DRC Violations:       434
1
```

- ❖ You can check WNS(Worst Negative Slack) first, and check TNS(Total Negative Slack) with NVE
- ❖ You can check Cell Area and number of DRC violations

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Ex) timing_report_min_cts_func.ss_125c**

```
Startpoint: I_ORCA_TOP/I_PARSER/sync_pcmd_out_valid_reg (rising edge-triggered flip-flop clocked by PCI_CLK)
Endpoint: I_ORCA_TOP/I_PARSER/r_pcmd_out_valid_reg (rising edge-triggered flip-flop clocked by pclk)
Mode: func
Corner: ss_125c
Scenario: func.ss_125c
Path Group: pclk
Path Type: min
```

```
Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**
- **Check Attributes and refer to the analysis**

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Ex) timing_report_min_cts_func.ss_125c (DAT)**

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	Attributes
clock PCI_CLK (rise edge)					0.000	0.000		
source latency					0.000	0.000		
I_CLOCK_GEN/I_PLL_PCI/CLK_1X (PLL)		0.029	1.000		0.000	0.000 r	(901.30,733.50)	s, n
I_CLOCK_GEN/net_pclk (net)	1							d
I_CLOCK_GEN/cts_trgdly_8807/A (INVX2_RVT)		0.029	1.000		0.000	0.000 r	(892.10,734.95)	
I_CLOCK_GEN/cts_trgdly_8807/Y (INVX2_RVT)		0.055	1.000		0.047	0.047 f	(892.23,735.05)	
I_CLOCK_GEN/cts16 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8811/A (INVX2_RVT)		0.055	1.000		0.001	0.047 f	(893.01,633.15)	
I_CLOCK_GEN/cts_trgdly_8811/Y (INVX2_RVT)		0.049	1.000		0.051	0.099 r	(893.14,633.05)	
I_CLOCK_GEN/cts19 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8813/A (INVX8_RVT)		0.049	1.000		0.000	0.099 r	(842.02,634.63)	
I_CLOCK_GEN/cts_trgdly_8813/Y (INVX8_RVT)		0.029	1.000		0.022	0.121 f	(841.89,634.73)	
I_CLOCK_GEN/cts21 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8812/A (INVX8_RVT)		0.029	1.000		0.000	0.121 f	(890.96,637.97)	
I_CLOCK_GEN/cts_trgdly_8812/Y (INVX8_RVT)		0.022	1.000		0.023	0.145 r	(890.83,638.08)	
I_CLOCK_GEN/cts20 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8810/A (INVX2_RVT)		0.022	1.000		0.001	0.145 r	(893.01,733.47)	
I_CLOCK_GEN/cts_trgdly_8810/Y (INVX2_RVT)		0.018	1.000		0.017	0.163 f	(893.14,733.37)	
I_CLOCK_GEN/cts17 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8806/A (INVX2_RVT)		0.018	1.000		0.000	0.163 f	(889.06,734.95)	
I_CLOCK_GEN/cts_trgdly_8806/Y (INVX2_RVT)		0.016	1.000		0.018	0.181 r	(889.19,735.05)	
I_CLOCK_GEN/cts14 (net)	1							d
I_CLOCK_GEN/macro_guide_buf_2/A (NBUFFX2_RVT)		0.016	1.000		0.000	0.181 r	(896.52,734.96)	
I_CLOCK_GEN/macro_guide_buf_2/Y (NBUFFX2_RVT)		0.020	1.000		0.039	0.219 r	(896.00,735.05)	
I_CLOCK_GEN/guide_buf_2 (net)	1							d
I_CLOCK_GEN/U10/A3 (AO22X1_RVT)		0.020	1.000		0.000	0.219 r	(894.70,734.06)	s
I_CLOCK_GEN/U10/Y (AO22X1_RVT)		0.044	1.000		0.075	0.294 r	(894.26,733.36)	s
I_CLOCK_GEN/o_pclk (net)	1							d
cto_inv_drc_8579/A (INVX4_RVT)		0.044	1.000		0.000	0.294 r	(896.28,740.16)	
cto_inv_drc_8579/Y (INVX4_RVT)		0.033	1.000		0.029	0.323 f	(896.15,740.06)	
cts1 (net)	1							d
cto_inv_drc_8578/A (INVX16_RVT)		0.033	1.000		0.000	0.324 f	(894.76,738.29)	
cto_inv_drc_8578/Y (INVX16_RVT)		0.040	1.000		0.032	0.356 r	(894.63,738.40)	
cts0 (net)	6							d
I_ORCA_TOP/ZCTSINV_6986_7062/A (INVX8_RVT)		0.042	1.000		0.008	0.364 r	(685.38,751.67)	
I_ORCA_TOP/ZCTSINV_6986_7062/Y (INVX8_RVT)		0.043	1.000		0.037	0.401 f	(685.51,751.77)	
I_ORCA_TOP/ZCTSNET_6 (net)	10							d
I_ORCA_TOP/ZCTSINV_6694_7058/A (INVX2_RVT)		0.043	1.000		0.001	0.403 f	(690.25,663.25)	
I_ORCA_TOP/ZCTSINV_6694_7058/Y (INVX2_RVT)		0.052	1.000		0.051	0.454 r	(690.37,663.15)	
I_ORCA_TOP/ZCTSNET_5 (net)	8							d
I_ORCA_TOP/I_PARSER/sync_pcmd_out_valid_reg/CLK (SDFFX2_LVT)		0.052	1.000		0.000	0.454 r	(685.45,622.02)	s, n
I_ORCA_TOP/I_PARSER/sync_pcmd_out_valid_reg/CLK (SDFFX2_LVT)		0.052	1.000		0.000	0.454 r	(685.45,622.02)	s, n
I_ORCA_TOP/I_PARSER/sync_pcmd_out_valid_reg/Q (SDFFX2_LVT)		0.061	1.000		0.170	0.625 r	(688.56,621.36)	s, n
I_ORCA_TOP/I_PARSER/dftopt52 (net)	2							
I_ORCA_TOP/I_PARSER/r_pcmd_out_valid_reg/D (SDFFX1_RVT)		0.061	1.000		0.002	0.626 r	(616.89,693.34)	s, n
data arrival time						0.626		

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CTS : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **Ex) timing_report_min_cts_func.ss_125c (DRT & slack)**

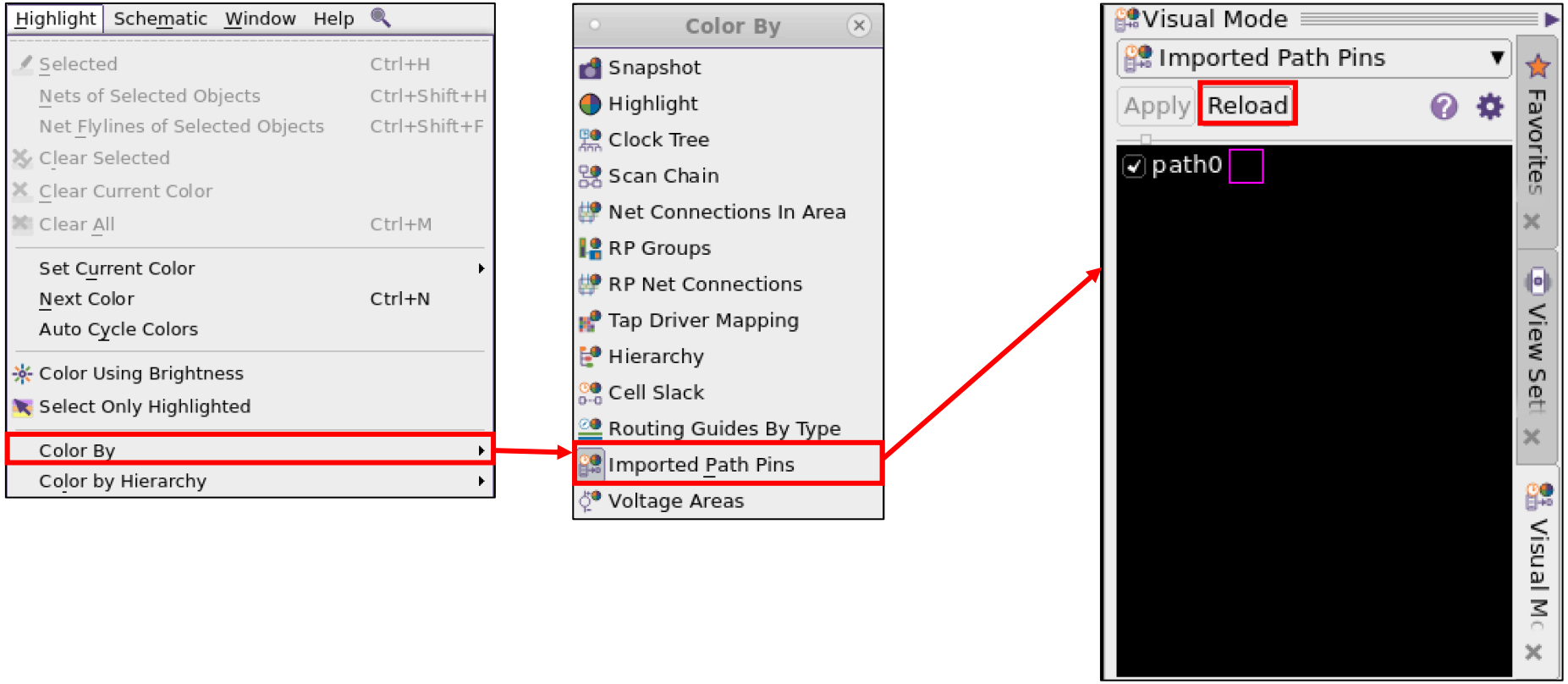
```
clock pclk (rise edge)                                                                    0.000      0.000
source latency                                                                            0.000      0.000
pclk (in)                                                    0.500      1.000         0.000      0.000 r    (1437.90,740.00)
pclk (net)                                        1                                                                          d
pclk_iopad/PADIO (I1025_NS)                                  0.594      1.000         0.000      0.000 r    (1437.90,740.00)  d
pclk_iopad/DOUT (I1025_NS)                                   0.205      1.000         0.494      0.494 r    (1240.42,736.55)  d
net_pclk (net)                                    2                                                                          d
I_CLOCK_GEN/U10/A1 (AO22X1_RVT)                              0.205      1.000         0.006      0.501 r    (895.16,733.45)   s
I_CLOCK_GEN/U10/Y (AO22X1_RVT)                               0.066      1.000         0.141      0.641 r    (894.26,733.36)   s
I_CLOCK_GEN/o_pclk (net)                          1                                                                          d
cto_inv_drc_8579/A (INVX4_RVT)                               0.066      1.000         0.000      0.641 r    (896.28,740.16)
cto_inv_drc_8579/Y (INVX4_RVT)                               0.048      1.000         0.041      0.682 f    (896.15,740.06)
cts1 (net)                                        1                                                                          d
cto_inv_drc_8578/A (INVX16_RVT)                              0.048      1.000         0.000      0.682 f    (894.76,738.29)
cto_inv_drc_8578/Y (INVX16_RVT)                              0.050      1.000         0.040      0.722 r    (894.63,738.40)
cts0 (net)                                        6                                                                          d
I_ORCA_TOP/ZCTSINV_4833_7120/A (INVX16_RVT)                  0.053      1.000         0.011      0.733 r    (593.04,733.47)
I_ORCA_TOP/ZCTSINV_4833_7120/Y (INVX16_RVT)                  0.035      1.000         0.029      0.762 f    (592.91,733.37)
I_ORCA_TOP/ZCTSNET_16 (net)                       8                                                                          d
I_ORCA_TOP/ZCTSINV_3986_7094/A (INVX4_RVT)                   0.035      1.000         0.000      0.762 f    (590.23,733.47)
I_ORCA_TOP/ZCTSINV_3986_7094/Y (INVX4_RVT)                   0.046      1.000         0.044      0.806 r    (590.36,733.37)
I_ORCA_TOP/ZCTSNET_10 (net)                       3                                                                          d
I_ORCA_TOP/cto_inv_drc_8663/A (INVX8_RVT)                    0.046      1.000         0.001      0.807 r    (625.80,734.95)
I_ORCA_TOP/cto_inv_drc_8663/Y (INVX8_RVT)                    0.047      1.000         0.042      0.849 f    (625.93,735.05)
I_ORCA_TOP/cts84 (net)                           11                                                                          d
I_ORCA_TOP/ZCTSINV_2784_7079/A (INVX2_RVT)                   0.047      1.000         0.001      0.849 f    (623.82,701.51)
I_ORCA_TOP/ZCTSINV_2784_7079/Y (INVX2_RVT)                   0.060      1.000         0.058      0.907 r    (623.95,701.61)
I_ORCA_TOP/ZCTSNET_8 (net)                       10                                                                          d
I_ORCA_TOP/I_PARSER/r_pcmd_out_valid_reg/CLK (SDFFX1_RVT)    0.060      1.000         0.000      0.907 r    (617.81,692.58)   s, n

clock reconvergence pessimism                                                        -0.000      0.907
clock uncertainty                                                                     0.100      1.007
library hold time                                                       1.000        -0.074      0.933
data required time                                                                              0.933
-------------------------------------------------------------------------------------------------------------------------------------------------------------
data required time                                                                              0.933
data arrival time                                                                              -0.626
-------------------------------------------------------------------------------------------------------------------------------------------------------------
slack (VIOLATED)                                                                               -0.307
```


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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **The "imported Path Pins" visual mode takes timing path data from a timing report and highlights the timing path in the Layout**
- **Paths can be viewed one-by-one**
- **Menu Location**
  - Highlight > Color By > Imported Path Pins



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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

1. Choose Highlight > color By > imported Path Pins to open the imported path pins visual Mode
2. Update the timing for your design with the update_timing command (**update_timing -full**)
3. Report the timing for a path with the report_timing command and save the result to a fil

```
report_timing  \
        -from   I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg \
        -to   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_6_ \
        -physical -nets -nosplit -input_pins -delay_type min > ./test_timing.rpt
```

4. Click the Reload button
5. Select Import Path Pins from Timing Report and click the file chooser button
6. Select the timing.rpt file you wrote out in step 3 and click ok
7. Choose the show path as option to specify the path type to show: Flylines, routes or both (default option is Flylines) and apply

❖ The Imported Path Pins Visual Mode panel displays a histogram bar for the path with the number of pins in the path

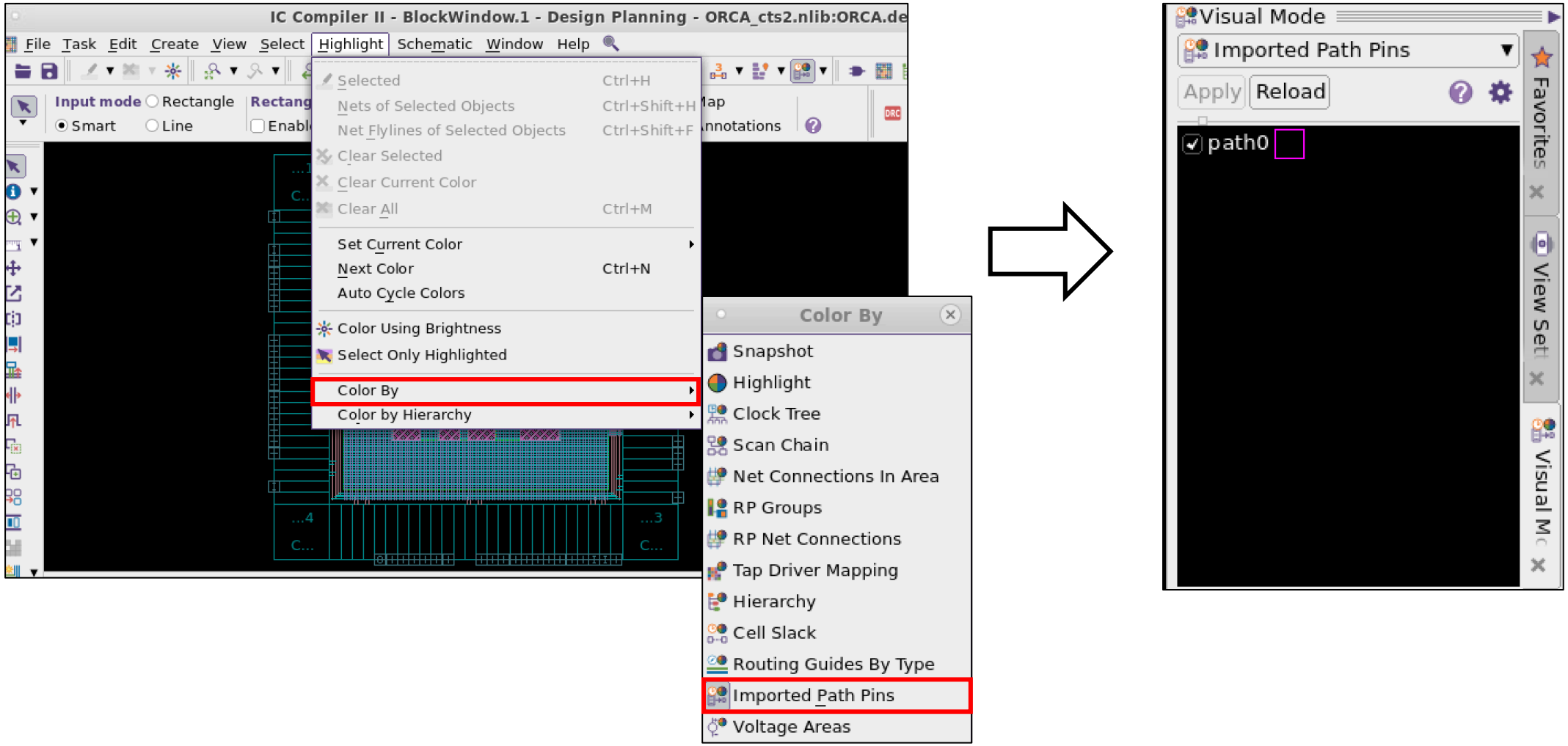
❖ Repeat steps 3 through 6 to import and display additional paths (Choose the Clear all buckets before importing new path pins option to delete the histogram before importing the new path)

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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

1. Choose Highlight > color By > imported Path Pins to open the imported path pins visual Mode



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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

2. Update the timing for your design with the update_timing command (**update_timing -full**)

```
icc2_shell> update_timing -full
```

3. Report the timing for a path with the report_timing command and save the result to a file

```
report_timing  \
        -from   I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg \
        -to   I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_6_  \
        -physical -nets -nosplit -input_pins -delay_type min > ./test_timing.rpt
```

4. Click the Reload button

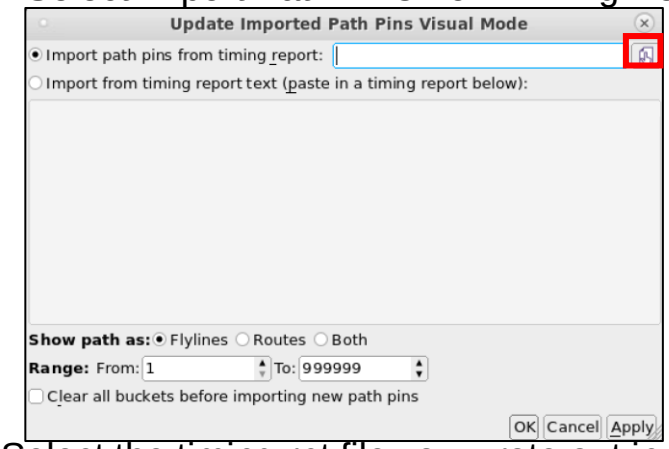


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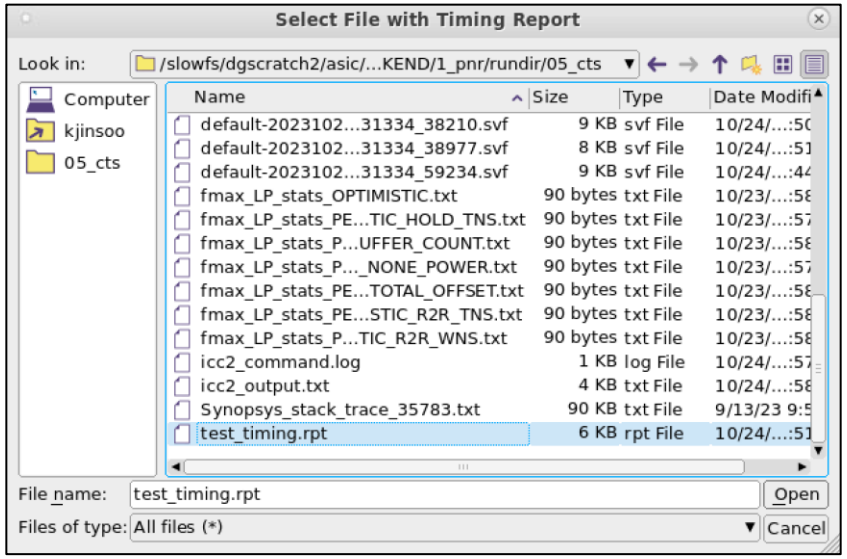
CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

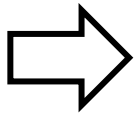
- **Load Timing Report**

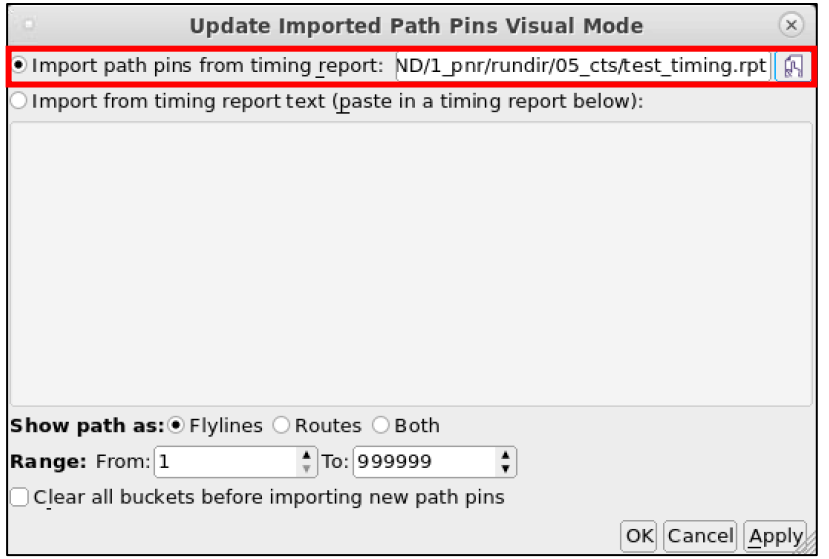
5. Select Import Path Pins from Timing Report and click the file chooser button



6. Select the timing.rpt file you wrote out in step 3 and click ok (test_timing.rpt)





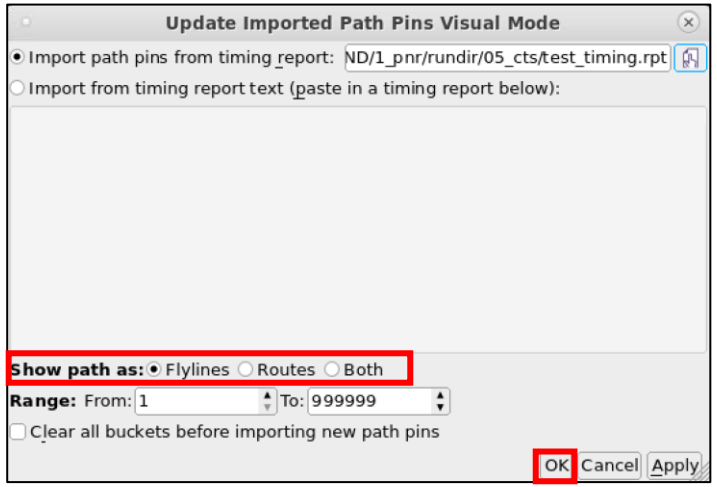


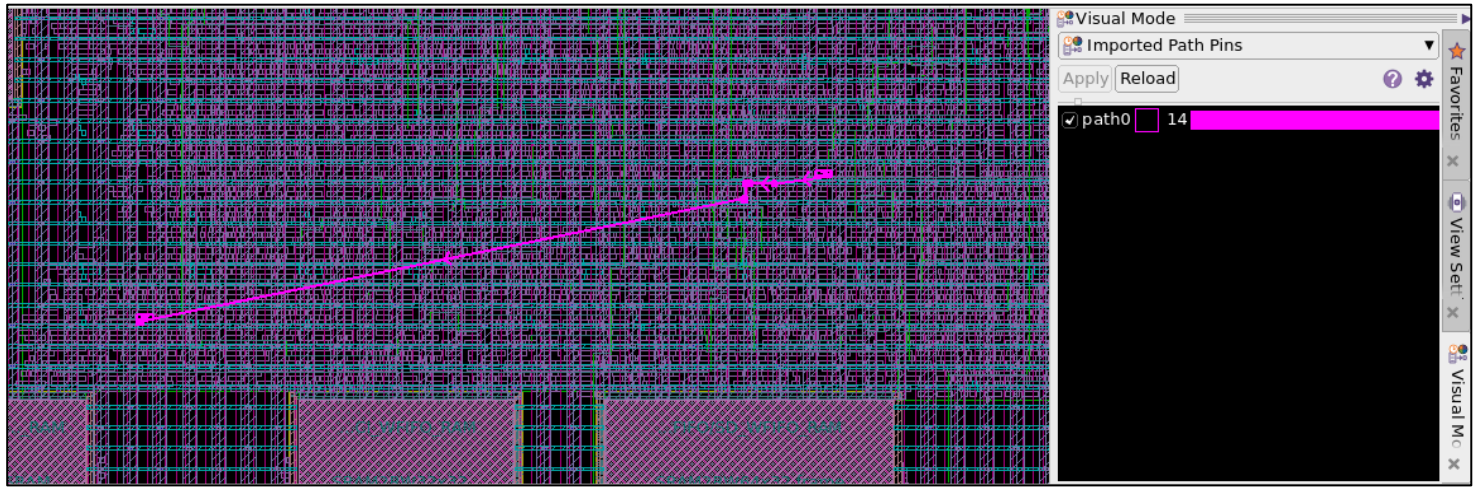
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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Load Timing Report**

7. Choose the show path as option to specify the path type to show: Flylines, routes or both (default option is Flylines)





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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Analyze Timing Violation using Imported_path_pins Visual Mode**

1. Open $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts/timing_report_min_cts_func.ff_125c
2. Find VIOLATED Path in timing reports
   - ❖ Startpoint: I_ORCA_TOP/I_PARSER/sync_pcmd_out_reg_1_ (rising edge-triggered flip-flop clocked by PCI_CLK)
   - ❖ Endpoint: I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_1_ (rising edge-triggered flip-flop clocked by pclk)
3. Open Imported_path_pins visual Mode and click reload button & select import from timing report text
4. Click ok button by copying and pasting DAT and DRT path of timing reports respectively
5. Analyze two path (DAT, DRT)

- **Open timing report & find VIOLATED PATH**

< timing_report_min_cts_func.ff_125c >

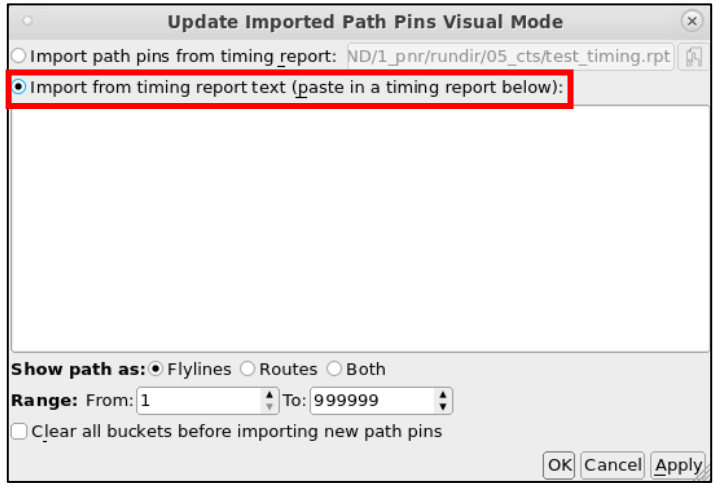
```
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:22 2023
****************************************
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_ORCA_TOP/I_PARSER/sync_pcmd_out_reg_1_ (rising edge-triggered flip-flop clocked by PCI_CLK)
  Endpoint: I_ORCA_TOP/I_PARSER/r_pcmd_out_reg_1_ (rising edge-triggered flip-flop clocked by pclk)
  Mode: func
  Corner: ff_125c
  Scenario: func.ff_125c
  Path Group: pclk
  Path Type: min
```


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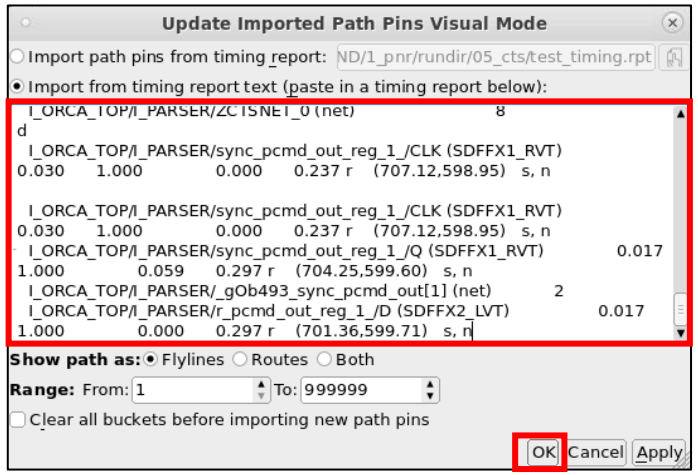
CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Analyze Timing Violation using Imported_path_pins Visual Mode**
  - Open Imported_path_pins visual Mode and click reload button & select import from timing report text

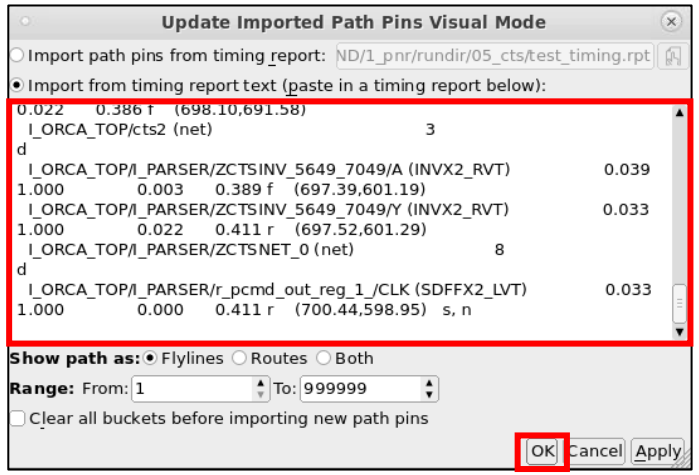


  - Click ok button by copying and pasting DAT and DRT path of timing reports respectively

< DAT >



< DRT >

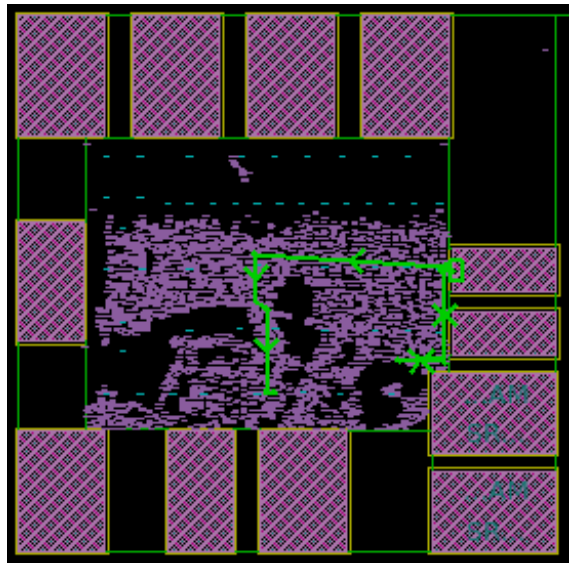


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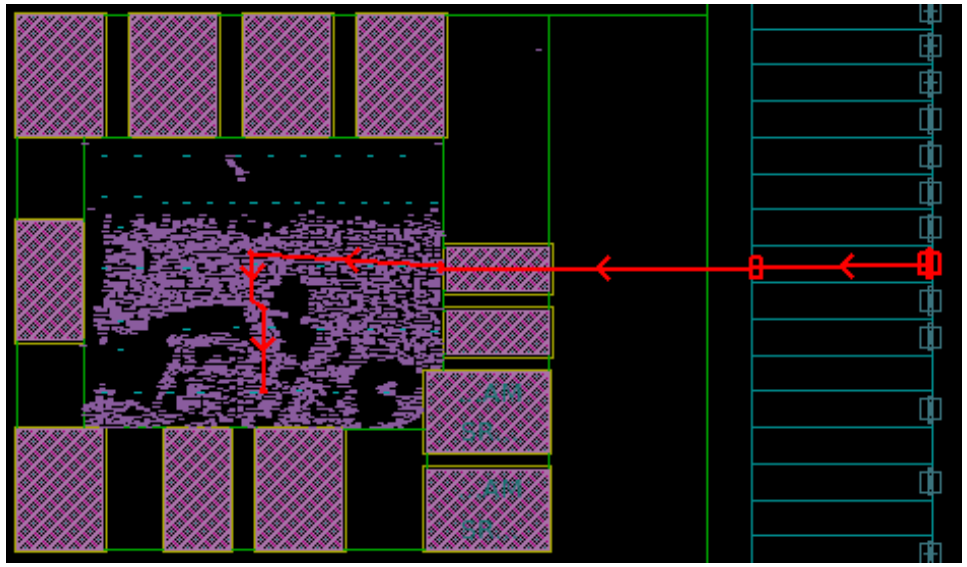
CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Analyze Timing Violation using Imported_path_pins Visual Mode**
  - Analyze two path (DAT, DRT)

< DAT >



< DRT >



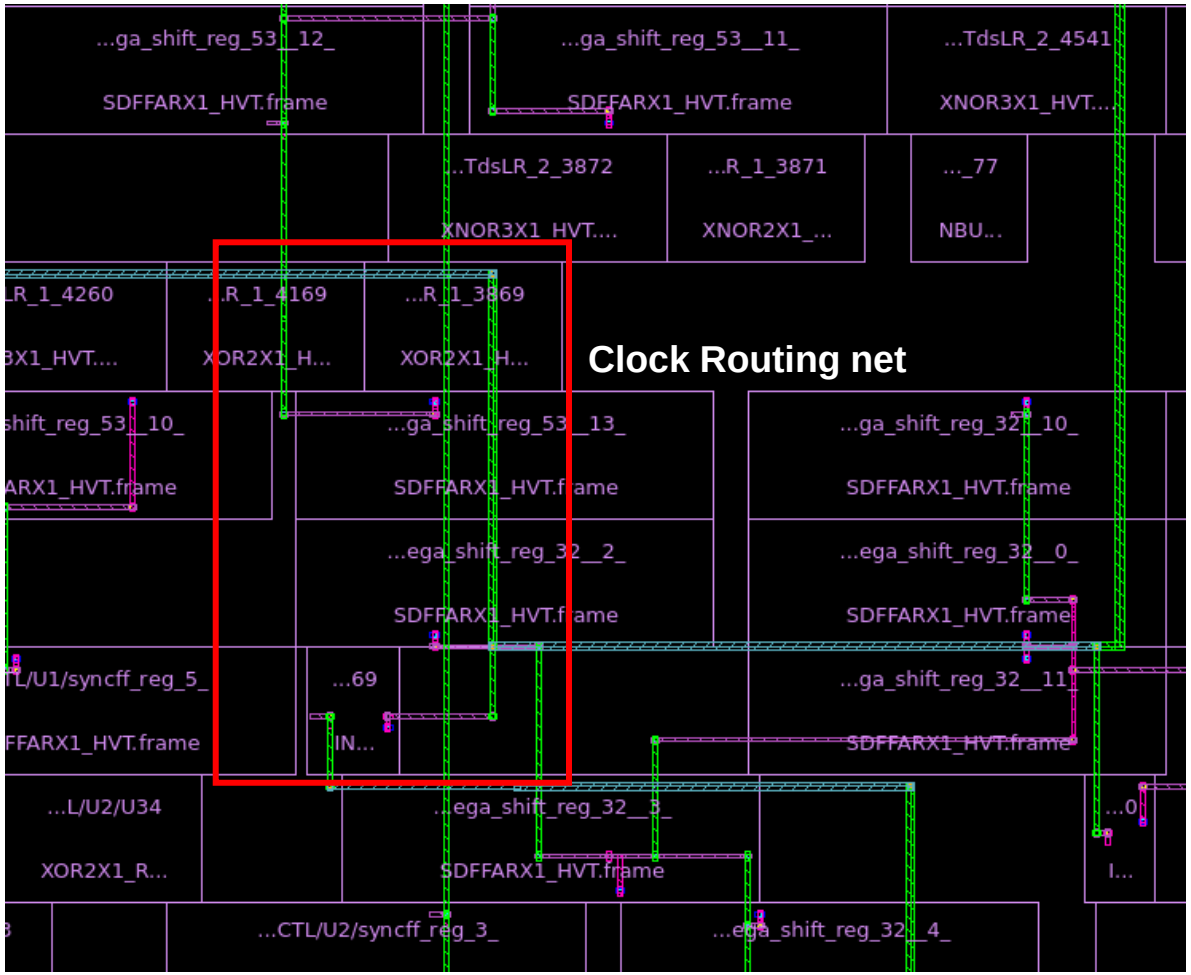
- **Setup Slack = DRT - DAT**
- **Hold Slack = DAT - DRT**

This violation occurred because DRT Path was longer than DAT Path

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CTS : Use ICC2 (IC Compiler II tool) Check Clock Routing

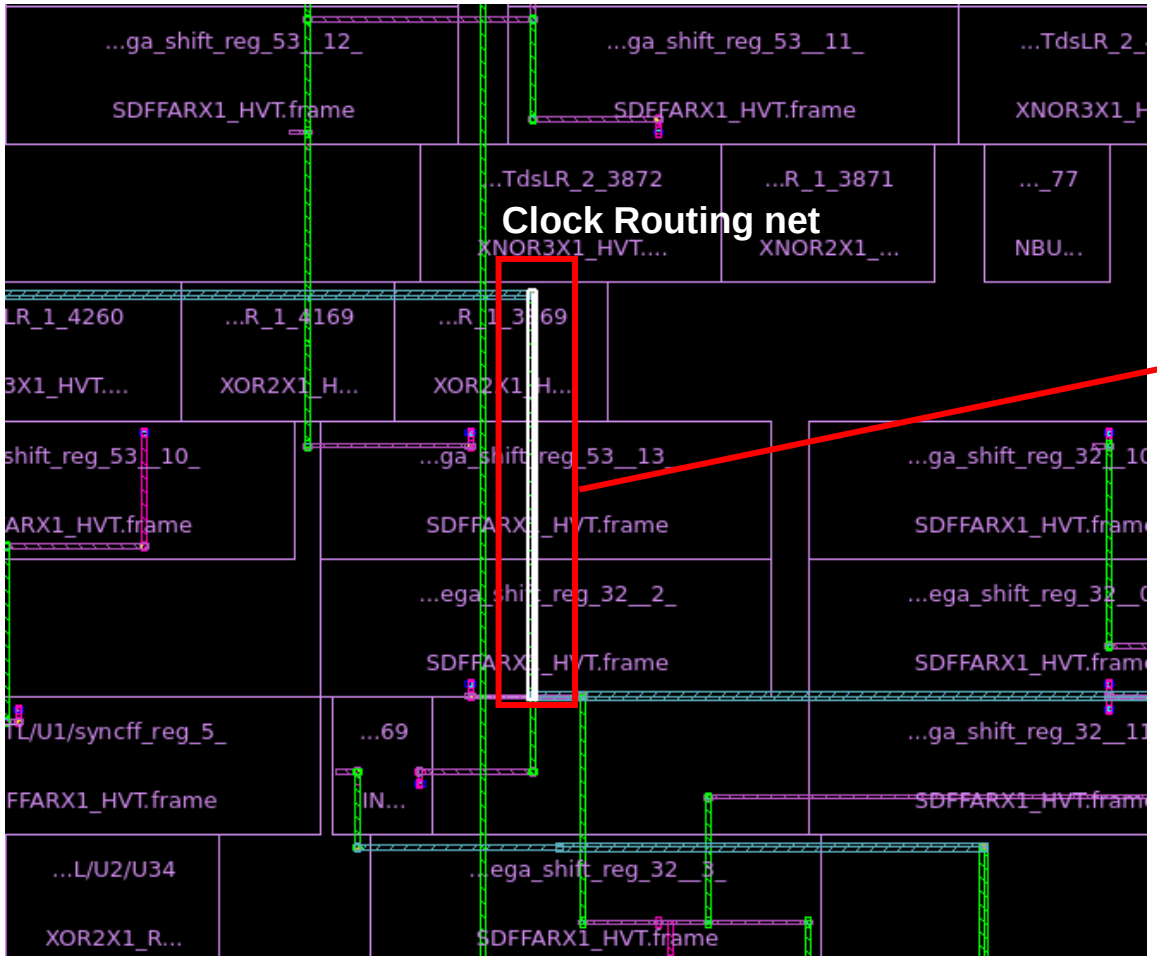
- **Clock Routing**
  - **Check the detail routing of the clock trees**

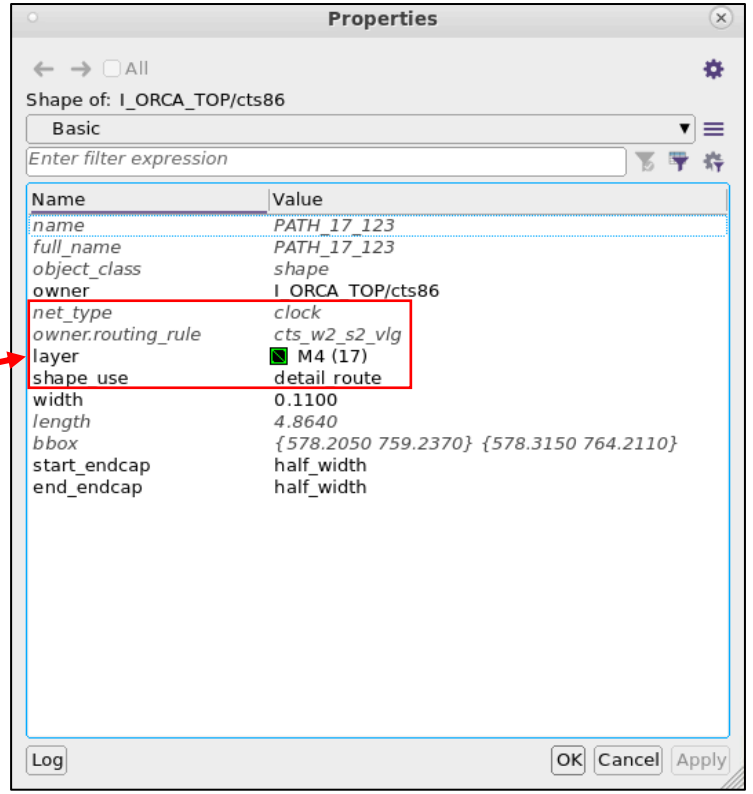


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CTS : Use ICC2 (IC Compiler II tool) Check Clock Routing

- **Clock Routing**
  - **Check the detail routing of the clock trees (NDR Rules)**



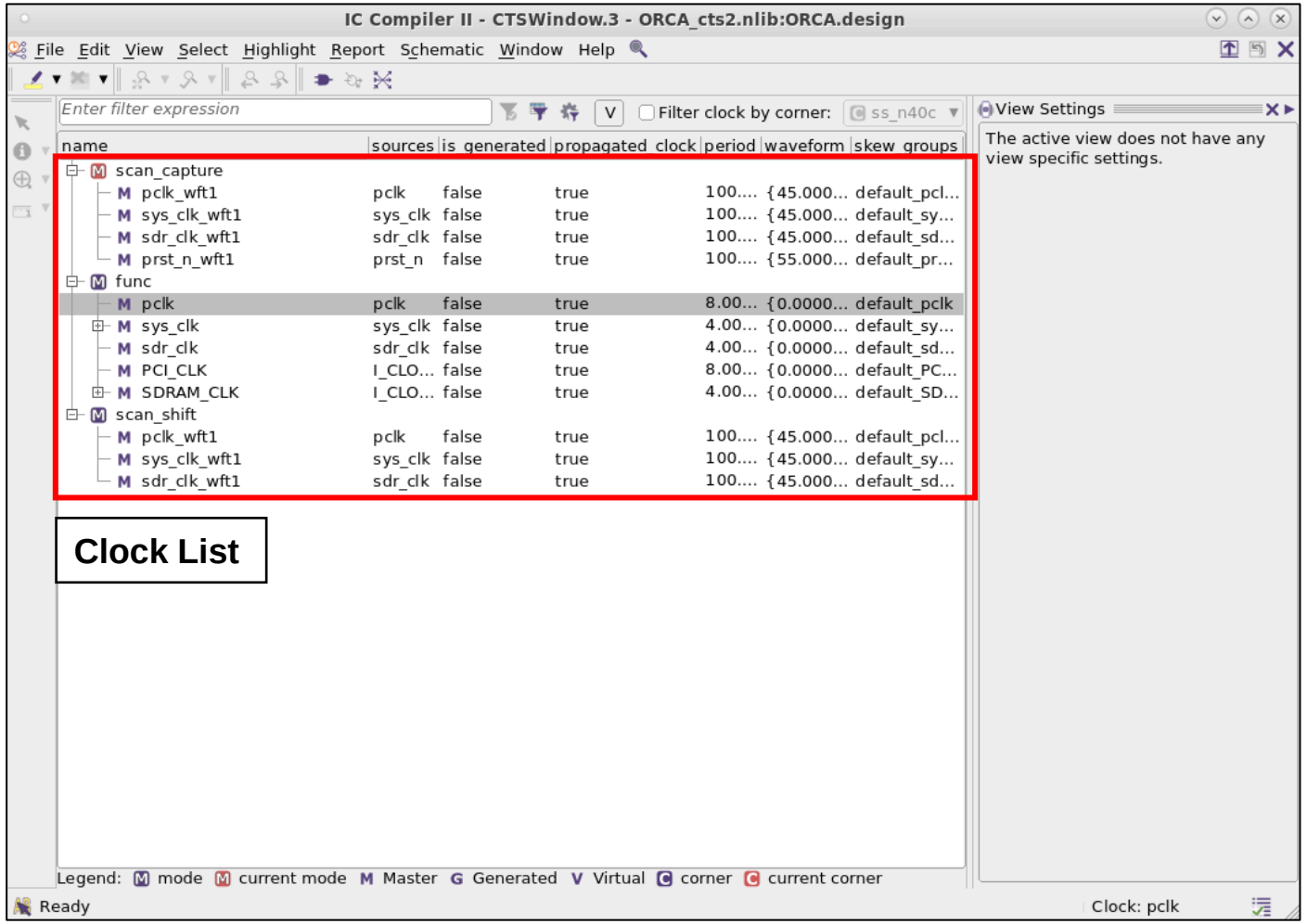


- **Open Properties and check (owner.routing_rule, shape_use, net_type)**

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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Open Clock Tree Analysis Window**



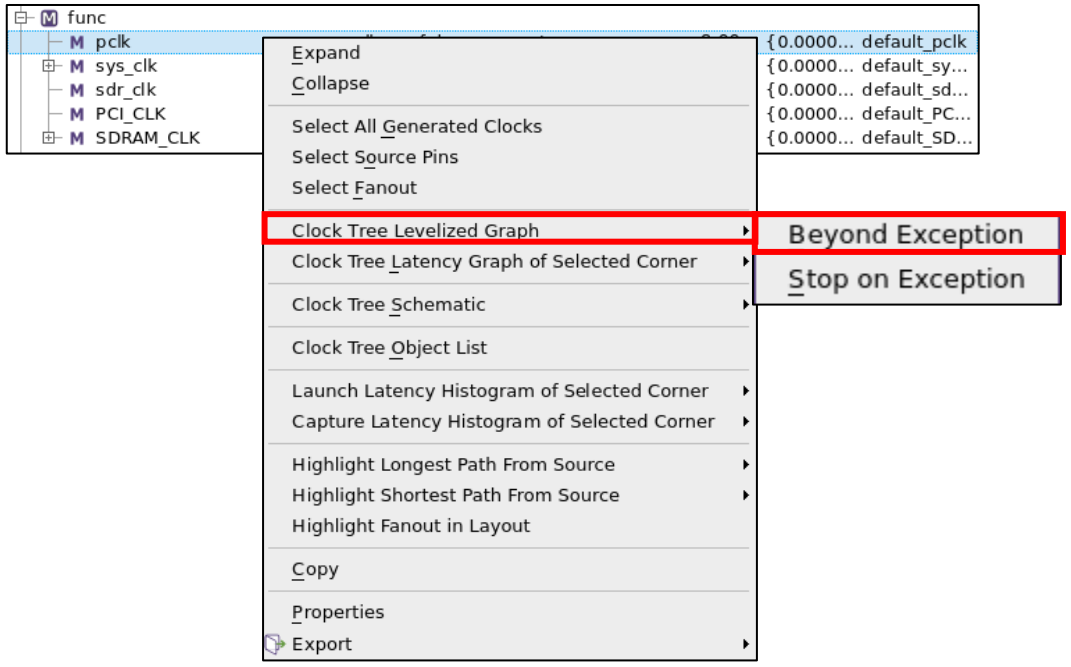
- **Open Clock Tree Analysis Window**

  Window > Clock Tree Analysis Window

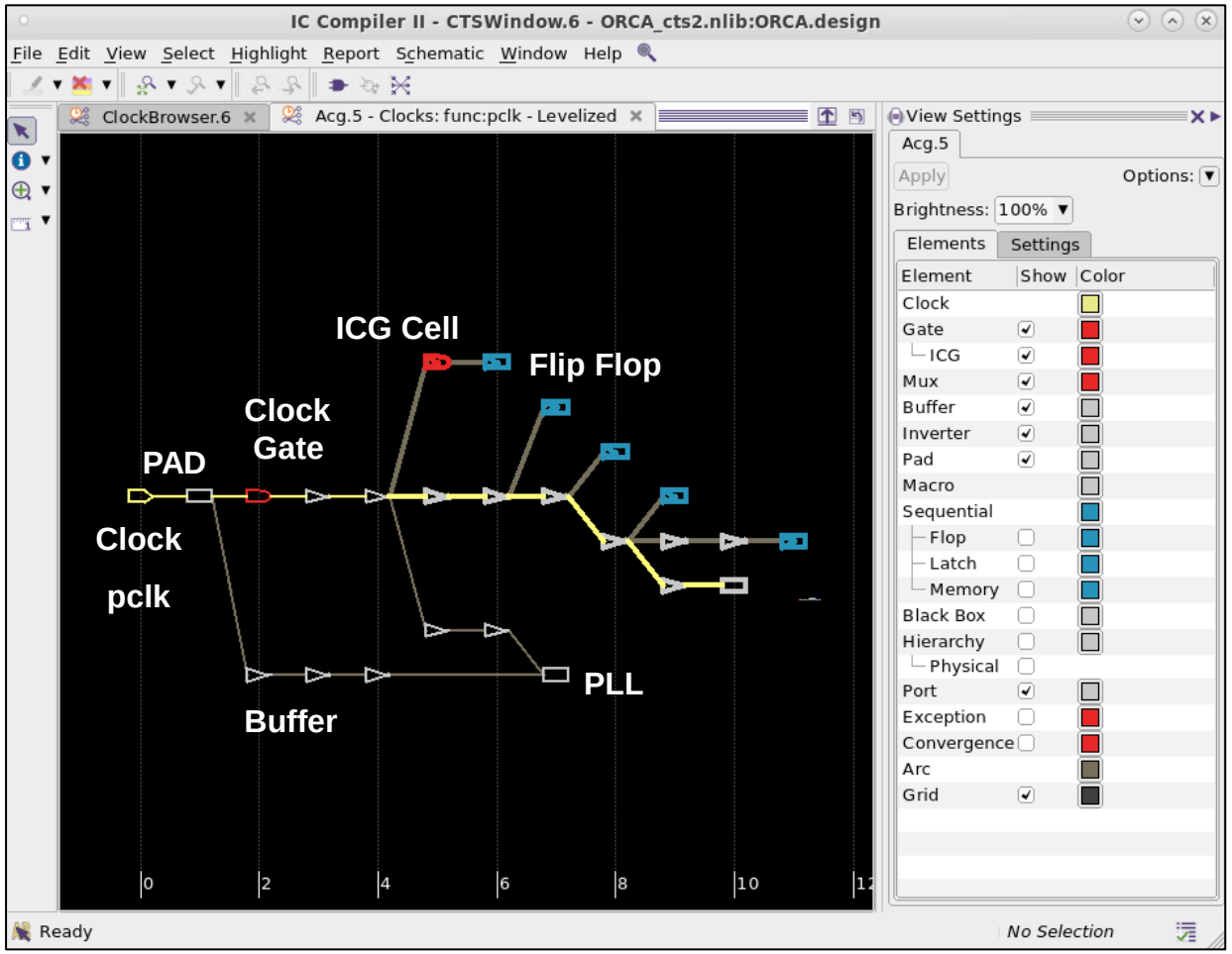
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Clock Tree Levelized Graph**



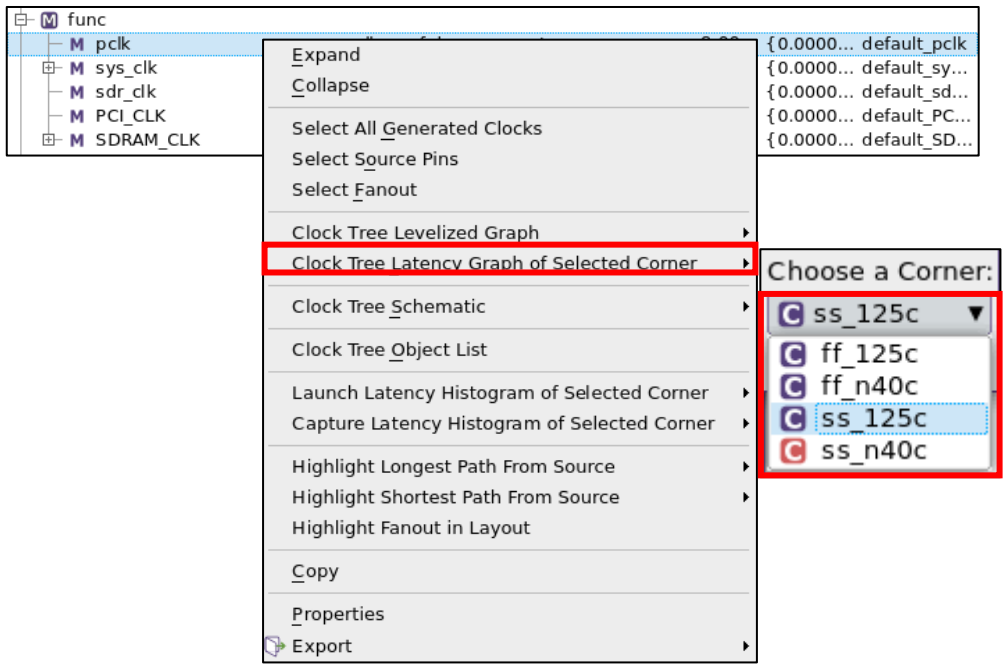
- **Click Right button on mouse > select Clock Tree Levelized Graph > Beyond Exception**



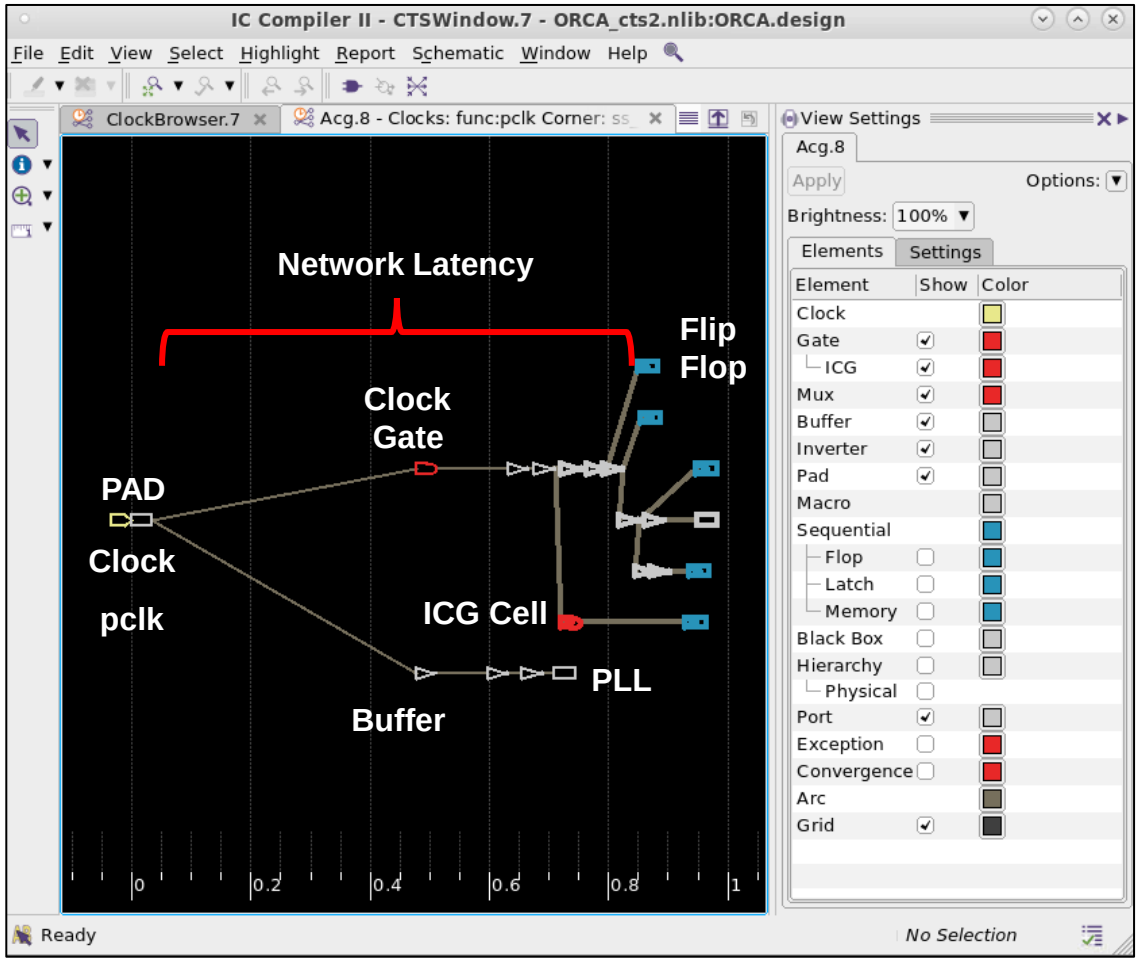
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Clock Tree Latency Graph of Selected Corner**



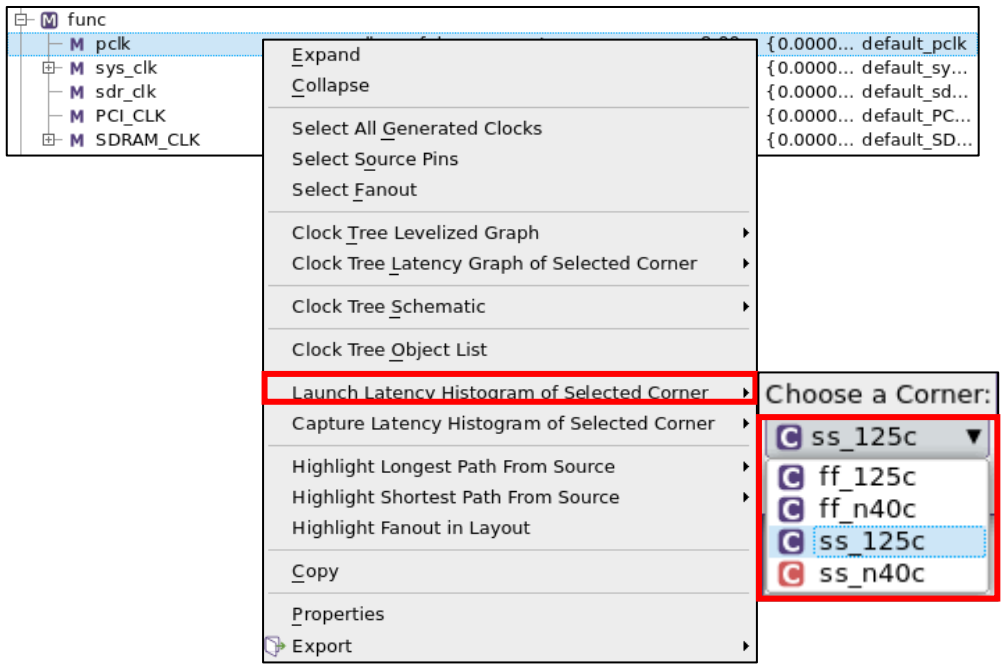
- **Click Right button on mouse > select Clock Tree Latency Graph of Selected Corner > Choose a Corner**



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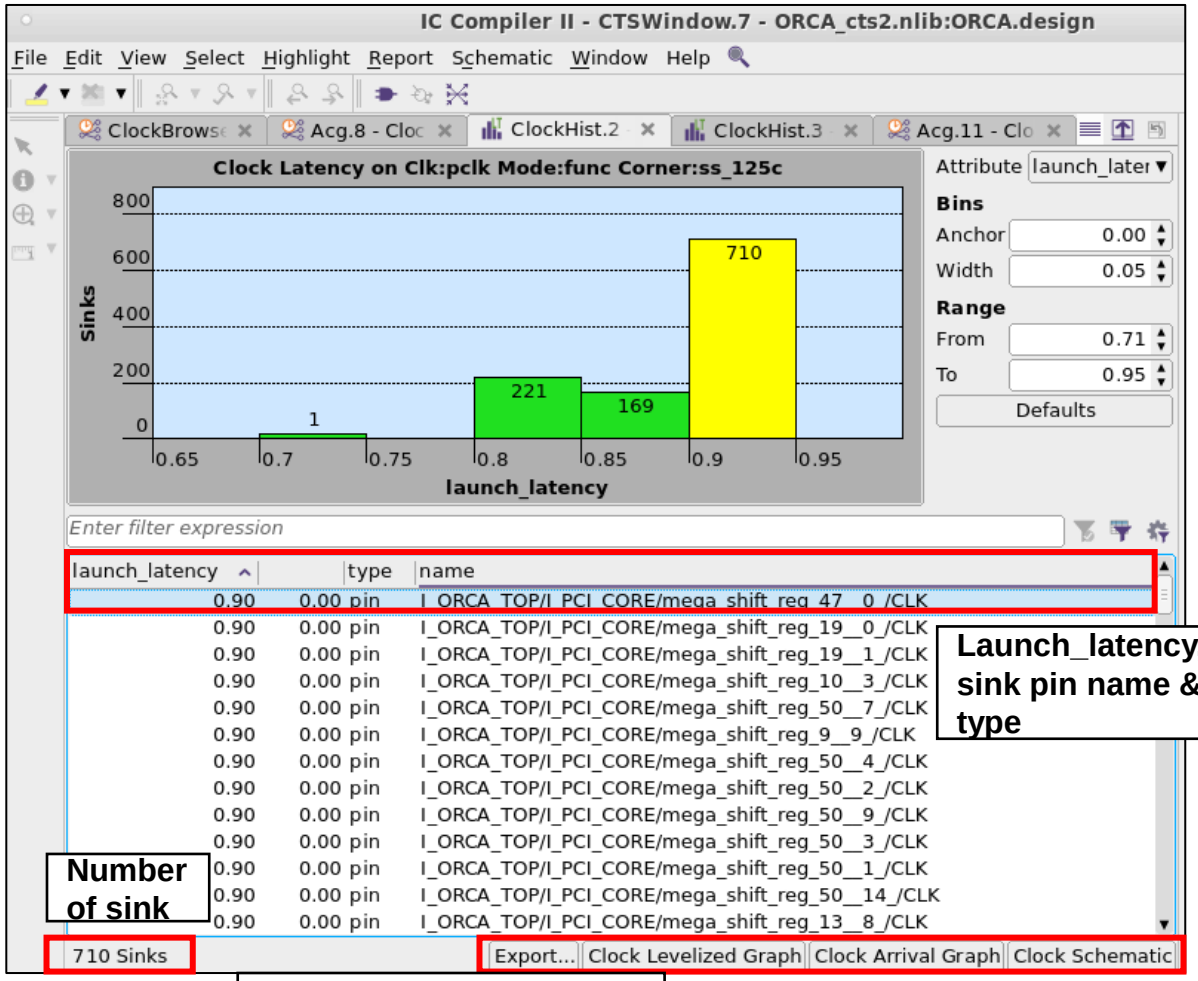
CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Launch Latency Histogram of Selected Corner**



- **Click Right button on mouse > select Clock Launch Latency Histogram of Selected Corner > Choose a Corner**

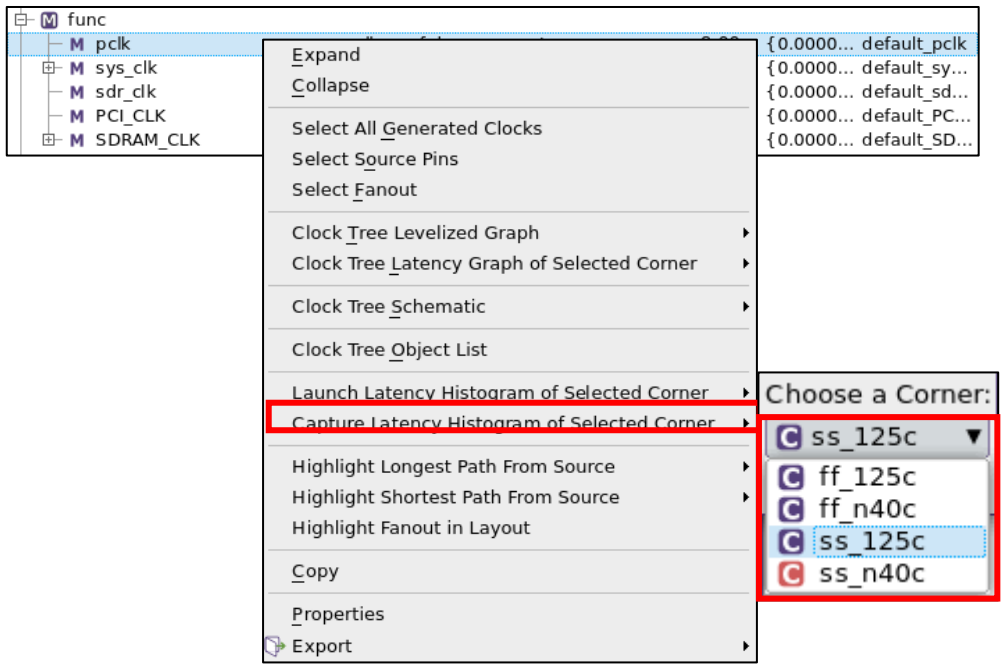
- **Launch path is launch clock path which is responsible for launching the data at launch flip flop**



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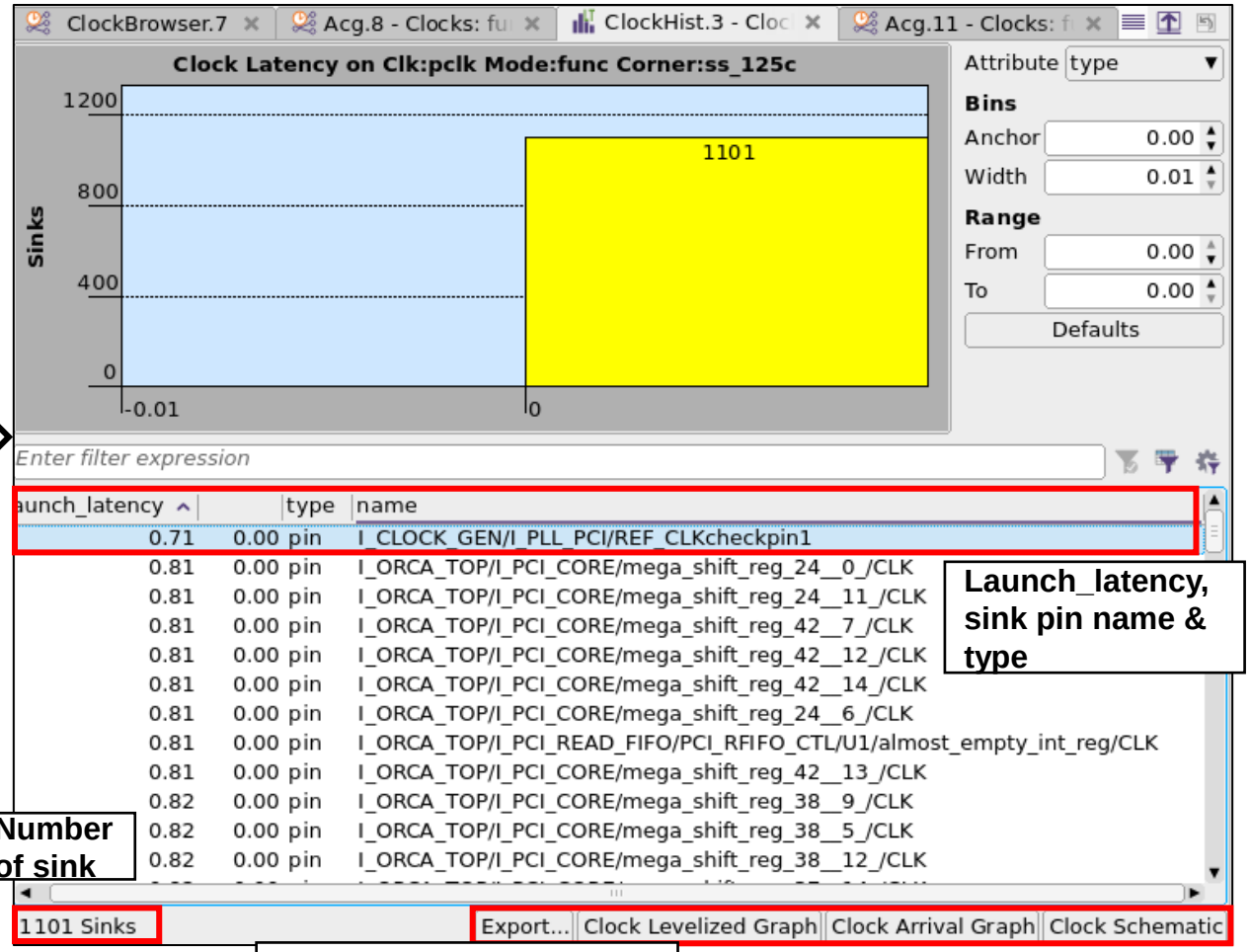
CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Capture Latency Histogram of Selected Corner**



- **Click Right button on mouse > select Clock Capture Latency Histogram of Selected Corner > Choose a Corner**

- **Capture path is capture clock path which is responsible for capturing the data at capture flip flop**

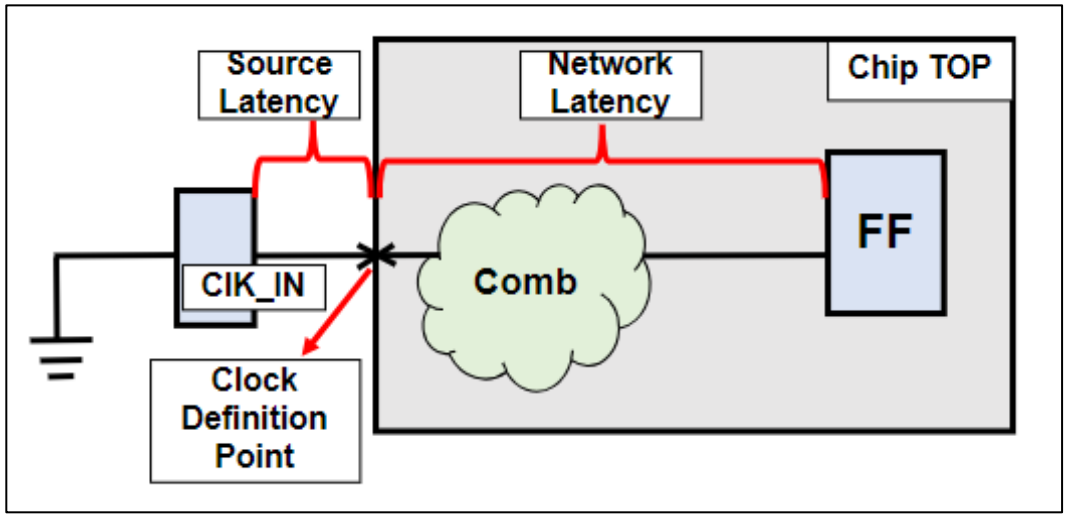


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Introduction to CTS : What is Clock Latency

- **Clock Latency**
  - Clock Latency is the time taken by the clock to reach the sink pin from the clock source
  - It is divided into two parts (clock source latency, clock network latency)
  - Clock source Latency defines the delay between the clock waveform origin point to the definition point
  - Clock network latency is the delay form the clock definition point to the sink pin
  - Clock Latency is also called Clock insertion Delay

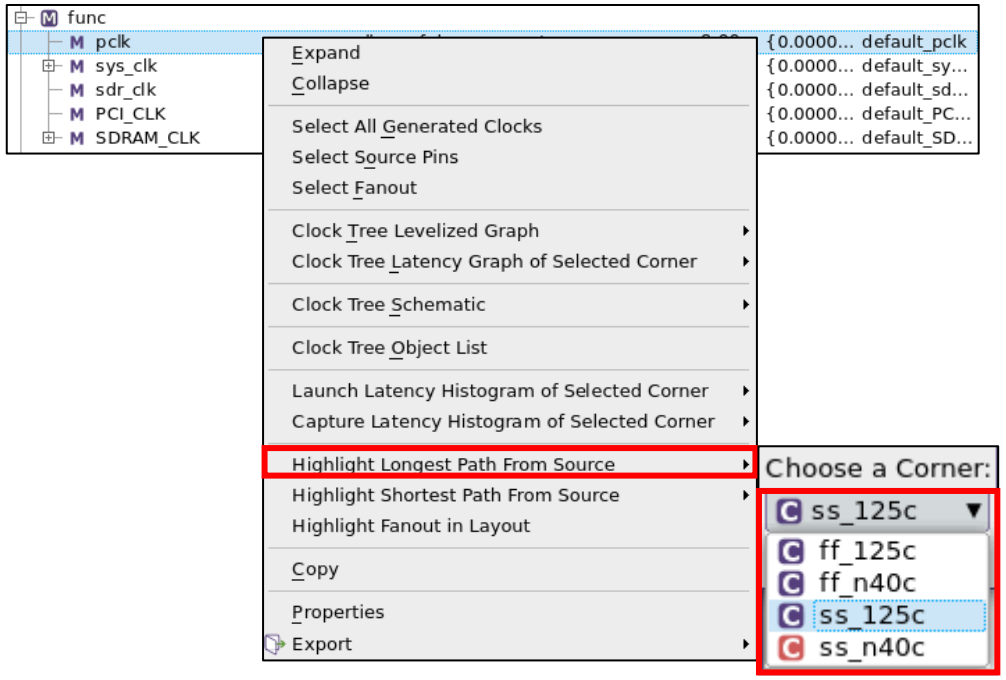
Clock Latency = Source latency + Network latency



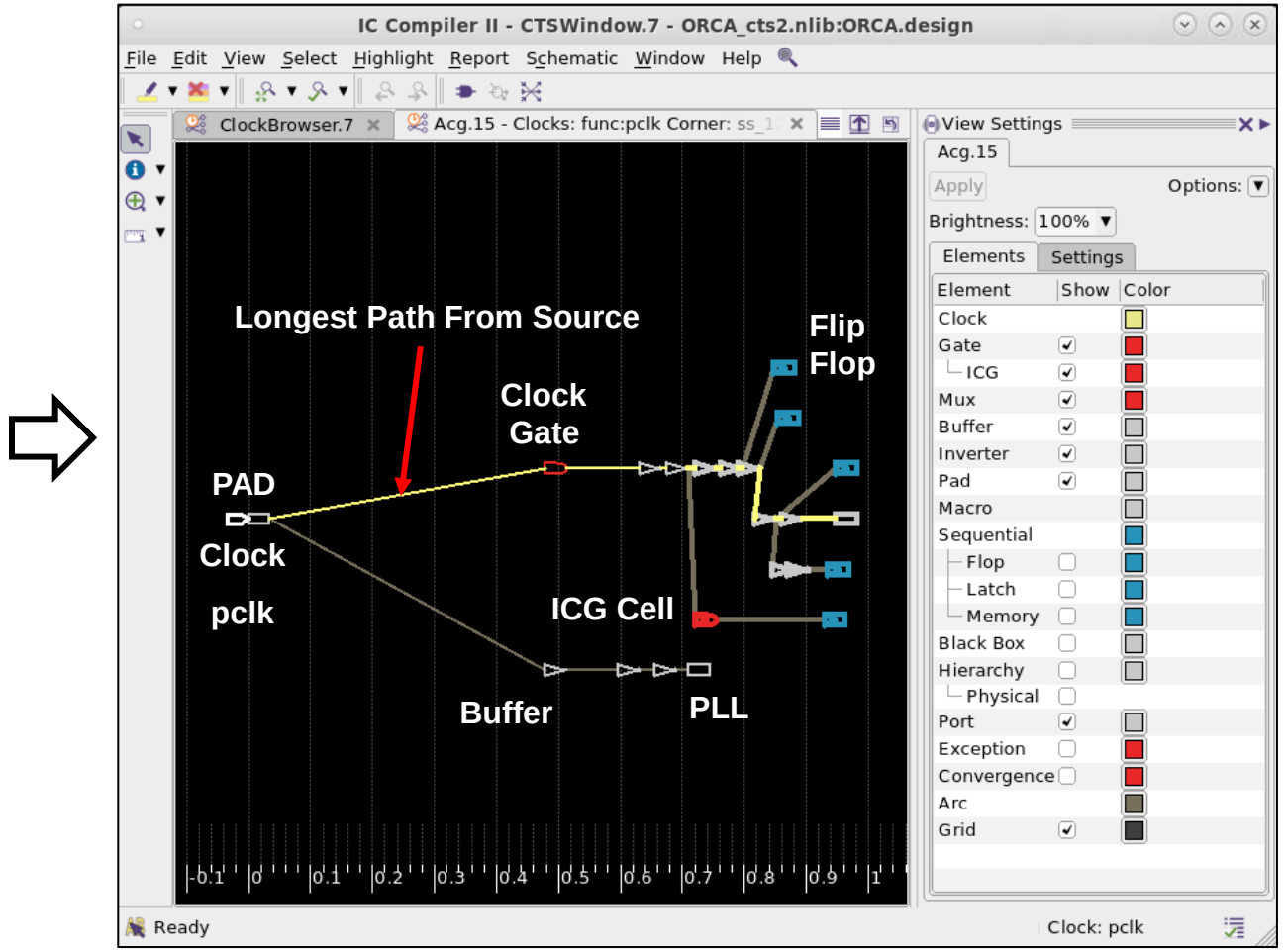
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Highlight Longest Path From Source**



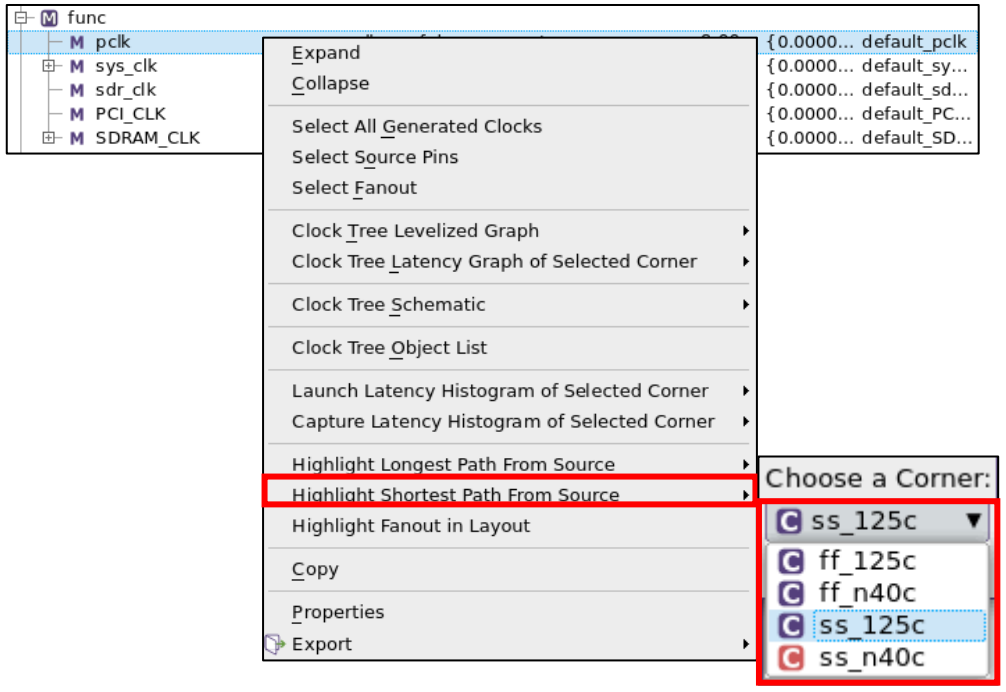
- **Click Right button on mouse > Highlight Longest Path From Source > Choose a Corner**



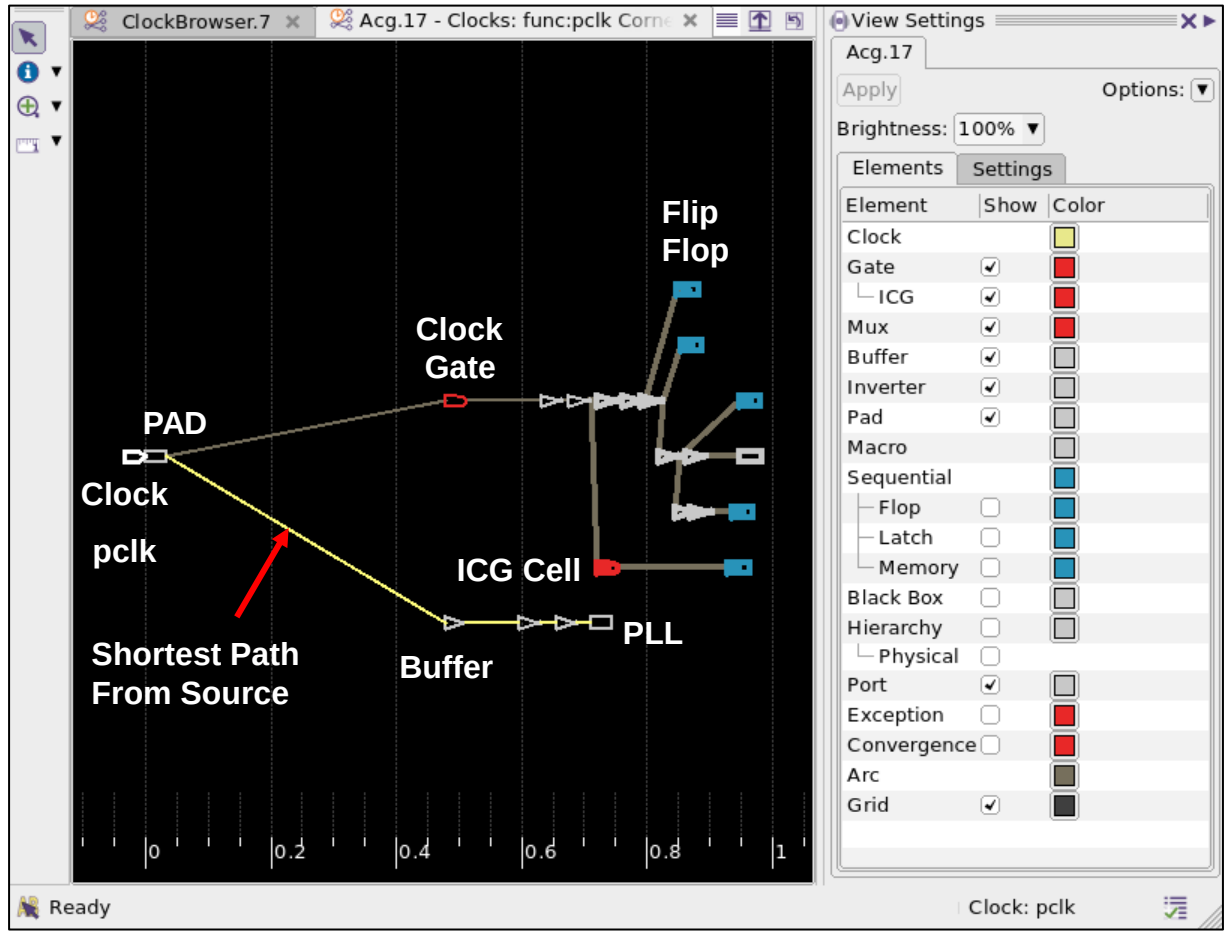
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Highlight Shortest Path From Source**



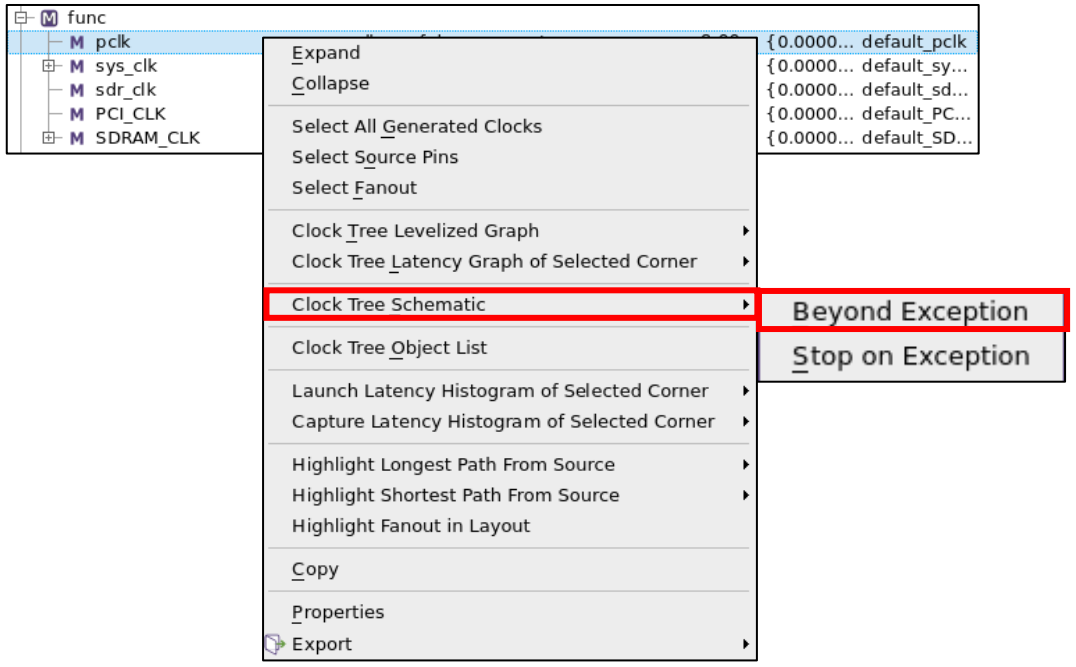
- **Click Right button on mouse > Highlight Shortest Path From Source > Choose a Corner**



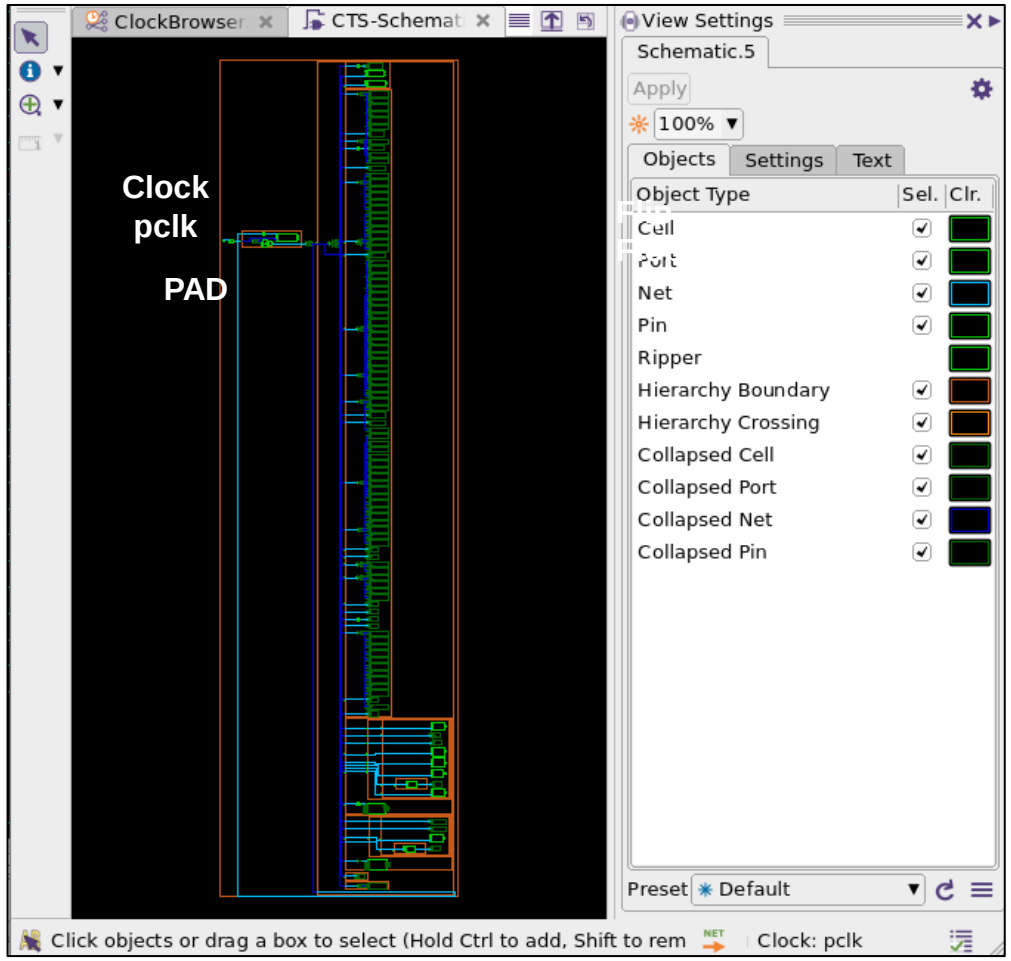
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Clock Tree Schematic**



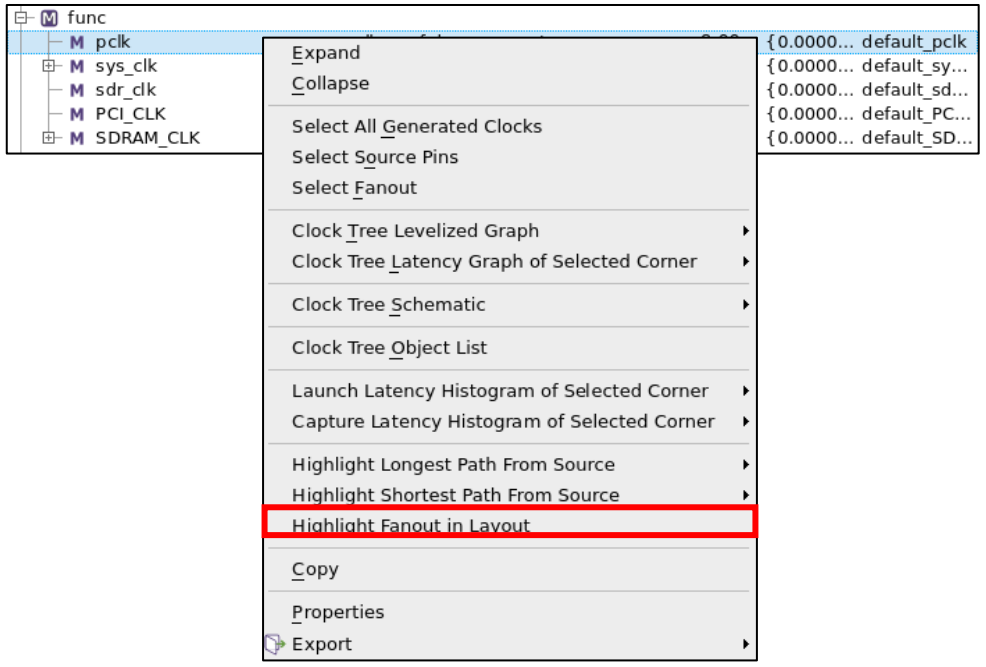
- **Click Right button on mouse > Clock Tree Schematic > Beyond Exception**



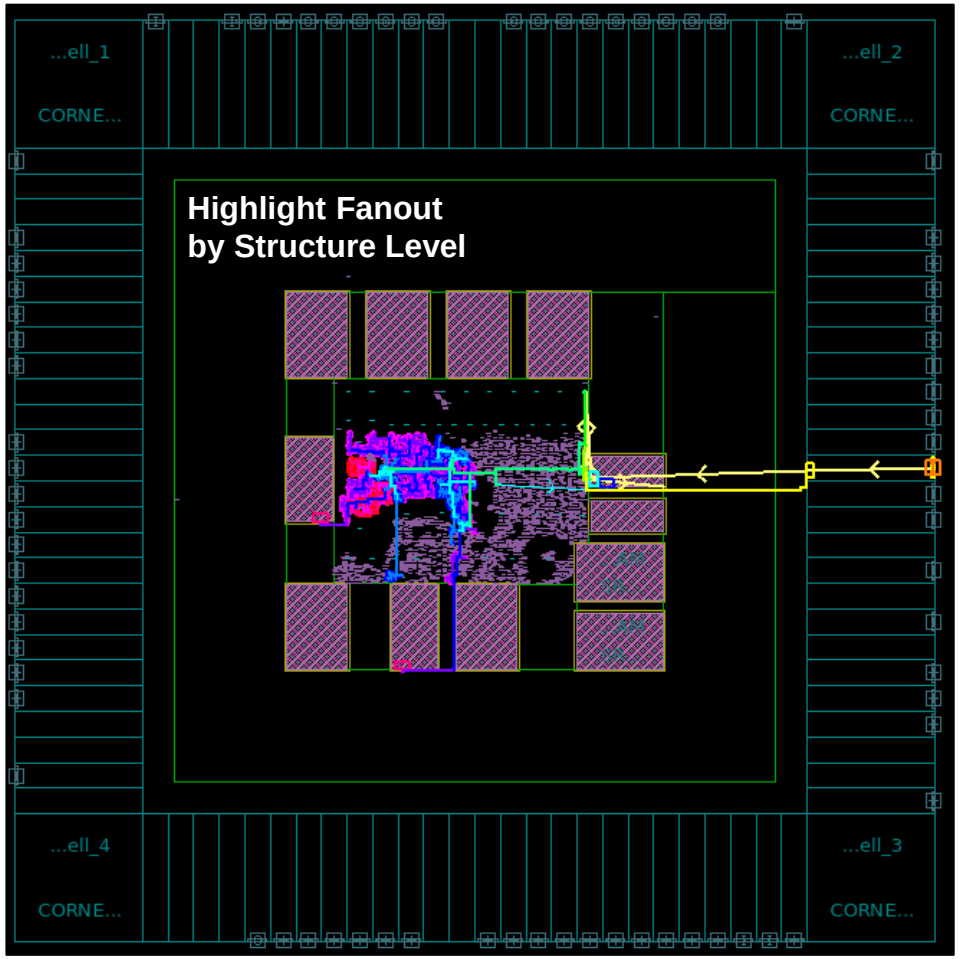
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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Highlight Fanout in Layout**



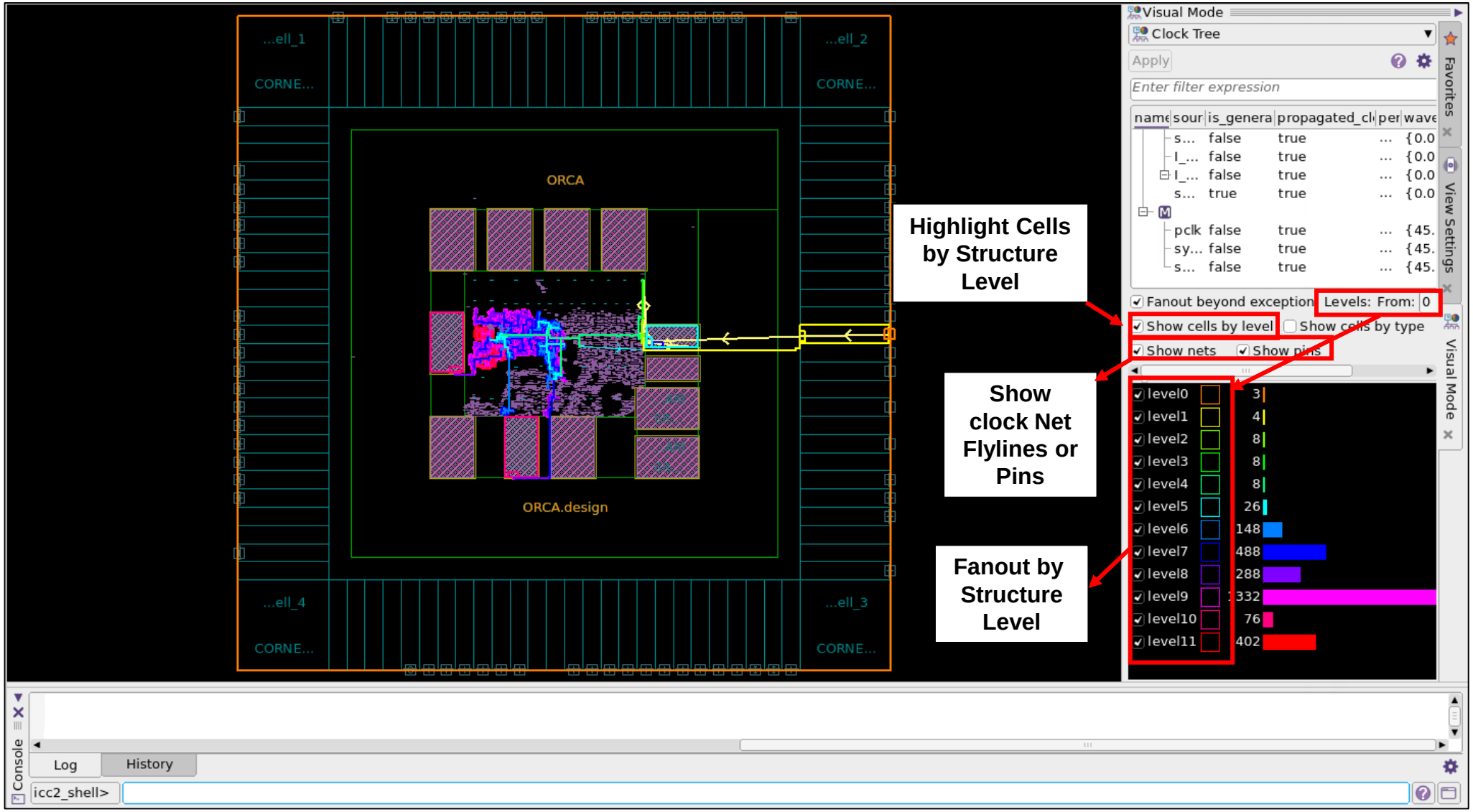
- **Click Right button on mouse > Highlight Fanout in Layout**



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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Highlight Fanout in Layout (you can also open this window Highlight > Color By > Clock Tree)**



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Introduction to CTS : What is Generated Clock & Virtual Clock

- **Generated Clock**
  - When a clock is derived from a master clock it is referred to as a generated clock
  - The master clock is a clock defined by using the create_clock command in SDC files
  - For example, if the generated clock is divided by 1 of the master clock, then the generated clock is defined in the output of t he circuit

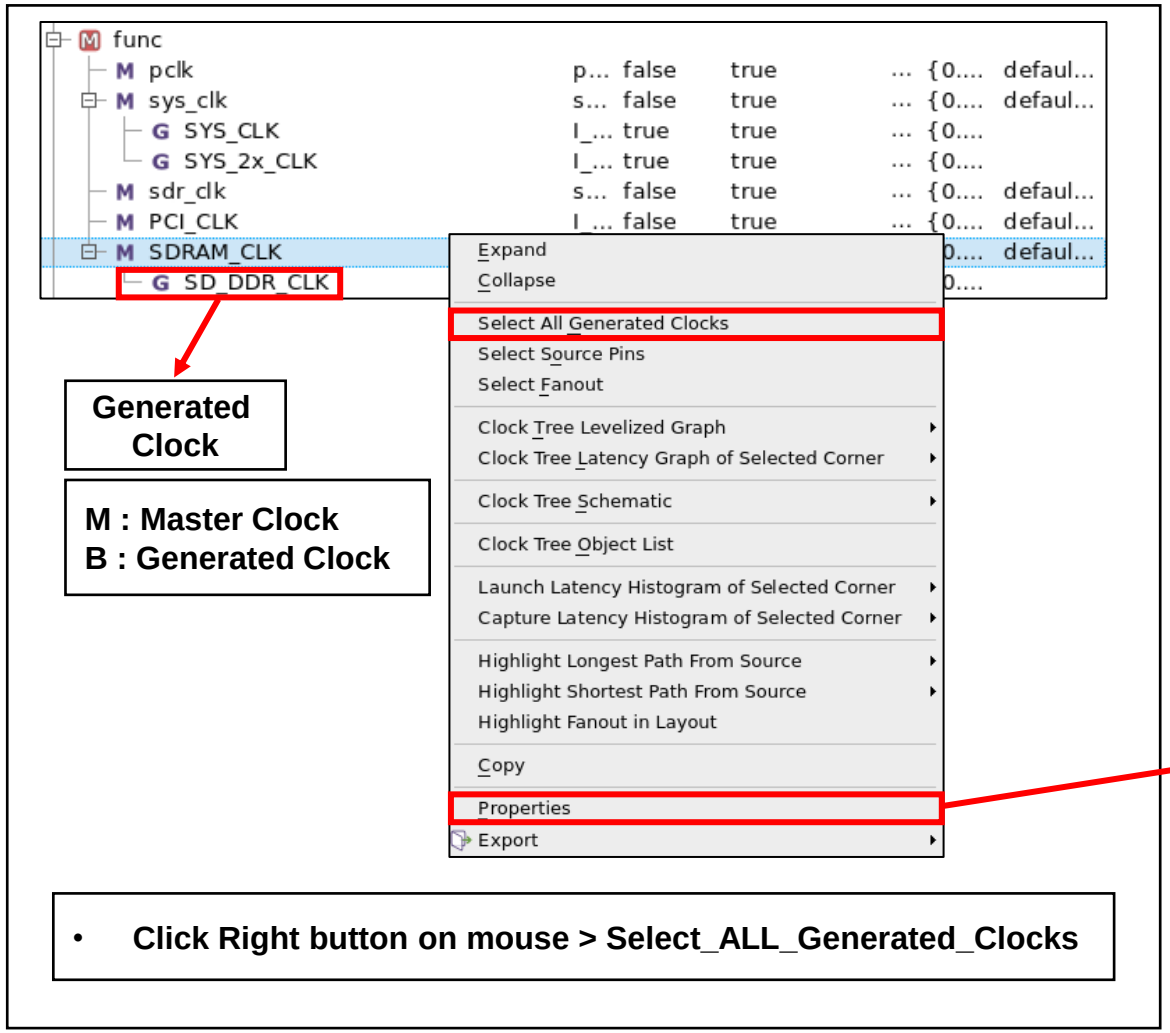
```
create_clock [get_pins I_CLOCK_GEN/I_PLL_SD/CLK_1X]  -name SDRAM_CLK  -period 4  -waveform {0 2}
set_clock_uncertainty -setup 0.3  [get_clocks SDRAM_CLK]
create_generated_clock [get_ports sd_CK]  -name SD_DDR_CLK  -source [get_pins I_CLOCK_GEN/I_PLL_SD/CLK_1X]  -divide_by 1
set_clock_uncertainty -setup 0.3  [get_clocks SD_DDR_CLK]
create_generated_clock [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_1X]  -name SYS_CLK  -source [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_IN]  -multiply_by 1
set_clock_uncertainty -setup 0.3  [get_clocks SYS_CLK]
set_clock_gating_check -rise -setup 0.2 [get_clocks SYS_CLK]
set_clock_gating_check -fall -setup 0.2 [get_clocks SYS_CLK]
set_clock_gating_check -rise -hold 0.2 [get_clocks SYS_CLK]
set_clock_gating_check -fall -hold 0.2 [get_clocks SYS_CLK]
create_generated_clock [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_2X]  -name SYS_2x_CLK  -source [get_pins I_CLOCK_GEN/I_CLKMUL/CLK_IN]  -multiply_by 2
```

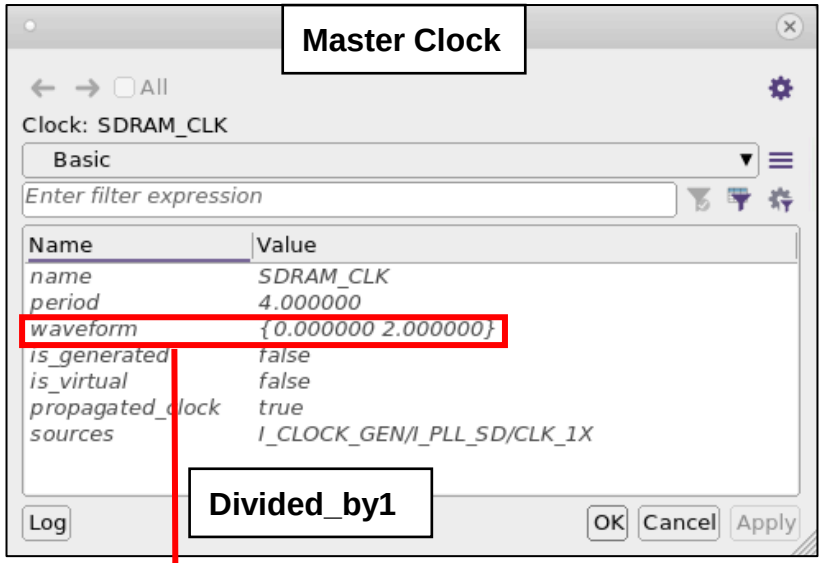
- **Virtual Clock**
  - Virtual clock is a clock definition without a specific port or pin assigned to the definition
  - It is used for constraining input or output delays for a block
  - Virtual clock is a clock that has been defined, but has not been associated with any pin/port

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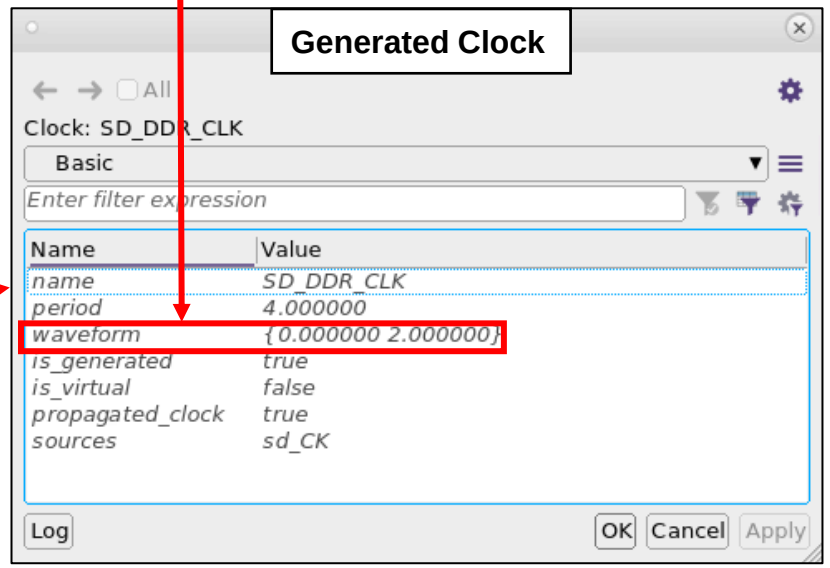
CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Highlight Longest Path From Source**





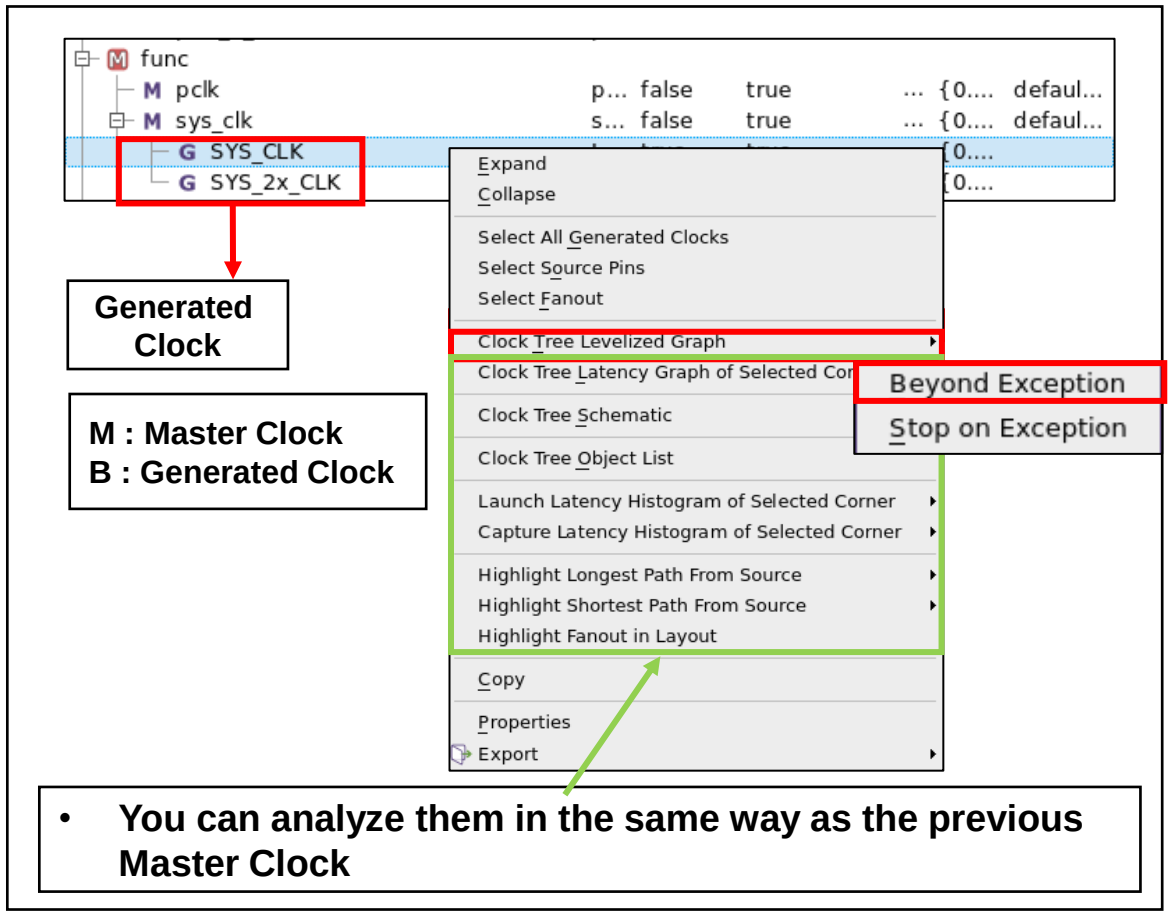
Divided_by1

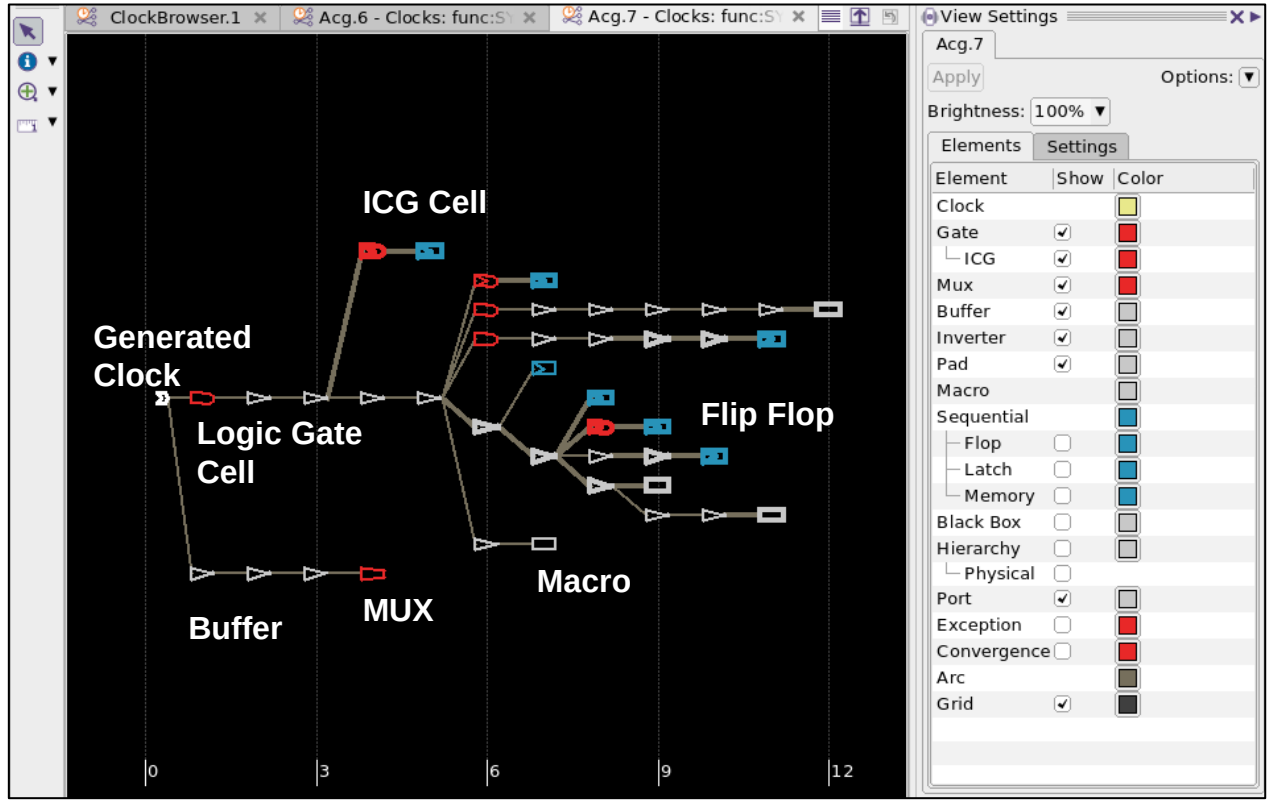


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CTS : Use ICC2 (IC Compiler II tool) Check Clock Tree

- **Check Clock Tree Analysis Window**
  - **Check Generated Clock Tree (Levelized Graph, Latency Graph, Schematic, Longest & Shortest Path, Fanout)**





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WEEK 10-2 : Run PSYN

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PSYN: Use ICC2 (IC Compiler II tool)

1. **Invoke tool icc2_shell**
   - **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./05_cts.log ;# (| tee -I ./05_psyn.log) is the command to output log

```
 < user@linux 05_cts >> icc2_shell -gui | tee -i ./06_psyn.log



                            IC Compiler II (TM)

                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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                         https://solvnetplus.synopsys.com)

Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
icc2_shell>
```

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

2. **Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_2_design_setup_for_PSYN.tcl**

   ❖ **If you want ccd_flow, source 01_4_design_setup_for_PSYN_ccd.tcl**

- **set step & source design_setup.tcl**

```
icc2_shell> set step psyn
psyn
icc2_shell> set before_step cts
cts
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```


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PSYN : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_2_design_setup_for_PSYN.tcl

- **If you want ccd_flow, source 01_4_design_setup_for_PSYN_ccd.tcl**

- **Open_lib & copy lib**

```
icc2_shell> source ./icc2_common_env/sec_runtime.tcl
icc2_shell> sec_StartTimer copy_lib
1
icc2_shell> open_lib ../04_placement/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/04_placement/nlib/ORCA_refine_opt.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_refine_opt.nlib}
icc2_shell>
icc2_shell> copy_lib -from ../04_placement/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_
${step}.nlib
Information: Copying from library "ORCA_refine_opt.nlib" to "./nlib/ORCA_cts.nlib".
{ORCA_cts.nlib}
```


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PSYN : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/01_2_design_setup_for_PSYN.tcl

- **If you want ccd_flow, source 01_4_design_setup_for_PSYN_ccd.tcl**

- **Close reference lib & open & link current block**

```
icc2_shell> close_lib -all
Closing all libraries...
1
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/05_cts/nlib/ORCA_cts.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/r
undir/00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_cts.nlib}
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
```

```
icc2_shell> #link_block
Using libraries: ORCA_psyn.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32rvt saed32sram
Visiting block ORCA_psyn.nlib:ORCA.design

Design 'ORCA' was successfully linked.
1
icc2_shell> link_block
icc2_shell> link_block -rebind
Warning: Block 'ORCA_psyn.nlib:ORCA.design' is already linked. (LNK-067)
0
```


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PSYN : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/02_mcmm.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> remove_scenarios -all
12
icc2_shell> remove_modes -all
3
icc2_shell> remove_corners -all
4
```

- **Read TLU+ files**

```
icc2_shell> read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
1
icc2_shell> read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
1
```

- **Source mcmm.tcl (create mcmm, set parasitic_parameters, set_voltage)**

```
icc2_shell> source -e ./icc2_common_env/mcmm.tcl
```

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/03_read_sdc.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> read_sdc $data(shift_sdc)
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> read_sdc $data(capture_sdc)
```

```
icc2_shell> current_mode func
{func}
icc2_shell> read_sdc $data(func_sdc)
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set_clock_balance_points**

```
icc2_shell> set all_modes "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell> foreach mode $all_modes {
?    current_mode $mode
?    set_clock_balance_points \
?       -consider_for_balancing true \
?       -balance_points [get_pins "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*/S0"]
? }
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
Information: The set_clock_balance_points command, when using "-consider_for_balancing true", becomes scenario-s
pecific and therefore expects either "-mode" or "-corner" or "-scenario" to be used. If none of these is specifi
ed then the balance point will be applied to the current_scenario. (TIM-035)
```

- **Set_size_only_cells**

```
icc2_shell> set_size_only [get_cells "I_ORCA_TOP/I_SDRAM_IF/*sd_mux_*"]
1
icc2_shell> set_size_only [get_cells "I_ORCA_TOP/I_RESET_BLOCK/sys_2x_rst_ff_reg"]
1
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set CTS_cell**

```
icc2_shell> set CTS_LIB_CELL_PATTERN_LIST "*/INVX2_RVT */INVX4_RVT */INVX8_RVT */INVX16_RVT"
*/INVX2_RVT */INVX4_RVT */INVX8_RVT */INVX16_RVT
icc2_shell> set CTS_CELLS [get_lib_cells $CTS_LIB_CELL_PATTERN_LIST]
{saed32rvt/INVX2_RVT saed32rvt/INVX4_RVT saed32rvt/INVX8_RVT saed32rvt/INVX16_RVT}
icc2_shell> set_dont_touch $CTS_CELLS false
1
icc2_shell> suppress_message ATTR-12
icc2_shell> set_lib_cell_purpose -exclude cts [get_lib_cells]
icc2_shell> set_lib_cell_purpose -include cts $CTS_CELLS
1
icc2_shell> unsuppress_message ATTR-12
```

- **Set ndr_rule (source ./icc2_common_env/ndr.tcl)**

```
icc2_shell> set CTS_NDR_MIN_ROUTING_LAYER "M4"
M4
icc2_shell> set CTS_NDR_MAX_ROUTING_LAYER "M6"
M6
icc2_shell> set CTS_LEAF_NDR_MIN_ROUTING_LAYER "M1"
M1
icc2_shell> set CTS_LEAF_NDR_MAX_ROUTING_LAYER "M6"
M6
icc2_shell> set CTS_NDR_RULE_NAME "cts_w2_s2_vlg"
cts_w2_s2_vlg
icc2_shell> set CTS_LEAF_NDR_RULE_NAME "cts_w1_s2"
cts_w1_s2
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**
- **Set ndr_rule (source ./icc2_common_env/ndr.tcl)**
- **Routing rules for routing layers**

```
icc2_shell> if {$CTS_NDR_RULE_NAME != ""} {
? remove_routing_rules $CTS_NDR_RULE_NAME > /dev/null
?
? create_routing_rule $CTS_NDR_RULE_NAME \
? -default_reference_rule \
? -widths { M1 0.1 M2 0.11 M3 0.11 M4 0.11 M5 0.11 M6 0.11 } \
? -spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1 }
?
? set_clock_routing_rules -rules $CTS_NDR_RULE_NAME \
? -min_routing_layer $CTS_NDR_MIN_ROUTING_LAYER \
? -max_routing_layer $CTS_NDR_MAX_ROUTING_LAYER
?
? }
Information: The command 'set_clock_routing_rules' cleared the undo history. (UNDO-016)
1
```

- **Routing rules for leaf**

```
icc2_shell> if {$CTS_LEAF_NDR_RULE_NAME != ""} {
? remove_routing_rules $CTS_LEAF_NDR_RULE_NAME > /dev/null
?
? create_routing_rule $CTS_LEAF_NDR_RULE_NAME \
? -default_reference_rule \
? -spacings { M2 0.16 M3 0.45 M4 0.45 M5 1.1 M6 1.1}
?
? set_clock_routing_rules -net_type sink -rules $CTS_LEAF_NDR_RULE_NAME \
? -min_routing_layer $CTS_LEAF_NDR_MIN_ROUTING_LAYER \
? -max_routing_layer $CTS_LEAF_NDR_MAX_ROUTING_LAYER
? }
Information: The command 'set_clock_routing_rules' cleared the undo history. (UNDO-016)
1
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Source cts_ex_ndr.tcl (set_balance_point, size_only_cell, CTS_cell, ndr rule, clock_uncertainty & Max Transition)**

- **Set clock uncertainty for all clocks in all scenarios**

```
icc2_shell> current_mode func
{func}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell>
icc2_shell> foreach_in_collection scen [all_scenarios] {
? current_scenario $scen
? set_clock_uncertainty 0.3 -setup [all_clocks]
? set_clock_uncertainty 0.1 -hold [all_clocks]
? }
```

- **Set max_transition for the clocks in all modes & corners**

```
icc2_shell> current_mode func
{func}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell> return 1
1
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_scenarios after create mcmm**

```
icc2_shell> get_scenario  -filter active&&hold
{func.ff_125c func.ff_n40c func.ss_125c func.ss_n40c scan_capture.ff_125c scan_capture.ff_n40c scan_capture.ss_125c scan_captu
re.ss_n40c scan_shift.ff_125c scan_shift.ff_n40c scan_shift.ss_125c scan_shift.ss_n40c}
```

```
icc2_shell> report_scenarios
****************************************
Report : scenario
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 11:56:19 2023
****************************************
```

```
                                                                Leakage  Dynamic                                Cell  Signa
l
Name            Mode            Corner          Active  Setup  Hold  Power    Power    Max_tran  Max_cap  Min_cap  EM    EM
-------------------------------------------------------------------------------------------------------------------------
-
func.ff_125c *  func            ff_125c         true    true   true  true     true     true      true     true     false false
func.ff_n40c *  func            ff_n40c         true    true   true  true     true     true      true     true     false false
func.ss_125c *  func            ss_125c         true    true   true  true     true     true      true     true     false false
func.ss_n40c *  func            ss_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_125c *
                scan_capture    ff_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ff_n40c *
                scan_capture    ff_n40c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_125c *
                scan_capture    ss_125c         true    true   true  true     true     true      true     true     false false
scan_capture.ss_n40c *
                scan_capture    ss_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_125c *
                scan_shift      ff_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ff_n40c *
                scan_shift      ff_n40c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_125c *
                scan_shift      ss_125c         true    true   true  true     true     true      true     true     false false
scan_shift.ss_n40c *
                scan_shift      ss_n40c         true    true   true  true     true     true      true     true     false false

1
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
report_clock_balance_points
****************************************
Report : clock tree balance points
Design : ORCA
Mode   : scan_shift
Version: S-2021.06-SP5
Date   : Tue Oct 24 11:58:52 2023
****************************************
```

```
Clock Independent:
Balance points:              ------------- DELAY ---------------    ------------- OFFSET ---------------
                           Early-    Early-     Late-     Late-    Early-    Early-     Late-     Late-
Point                       Rise      Fall       Rise      Fall     Rise      Fall       Rise      Fall     Corner
-------------------------------------------------------------------------------------------------------------------
I_ORCA_TOP/I_SDRAM_IF/genblk1_15__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_14__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_13__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_11__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_10__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_9__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_8__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_7__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_6__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_5__sd_mux_dq_out/S0
                            0.00      0.00       0.00      0.00       --        --         --        --    ss_125c
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
I_ORCA_TOP/I_SDRAM_IF/genblk1_4__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_3__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_2__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_1__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/genblk1_0__sd_mux_dq_out/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CKn/S0
                            0.00      0.00      0.00      0.00       --        --        --        --    ss_125c

Exclude points:
Point
--------------------------------------------------------------------------------------------------------------------
No exclude points.


1
```

- **Report_clock_routing_rules**

```
icc2_shell> report_clock_routing_rules
****************************************
 Report : clock routing rules
 Design : ORCA
 Date   : Tue Oct 24 12:01:49 2023
****************************************
   Design Base; Net Type: all;      Rule: cts_w2_s2_vlg; Min Layer: M4; Max Layer: M6
   Design Base; Net Type: sink;     Rule: cts_w1_s2; Min Layer: M1; Max Layer: M6
```


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PSYN : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/04_setup_for_ndr_and_cells.tcl

- **Report_clock_balance_points**

```
icc2_shell> set_lib_cell_purpose -exclude hold [get_lib_cells]
```

```
icc2_shell> set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
```

```
icc2_shell> set_lib_cell_purpose -exclude optimization [get_lib_cells]
```

```
icc2_shell> set_lib_cell_purpose -include optimization [get_lib_cells "*/NBUFFX2_*VT */NBUFFX4_*VT */NBUFFX8_*VT */NBUFFX1
6_*VT"]
```

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set_scenarios_status**

```
icc2_shell> set_scenario_status {*.ss*} -setup true -hold true \
?     -active true -max_transition true \
?     -max_capacitance true \
?     -min_capacitance false \
?             -leakage_power true \
?             -dynamic_power true
Scenario func.ss_125c (mode func corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario func.ss_n40c (mode func corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario scan_capture.ss_125c (mode scan_capture corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_capture.ss_n40c (mode scan_capture corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_shift.ss_125c (mode scan_shift corner ss_125c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
Scenario scan_shift.ss_n40c (mode scan_shift corner ss_n40c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
1
```


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PSYN : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set_scenarios_status**

```
icc2_shell> set_scenario_status {*.ff*} -setup true -hold true \
?                              -active true -max_transition true \
?              -max_capacitance true \
?     -min_capacitance false \
?      -leakage_power true \
?                     -dynamic_power true
Scenario func.ff_125c (mode func corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario func.ff_n40c (mode func corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_transition/max_c
apacitance analysis.
Scenario scan_capture.ff_125c (mode scan_capture corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_capture.ff_n40c (mode scan_capture corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_
transition/max_capacitance analysis.
Scenario scan_shift.ff_125c (mode scan_shift corner ff_125c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
Scenario scan_shift.ff_n40c (mode scan_shift corner ff_n40c) is active for setup/hold/leakage_power/dynamic_power/max_tran
sition/max_capacitance analysis.
1
```

- **Source remove_ideal_network_before_cts.tcl**

```
icc2_shell> source -e ./user_script/remove_ideal_network_before_cts.tcl
set mode "func scan_shift scan_capture"
foreach MODE $mode {
current_mode $MODE

echo "current mode is $MODE"
        switch $MODE {
                func {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
```


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PSYN : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Source remove_ideal_network_before_cts.tcl**

```
                scan_shift {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
                scan_capture {
                        set clock_sink [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK_*X"]
                        remove_ideal_network $clock_sink
                        remove_ideal_network [get_nets buf_sdram_clk]
                        remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]] ]
                        remove_clock_latency [get_clocks *]
                        remove_clock_latency [get_pins -hier -filter "is_clock_pin == true || full_name == I_CLOCK_GEN/I_*/CLK
_*X"]
                        set_propagated_clock [all_clocks]
                }
        }
}
current mode is func
current mode is scan_shift
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
current mode is scan_capture
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
```


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PSYN : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Source set_disable_timing.tcl**

```
source -e ./user_script/set_disable_timing.tcl
#set current_scenarios
current_scenario "func.ss_125c"
#set_disable_timing SDRAM_DQ_Bus_1__sdram_DQ_iopad ~ SDRAM_DQ_Bus_15__sdram_DQ_iopad from EN pin to PADIO pin
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_1__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_1__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_2__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_2__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_3__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_3__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_4__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_4__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_5__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_5__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_6__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_6__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_7__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_7__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_8__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_8__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_9__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_9__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_10__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_10__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_11__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_11__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_12__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_13__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_13__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_14__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_14__sdram_DQ_iopad/PADIO } ]
set_disable_timing [get_timing_arcs -from { SDRAM_DQ_Bus_15__sdram_DQ_iopad/EN } -to { SDRAM_DQ_Bus_15__sdram_DQ_iopad/PADIO } ]
1
```

- **Set target_skew & latency constraints for clock trees**

```
set mode "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell>
icc2_shell> foreach MODE $mode {
? current_mode $MODE
?
? echo "current mode is $MODE"
? switch $MODE {
? func {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
```


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PSYN : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/05_setup_for_skew_and_latency.tcl

- **Set target_skew & latency constraints for clock trees**

```
? scan_shift {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
?
? scan_capture {
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ss*]
? set_clock_tree_options -target_skew 0.3 -corners [get_corners ff*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ss*]
? set_clock_tree_options -target_latency 0.5 -corners [get_corners ff*]
? }
? }
? }
current mode is func
current mode is scan_shift
current mode is scan_capture
```

- **Report clock_tree_options**

```
icc2_shell> report_clock_tree_options
****************************************
 Report : target skew and latency
 Design : ORCA
 Date   : Tue Oct 24 13:25:50 2023
****************************************

##Target Skew

Clock                        Corner                       Target
------------------------------------------------------------------
All clocks                   ss_125c                      0.3
All clocks                   ss_n40c                      0.3
All clocks                   ff_n40c                      0.3
All clocks                   ff_125c                      0.3
```

```
##Target Latency

Clock                        Corner                       Target
------------------------------------------------------------------
All clocks                   ss_125c                      0.5
All clocks                   ss_n40c                      0.5
All clocks                   ff_n40c                      0.5
All clocks                   ff_125c                      0.5
```


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PSYN : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/06_1_app_option_for_cts.tcl

❖ **If you want ccd_flow, source 06_2_app_option_for_cts_ccd.tcl**

- **Set app_options for PSYN**

Non-CCD PSYN

```
set_app_options -name clock_opt.flow.enable_ccd -value false ;
clock_opt.flow.enable_ccd false
icc2_shell> set_app_options -name opt.dft.clock_aware_scan_reorder -value true
opt.dft.clock_aware_scan_reorder true
icc2_shell> set_app_options -name time.remove_clock_reconvergence_pessimism -value true
time.remove_clock_reconvergence_pessimism true
icc2_shell> set_app_options -list {opt.tie_cell.max_fanout {16}}
opt.tie_cell.max_fanout 16
icc2_shell> set_app_options -list {cts.common.max_fanout {16}}
cts.common.max_fanout 16
icc2_shell> set_app_options -list {opt.common.max_fanout {16}}
opt.common.max_fanout 16
```

CCD_flow PSYN

```
icc2_shell> set_app_options -name ccd.optimize_boundary_timing -value false
ccd.optimize_boundary_timing false
icc2_shell> set_app_options -name ccd.ignore_scan_reset_for_boundary_identification -value true
ccd.ignore_scan_reset_for_boundary_identification true
icc2_shell> set_app_options \
? -name ccd.ignore_ports_for_boundary_identification \
? -value {scan_en prst_n}
ccd.ignore_ports_for_boundary_identification {scan_en prst_n}
icc2_shell> set_app_options -name ccd.hold_control_effort -value high
ccd.hold_control_effort high
icc2_shell> set_app_options -name clock_opt.flow.enable_ccd -value true
clock_opt.flow.enable_ccd true
icc2_shell> set_app_options -name opt.dft.clock_aware_scan_reorder -value true
opt.dft.clock_aware_scan_reorder true
icc2_shell> set_app_options -name time.remove_clock_reconvergence_pessimism -value true
time.remove_clock_reconvergence_pessimism true
icc2_shell> set_app_options -name clock_opt.hold.effort -value high
clock_opt.hold.effort high
icc2_shell> set_app_options -list {opt.tie_cell.max_fanout {16}}
opt.tie_cell.max_fanout 16
icc2_shell> set_app_options -list {cts.common.max_fanout {16}}
cts.common.max_fanout 16
icc2_shell> set_app_options -list {opt.common.max_fanout {16}}
opt.common.max_fanout 16
```


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PSYN : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/06_1_app_option_for_cts.tcl

- **Report_ideal_network before PSYN**

<func>

```
icc2_shell> report_ideal_network -modes {func scan_capture scan_shift}
****************************************
Report : ideal_network
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 13:42:54 2023
****************************************

Ideal network data for scenario 'func.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'func.ss_n40c':
No ideal source pins or ports.
```

<scan_capture>

```
Ideal network data for scenario 'scan_capture.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_capture.ss_n40c':
No ideal source pins or ports.
```

<scan_shift>

```
Ideal network data for scenario 'scan_shift.ff_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ff_n40c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ss_125c':
No ideal source pins or ports.


Ideal network data for scenario 'scan_shift.ss_n40c':
No ideal source pins or ports.

1
```


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PSYN : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/09_1_run_PSYN_and_save.tcl

❖ **If you want ccd flow, source 07_1_run_PSYN_ccd_and_save.tcl**

- **Run CTS (to route clock)**

```
icc2_shell> sec_StartTimer psyn
1
icc2_shell> clock_opt -from final_opto
```

❖ **check PVT mismatches in log**

```
Information: Corner ss_n40c: no PVT mismatches. (PVT-032)
Information: Corner ff_125c: no PVT mismatches. (PVT-032)
Information: Corner ss_125c: no PVT mismatches. (PVT-032)
Information: Corner ff_n40c: no PVT mismatches. (PVT-032)
```

- **Save libraries after CTS**

```
icc2_shell> sec_ReportResourceUtil cts
icc2_shell> save_lib -all
Saving all libraries...
1
```

- **Update timing before report Timing & QoR**

```
icc2_shell> update_timing -full
```


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PSYN : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/09_1_run_PSYN_and_save.tcl

❖ **If you want ccd flow, source 09_2_run_PSYN_ccd_and_save.tcl**

- **Report_timing**

```
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Oct 25 11:22:19 2023
****************************************
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_ORCA_TOP/I_BLENDER/s4_op2_reg_0_ (rising edge-triggered flip-flop clocked by sys_clk)
  Endpoint: I_ORCA_TOP/I_BLENDER/s5_result_reg_31_ (rising edge-triggered flip-flop clocked by sys_clk)
  Mode: func
  Corner: ss_n40c
  Scenario: func.ss_n40c
  Path Group: sys_clk
  Path Type: max
```

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**

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PSYN : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/09_1_run_PSYN_and_save.tcl

❖ **If you want ccd flow, source 09_2_run_PSYN_ccd_and_save.tcl**

- **Report_timing**

DAT Path

```
Point                                                   Incr       Path
-----------------------------------------------------------------------------
clock sys_clk (rise edge)                            0.00000    0.00000
clock network delay (propagated)                     1.08582    1.08582

I_ORCA_TOP/I_BLENDER/s4_op2_reg_0_/CLK (SDFFX1_RVT)
                                                     0.00000    1.08582 r
I_ORCA_TOP/I_BLENDER/s4_op2_reg_0_/Q (SDFFX1_RVT)
                                                     0.20121    1.28702 r
I_ORCA_TOP/I_BLENDER/add_215/U2/Y (AND2X1_HVT)       0.11349    1.40051 r
I_ORCA_TOP/I_BLENDER/add_215/U1_1/CO (FADDX1_RVT)
                                                     0.11183    1.51234 r
I_ORCA_TOP/I_BLENDER/add_215/U1_2/CO (FADDX1_RVT)
                                                     0.11417    1.62651 r
I_ORCA_TOP/I_BLENDER/add_215/U1_3/CO (FADDX1_RVT)
                                                     0.11395    1.74046 r
I_ORCA_TOP/I_BLENDER/add_215/U1_4/CO (FADDX1_RVT)
                                                     0.11299    1.85345 r
I_ORCA_TOP/I_BLENDER/add_215/U1_5/CO (FADDX1_RVT)
                                                     0.11553    1.96898 r
I_ORCA_TOP/I_BLENDER/add_215/U1_6/CO (FADDX1_RVT)
                                                     0.11271    2.08168 r
I_ORCA_TOP/I_BLENDER/add_215/U1_7/CO (FADDX1_RVT)
                                                     0.11221    2.19389 r
I_ORCA_TOP/I_BLENDER/add_215/U1_8/CO (FADDX1_RVT)
                                                     0.11389    2.30778 r
I_ORCA_TOP/I_BLENDER/add_215/U1_9/CO (FADDX1_RVT)
                                                     0.11410    2.42188 r
I_ORCA_TOP/I_BLENDER/add_215/U1_10/CO (FADDX1_RVT)
                                                     0.11253    2.53441 r
I_ORCA_TOP/I_BLENDER/add_215/U1_11/CO (FADDX1_RVT)
                                                     0.11272    2.64713 r
I_ORCA_TOP/I_BLENDER/add_215/U1_12/CO (FADDX1_RVT)
                                                     0.11513    2.76226 r
I_ORCA_TOP/I_BLENDER/add_215/U1_13/CO (FADDX1_RVT)
                                                     0.11484    2.87710 r
I_ORCA_TOP/I_BLENDER/add_215/U1_14/CO (FADDX1_RVT)
                                                     0.11497    2.99208 r
I_ORCA_TOP/I_BLENDER/add_215/U1_15/CO (FADDX1_RVT)
                                                     0.11580    3.10787 r
I_ORCA_TOP/I_BLENDER/add_215/U1_16/CO (FADDX1_RVT)
                                                     0.11616    3.22403 r
I_ORCA_TOP/I_BLENDER/add_215/U1_17/CO (FADDX1_LVT)
                                                     0.07805    3.30208 r
I_ORCA_TOP/I_BLENDER/add_215/U1_18/CO (FADDX1_LVT)
                                                     0.07565    3.37772 r
I_ORCA_TOP/I_BLENDER/add_215/U1_19/CO (FADDX1_LVT)
                                                     0.07240    3.45013 r
I_ORCA_TOP/I_BLENDER/add_215/U1_20/CO (FADDX1_RVT)
                                                     0.10798    3.55810 r
I_ORCA_TOP/I_BLENDER/add_215/U1_21/CO (FADDX1_RVT)
                                                     0.11314    3.67125 r
I_ORCA_TOP/I_BLENDER/add_215/U1_22/CO (FADDX1_RVT)
                                                     0.11097    3.78222 r
I_ORCA_TOP/I_BLENDER/add_215/U1_23/CO (FADDX1_RVT)
                                                     0.11120    3.89341 r
I_ORCA_TOP/I_BLENDER/add_215/U1_24/CO (FADDX1_RVT)
                                                     0.11240    4.00581 r
I_ORCA_TOP/I_BLENDER/add_215/U1_25/CO (FADDX1_RVT)
                                                     0.11377    4.11959 r
I_ORCA_TOP/I_BLENDER/add_215/U1_26/CO (FADDX1_RVT)
                                                     0.11501    4.23460 r
I_ORCA_TOP/I_BLENDER/add_215/U1_27/CO (FADDX1_RVT)
                                                     0.11299    4.34759 r
I_ORCA_TOP/I_BLENDER/add_215/U1_28/CO (FADDX1_RVT)
                                                     0.11276    4.46035 r
I_ORCA_TOP/I_BLENDER/add_215/U1_29/CO (FADDX1_RVT)
                                                     0.11118    4.57153 r
I_ORCA_TOP/I_BLENDER/add_215/U1_30/CO (FADDX1_RVT)
                                                     0.11341    4.68494 r
I_ORCA_TOP/I_BLENDER/add_215/U1_31/S (FADDX1_RVT)
                                                     0.16718    4.85212 f
I_ORCA_TOP/I_BLENDER/s5_result_reg_31_/D (SDFFX1_RVT)
                                                     0.00002    4.85214 f
data arrival time                                               4.85214
```

❖ **When analyzing this Timing report, Timing violation(Slack) occurs because the length of the DRT path is shorter than that of the DAT path**

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

DRT Path

```
clock sys_clk (rise edge)                            4.00000    4.00000
clock network delay (propagated)                     0.99634    4.99634
clock reconvergence pessimism                        0.05360    5.04994
I_ORCA_TOP/I_BLENDER/s5_result_reg_31_/CLK (SDFFX1_RVT)
                                                     0.00000    5.04994 r
clock uncertainty                                   -0.30000    4.74994
library setup time                                  -0.13586    4.61408
data required time                                              4.61406
```

DRT – RAT = Slack

```
data required time                                              4.61406
data arrival time                                              -4.85214
-----------------------------------------------------------------------------
slack (VIOLATED)                                               -0.23808
```


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PSYN : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/07_1_run_cts_and_save.tcl

❖ **If you want ccd flow, source 07_2_run_cts_ccd_and_save.tcl**

- **Report_qor -summary**

```
Information: Timer using 'CRPR'. (TIM-050)

Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)             0.94            0.00               0
func.ff_n40c       (Setup)             1.14            0.00               0
func.ss_125c       (Setup)             0.00            0.00               0
func.ss_n40c       (Setup)            -0.24           -1.33              12
scan_capture.ff_125c
                   (Setup)             9.27            0.00               0
scan_capture.ff_n40c
                   (Setup)             9.35            0.00               0
scan_capture.ss_125c
                   (Setup)             8.69            0.00               0
scan_capture.ss_n40c
                   (Setup)             8.52            0.00               0
scan_shift.ff_125c (Setup)             9.59            0.00               0
scan_shift.ff_n40c (Setup)             9.60            0.00               0
scan_shift.ss_125c (Setup)             9.38            0.00               0
scan_shift.ss_n40c (Setup)             9.27            0.00               0
Design             (Setup)            -0.24           -1.33              12

func.ff_125c       (Hold)             -0.19           -3.83              83
func.ff_n40c       (Hold)             -0.17           -3.71              74
func.ss_125c       (Hold)             -0.28           -6.66              35
func.ss_n40c       (Hold)             -0.18           -0.90              13
scan_capture.ff_125c
                   (Hold)             -0.00           -0.00               1
scan_capture.ff_n40c
                   (Hold)             -0.01           -0.04               6
scan_capture.ss_125c
                   (Hold)              0.17            0.00               0
scan_capture.ss_n40c
                   (Hold)              0.18            0.00               0
scan_shift.ff_125c (Hold)             -0.05           -0.05               1
scan_shift.ff_n40c (Hold)             -0.05           -0.08              28
scan_shift.ss_125c (Hold)              0.05            0.00               0
scan_shift.ss_n40c (Hold)              0.04            0.00               0
Design             (Hold)             -0.28           -8.40             150
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                         767620.54
Cell Area (netlist and physical only):      1191620.53
Nets with DRC Violations:       45
1
```

❖ **You can check WNS(Worst Negative Slack) first, and check TNS(Total Negative Slack) with NVE**

❖ **You can check Cell Area and number of DRC violations**

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PSYN : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/script/10_1_report_PSYN_and_exit.tcl

❖ **If you want ccd flow, source 10_2_report_PSYN_ccd_and_exit.tcl**

- **Update timing before report**

```
icc2_shell> update_timing -full
```

- **Report CTS & exit**

```
icc2_shell> source ./user_script/psyn_rpt.tcl
icc2_shell> sec_ReportResourceUtil report
icc2_shell> exit
```

- **Analyze Reports about CTS**

```
  < user@linux psyn >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn
  < user@linux psyn >> ls
psyn.clocks.rpt      psyn.structure.rpt                          timing_report_max_psyn_scan_capture.ss_n40c  timing_report_min_psyn_scan_capture.ff_125c
psyn.clocks.vio.rpt  psyn.util.rpt                               timing_report_max_psyn_scan_shift.ff_125c    timing_report_min_psyn_scan_capture.ff_n40c
psyn.congestion.rpt  timing_report_max_psyn_func.ff_125c         timing_report_max_psyn_scan_shift.ff_n40c    timing_report_min_psyn_scan_capture.ss_125c
psyn.fanout.rpt      timing_report_max_psyn_func.ff_n40c         timing_report_max_psyn_scan_shift.ss_125c    timing_report_min_psyn_scan_capture.ss_n40c
psyn.latnecy.rpt     timing_report_max_psyn_func.ss_125c         timing_report_max_psyn_scan_shift.ss_n40c    timing_report_min_psyn_scan_shift.ff_125c
psyn.local.skew.rpt  timing_report_max_psyn_func.ss_n40c         timing_report_min_psyn_func.ff_125c          timing_report_min_psyn_scan_shift.ff_n40c
psyn.max_tran.rpt    timing_report_max_psyn_scan_capture.ff_125c  timing_report_min_psyn_func.ff_n40c          timing_report_min_psyn_scan_shift.ss_125c
psyn.qor.snap.rpt    timing_report_max_psyn_scan_capture.ff_n40c  timing_report_min_psyn_func.ss_125c          timing_report_min_psyn_scan_shift.ss_n40c
psyn.runtime.rpt     timing_report_max_psyn_scan_capture.ss_125c  timing_report_min_psyn_func.ss_n40c
```


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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **Psyn.clock.rpt (you can check for Hold & Setup Uncertainty for All clocks)**

Object	Min Rise Delay	Min Fall Delay	Max Rise Delay	Max Fall Delay	Hold Uncertainty	Setup Uncertainty	Related Clock	Scenario	Origin
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
PCI_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SDRAM_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SD_DDR_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SYS_2x_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ff_125c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ss_125c	user
SYS_CLK	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
pclk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
pclk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
pclk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
pclk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
sdr_clk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ff_125c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ff_n40c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ss_125c	user
sys_clk	-	-	-	-	0.100	0.300	--	func.ss_n40c	user

1

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/cts or cts_ccd/* reports**
  - **psyn.structure.rpt**

```
****************************************
Report : report clock qor
         -type structure
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:09 2023
****************************************

C3 is corner ff_125c
C1 is corner ff_n40c
C2 is corner ss_125c
C4 is corner ss_n40c

Printing structure of pclk_wft1 (mode scan_capture) at root pin pclk:
(0) pclk [in Port] [Location (1437.90, 740.00)] [Net: pclk] [Fanout: 1]
 (1) pclk_iopad:PADIO->DOUT [Ref: saed32io_fc/I1025_NS] [Location (1440.00, 720.00)] [Net: net_pclk] [Fanout: 2]
  (2) I_CLOCK_GEN/macro_guide_buf_1:A->Y [Ref: saed32rvt/NBUFFX2_RVT] [Location (896.85, 724.18)] [Net: I_CLOCK_GEN/guide_buf_1] [Fanout: 1]
   (3) I_CLOCK_GEN/cto_dtrdly_8793:A->Y [Ref: saed32rvt/INVX2_RVT] [Location (895.03, 774.34)] [Net: I_CLOCK_GEN/cts15] [Fanout: 1]
    (4) I_CLOCK_GEN/cto_dtrdly_8798:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (893.36, 857.94)] [Net: I_CLOCK_GEN/cts18] [Fanout: 1]
     (5) I_CLOCK_GEN/I_PLL_PCI:REF_CLK->REF_CLKcheckpin1 [Ref: saed32pll/PLL] [Location (1014.46, 709.92)] [INTERNAL PIN] [SINK PIN]
  (2) I_CLOCK_GEN/U10:A1->Y [Ref: saed32rvt/AO22X1_RVT] [Location (895.48, 734.21)] [Net: I_CLOCK_GEN/o_pclk] [Fanout: 1]
   (3) cto_inv_drc_8579:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (896.85, 740.90)] [Net: cts1] [Fanout: 1]
    (4) cto_inv_drc_8578:A->Y [Ref: saed32rvt/INVX16_RVT] [Location (896.24, 737.56)] [Net: cts0] [Fanout: 6]
     (5) I_ORCA_TOP/ZCTSINV_6986_7062:A->Y [Ref: saed32rvt/INVX8_RVT] [Location (684.51, 750.93)] [Net: I_ORCA_TOP/ZCTSNET_6] [Fanout: 10]
      (6) I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSINV_6173_7037:A->Y [Ref: saed32rvt/INVX4_RVT] [Location (718.86, 714.15)] [Net: I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSNET_8] [Fanout: 8]
       (7) I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSINV_6164_7036:A->Y [Ref: saed32rvt/INVX8_RVT] [Location (714.91, 714.15)] [Net: I_ORCA_TOP/U_SNPS_DFT_LOGIC/ZCTSNET_7] [Fanout: 7]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP14/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (710.67, 715.60)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP19/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.66, 715.60)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP15/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.66, 716.04)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP17/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (720.40, 718.94)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (716.75, 718.94)] [SINK PIN]
        (8) I_ORCA_TOP/U_SNPS_DFT_LOGIC/LOCKUP13/CLK [Ref: saed32hvt/LATCHX1_HVT] [Location (715.23, 716.04)] [SINK PIN]
```

- **The structure report is a very detailed representation of the clock tree structure**
- **The entire fanout of the clock tree is represented in this report, with indentation and numbering indicating different levels of the clock tree**
- **Clock balance points and ignore points and also represented in this report**
- **It can accept hierarchical pin as input to –to, or –through options**

❖ **Structure Level**

Number of buffers in clock tree divided from source

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **psyn.congestion.rpt (report about congestion)**

```
****************************************
Report : congestion
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Oct 25 11:33:16 2023
****************************************

Layer     |    overflow     |              # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |     161 |     3 |     155  ( 0.01%) |       2
H routing |     144 |     2 |     142  ( 0.02%) |       2
V routing |      17 |     3 |      13  ( 0.00%) |       2

1
```

  - **psyn.fanout.rpt (report high fanout)**

```
****************************************
Report : net_fanout
        -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Tue Oct 24 14:17:05 2023
****************************************
1
```

❖ **There are no high fanout in this design**

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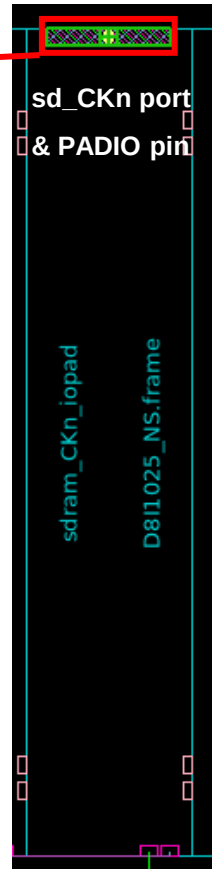
PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **psyn.max_tran.rpt (report about congestion)**

```
Mode: scan_capture Corner: ss_125c
Scenario: scan_capture.ss_125c
max_transition
                          Required        Actual
Net                      Transition     Transition        Slack  Violation
-------------------------------------------------------------------------
sd_CKn                         0.50           0.90        -0.40  (VIOLATED)
  PIN : sdram_CKn_iopad/PADIO
                               0.50           0.90        -0.40  (VIOLATED)
  PORT: sd_CKn                 0.50           0.90        -0.40  (VIOLATED)

sd_CK                          0.50           0.90        -0.40  (VIOLATED)
  PIN : sdram_CK_iopad/PADIO
                               0.50           0.90        -0.40  (VIOLATED)
  PORT: sd_CK                  0.50           0.90        -0.40  (VIOLATED)
```





❖ **Required Transition – Actual Transition = Slack**

❖ **Each PAD has a different max transition rule**

❖ **Violation occurs in the virtual port because the max transition rules of (sdram_CK_iopad, sdram_CKn_iopad) are different from other PADs**

❖ **Therefore, Max Transition violations occurring in virtual PORT of signal PAD can be ignored**

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **psyn.qor_snap.rpt**

```
****************************************
Report : qor
        -summary
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Oct 25 11:33:16 2023
****************************************
Information: Timer using 'CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------------
func.ff_125c       (Setup)            0.945           0.000               0
func.ff_n40c       (Setup)            1.136           0.000               0
func.ss_125c       (Setup)            0.003           0.000               0
func.ss_n40c       (Setup)           -0.238          -1.326              12
scan_capture.ff_125c (Setup)          9.268           0.000               0
scan_capture.ff_n40c (Setup)          9.349           0.000               0
scan_capture.ss_125c (Setup)          8.686           0.000               0
scan_capture.ss_n40c (Setup)          8.518           0.000               0
scan_shift.ff_125c (Setup)            9.587           0.000               0
scan_shift.ff_n40c (Setup)            9.598           0.000               0
scan_shift.ss_125c (Setup)            9.377           0.000               0
scan_shift.ss_n40c (Setup)            9.266           0.000               0
Design             (Setup)           -0.238          -1.326              12

func.ff_125c       (Hold)            -0.186          -3.834              83
func.ff_n40c       (Hold)            -0.173          -3.713              74
func.ss_125c       (Hold)            -0.279          -6.661              35
func.ss_n40c       (Hold)            -0.181          -0.904              13
scan_capture.ff_125c (Hold)          -0.002          -0.002               1
scan_capture.ff_n40c (Hold)          -0.011          -0.042               6
scan_capture.ss_125c (Hold)           0.174           0.000               0
scan_capture.ss_n40c (Hold)           0.185           0.000               0
scan_shift.ff_125c (Hold)            -0.046          -0.046               1
scan_shift.ff_n40c (Hold)            -0.047          -0.078              28
scan_shift.ss_125c (Hold)             0.049           0.000               0
scan_shift.ss_n40c (Hold)             0.044           0.000               0
Design             (Hold)            -0.279          -8.398             150
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                     767620.535
Cell Area (netlist and physical only):   1191620.535
Nets with DRC Violations:       45
1
```

- ❖ You can check WNS(Worst Negative Slack) first, and check TNS(Total Negative Slack) with NVE
- ❖ You can check Cell Area and number of DRC violations

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **Ex) timing_report_min_psyn_func.ss_125c**

```
Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg (rising edge-triggered flip-flop clocked by SDRAM_CLK)
Endpoint: I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U2/syncff_reg_5_ (removal check against rising-edge clock clocked by sys_clk)
Mode: func
Corner: ff_125c
Scenario: func.ff_125c
Path Group: sys_clk
Path Type: min
```

```
Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**
- **Check Attributes and refer to the analysis**

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **Ex) timing_report_min_psyn_func.ss_125c (DAT)**

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	A
clock SDRAM_CLK (rise edge)					0.000	0.000		
source latency					0.000	0.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.013	1.000		0.000	0.000 r	(901.30,664.39)	s,
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/cts_trgdly_8724/A (INVX2_RVT)		0.013	1.000		0.000	0.000 r	(895.52,666.59)	
I_CLOCK_GEN/cts_trgdly_8724/Y (INVX2_RVT)		0.029	1.000		0.021	0.021 f	(895.39,666.49)	
I_CLOCK_GEN/cts9 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8728/A (INVX2_RVT)		0.029	1.000		0.001	0.022 f	(895.75,751.67)	
I_CLOCK_GEN/cts_trgdly_8728/Y (INVX2_RVT)		0.030	1.000		0.022	0.044 r	(895.88,751.77)	
I_CLOCK_GEN/cts11 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8730/A (INVX8_RVT)		0.030	1.000		0.001	0.044 r	(939.07,765.04)	
I_CLOCK_GEN/cts_trgdly_8730/Y (INVX8_RVT)		0.019	1.000		0.009	0.054 f	(939.20,765.15)	
I_CLOCK_GEN/cts13 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8729/A (INVX8_RVT)		0.019	1.000		0.000	0.054 f	(893.93,748.32)	
I_CLOCK_GEN/cts_trgdly_8729/Y (INVX8_RVT)		0.014	1.000		0.008	0.062 r	(894.05,748.43)	
I_CLOCK_GEN/cts12 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8727/A (INVX2_RVT)		0.014	1.000		0.000	0.062 r	(894.84,668.07)	
I_CLOCK_GEN/cts_trgdly_8727/Y (INVX2_RVT)		0.011	1.000		0.008	0.070 f	(894.97,668.17)	
I_CLOCK_GEN/cts10 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8723/A (INVX2_RVT)		0.011	1.000		0.000	0.070 f	(892.48,666.59)	
I_CLOCK_GEN/cts_trgdly_8723/Y (INVX2_RVT)		0.009	1.000		0.006	0.076 r	(892.35,666.49)	
I_CLOCK_GEN/cts8 (net)	1							d
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)		0.009	1.000		0.000	0.076 r	(896.52,664.73)	
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)		0.014	1.000		0.021	0.097 r	(896.00,664.83)	
I_CLOCK_GEN/guide_buf_5 (net)	1							d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)		0.014	1.000		0.000	0.097 r	(898.80,697.28)	s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)		0.030	1.000		0.037	0.134 r	(898.37,696.57)	s
I_CLOCK_GEN/o_sdram_clk (net)	2							d
I_ORCA_TOP/ZCTSINV_10599_7208/A (INVX8_RVT)		0.030	1.000		0.000	0.134 r	(901.98,698.16)	
I_ORCA_TOP/ZCTSINV_10599_7208/Y (INVX8_RVT)		0.019	1.000		0.009	0.143 f	(902.11,698.27)	
I_ORCA_TOP/ZCTSNET_40 (net)	1							d
I_ORCA_TOP/ZCTSINV_9646_7207/A (INVX16_RVT)		0.019	1.000		0.000	0.143 f	(899.55,698.16)	
I_ORCA_TOP/ZCTSINV_9646_7207/Y (INVX16_RVT)		0.018	1.000		0.009	0.153 r	(899.68,698.27)	
I_ORCA_TOP/ZCTSNET_39 (net)	3							d
I_ORCA_TOP/ZCTSINV_6080_7186/A (INVX8_RVT)		0.019	1.000		0.002	0.155 r	(829.63,691.48)	
I_ORCA_TOP/ZCTSINV_6080_7186/Y (INVX8_RVT)		0.016	1.000		0.011	0.166 f	(829.76,691.58)	
I_ORCA_TOP/ZCTSNET_31 (net)	3							d
I_ORCA_TOP/ZCTSINV_2026_f_7145/A (INVX8_RVT)		0.016	1.000		0.000	0.166 f	(819.29,706.72)	
I_ORCA_TOP/ZCTSINV_2026_f_7145/Y (INVX8_RVT)		0.012	1.000		0.007	0.173 r	(819.42,706.62)	
I_ORCA_TOP/ZCTSNET_30 (net)	2							d
I_ORCA_TOP/cto_inv_drc_8385/A (INVX8_RVT)		0.012	1.000		0.000	0.173 r	(808.20,711.54)	
I_ORCA_TOP/cto_inv_drc_8385/Y (INVX8_RVT)		0.013	1.000		0.010	0.183 f	(808.33,711.64)	
I_ORCA_TOP/cts4 (net)	4							d
I_ORCA_TOP/cto_inv_drc_8399/A (INVX8_RVT)		0.013	1.000		0.000	0.183 f	(812.76,714.88)	
I_ORCA_TOP/cto_inv_drc_8399/Y (INVX8_RVT)		0.014	1.000		0.011	0.194 r	(812.89,714.99)	

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **Ex) timing_report_min_psyn_func.ss_125c (DAT)**

I_ORCA_TOP/cts7 (net)	14						d
I_ORCA_TOP/cto_inv_drc_8436/A (INVX16_RVT)		0.014	1.000	0.001	0.195 r	(810.78,698.16)	
I_ORCA_TOP/cto_inv_drc_8436/Y (INVX16_RVT)		0.011	1.000	0.007	0.201 f	(810.91,698.27)	
I_ORCA_TOP/cts32 (net)	2						d
I_ORCA_TOP/cto_inv_drc_8434/A (INVX8_RVT)		0.011	1.000	0.000	0.202 f	(793.76,681.44)	
I_ORCA_TOP/cto_inv_drc_8434/Y (INVX8_RVT)		0.010	1.000	0.007	0.209 r	(793.89,681.55)	
I_ORCA_TOP/cts31 (net)	1						d
I_ORCA_TOP/cto_inv_drc_8460/A (INVX16_RVT)		0.010	1.000	0.000	0.209 r	(789.50,671.41)	
I_ORCA_TOP/cto_inv_drc_8460/Y (INVX16_RVT)		0.009	1.000	0.007	0.216 f	(789.63,671.52)	
I_ORCA_TOP/cts34 (net)	5						d
I_ORCA_TOP/cto_inv_drc_8458/A (INVX2_RVT)		0.009	1.000	0.000	0.216 f	(777.95,658.04)	
I_ORCA_TOP/cto_inv_drc_8458/Y (INVX2_RVT)		0.012	1.000	0.011	0.227 r	(778.08,658.14)	
I_ORCA_TOP/cts33 (net)	1						d
I_ORCA_TOP/cto_inv_drc_8472/A (INVX8_RVT)		0.012	1.000	0.000	0.227 r	(776.89,651.35)	
I_ORCA_TOP/cto_inv_drc_8472/Y (INVX8_RVT)		0.015	1.000	0.011	0.238 f	(777.01,651.45)	
I_ORCA_TOP/cts37 (net)	4						d
I_ORCA_TOP/cto_inv_drc_8477/A (INVX4_RVT)		0.015	1.000	0.000	0.239 f	(764.27,634.63)	
I_ORCA_TOP/cto_inv_drc_8477/Y (INVX4_RVT)		0.015	1.000	0.012	0.250 r	(764.40,634.73)	
I_ORCA_TOP/cts40 (net)	10						d
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/CLK (SDFFARX1_HVT)		0.015	1.000	0.000	0.250 r	(764.88,632.05)	s, n

I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/CLK (SDFFARX1_HVT)		0.015	1.000	0.000	0.250 r	(764.88,632.05)	s, n
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/Q (SDFFARX1_HVT)		0.020	1.000	0.075	0.325 r	(761.70,631.39)	s, n
I_ORCA_TOP/I_RESET_BLOCK/test_so2 (net)	2						
I_ORCA_TOP/I_RESET_BLOCK/U14/A1 (AND2X1_LVT)		0.020	1.000	0.000	0.325 r	(766.71,629.64)	
I_ORCA_TOP/I_RESET_BLOCK/U14/Y (AND2X1_LVT)		0.014	1.000	0.018	0.343 r	(766.19,629.71)	
I_ORCA_TOP/I_RESET_BLOCK/N4 (net)	1						
I_ORCA_TOP/I_RESET_BLOCK/C52/A1 (OR2X1_LVT)		0.014	1.000	0.000	0.343 r	(764.89,628.01)	
I_ORCA_TOP/I_RESET_BLOCK/C52/Y (OR2X1_LVT)		0.015	1.000	0.023	0.365 r	(764.14,628.07)	
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n (net)	1						
I_ORCA_TOP/HFSINV_1862_2285/A (INVX2_HVT)		0.015	1.000	0.000	0.365 r	(763.74,626.47)	
I_ORCA_TOP/HFSINV_1862_2285/Y (INVX2_HVT)		0.016	1.000	0.013	0.378 f	(763.61,626.36)	
I_ORCA_TOP/net_net_1531 (net)	2						
I_ORCA_TOP/HFSINV_1278_2281/A (INVX4_HVT)		0.016	1.000	0.000	0.378 f	(763.28,627.94)	
I_ORCA_TOP/HFSINV_1278_2281/Y (INVX4_HVT)		0.010	1.000	0.006	0.383 r	(763.15,628.04)	
I_ORCA_TOP/HFSNET_234 (net)	1						
I_ORCA_TOP/ZBUF_798_inst_9488/A (NBUFFX16_LVT)		0.010	1.000	0.000	0.383 r	(762.75,626.47)	
I_ORCA_TOP/ZBUF_798_inst_9488/Y (NBUFFX16_LVT)		0.020	1.000	0.021	0.405 r	(761.02,626.36)	
I_ORCA_TOP/ZBUF_798_101 (net)	7						
I_ORCA_TOP/HFSBUF_682_2276/A (NBUFFX4_HVT)		0.023	1.000	0.007	0.412 r	(557.47,581.12)	
I_ORCA_TOP/HFSBUF_682_2276/Y (NBUFFX4_HVT)		0.017	1.000	0.026	0.438 r	(558.13,581.23)	
I_ORCA_TOP/HFSNET_231 (net)	5						
I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U2/syncff_reg_5_/RSTB (SDFFARX1_HVT)		0.017	1.000	0.000	0.438 r	(552.28,583.88)	s, n
data arrival time					0.438		

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PSYN : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn or psyn_ccd/* reports**
  - **Ex) timing_report_min_psyn_func.ss_125c (DRT)**

```
clock sys_clk (rise edge)                                                
```

PSYN: Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Analyze Timing Violation using Imported_path_pins Visual Mode**

1. Open $WORK_DIR/2_BACKEND/1_pnr/rundir/05_cts/reports/psyn/timing_report_min_cts_func.ff_125c
2. Find VIOLATED Path in timing reports
   - ❖ Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg (rising edge-triggered flip-flop clocked by SDRAM_CLK)
   - ❖ Endpoint: I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U2/syncff_reg_5_ (removal check against rising-edge clock cl ocked by sys_clk)
3. Open Imported_path_pins visual Mode and click reload button & select import from timing report text
4. Click ok button by copying and pasting DAT and DRT path of timing reports respectively
5. Analyze two path (DAT, DRT)

- **Open timing report & find VIOLATED PATH**

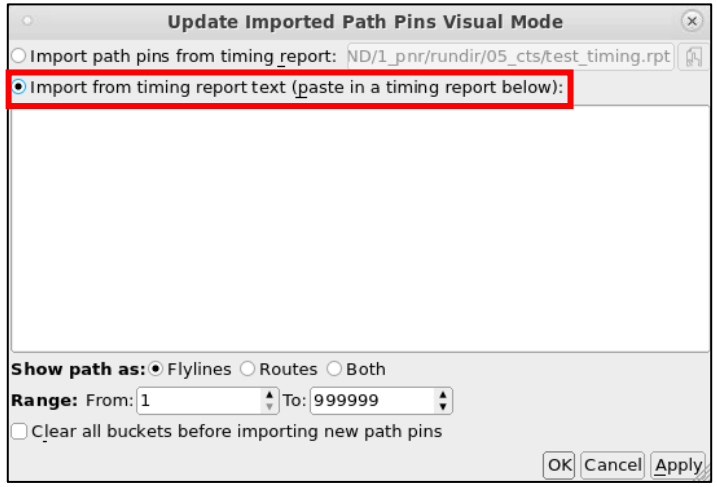
< timing_report_min_cts_func.ff_125c >

```
Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg (rising edge-triggered flip-flop clocked by SDRAM_CLK)
Endpoint: I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U2/syncff_reg_5_ (removal check against rising-edge clock clocked by sys_clk)
Mode: func
Corner: ff_125c
Scenario: func.ff_125c
Path Group: sys_clk
Path Type: min
```


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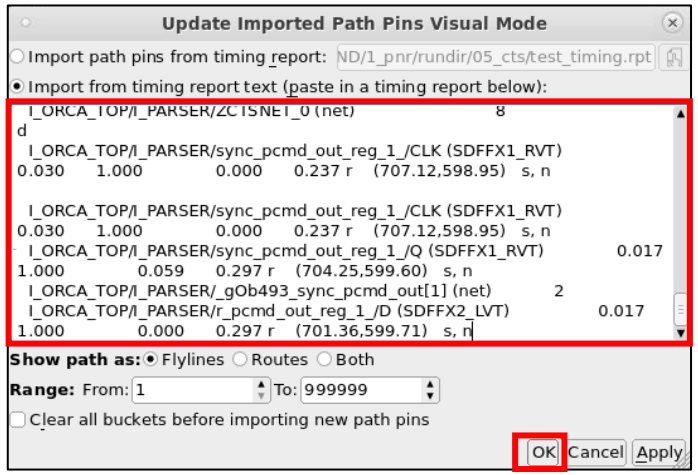
PSYN: Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

- **Analyze Timing Violation using Imported_path_pins Visual Mode**
  - Open Imported_path_pins visual Mode and click reload button & select import from timing report text

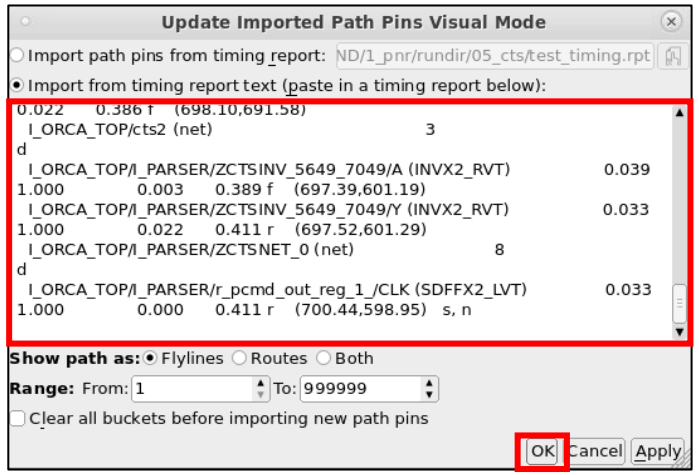


  - Click ok button by copying and pasting DAT and DRT path of timing reports respectively

< DAT >



< DRT >



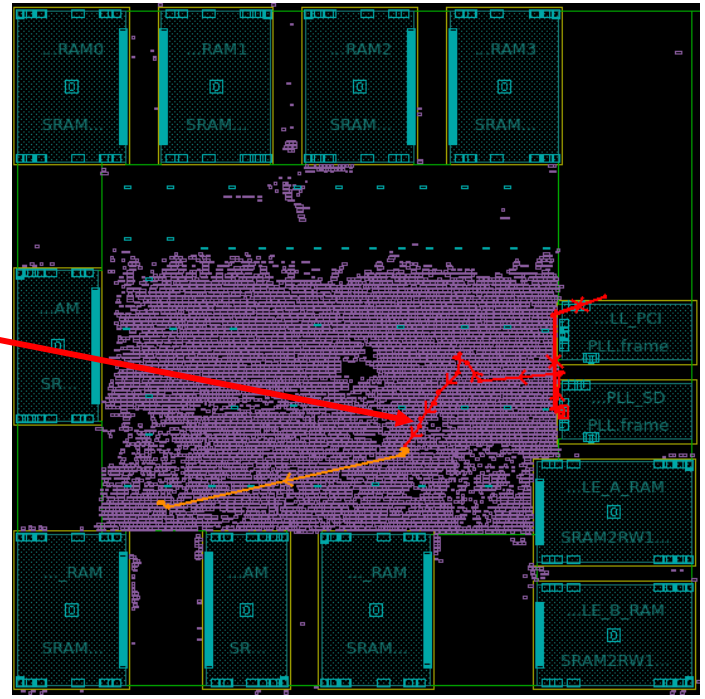
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CTS : Use ICC2 (IC Compiler II tool) Imported_path_pins Visual Mode

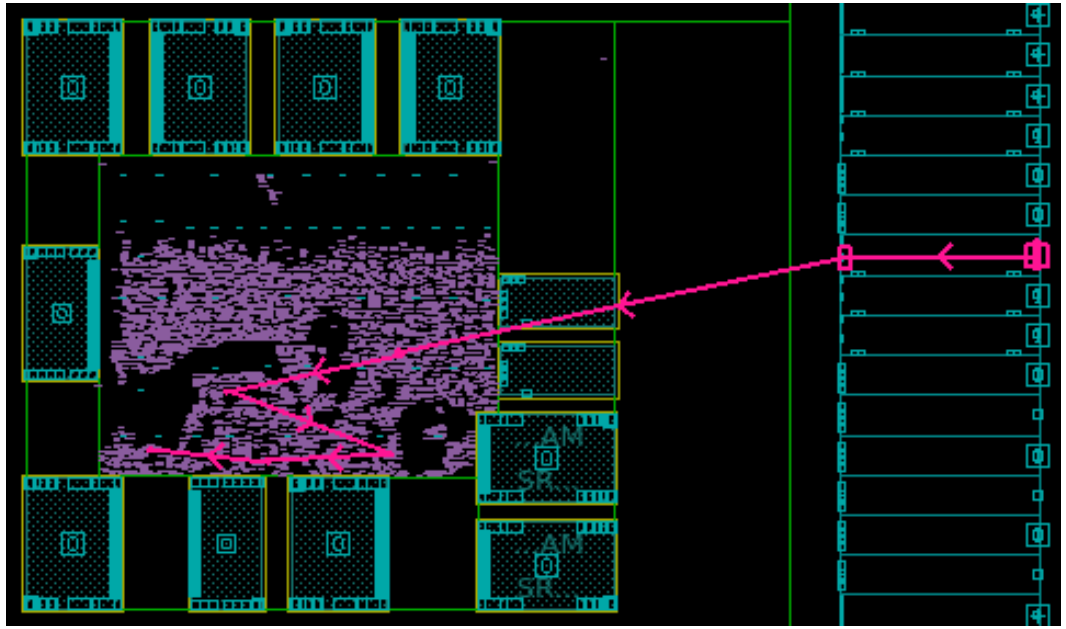
- **Analyze Timing Violation using Imported_path_pins Visual Mode**
  - Analyze two path (DAT, DRT)

Insert Buffer or inverter pairs at Data Path

< DAT >



< DRT >



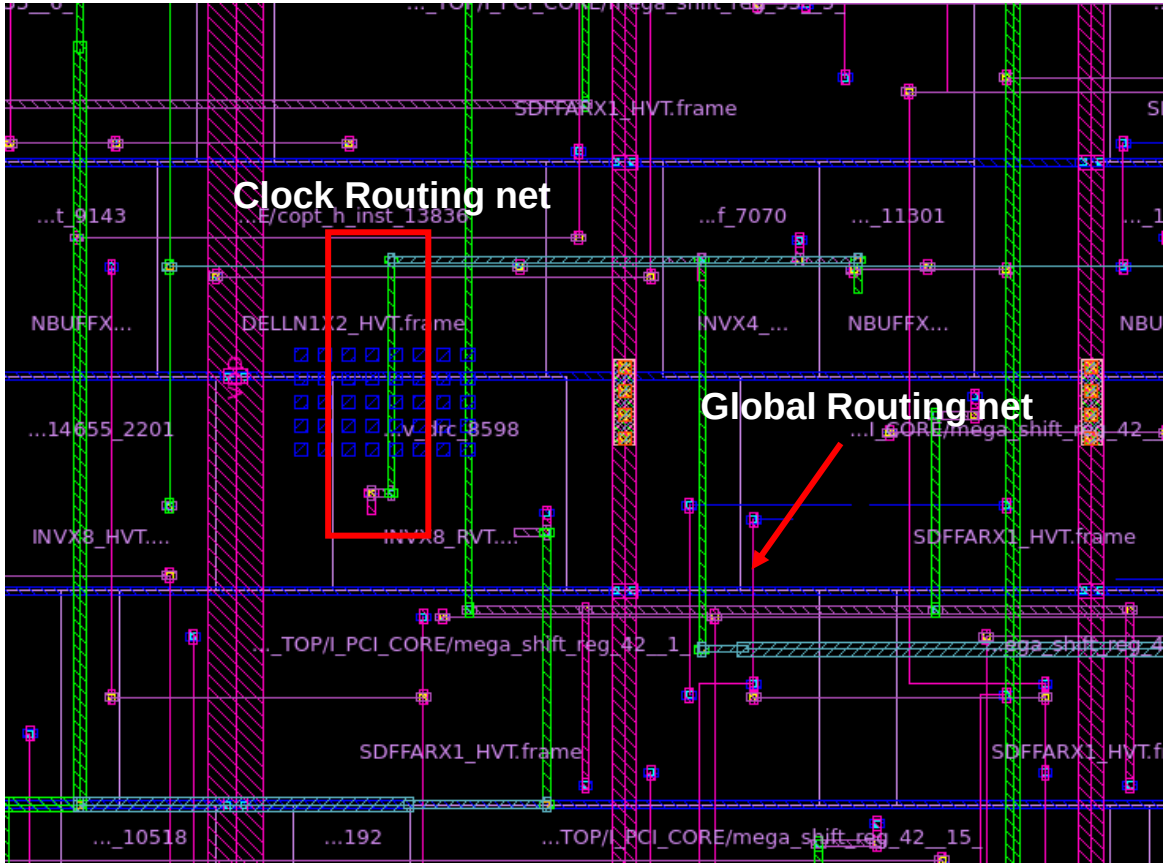
- ❖ **Setup Slack = DRT - DAT**
- ❖ **Hold Slack = DAT - DRT**

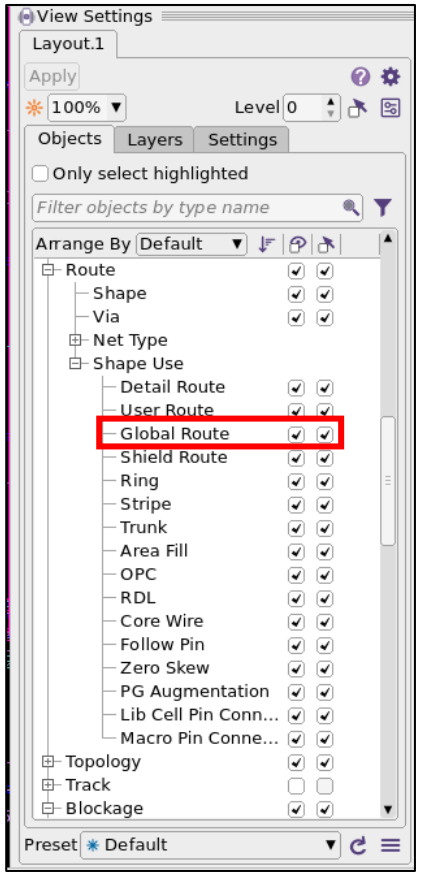
- **This violation occurred because DRT Path was longer than DAT Path**
- ***You can fix this violation (Cell sizing, Increase the delay in the data path, insert buffer or inverter pairs) in ECO step after STA***

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PSYN : Use ICC2 (IC Compiler II tool) GR (Global Route)

- **Clock Routing & Global Routing**
  - **Check the detail routing of the clock trees & Global Routing**

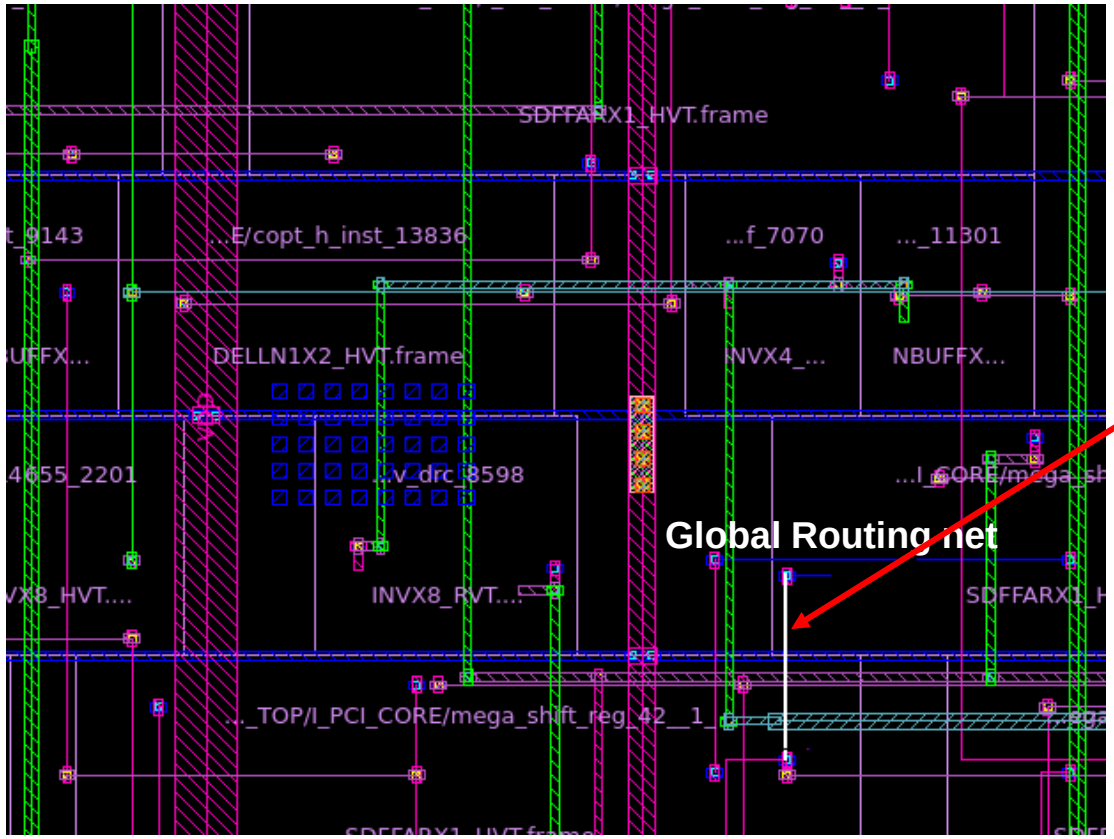


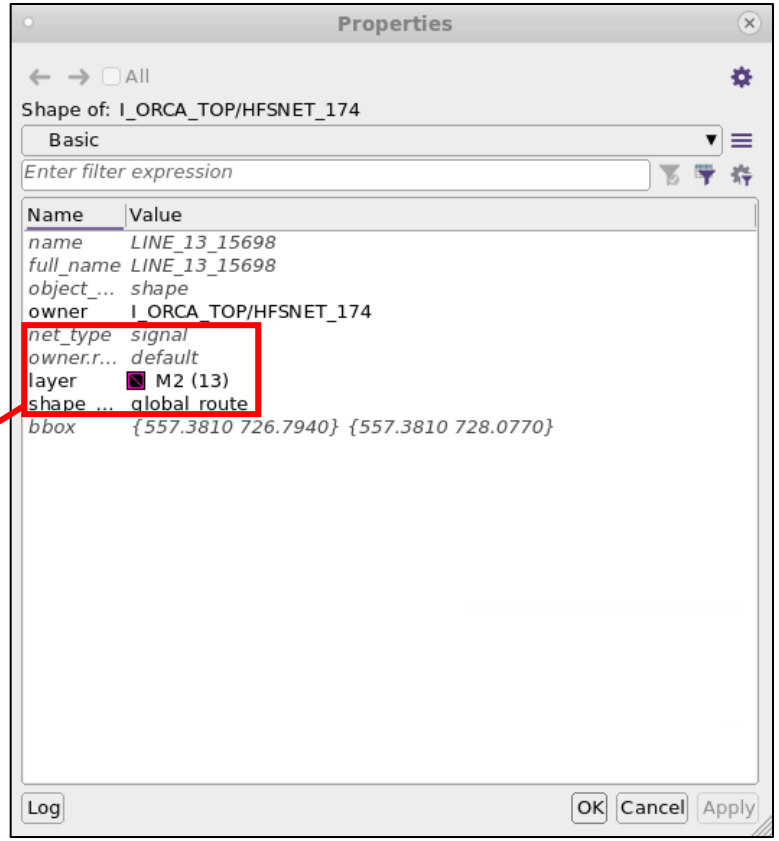


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PSYN : Use ICC2 (IC Compiler II tool) GR (Global Route)

- **Global Routing**
  - **Check the global routing of the signal nets**





- **Open Properties and check (owner.routing_rule, shape_use, net_type)**

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WEEK 11 : Exercise for CTS & PSYN

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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **In the "Exercise for placement", Run the CTS & PSYN under several case, and check the results"**

  - **Using script & refine_opt.nlib**

    placement script - ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/script/*

    Refine_opt.nlib - ${WORK_DIR}/2_BACKEND/1_pnr/04_placement/nlib/ORCA_refine_opt.nlib

    Refine_opt.nlib - ${WORK_DIR}/2_BACKEND/1_pnr/04_placement/nlib/ORCA_refine_opt_homework.nlib

  - **Guidelines**

  - Exercise 1 – Compare Two Timing Path in CTS & PSYN
  - Exercise 2 – Fix max_transition violation at CTS Step (Manual)
  - Exercise 3 – Analyze the results between ccd and non-ccd flow
  - Exercise 4 – run CTS & PSYN with ORCA_refine_opt_homework.nlib
  - Exercise 5 – Compare the results between ORCA_psyn_homework.nlib and ORCA_psyn.nlib

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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 1 – Compare Two Timing Path in CTS & PSYN**

  - **Analyze violated paths in CTS first**

    Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts/

  - **Timing path(VIOLATED) to compare**

```
Start_point - I_ORCA_TOP/I_PCI_CORE/d_out_p_bus_reg_0_
End_point - I_ORCA_TOP/I_PCI_CORE/pad_en_reg
```

  - **Use report_timing command to compare two paths (use –from –to option) after PSYN**

```
report_timing -scenarios func.ss_125c \
              -significant_digits 3 -delay_type min \
              -input_pins -physical -attributes -nosplit \
              -derate -crosstalk_delta -path_type full_clock \
              -from $StartPoint -to $EndPoint \
             > cts_timing.rpt
```

  - **Compare Timing Path use "imported path pins" in ICC2 (use ./cts_timing.rpt)**
  - **Analyze how the violated path changed after PSYN**
  - **IS this Path Optimized in PSYN ?**

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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 2 – Fix max_transition violation at CTS Step (Manual)**

  - **Analyze violated paths in CTS & PSYN first**

    Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts/cts.max_tran.rpt

  - **Timing path(VIOLATED) to compare**

```
                          Required        Actual
Net                      Transition     Transition        Slack  Violation
-------------------------------------------------------------------------
net_sdram_DQ_out[0]            0.19           0.26        -0.07  (VIOLATED)
  PIN : I_ORCA_TOP/I_SDRAM_IF/genblk1_0__sd_mux_dq_out/Y
                                 0.19           0.26        -0.07  (VIOLATED)
```

  - **Insert buffer in front of *"SDRAM_DQ_Bus_0__sdram_DQ_iopad/DIN"* pin**

Pin : SDRAM_DQ_Bus_0__sdram_DQ_iopad/DIN

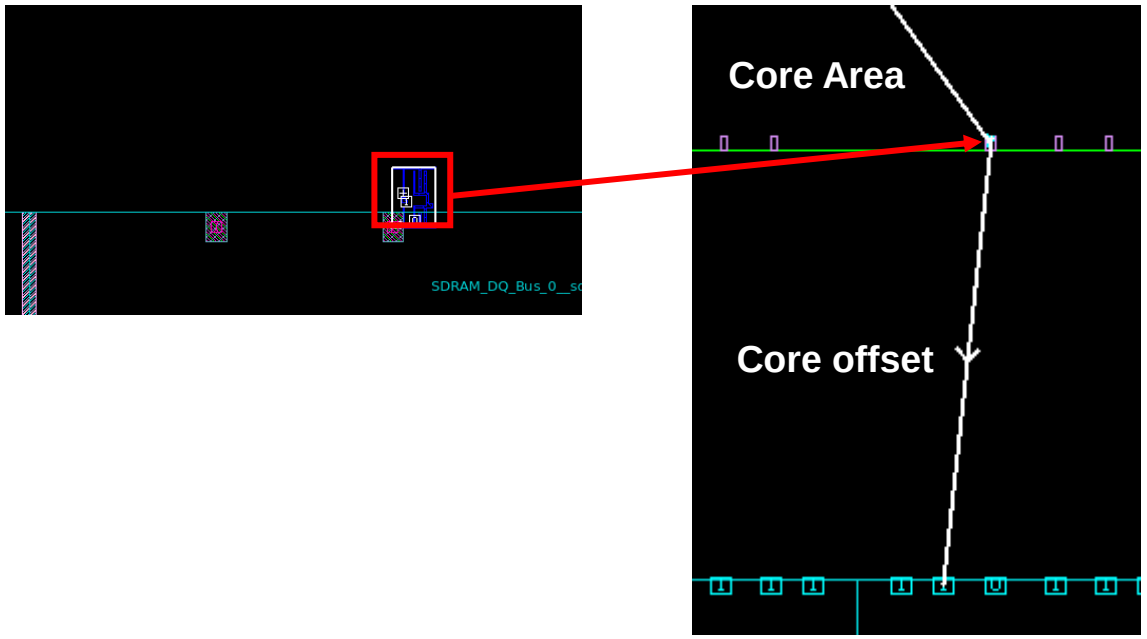
Buffer : saed32rvt/NBUFFX2_RVT

```
add_buffer $pin -lib_cell $lib_cell -new_cell_names Max_Trn_fix_buf
```


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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 2 – Fix max_transition violation at CTS Step (Manual)**
  - **Move buffer in Core Area**



  - **Report max_transition and compare with "cts.max_tran.rpt"**

```
report_constraints -max_transition -all_violators -corners * > homework_Max_Trn.rpt
```


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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 3 – Analyze the results between ccd and non-ccd flow**

  - **Run CCD Flow & report CTS & PSYN**

    Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts_ccd or psyn_ccd/*

  - **Compare Timing reports non-ccd and ccd flow**

    non-CCD Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts or psyn/*

    CCD Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts_ccd or psyn_ccd/*

  - **Is CCD Flow is better than non-CCD Flow in this design?**

    ❖ CCD Flow does not necessarily have good results

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CTS & PSYN : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 4 – run CTS & PSYN with ORCA_refine_opt_homework.nlib**

  - **Run CTS & PSYN with ORCA_refine_opt_homework.nlib**

    Refine_opt nlib path **:** ${WORK_DIR}/2_BACKEND/1_pnr/04_placement/nlib/ORCA_refine_opt_homework.nlib

    CTS & PSYN nlib path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/nlib/*

    Save nlib name **:** ORCA_cts_homework.nlib & ORCA_psyn_homework.nlib

- **Exercise 5 – Compare the results between ORCA_psyn_homework.nlib and ORCA_psyn.nlib**

  - **Compare reports after CTS & PSYN**

    Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/reports/cts or psyn/*

  ❖ *Please Backup the report files, report during LAB before proceeding with Homeworks separately*

  - **Which floorplan has better results?**

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Thank You

